Foundations of Sensitivity Analysis

From Local Sensitivity to Global Uncertainty

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Preface

This book develops sensitivity analysis (SA) as a mathematical discipline, not merely as a collection of computational recipes. SA asks the foundational question of quantitative modeling: which parameters most strongly determine the model's predictions?

Four ideas run through the book:

1. The sensitivity index is a normalized derivative.
2. The adjoint is the transpose, in whatever space you work.
3. Local SA is a linearization, with all the limitations that implies.
4. A sensitivity index describes a model, not the world.

Students who understand these four ideas (not just their definitions but their implications and connections) will be equipped to design, implement, and critique SA studies in any application domain.

The SIR epidemic model serves as the running example throughout. Most methods are demonstrated on this one model, so students can track increasing mathematical sophistication against a fixed biological context. All computational examples use MATLAB; each chapter after the first includes a structured laboratory, and most include a reusable code template. Companion Colab notebooks, one per chapter, recompute examples from the text in Python and assert selected results against fresh computation. The MATLAB reference library (eighteen functions across four modules, with an automated test suite) is available alongside them. A solutions manual covers every exercise and is written for self-study as much as for use as an instructor's key. The notebooks, the library, and the manual are hosted at:

`https://github.com/machyman/arriola2026sa`

The adjoint method (Chapters 5–7) is the conceptual centerpiece of the book. It is the same idea encountered three times in three different mathematical spaces: the matrix transpose in linear algebra, the reversed computation graph in algorithmic differentiation, and a backward ODE in dynamical systems. Chapter 8, *Caveat Emptor*, asks when sensitivity anal-

ysis can mislead, and is the chapter students most frequently cite as the one that changed how they think about mathematical models.

Relationship to existing literature. Cacuci (1981) established the adjoint formalism for nonlinear systems; Giles and Pierce (2000) developed it for aeronautical design; Cao et al. (2002) gave the computational treatment for ODE systems; and Griewank and Walther (2008) treat the computation-graph view comprehensively. The present book differs from each of these in its pedagogical aim: it is one of the few texts, and to our knowledge the first at this level, to introduce all three forms of the adjoint (matrix, graph, and ODE) as instances of a single unifying idea, and that teaches students when *not* to trust the results they compute.

The last of those is where this book departs furthest from its neighbors. Chapter 9 closes with a short protocol for auditing a sensitivity analysis someone else performed. Six questions establish which decision the result informs, whether the index reported answers that question, and whether the method's assumptions were met. We know of no comparable treatment in the existing texts, and it is the part of the book we would point a referee to first.

The closest neighbor is Smith's *Uncertainty Quantification* (2024), which SIAM offers as a one-semester sensitivity analysis course text, and the overlap is larger than the titles suggest. The difference is level and route rather than territory. Smith assumes a graduate background and reaches sensitivity analysis through the full uncertainty-quantification apparatus. This book is pitched at a more elementary audience: it assumes calculus, linear algebra, and ODEs, and reaches sensitivity analysis through one worked epidemic model. It can be read first, or in parallel with the sensitivity analysis chapters there.

A note on register. Results are developed in the running text rather than stated as numbered theorems. The book contains two numbered theorems and one numbered proposition, and the sparseness is deliberate: at this level the derivation is what carries the understanding, and a formal statement placed beside it tends to be read instead of the argument rather than alongside it. Where a result needs to be quotable it is displayed and labeled.

To the Instructor

The book supports a one-semester course for advanced undergraduates and beginning graduate students. Students need calculus, linear algebra, and ODEs; no measure theory or functional analysis is assumed. A minimal one-semester course can omit the computational AD material in Chapter 7

and Appendices A to E without loss of continuity. Section 8.3 uses the singular values defined in Appendix C.2, so a course that keeps Chapter 8 should keep that section. Appendix F is the probability that Chapter 9 assumes, and a course that reaches Chapter 9 should keep it. Chapters 1–4 and 8–9 form a self-contained global-SA course for students who will not use the adjoint in their research. Exercises tagged [Bloom L5] and [L6] are designed for semester projects and take-home examinations; those tagged [L2]–[L4] are suitable for weekly problem sets. Every chapter but the first closes with a Computational Laboratory. Chapter 1 has none by design: it orients the reader before any method has been introduced, so there is nothing yet to compute. Chapter 9 is twice the length of a typical chapter and is best given two weeks; the others take one.

One further word on assessment. Students now have capable computational assistants, and the exercises in this book are not equally exposed to them. An assistant will produce a competent answer to most [L2] and [L3] exercises, which are recall and routine application, so those are best used for practice rather than for grading, or set under supervision. The [L5] and [L6] project exercises are far less exposed, because they require judgments about a specific model that an assistant cannot make on the student's behalf. The audit exercises introduced in this edition, marked as such in their statements, are deliberately resistant for a different reason: each presents a plausible finished analysis and asks what is wrong with it. Producing a confident answer is easy, and recognizing that a confident answer addresses the wrong question is the skill being assessed. Worked solutions to every exercise are public, in the solutions manual posted with the notebooks, so an exercise used for grading is best varied from its printed form.

Prerequisites. Students should have completed: (1) calculus through multivariable (partial derivatives, chain rule, change of variables); (2) linear algebra through eigenvalues and the matrix transpose; (3) a first course in ordinary differential equations (existence and uniqueness, systems of ODEs, basic phase-plane analysis); and (4) introductory MATLAB or a comparable computing environment (the first laboratory session introduces every MATLAB construct the book uses, so prior MATLAB experience is helpful but not required). No course in probability is assumed. The probability the book uses (expectation, variance, the uniform distribution, conditional expectation and correlation) is collected in Appendix F, which a reader new to it should work through before Chapter 9. No measure theory or functional analysis is assumed.

For students. You will almost certainly have a computational assistant available while you work through this book, and you should use it. It will differentiate an expression, set up a sampling scheme, or write a Sobol

estimator faster and more reliably than you can, and there is nothing to be gained by pretending otherwise.

What it cannot do is tell you which sensitivity index answers your question. That choice depends on what you intend to do with the answer, on which simplifications your model already contains, and on whether the assumptions behind a method hold for the system in front of you. None of that is visible in the expression you hand over. An assistant asked which parameter matters most will return a ranking, and it will return one whether or not the question was well posed. Table 9.4 in Chapter 9 catalogs the answers that follow from asking the wrong version of that question, and it was written from the published literature rather than from anything a machine produced. The errors are ours first.

The book therefore asks you to develop a habit that is useful well beyond this subject: treat any result you did not derive yourself as provisional until you have checked what question it answers and what it assumes. Section 9.10 sets out how to do that, and Section 1.5 gives a first form you can use from the start. Several exercises give you finished analyses to audit rather than problems to solve.

Each chapter and appendix contains short self-checks. Their answers are collected at the end of the chapter or appendix, so you can commit to an answer of your own before you read ours.

In memory of Leon M. Arriola (1957–2026).

Leon Arriola and I began working on sensitivity analysis together more than twenty-five years ago. What started as a shared mathematical project became, over time, one of the sustaining collaborations of my life. Leon brought deep insight, patience, humor, and generosity to the work. He cared deeply about helping others learn, and every page of this book reflects his commitment to making hard ideas genuinely accessible.

Leon passed away in April 2026, before he could see the manuscript reach its final form. Completing this book, and offering it to the mathematical community in his name, is one way I know to carry forward what he taught me, and what he gave so freely to his students and colleagues over a lifetime of teaching.

James M. Hyman, Tulane University
New Orleans, Louisiana
April 2026

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Declaration on the use of AI tools. Generative AI tools were used in preparing this book: in drafting and revising narrative text, in writing and testing the companion code, in checking algebraic and numerical results against independent computation, and in literature search. This assistance was subject to the following verification. The companion notebooks, which readers can run for themselves, recompute examples from each chapter and assert values against fresh computation rather than merely printing them. A ledger checked against the source before each release accounts for every decimal value and every computed count that the chapters and appendices quote: each is reproduced by a notebook check, verified as data the text supplies or as printed arithmetic, or classified with its reason recorded. Values quoted from a figure are reproduced instead by the generator of that figure. Every figure generated from computation is produced by a generator included in the source distribution, and those generators are byte-reproducible. Every bibliographic entry has been checked against a primary source, and the checks are recorded in the `annote` field of each entry in the `.bib` file; that process corrected four incorrect DOIs and errors of volume, number, journal or author list in thirteen further entries. This work was carried out after Leon Arriola's death, and James Hyman alone is accountable for it. The authors assume responsibility for all content.

The companion code was developed with support from the Tulane University Department of Mathematics.

Dedication.

To Leon, who taught us to ask which parameters matter
— and to our students, past and future,
who will carry the question forward.

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Notation and Conventions

This book uses one running model and a compact, consistent notation. The tables below collect the symbols that recur across chapters, so a reader who meets a symbol mid-derivation can find its meaning without hunting for its first appearance. A symbol defined and used only within a single example is not listed here.

Three typographic conventions hold throughout. Vectors are set with an arrow, $\vec{u}$; matrices are uppercase upright or italic by context, A, J_F; and scalars are italic, c, β. A subscript on a rate or time constant names the compartment or transition it governs (γ_R, τ_m), never a running index. A sensitivity index is written $\mathcal{S}$ with the parameter as subscript and the response as superscript: $\mathcal{S}_\beta^J$ is the sensitivity of the response J to the parameter β.

The objects the framework is stated in terms of recur throughout and are collected here.

Symbol	**Meaning**
$\vec{u}(t)$	the model state vector; for the SIR model, (S, I, R)
p	a generic parameter (parameter of interest, POI)
ω_{ij}	coupling coefficient, $\Delta p_j/p_j \approx \omega_{ij}\,\Delta p_i/p_i$ (Chapter 3)
q	a generic response (quantity of interest, QOI)
J	a named response, often a functional such as $\int_0^T I\,dt$
Y	the response in the global-SA chapter
Q_i, q_i	input i as a random variable, and a value it takes (Chapters 8 and 9)
T	terminal time of the adjoint / optimal-control problem
A	a generic matrix (eigenvalue and SVD sections)
D, D_i	total and partitioned output variance in the Sobol decomposition
$\mathbb{E}[\cdot]$, $\mathrm{Var}(\cdot)$	expectation and variance (Appendix F)
$\mathbb{E}[Y \mid Q_i]$	conditional expectation of Y with Q_i held fixed
Cov, ρ	covariance and correlation coefficient
$\mathcal{U}(a, b)$	uniform distribution on $[a, b]$

The sensitivity index

The central object is the normalized (relative) sensitivity index. For a scalar response q and a parameter p,

$$\mathcal{S}_p^q \;=\; \frac{p}{q}\,\frac{\partial q}{\partial p},$$

the fractional change in q per fractional change in p. It is dimensionless, and $\mathcal{S}_p^q = 1$ means unit elasticity: a one-percent change in p produces a one-percent change in q. The book writes *parameter of interest* (POI) for the input under study and *quantity of interest* (QOI) for the response.

Symbol	Meaning
$\mathcal{S}_p^q$	normalized sensitivity index of response q to parameter p
S_i	first-order Sobol index of parameter i
S_{T_i}	total-effect Sobol index of parameter i
S_i^σ	σ-normalized (uncertainty-weighted) index of parameter i
μ_i^*	Morris mean of absolute elementary effects for parameter i
δ_i	Borgonovo (moment-independent) delta index
η^2	correlation ratio / first-order effect measure

The distinctions among S_i, S_{T_i}, S_p^σ, and μ_i^*, which question each answers, and which error each guards against, are set out in Table 9.4 in Chapter 9. That table, not this one, is the place to resolve which index a given task requires.

The adjoint and operator symbols

Symbol	Meaning
$\vec{\lambda}$	adjoint variable (costate); components $\lambda_S, \lambda_I, \lambda_R$
$\vec{c}$	weights of a linear response $J = \vec{c}^{\top}\vec{u}$ (Chapters 5 and 6); always with the arrow, which separates it from the contact rate c
J_F	Jacobian of the model right-hand side F
F_p	partial derivative of F with respect to a parameter p
g_u	partial derivative of the integrand g with respect to the state
$L,\ L^\dagger$	a linear operator and its adjoint
κ	condition number ($\kappa_2 = \sigma_1/\sigma_n$)
$\Phi(t)$	fundamental matrix (Appendix D)
Φ_j	normalized Shapley effect of parameter j (Chapter 9)
N	sample size, as in the Saltelli cost $N(m+2)$ (Chapter 9)

The SIR model and its parameters

Most methods in the book are demonstrated on one susceptible–infectious–recovered model. Its parameters are the following.

Symbol	Meaning	Nominal
S, I, R	susceptible, infectious, recovered counts	–
N	total population, $S + I + R = N$	–
c	contacts per individual per day	5
β	transmission probability per contact	0.06
τ_R	mean infectious period (days)	7
γ_R	recovery rate, $\gamma_R = 1/\tau_R$	–
τ_m	mean time in the population (days)	10,000
ν_m	demographic rate, $\nu_m = 1/\tau_m$	–
$\mathcal{R}_0$	basic reproduction number, $c\beta\tau_R$	2.1
$\lambda(t)$	force of infection, $c\beta I(t)/N$	–

A note on λ

One Greek letter carries more than one meaning in this book, by established convention in the fields it borrows from, and the reader who knows which is which will not be misled.

In the epidemic model, $\lambda(t) = c\beta I(t)/N$ is the *force of infection*, the instantaneous rate at which a susceptible individual becomes infected. This is the universal notation of mathematical epidemiology, and the book keeps it. It is written as λ only in the modeling chapters (Chapters 2 and 3), where no adjoint is present.

In the sensitivity chapters, $\vec{\lambda}$ is the *adjoint variable*, or costate. It is a vector, carrying one component per state equation (λ_S, λ_I, λ_R), and the book writes it with an arrow wherever it is a vector and with a subscript for a named component. Chapter 5 first uses it at the nodes of a computation graph, as $\lambda_i = \partial J/\partial v_i$ (Section 5.6), and previews the adjoint ODE for $\vec{\lambda}(t)$. Chapter 6 introduces that ODE first for a single-state equation, where the adjoint is genuinely a scalar $\lambda(t)$; from the vector case onward it carries the arrow. The two meanings of λ never appear in the same equation. When an adjoint chapter needs the force of infection, it writes the product $c\beta I(t)/N$ in full rather than $\lambda(t)$.

Two further uses are confined to the appendices and are standard there. In Appendix C, λ_k is an *eigenvalue*. In Appendix E, $\vec{\lambda}$ is the vector of *Karush–Kuhn–Tucker multipliers* of a constrained optimization problem. That the multiplier and the adjoint share a symbol is not a coincidence: Appendix E shows they are the same object, the sensitivity of an optimum

to a constraint, seen from two directions.

A note on σ

Like λ, the letter σ carries more than one standard meaning, and position and subscript make clear which is intended. As a *standard deviation*, σ_i is the uncertainty in parameter i and appears in the σ-normalized index S_i^{σ} (Chapter 9). As *biting rates*, σ_v and σ_h are model parameters in the vector-borne example (Chapter 3). As *singular values*, $\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_n$ are the diagonal entries of the SVD and define the condition number $\kappa_2 = \sigma_1/\sigma_n$ (Chapter 8, Appendix C). And $\sigma(\cdot)$ with a function argument is the *sigmoid* used to restore differentiability at a switch (Chapter 7). These uses do not overlap within any one expression.

A note on N

The capital N is the third symbol in this book to carry two standard meanings, and the two are separated by chapter rather than by decoration. In the modeling chapters, N is the *total population*, $S + I + R = N$, which is the universal notation of mathematical epidemiology and appears in the force of infection $c\beta I(t)/N$.

In Chapter 9, N is the *sample size*: the number of parameter vectors drawn from the joint distribution, and the base sample size in the Saltelli cost $N(m + 2)$. This is equally standard in the sensitivity literature, where $N(m+2)$ is the customary way to state the price of a Sobol' analysis, and the book keeps it so that a reader moving between this text and that literature meets the same expression.

The two never appear together. Chapter 9 carries no population N: its epidemic examples are written in terms of $\mathcal{R}_0 = c\beta/\gamma_R$ and of sample estimates, not of the force of infection. A reader who keeps the chapter in mind will not be misled, and a subscripted N always denotes a population subgroup, as in the vector and host populations N_v and N_h of the malaria example (Chapter 3).

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Part I

The Sensitivity Question

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Chapter 1
Introduction and Overview

Sensitivity analysis begins when a model is forced to say what matters.

An infectious disease has just entered a new country. The public health response team has built a compartmental model with fourteen parameters: transmission rates, recovery rates, contact patterns, immunity duration, disease-induced mortality, and more. The team has funding to deploy exactly two interventions and time to measure exactly three parameters with high precision before a policy decision must be made.

The question due by tomorrow morning is this: *which three parameters are worth measuring, and which two interventions will most effectively reduce the spread?*

This is the problem that sensitivity analysis is designed to answer. Before the mathematics, we need a language for asking the question precisely. This chapter builds that language and does not yet answer the question. Chapter 2 gives the first tool for ranking the parameters, and Exercise 1.4 at the end of this chapter returns to the question for a dengue model.

1.1 ▪ The Central Question

Every mathematical model takes inputs and produces outputs. The inputs include parameters whose exact values may be uncertain or whose values might be changed through intervention. The outputs include whatever the model is designed to predict. Sensitivity analysis asks: *which inputs have the most effect on the outputs we care about?*

To answer this precisely, we adopt two terms used throughout the book.

A *Parameter of Interest* (POI) is any input to the model whose variation is worth studying. In practice, a parameter becomes a POI for one of three reasons, and the reason matters for how you interpret the sensitivity index.

1. *Epistemic uncertainty* (uncertainty from lack of knowledge). The parameter is fixed by nature but imperfectly known: different studies report different estimates, the data are sparse, or the measurement process is imprecise. Epistemic uncertainty is *reducible*: better data, improved experiments, or longer observation windows can sharpen the estimate. Sensitivity analysis identifies which epistemically uncertain parameters most affect the output, directing limited data-collection resources toward the measurements that would reduce output uncertainty the most. In the disease example, the transmission probability per contact is epistemically uncertain: studies estimate it from contact-tracing data, but the data are sparse, and more tracing would narrow the estimate.

2. *Aleatory uncertainty* (inherent stochastic variability). The parameter genuinely varies from individual to individual, place to place, or time to time, and this variation is *irreducible*: it persists even with unlimited data. More observations characterize the distribution more precisely but do not eliminate the randomness. In the disease example, the contact rate varies stochastically across the population: individuals differ in how many contacts they make per day. Sensitivity analysis with aleatory uncertainty typically asks how much of the output variance is attributable to each stochastic source, motivating the global SA methods of Chapter 9. Appendix F reviews variance and the other probability those methods use.

3. *Control variable* (a lever the investigator can move). The parameter is known and could, in principle, be changed by a policy decision, intervention, or design choice. Sensitivity analysis in this context answers the question: which intervention has the most leverage over the outcome I care about? In the disease example, the contact rate can be reduced by social distancing policies, and the transmission probability can be reduced by mask requirements or vaccination. The sensitivity index tells the policy maker which reduction, contacts or transmission, produces the larger reduction in the basic reproduction number $\mathcal{R}_0$ (see the Glossary), or else, as happens for $\mathcal{R}_0$ in Chapter 2, that the two are equally leveraged and the choice has to be made on feasibility and cost instead.

A test sorts the first two roles. Ask whether more data would shrink the spread in the parameter's value. If it would, the uncertainty is epistemic; if the spread would remain however much data arrive, it is aleatory.

Many parameters fall into more than one category simultaneously: the contact rate is both uncertain (its current value is imprecisely known) and

controllable (it can be targeted by policy). In practice, the interpretation of a sensitivity index should always be read in light of which role the POI is playing. A large sensitivity index for an epistemically uncertain parameter is a call to collect better data; for an aleatory parameter, it is a call to characterize the distribution more carefully; for a control variable, it is a call to prioritize that intervention.

In the disease example above, the transmission probability per contact is primarily a POI because it is epistemically uncertain (different studies report different values) and because it can be targeted by an intervention (vaccines reduce it). The contact rate is a POI for three reasons: individuals differ in how many contacts they make (aleatory uncertainty), its current mean is imprecisely known (epistemic uncertainty), and it can be reduced through social distancing policies (control variable).

A *Quantity of Interest* (QOI) is any model output that the investigator cares about predicting or controlling. The peak number of infectious individuals, the total number of deaths over a six-month period, and the basic reproduction number $\mathcal{R}_0$ are all plausible QOIs for an epidemic model. The choice of QOI is not neutral: different choices lead to different rankings of parameter importance. Figure 4.1 in Chapter 4 shows one case: the ranking of the POIs changes with the time at which $I(t)$ is read.

The relationship between POIs and QOIs is the subject of this book. Specifically, we want to know how much each POI affects each QOI, so that we can rank parameters by importance, design interventions efficiently, and allocate measurement resources wisely.

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For Readers with a Statistics Background

The POI/QOI vocabulary will feel familiar: a POI resembles a predictor variable and a QOI resembles a response variable from regression. The sensitivity index introduced in Chapter 2 is the same quantity known as an *elasticity* in econometrics (the percentage change in output per percentage change in input); a dimensionless measure of proportional change. In Chapter 8 the sigma-normalized sensitivity index will turn out to be identical to the standardized regression coefficient reported by most statistical software. In Chapter 9 the first-order Sobol index is the proportion of output variance explained by one parameter acting alone, the same quantity as η^2 (eta-squared) in a classical ANOVA (the ratio of between-group variance to total variance), and the same idea behind permutation feature importance and Shapley-value attribution (a game-theoretic method that allocates a model's output among its inputs) in machine learning.

For readers from machine learning: Forward sensitivity analysis is the local, interpretable cousin of computing a gradient; it measures how the output changes when one input is perturbed. *Forward-mode* automatic differentiation (AD) does this one parameter at a time, propagating dual numbers through the computation; it is the analogue of the forward sensitivity equations (FSE) of Chapter 4. `torch.autograd` and similar frameworks, however, use *reverse-mode* AD (backpropagation) by default, which computes gradients with respect to all inputs in a single backward pass. This is the adjoint method of Chapters 5–7, applied to a computation graph rather than an ODE. The adjoint method (Chapters 5–7) is, mathematically, the same operation as backpropagation in a neural network: the chain rule applied in reverse order, computing the sensitivity of one output with respect to all inputs in a single backward pass. If you have trained a neural network, you have already used the adjoint method under a different name.
What this book adds beyond regression and ML is threefold: how to compute these quantities for models defined by equations rather than data; when the linearization underlying all of these quantities breaks down and global variance-based methods are required; and the mathematical unification showing that gradient descent, backpropagation, and sensitivity analysis are facets of the same idea.

One of the most persistent misconceptions in mathematical modeling is the assumption that all parameters matter equally until proven otherwise.

They do not. In virtually every model studied in this book, a small number of POIs account for most of the variation in the QOIs, while the remaining parameters have negligible effect. Identifying that small number is one of the primary goals of sensitivity analysis, and the answer is rarely obvious from the model equations alone.

Self-check 1.1. In the outbreak scenario above, classify each of the following as a POI, a QOI, or neither: the transmission probability per contact; the peak number of infectious individuals; the day the first case was reported; the mean number of days a patient is infectious.

Figure 1.1 illustrates the basic structure. The forward problem takes nominal POI values as input and produces the QOI as output. Sensitivity analysis adds one more step: it asks how the QOI changes as the POIs vary.

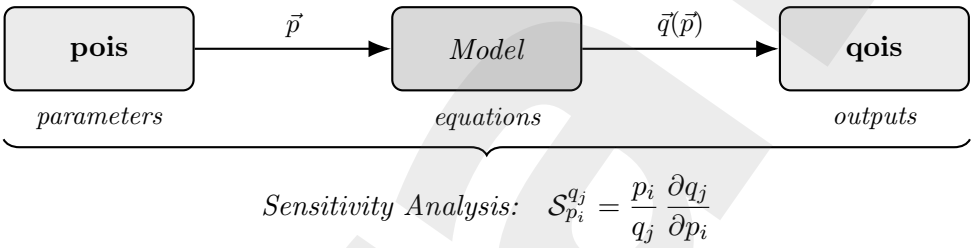


Figure 1.1. *The forward problem takes nominal values for the input* POIs *and produces the associated output* QOI. *Sensitivity analysis quantifies how changes in the* POIs *propagate to the* QOI *by computing the normalized partial derivative* $\mathcal{S}_{p_i}^{q_j}$ *for each* POI–QOI *pair.*

1.2 ▪ Two Directions: Forward and Adjoint Sensitivity

There are two ways to compute the same sensitivities. They return the same derivatives and differ in the end of the model they start from, and so in what they cost.

Figure 1.2 captures the distinction geometrically. In panel (a), a perturbation to one input POI, p_j, spreads forward through the model and changes the output QOI. This is *forward sensitivity analysis*. Panel (b) reverses the direction: it starts from one QOI and carries its derivative backward through the model to every POI at once. This is *adjoint sensitivity analysis*.

This book counts the cost of a method in solves, and the baseline solve at the nominal values always counts. Perturbing each of m POIs once then costs $m + 1$ solves. For one QOI the adjoint costs two: the baseline forward solve and one backward adjoint solve. For r QOIs it costs $r + 1$: the same forward solve plus one adjoint solve per QOI. Both directions produce the

same derivatives $\partial q_j/\partial p_i$, so the choice between them is a matter of cost. Forward SA is cheaper when there are fewer POIs than QOIs ($m < r$), and the adjoint when there are fewer QOIs ($r < m$).

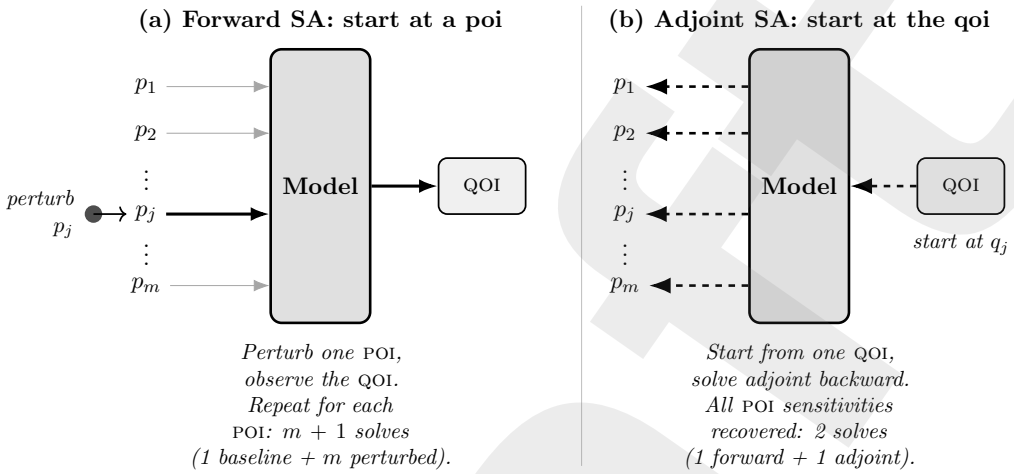


Figure 1.2. *(a) Forward sensitivity analysis starts at an input. A perturbation to one input* POI p_j *propagates forward through the model and produces a change in the output* QOI *. Because the model must be solved once at the baseline and once for each* POI*, forward sensitivity requires $m+1$ solves for m* POI*s. (b) Adjoint sensitivity analysis starts at the output and computes the same derivatives. The derivative of the output* QOI q_j *enters the model from the right and is propagated* backward *through it. A single adjoint solve simultaneously recovers the sensitivity of q_j with respect to every* POI $p_1, \ldots, p_m$*. The adjoint is developed in Chapters 5 and 6; for now, the key contrast is computational: forward SA costs $m+1$ solves, adjoint SA costs* 2 *for one* QOI*.*

The two directions are two computational strategies for one set of derivatives, not answers to two different questions. Forward SA builds the derivatives $\partial q_j/\partial p_i$ one POI at a time; the adjoint builds the same derivatives one QOI at a time. Neither says what caused an observed output. Both report how the model's output responds to its inputs.

1.2.1 ▪ Forward Sensitivity Analysis: Starting from an Input

Consider a disease model for an emerging outbreak. The transmission probability β is estimated from early case data, but the estimate carries significant uncertainty. A public health officer asks: *if our estimate of β is wrong by 10%, how much does our predicted peak infection count change?*

Forward SA handles it by tracing the effect of a change in an input POI (the transmission probability) forward to an output QOI (the peak infection count). The answer is a sensitivity index: a number that tells us, for a small percentage change in a POI, what percentage change to expect in the QOI.

Connection to gradient computation (for readers who know machine learning). Forward SA is the interpretable, application-oriented cousin of computing a gradient. When a machine-learning framework computes the derivative $\partial\mathcal{L}/\partial\theta$ of a training loss $\mathcal{L}$ with respect to one model weight θ using forward-mode automatic differentiation, or when you perturb one input and measure the resulting output change (an ablation study, in machine-learning terms), you are performing forward sensitivity analysis. The sensitivity index normalizes that gradient into a dimensionless, unit-free quantity that can be compared across parameters measured on different scales. Figure 1.3 contrasts the two directions on a single small computation graph: one forward pass per input, against one backward pass per output.

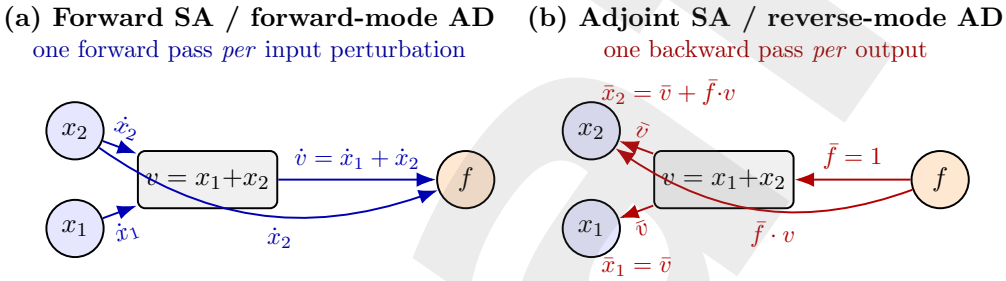


Figure 1.3. *Computation graph for $f(x_1, x_2) = (x_1 + x_2) \cdot x_2$, with intermediate node $v = x_1 + x_2$, illustrating the two modes of sensitivity analysis. A dot marks a derivative carried forward from the seeded input (a tangent); a bar marks the derivative of f with respect to a node, carried backward (an adjoint). Section 7.1 works both by hand. (a) Forward SA propagates tangent vectors $(\dot{x}_1, \dot{x}_2)$ from each input to the output; one forward pass per input (POI). Each pass returns $\dot{f} = \dot{v}\, x_2 + v\, \dot{x}_2 = \dot{x}_1 x_2 + \dot{x}_2(x_1 + 2x_2)$: the seed $(\dot{x}_1, \dot{x}_2) = (1, 0)$ gives $\partial f/\partial x_1 = x_2$, and the seed $(0, 1)$ gives $\partial f/\partial x_2 = x_1 + 2x_2$. This is identical to forward-mode automatic differentiation. (b) Adjoint SA propagates the adjoint $\bar{f} = 1$ backward from the output to all inputs simultaneously, one backward pass per output (QOI). Edge weights are local partial derivatives: $\bar{v} = \bar{f} \cdot (\partial f/\partial v) = x_2$; the direct $f \to x_2$ edge carries $\bar{f} \cdot v = x_1 + x_2$. Final gradients: $\bar{x}_1 = \bar{v} = x_2$; $\bar{x}_2 = \bar{v} + \bar{f} \cdot v = x_2 + (x_1 + x_2) = x_1 + 2x_2$. This is identical to backpropagation in a neural network. The cost advantage: forward mode costs m passes (one per input, each also evaluating f); adjoint mode costs two passes, one forward and one backward, regardless of the number of inputs.*

A sensitivity index of $+0.8$ would mean that a 10% increase in β produces approximately an 8% increase in the peak infection count. A sensitivity index of -1.0 for the recovery rate would mean that a 10% increase in the recovery rate produces a 10% decrease in the peak. The sign tells you the direction; the magnitude tells you the amplification. Chapter 2 develops this definition carefully and shows how to compute these indices for a wide range of models; indices of outputs that change in time, such as $I(t)$, follow in Chapter 4.

Self-check 1.2. A forward sensitivity analysis gives $S_{\tau_R}^{\mathcal{R}_0} = +1.0$ (the notation is defined formally in Chapter 2), where τ_R is the mean infectious period (in days) and $\mathcal{R}_0$ is the basic reproduction number. If the mean infectious period increases by 5%, approximately what happens to $\mathcal{R}_0$? Is this good news or bad news for disease control?

1.2.2 ▪ Adjoint Sensitivity Analysis: Starting from an Output

Now consider a different problem. City planners want to site a new industrial facility near a residential area. The facility emits pollutants, and regulators have set a standard: the average pollutant concentration at a specified monitoring station must not exceed a given threshold. A transport model predicts whether any candidate site meets this standard.

The planner needs the same kind of derivative as the public health officer, arranged differently. One output matters: the average concentration at the station. Many inputs feed it, among them wind speed, emission rate and atmospheric stability. The planner wants *the derivative of that one output with respect to every input.* These derivatives show which inputs the concentration is most sensitive to and, to first order, how far each can move before the threshold is crossed.

The adjoint computes all of them with one backward solve after the baseline run. It starts from the QOI and carries its derivative backward through the model to every POI together. Forward SA would return the same numbers, one POI at a time. For models with many POIs and only a few QOIs, the adjoint is therefore far cheaper: $r + 1$ solves against $m + 1$. Chapters 5 and 6 explain why and show how to exploit it.

Self-check 1.3. An environmental regulator's transport model has 40 weather inputs and one output: the pollutant concentration at a monitoring station, which exceeds the legal threshold. She wants the sensitivity of that concentration to every weather input. Which direction computes these derivatives with fewer solves? Do the derivatives tell her which assumption caused the exceedance?

1.3 ▪ Global Sensitivity: When Local Is Not Enough

Both forward and adjoint sensitivity analysis are *local*: they measure the effect of small perturbations to the POIs near a fixed nominal value. This is powerful, but it assumes that the model behaves approximately linearly in the vicinity of that nominal value.

Many real models are not linear, and many POIs carry substantial uncertainty that cannot be called "small." In these cases, local sensitivity indices

can be misleading. If the model is nonlinear in a POI, a linear prediction built from its local index can badly misjudge the effect of varying that POI across its full range of plausible values.

Example 1.1 (A linear prediction over a large range). Consider $q(p) = p^2$ and suppose the nominal value is $p^* = 0.1$. The local sensitivity index is $S_p^q = (p/q)\,(dq/dp) = (p/p^2)(2p) = 2$. But if p is uncertain and could range from 0.1 to 1.0, the QOI ranges from 0.01 to 1.0, a hundredfold change. The linear prediction from $p^* = 0.1$ is $q(1) \approx 0.01 + 0.2 \times 0.9 = 0.19$, short of the true value by a factor of more than five. The index itself is not at fault. It equals 2 at every p, and read as the slope of the log-log plot of q against p (2.6), it gives the hundredfold change exactly: $(1.0/0.1)^2 = 100$. What fails is the straight-line extrapolation from one point. When the index also changes across the range, no single local value describes the whole range. Global SA, introduced in Chapter 9, is designed to capture exactly this kind of large-range nonlinear behavior.

Chapter 9 introduces *global sensitivity analysis*, which asks how the output varies when the POIs are varied across their entire plausible ranges simultaneously. That joint variation over whole ranges, set against a local analysis at one nominal point, is what makes an analysis global. The other contrasts in this section, large changes against small ones and variances against derivatives, follow from it. The primary tool there, the Sobol index, measures how much of the total output variance is attributable to each POI. Global SA does not replace local SA; it complements it, answering questions that local SA cannot.

Readers from statistics and machine learning will recognize this question. The box in Section 1.1 names the measures from those fields that answer it, and Chapter 9 makes these connections precise and shows exactly when each method is appropriate.

1.4 ▪ Sensitivity Analysis and Uncertainty Quantification

Sensitivity analysis and uncertainty quantification are related but distinct enterprises.

Sensitivity analysis, as developed in this book, begins as a *deterministic* enterprise: we assign nominal values to the POIs and compute how the QOI changes when those values are perturbed. The POIs are treated as numbers with specific values, not as random variables. The sensitivity index is a derivative: a precise mathematical object that exists when the model is differentiable. This local deterministic form is the focus of Chapters 2–8. Chapter 9 extends sensitivity analysis to the global setting, where the POIs are treated as random variables drawn from specified distributions, and the

goal shifts from linearized local derivatives to variance-based attributions of output uncertainty.

Uncertainty quantification treats the POIs as random variables with associated probability distributions. The goal is to determine the distribution of the QOI given the distributions of the POIs. This is a statistical enterprise, and its primary tools (Monte Carlo methods, Bayesian inference, polynomial chaos) are developed in depth in Smith's *Uncertainty Quantification: Theory, Implementation, and Applications* [60], which complements the present book. Smith's Chapters 8 and 9 develop local and global sensitivity analysis, extending the material in this book with functional-analytic and measure-theoretic foundations. Smith's Chapter 7 treats parameter identifiability and influence using the sensitivity Jacobian introduced computationally in our Chapter 3 and developed analytically in Chapter 4, connecting it to the surrogate-model literature. The SA foundations developed here follow Arriola and Hyman [2, 3]. For the broader development of SA as a formal discipline, see Saltelli et al. [52], Razavi et al. [47], and the annotated historical timeline in Tarantola et al. [64].

The connection between SA and uncertainty quantification is real and important. Global SA, covered in Chapter 9, bridges the gap: Sobol indices are defined in terms of variances, which requires probabilistic inputs, but they can be estimated by methods that build directly on the ideas in this book.

1.5 ▪ Four Ideas That Run Through the Book

Every chapter of this book revisits one or more of four fundamental ideas. We name them here so that students can recognize them when they reappear, each time in a new setting.

Unifying Idea 1: The sensitivity index is a normalized derivative.

At its core, SA answers a single question: if this input changes by a small percentage, by what percentage does the output change? That ratio is the sensitivity index. It is not a mysterious new object: it is the derivative, normalized so that its magnitude is independent of units and scales.

This idea appears first in Chapter 2, where we define the sensitivity index formally. It reappears in Chapter 3 as a finite-difference approximation and in Chapters 4–6 as an analytic derivative. In Chapter 9 it appears inside the variance-based formulas, whenever the response is close to linear over the input range. Every local sensitivity index in this book is an instance of this one idea. Chapter 9 also uses global measures that are built from other quantities and contain no local index.

Unifying Idea 2: The adjoint is the transpose in whatever space

you are working in.

The adjoint method computes the same derivatives as forward SA, starting from the output instead of the input. For one QOI and hundreds of POIs it needs two solves instead of hundreds, and that saving rests on a single mathematical operation: transposition. For a linear system $A\vec{u} = \vec{b}$, the adjoint involves the transposed matrix A^T. For a computation graph (the structure underlying every computer program), the adjoint is the chain rule applied in reverse order, which is the operation machine learning calls backpropagation. For an ordinary differential equation, the adjoint is a second ODE running backward in time.

These three forms look different on the surface. They are the same idea.

Chapter 5 introduces the adjoint through the matrix transpose, which is the simplest and most familiar form, works a 2×2 example in Section 5.1, and turns to the computation graph in Section 5.6. Chapter 6 extends it to differential equations, and Chapter 7 puts the graph form to work in algorithmic differentiation. The Glossary entry *Adjoint operator* states the identity behind all three forms. At each transition, the book explicitly identifies the connection back to this unifying idea.

Unifying Idea 3: Local SA is a linearization.

Every sensitivity index in Chapters 2–8 is based on a derivative, and a derivative is a linear approximation to a generally nonlinear function. This approximation is valid when the perturbation is small and the model is smooth. It fails near bifurcation points, where the model changes qualitative behavior (Section 8.4.1 sorts the ways it fails there). It fails when the POI uncertainty is large. It fails when the model is strongly nonlinear in the POI of interest.

Chapter 8 (*Caveat Emptor*) documents the conditions under which local SA breaks down. Chapter 9 provides the global alternative. Both chapters are applications of this one principle: local SA is a linearization, and linearizations have limits.

Unifying Idea 4: A sensitivity index describes a model, not the world.

Every index in this book is computed from a model's equations. It reports how that model responds, and it is silent on whether the model's simplifications are faithful to the system the model stands for. Holding those two claims apart is the most easily overlooked discipline in applied SA, and the most consequential when a result informs a decision.

Box put the underlying point in a sentence: "Essentially, all models are wrong, but some are useful" [8]. The useful ones are wrong in ways that do not matter for the question being asked. Sensitivity analysis is how you find out which of the simplifications a model contains matter, which parts of the

simplification are load-bearing. Picasso said something close about his own trade in 1923: "Art is a lie that makes us realize truth, at least the truth that is given us to understand."[1] A model is a deliberate lie about a system, told so that something true about the system becomes visible. The indices in this book measure the lie: they report how the model, simplifications and all, responds to its inputs, not how the system does.

Two failure modes follow, and they point in opposite directions. A POI can dominate a model's output and matter little in the field, if the model's structure hands it more leverage than the system does. And a factor that is absent from the model returns a sensitivity of exactly zero, which is not evidence that the factor is unimportant, only that it was not modeled. A closed epidemic model with no births or deaths returns exactly zero sensitivity to the mean residence time in the population, because that parameter never appears on the right-hand side. Nothing whatever has been established about demography. The demographic model used from Chapter 4 onward, and in the Chapter 6 laboratory, does carry that parameter, and the contrast is worth holding onto. A mean residence time of 10^4 days set against a 90-day outbreak means almost nobody enters or leaves the population while the epidemic runs. The computed index of the total burden $J = \int_0^{90} I\,dt$ with respect to the mean residence time is -7.9×10^{-4}, three orders of magnitude below the indices of c, β and τ_R, which are of order one (Exercise 6.3 computes all four). Small for a reason and zero by construction look alike in a table, and only one of them says anything about demography.

This distinction is hardest to hold when the result was produced by someone, or something, other than you. In that case whoever computed the index did not make the modeling decisions, and so cannot tell you which of the model's simplifications are load-bearing for your question. What arrives is a number and a ranking, with the assumptions that generated them left behind. A borrowed result is therefore a claim about a model whose construction you may not have seen. Reading it as a claim about the world adds that gap to the one between the model and the world. Section 9.10 returns to this once the global methods are in place, since that is where the temptation is strongest.

Saltelli, Tarantola and Campolongo argued on these grounds that sensitivity analysis belongs inside the modeling cycle rather than appended to it [55]. The reminders placed through the chapters that follow are there to keep the distinction visible at the points where it is easiest to lose.

[1] "Picasso Speaks: A Statement by the Artist," *The Arts*, New York, May 1923, p. 315.

1.6 ▪ A Note on Parameterization

One practical decision that profoundly affects the usefulness of SA results deserves mention before Chapter 2.

The sensitivity index depends on how a POI is defined. In a compartmental epidemic model, the recovery process can be parameterized either by the *recovery rate* γ_R (units: per day) or by the *mean infectious period* $\tau_R = 1/\gamma_R$ (units: days). These two parameterizations are mathematically equivalent, each determining the other. But their sensitivity indices are not equal: they have the same magnitude and opposite signs.

Which one should you use? The answer is: whichever one you can discuss in ordinary language without confusion. People who work with epidemic models (clinicians, public health officials, and policy makers) think naturally in terms of days. A clinician understands immediately what it means to say "the mean infectious period is 7 days." They do not think in recovery rates. Reporting $\mathcal{S}_{\gamma_R}^{\mathcal{R}_0} = -1.0$ requires the reader to invert γ_R mentally before the result makes intuitive sense. Reporting $\mathcal{S}_{\tau_R}^{\mathcal{R}_0} = +1.0$ is immediately clear: a longer infectious period increases transmission.

Throughout this book, we choose POIs that can be understood in everyday terms: durations rather than rates, probabilities rather than their complements, counts rather than densities when the distinction matters. This is not just a matter of aesthetics. Sensitivity indices that require mental inversion to interpret are sensitivity indices that will be misread.

Chapter 2 formalizes this principle and shows how to convert between parameterizations. Readers who find that their model's natural parameters are rates will learn how to restate their results in terms of the corresponding durations before reporting them.

1.7 ▪ Structure of the Book

The book is organized in five parts. Each part builds on the previous, and the SIR epidemic model [27] serves as the recurring example throughout, so that students can track a single application as the mathematical tools become progressively more powerful. The SIR model divides a population into three compartments: Susceptible (S), Infectious (I), and Recovered (R); it is derived from first principles in Section 2.5.

Part I: The Sensitivity Question (Chapters 1–2) establishes the central problem and introduces the sensitivity index as the primary tool. Chapter 2 develops the formal definition, the sign interpretation, the tornado plot as a display format, and the principle of interpretable parameterization. By the end of Part I, students can compute and interpret sensitivity indices for

any differentiable function of several variables.

Part II: Computing Sensitivity in Practice (Chapter 3) gives students an immediately usable computational method for any model, including models that cannot be differentiated analytically. Finite-difference approximations, step-size selection, the sensitivity Jacobian, and the treatment of correlated POIs are developed here. Students who work through Chapter 3 can perform SA on any model they can run, including inherited code and black-box simulation software.

Part III: Analytic Forward Sensitivity Analysis (Chapter 4) shows how to differentiate model equations directly to obtain exact sensitivity information at lower computational cost. The forward sensitivity equation is derived for algebraic systems, difference equations, and ordinary differential equations. The chapter ends by quantifying the computational bottleneck that motivates the adjoint approach.

Part IV: The Adjoint Approach (Chapters 5–7) introduces adjoint sensitivity analysis through three progressively more abstract settings: linear systems in Chapter 5, dynamical systems in Chapter 6, and computational tools including algorithmic differentiation in Chapter 7. The central theme is Unifying Idea 2: the adjoint is the transpose in whatever space you are working in.

Part V: Limitations and Extensions (Chapters 8–9) addresses the boundaries of local SA. Chapter 8 documents the conditions under which sensitivity indices give misleading results: bifurcation points, ill-conditioned systems, and method-question mismatch. Chapter 9 introduces global SA through variance decomposition, Sobol indices, and partial rank correlation coefficients. It closes with connections to distribution-based uncertainty quantification, developed at length in Smith's *Uncertainty Quantification* [60].

Computing environment. Every chapter after this one includes a structured computational laboratory that implements the chapter's core idea, in most chapters on the SIR epidemic model. Students who complete these laboratories will leave the course with a portable toolkit for performing SA on any model they encounter in research or practice. Companion Colab notebooks recompute examples from each chapter in Python and assert results against fresh computation, and a ledger accounts for every decimal value and computed count that the chapters and appendices quote. The MATLAB reference library (eighteen functions across four modules, with an automated test suite) is available alongside them. All companion materials are hosted at:

`https://github.com/machyman/arriola2026sa`

Each function in that library will match a concept or algorithm in the text. `sir_nominal.m` will define the SIR parameters used in every chapter; `sensitivity_jacobian.m` the centered-difference Jacobian from Chapter 3; `sobol_jansen.m` the Jansen estimators from Chapter 9. The repository is organized to mirror the book's structure, so the code for any chapter is found without searching.

Notation. Throughout the book, the sensitivity index of a QOI q with respect to a POI p is written $\mathcal{S}_p^q$. POIs are always chosen to be interpretable in everyday terms: durations, probabilities, counts. The relationship to the mathematical parameters in the model equations is made explicit wherever the two differ.

Scope of This Book: Scalar QOIs

Every sensitivity index in this book is computed for a *scalar* QOI, a single number such as the peak infection count, the total disease burden, or the value of $\mathcal{R}_0$. Real applications sometimes require sensitivity of an entire output *trajectory*: how does the full time series $I(t)$ change when β changes? Extending SA to functional (trajectory-valued) QOIs requires either reducing the trajectory to a scalar (which loses information) or computing a time-dependent sensitivity index at each time point (as in Chapter 4's Figure 4.1). A rigorous treatment using the adjoint sensitivity analysis procedure (ASAP) for both algebraic and differential systems is developed in Chapter 8 of Smith [60], and is a natural next step after mastering the scalar case treated here.

1.8 ▪ Exercises

Each exercise carries a Bloom's Taxonomy tag indicating cognitive level: *L2* (understand), *L3* (apply), *L4* (analyze), *L5* (evaluate), *L6* (create and synthesize). Higher-level exercises require constructing an argument or designing a procedure, not just applying a formula.

Exercise 1.1 *(Bloom L2).* Explain in your own words the difference between a POI and a QOI. For a model describing the spread of antibiotic-resistant bacteria in a hospital, give two examples of each. Then explain, in one sentence, what sensitivity analysis would tell you about this model that simply running the model would not.

Exercise 1.2 *(Bloom L2).* Forward and adjoint sensitivity analysis compute the same derivatives. What, then, is the difference between them? For each question below, name the derivatives it needs (which QOIs, with respect to

which POIs). Then say which direction computes them with fewer solves, using the counts of Section 1.2.

(a) If the mean infectious period increases by 10%, how does the peak infection count change?

(b) The peak infection count is higher than expected. To which parameters is the peak most sensitive?

(c) Among all parameters in the model, which one has the largest effect on the basic reproduction number?

Exercise 1.3 *(Bloom L4).* For each of the following modeling scenarios, state whether forward SA, adjoint SA, or global SA is most appropriate. Justify each choice in one or two sentences.

(a) A climate model has 500 tunable parameters. The investigator wants to know which 10 parameters most affect the mean surface temperature over a 100-year run. Each model run takes 8 hours.

(b) A pollution transport model has 6 input parameters. The predicted pollution at a specific monitoring station exceeds the legal threshold. The regulator wants to know to which input assumptions that prediction is most sensitive.

(c) A pharmacokinetic model has 4 parameters, each of which is known only to within a factor of 2. The investigator wants to know the range of predicted drug concentrations consistent with this uncertainty.

Exercise 1.4 *(Bloom L5).* You are advising a team modeling the transmission of dengue fever. The model has 9 parameters. The team has funding to measure exactly 3 parameters precisely in the field and to implement exactly 2 vector-control interventions. Write a one-page plan describing how you would use SA to guide these decisions. Your plan should specify: what QOI you would focus on and why; how you would determine which 3 parameters to measure; what additional information (if any) you would need beyond the SA results to recommend interventions; and one important limitation of your approach that the team should be aware of.

Exercise 1.5 *(Bloom L6).* A published report on a disease model states: "Sensitivity analysis was performed on all 22 model parameters simultaneously, demonstrating that the model is robust to parameter uncertainty."

The analysis was conducted by computing local sensitivity indices at the best-fit parameter values.

Identify at least two substantive problems with this description or its conclusions. For each problem you identify, explain specifically what the authors should have done differently.

Notes

The sensitivity index as a normalized derivative appears throughout applied mathematics under different names. Relative sensitivity, elasticity (in economics), and logarithmic derivative all refer to the same quantity $p\,\partial q/\partial p \cdot q^{-1}$. The term *sensitivity index* as used here follows Arriola and Hyman [2, 3], who defined the normalized sensitivity index and discussed how its sign relates to the signs of the changes in input and output.

The SIR compartmental model originates with Kermack and McKendrick [27], whose 1927 paper remains a landmark of mathematical epidemiology. Anderson and May [1] developed the framework extensively, including the derivation of $\mathcal{R}_0 = c\beta/\gamma_R$ and its role as the threshold parameter for epidemic invasion. The basic reproduction number $\mathcal{R}_0$ and its precise definition for structured population models is treated rigorously by Diekmann, Heesterbeek, and Metz [16] and by van den Driessche and Watmough [65].

For the global methods of sensitivity analysis, the Primer by Saltelli et al. [53] is a manual written for students. The recent review by Razavi et al. [47] assesses the state of the field and identifies open research problems. Tarantola et al. [64] provide an annotated timeline of the major methodological developments. Smith [60] develops uncertainty quantification as a complementary topic, with particular strength on Bayesian and surrogate-model approaches. In Chapter 9 of that book, on global sensitivity analysis, Smith develops normalized derivative-based global sensitivity indices, connecting the local framework of this chapter to the global variance- and derivative-based methods covered in our Chapter 9.

Answers to the self-checks

1.1. POI: transmission probability (uncertain; can be reduced by vaccines), mean infectious period (uncertain; can be reduced by treatment). QOI: peak number of infectious individuals (model output we care about). Neither: day of first case. It is a historical fact, recorded rather than modeled: here it is neither an input whose variation we study nor an output the model predicts. Initial conditions, such as the number infectious on the first day, can be POIs: they are model inputs, and Chapter 4 shows how their variation enters the sensitivity equations. Note that the mean infectious period could also be a QOI in a different study; this classification depends on the question being asked.

1.2. $\mathcal{R}_0$ increases by $5\% \times 1.0 = 5\%$. The change is exact here, because $\mathcal{R}_0 = c\beta\tau_R$ is proportional to τ_R. This is bad news: a larger $\mathcal{R}_0$ means more transmission. The positive sign tells us that POI and QOI move in the same direction, so longer infectious periods lead to more disease spread.

1.3. The adjoint. With one QOI and $m = 40$ POIs it needs $r + 1 = 2$ solves, while forward SA needs $m+1 = 41$ to compute the same 40 derivatives. Neither direction identifies a cause. The derivatives rank how strongly each weather input moves the modeled concentration, which shows where the result is most sensitive, not which assumption was wrong.

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Chapter 2
Measuring Local Sensitivity

A sensitivity index is a normalized derivative with a practical job.

Calibration tells you which parameter values fit the data.
Sensitivity analysis tells you which parameters matter to the answer.

Two studies of the same epidemic model arrive at the following conclusions.

Study A reports: "Increasing the recovery rate by 1% decreases $\mathcal{R}_0$ by 1%."

Study B reports: "Increasing the mean infectious period by 1% increases $\mathcal{R}_0$ by 1%."

Both statements are mathematically correct. The recovery rate and the mean infectious period are reciprocals of each other, so the two studies analyzed the same model, used the same mathematics, and obtained the same numerical result. Yet they convey opposite signs and, more importantly, very different intuitions. Study A requires the reader to mentally invert a parameter before the public health implication becomes clear. Study B is immediately interpretable: patients who are infectious longer spread the disease more.

This chapter introduces the sensitivity index, the central quantitative tool of this book, and develops the conventions that make sensitivity analysis results readable rather than merely correct.

2.1 ▪ Ratios as Precursors to Derivatives

Comparison by ratio is one of the most fundamental quantitative habits in applied science. In finance, the price-to-earnings ratio tells an investor how many dollars they pay for one dollar of annual profit, quoted by convention as a pure multiple although it counts years of earnings. In aeronautical

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engineering, the lift-to-drag ratio determines whether a wing design is viable. In experimental physics, the signal-to-noise ratio separates measurement from artifact. None of these ratios requires calculus. Each one gives a dimensionless number that can be compared across different systems and used to make decisions.

The sensitivity index is a ratio too, but of two changes rather than two levels: it measures how much the output changes, per unit change in the input, normalized so that the result is dimensionless and comparable across parameters with different units and scales. The relative change in earnings computed below is the step the index copies, once for the output and once for the input.

To see why normalization matters, consider the price-to-earnings ratio P/E. Suppose a stock trades at \$50 per share and has earned \$2.50 over the past year. The P/E ratio is $50/2.50 = 20$: you pay \$20 for each dollar of annual earnings. Now suppose earnings increase by \$0.50 to \$3.00. A raw comparison of the two earnings figures gives $\$3.00 - \$2.50 = \$0.50$, which tells you nothing useful without knowing the original value. The normalized comparison is $(3.00 - 2.50)/2.50 = 20\%$, a 20% increase in earnings, regardless of whether the numbers are in dollars, euros, or any other currency.

A derivative, in the same spirit, measures the rate of change of the output per unit change in the input. The sensitivity index normalizes this derivative so that it answers a specifically useful question: if the input changes by a small percentage, by what percentage does the output change?

Self-check 2.1. A second stock trades at \$45 per share and has earned \$3.00 per year. Calculate the P/E ratio for each stock, using the first stock's original earnings of \$2.50. Which stock is a better value by this measure, and why?

2.2 ▪ Absolute, Relative, and Regularized Change

Before defining the sensitivity index, we need precise language for describing how quantities change. Three measures of change appear throughout the book, each appropriate in different circumstances.

2.2.1 ▪ Absolute Change

Let u denote the solution to a model, and suppose a perturbation δu is applied. The *absolute change* is simply the difference between the perturbed and original solutions:

$$\Delta u_{\text{Abs}} := \delta u = \hat{u} - u, \tag{2.1}$$

where $\hat{u} = u + \delta u$ is the perturbed solution. A 1-inch change in the length of a string has absolute change 1 inch, regardless of the original length.

2.2.2 ▪ Relative Change

The 1-inch change means something very different if the string is 6 inches long versus 1 mile long. The *relative change* accounts for this by normalizing against the original value:

$$\Delta u_{\mathrm{Rel}} := \frac{\delta u}{u} = \frac{\hat{u} - u}{u}. \tag{2.2}$$

For the 6-inch string, the relative change is $1/6 \approx 16.7\%$. For the 1-mile string (63,360 inches), it is $1/63{,}360 \approx 0.0016\%$. The same absolute change has a very different significance depending on context, and relative change captures this.

2.2.3 ▪ Regularized Change

Relative change has one practical problem: if $u \approx 0$, the denominator in (2.2) is near zero and the ratio becomes unreliable. In dynamical models, the solution can pass through or near zero during its evolution: the infectious compartment $I(t)$ in an epidemic approaches zero as the outbreak resolves, for instance. The *regularized change* handles this case by adding a reference scale to the denominator:

$$\Delta u_{\mathrm{Reg}} := \frac{\delta u}{u_{\mathrm{ref}} + u}. \tag{2.3}$$

Here $u_{\mathrm{ref}} > 0$ is a reference scale carrying the same units as u, chosen from the problem: a typical or nominal magnitude of the state, or the smallest value we care to resolve. The scale cannot be dropped. Writing $1 + u$ in its place adds a dimensionless 1 to a dimensional quantity, and the resulting index would change whenever the units change. When $u \ll u_{\mathrm{ref}}$, the regularized change approximates absolute change measured in units of u_{ref}; when $u \gg u_{\mathrm{ref}}$, it approximates relative change $\delta u/u$. It interpolates smoothly between the two regimes, and the crossover sits where the choice of u_{ref} puts it, which is why that choice should be stated whenever the index is reported. (For models where u can be negative, use $u_{\mathrm{ref}} + |u|$; all SIR state variables satisfy $u \geq 0$.)

Looking more carefully at regularized change reveals a pattern that recurs throughout applied mathematics: when a formula fails at a special value, adding a stabilizing term to the denominator restores well-definedness and changes the result little wherever that term is small beside the original

denominator. The same idea appears in numerical linear algebra (Tikhonov regularization of ill-conditioned systems) and in Bayesian statistics (pseudo-counts in categorical models). The regularized sensitivity index is one more instance. Chapter 3 leaves its reference scale to the analyst, since only the problem can supply u_{ref}.

Self-check 2.2. The infectious fraction of the population in a disease model drops from $I = 0.04$ to $\hat{I} = 0.03$. Calculate the absolute, relative, and regularized change. Which measure gives the most informative description of this transition, and why?

2.3 ▪ The Sensitivity Index

We now define the central tool of local sensitivity analysis.

2.3.1 ▪ The Sign Rule

Before any formal definition, the most important thing to know about the sensitivity index is what its sign means.

The sign rule: A positive sensitivity index means that the QOI and POI move in the same direction: increase the POI and the QOI increases. A negative sensitivity index means they move in opposite directions: increase the POI and the QOI decreases. The magnitude tells you by how much: a sensitivity index of 2 means that a 1% change in the POI produces approximately a 2% change in the QOI.

Important: the sign belongs to the *parameterization*, not to the underlying biology. The recovery rate γ_R and the mean infectious period $\tau_R = 1/\gamma_R$ describe the same biological process, yet $\mathcal{S}_{\gamma_R}^{\mathcal{R}_0} < 0$ while $\mathcal{S}_{\tau_R}^{\mathcal{R}_0} > 0$. When two papers report opposite signs for what appears to be the same parameter, check whether they are using rate versus duration parameterizations before concluding there is a contradiction. Section 2.5 illustrates this with the SIR model.

Students sometimes interpret a negative sensitivity index as indicating an error in the calculation. It is not. A negative index can be good news. The index of $\mathcal{R}_0$ with respect to the recovery rate is negative: faster recovery reduces transmission, so a treatment that speeds recovery lowers $\mathcal{R}_0$. The same fact stated with the mean infectious period has a positive index: longer illness leads to more transmission. Both statements are correct. The sign tells you which direction the influence runs, and whether that is good news depends on which way the POI can be moved. With this intuition established, the formal definition follows naturally.

2.3.2 ▪ Formal Definition

Definition 2.1 (Sensitivity Index)**.** *Let q denote a* QOI *that depends differentiably on a* POI *p. The* normalized sensitivity index *(SI for short) of q with respect to p is*

$$\mathcal{S}_p^q := \lim_{\delta p \to 0} \frac{\left(\frac{\delta q}{q}\right)}{\left(\frac{\delta p}{p}\right)} = \frac{p}{q}\frac{\partial q}{\partial p}, \qquad q, p \neq 0. \tag{2.4}$$

This quantity is called an elasticity *in economics and econometrics, where it measures the proportional responsiveness of one variable to another. The sensitivity index is its counterpart in mathematical modeling.*

Remark 2.2 (Near-zero QOI). When $q = 0$ at the nominal parameter values, the normalized SI is undefined (division by zero). In this case, use the absolute sensitivity $\partial q/\partial p$ directly, or the regularized form $S^q_{p,\mathrm{reg}} = \frac{p}{q_{\mathrm{ref}}+|q|}\frac{\partial q}{\partial p}$, where $q_{\mathrm{ref}} > 0$ carries the units of q and is chosen from the problem as in (2.3). Throughout this book we assume $q \neq 0$ and $p \neq 0$ unless stated otherwise; this holds for all QOIs and POIs in the SIR examples.

Practical switching rule. In dynamical models, a QOI may approach zero even when it is nonzero at the nominal point, for example, $I(t)$ becomes small near $t = T$ in epidemic models with a finite observation window. Use the normalized SI when $|q^*| \geq 0.01 \times \max_t |q(t)|$; switch to the absolute sensitivity $\partial q/\partial p$ when $|q^*|$ is smaller than this threshold. The 1% cutoff is a rule of thumb, not a derived bound: below it the factor p/q turns small absolute errors in q into large relative ones. If the timing of a near-zero crossing itself matters (for example, the epidemic end date), redefine the QOI to be the crossing time rather than the compartment value.

Three equivalent forms of this definition are in common use. All three give the same number; which one is most convenient depends on the problem at hand.

Ratio form (Definition 2.1): the ratio of relative changes in the QOI and the POI. This form repeats the relative-change computation of Section 2.1, once for the QOI and once for the POI.

Derivative form: multiply the partial derivative by the ratio of nominal values:

$$\mathcal{S}_p^q = \frac{p}{q}\frac{\partial q}{\partial p}. \tag{2.5}$$

This is the form used most often in computation. The factor p/q does the normalization work: it converts $\partial q/\partial p$, which has units of $[Q]/[P]$ and

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depends on the choice of units, into a dimensionless percent-for-percent ratio. Concretely, if $\mathcal{S}_p^q = 2$, then a 1% increase in p produces approximately a 2% increase in q, regardless of whether p is measured in days, kilograms, or dollars.

Logarithmic form: for positive POIs and QOIs,

$$\mathcal{S}_p^q = \frac{\partial \ln q}{\partial \ln p}. \tag{2.6}$$

This form reveals that the sensitivity index is the slope of the log-log plot of q versus p, a fact that becomes useful when comparing SI values across very different scales. It also gives the *monomial rule*: if $q = k\, p_1^{a_1} p_2^{a_2} \cdots$ with $k > 0$ and every $p_i > 0$, then $\ln q = \ln k + \sum_i a_i \ln p_i$, so $\mathcal{S}_{p_i}^q = a_i$. The index is the exponent, whatever the constant k.

The three forms are equivalent whenever q and p are positive and q is differentiable with respect to p. The chain rule gives the link to the logarithmic form in one line: $\partial \ln q / \partial \ln p = (\partial \ln q / \partial q)(\partial q / \partial p)(\partial p / \partial \ln p) = (1/q)(\partial q / \partial p)\, p$.

Remark 2.3. When p or q may be negative, but neither is zero, use the derivative form (2.5) directly; the logarithmic form (2.6) requires $p > 0$ and $q > 0$. Remark 2.2 treats a zero q. In compartmental epidemic models this holds wherever the compartments and rates are strictly positive, as at a standard endemic equilibrium; it can fail where a compartment or rate vanishes, such as a disease-free equilibrium or a switched-off intervention, and there the logarithmic form does not apply.

Unifying Idea 1 (the sensitivity index is a normalized derivative; Section 1.5). The sensitivity index is the central object of this book. It appears here as a ratio of relative changes. In Chapters 3 and 4 it reappears as a finite-difference approximation and as the solution of a sensitivity equation. In Chapters 5 and 6, it appears as the output of the adjoint calculation. Every one of these is the same normalized derivative, asked in the same way: *if the input changes by x%, by what percentage does the output change?* In Chapter 9, it reappears inside the variance-based formulas whenever the response is close to linear over the input range.

Why normalization is not optional. Without the p/q factor, comparing raw derivatives across parameters is not meaningful. In the SIR model, $\partial \mathcal{R}_0 / \partial c$ has units of days (since c has units day^{-1}), while $\partial \mathcal{R}_0 / \partial \beta$ has different units entirely. Asking which raw derivative is larger is like asking whether 1 pound is larger than 2 inches: the numbers may have any magnitude, but the comparison is incommensurable because the units differ. The normalized SI eliminates this problem. Both $\mathcal{S}_c^{\mathcal{R}_0}$ and $\mathcal{S}_\beta^{\mathcal{R}_0}$ equal $+1$; dimensionless, unit-free, so ranking them is legitimate and the tornado plot is an

honest comparison. This is why the tornado plot sorts by $|\mathcal{S}_p^q|$ rather than by $|\partial q/\partial p|$: only the normalized index gives a ranking that is independent of how parameters happen to be measured.

Self-check 2.3. Use each of the three forms to compute the sensitivity index of the area $A = \pi r^2$ with respect to the radius r. Verify that all three give the same answer.

2.3.3 ▪ Interpreting the Magnitude

The sensitivity index answers a specific quantitative question: if the POI changes by $x\%$, the QOI changes by approximately $x \cdot \mathcal{S}_p^q\%$. For small x, this estimate is accurate to first order. For larger x, second-order corrections become relevant, and Chapter 3 discusses how to assess whether those corrections matter. Exercise 2.2 applies the estimate when p doubles, a change large enough that it can be compared with the exact value.

As a practical interpretive guide:

- $|\mathcal{S}_p^q| > 1$: the QOI is more sensitive than the POI (the change in QOI exceeds the change in POI, percentage-for-percentage).
- $|\mathcal{S}_p^q| = 1$: the QOI changes in exact proportion to the POI.
- $|\mathcal{S}_p^q| < 1$: the QOI is less sensitive than the POI (the change in QOI is smaller than the change in POI).
- $|\mathcal{S}_p^q| \approx 0$: the QOI is essentially insensitive to this POI near the nominal value.

2.4 ▪ Worked Examples

The following examples build in complexity, beginning with purely geometric functions and ending with the epidemic model that serves as the book's primary running example.

2.4.1 ▪ Volume of a Cylinder

The volume of a cylinder with radius r and height h is $V = \pi r^2 h$. We want to know which dimension (radius or height) has the greater effect on the volume.

Using the derivative form (2.5):

$$\mathcal{S}_r^V = \frac{r}{V}\frac{\partial V}{\partial r} = \frac{r}{\pi r^2 h} \cdot 2\pi r h = 2. \tag{2.7}$$

$$\mathcal{S}_h^V = \frac{h}{V}\frac{\partial V}{\partial h} = \frac{h}{\pi r^2 h} \cdot \pi r^2 = 1. \tag{2.8}$$

The volume is twice as sensitive to radius as to height. A 1% increase in radius produces a 2% increase in volume; the same percentage increase in height produces only a 1% increase. This follows from the fact that r appears squared in the volume formula while h appears linearly.

For a designer building a cylindrical tank, this result has a direct implication: a manufacturing error of a given percentage in the radius changes the volume twice as much as the same percentage error in the height.

Self-check 2.4. For a cone with volume $V = \frac{1}{3}\pi r^2 h$, compute $\mathcal{S}_r^V$ and $\mathcal{S}_h^V$. Compare with the cylinder results. Explain why the answer is the same despite the different formula.

2.4.2 ▪ Arc Length of a Line

Let $u(t) = mt + b$ be a line with slope m and intercept b, defined on $[0, a]$. The arc length is

$$J(m, b) = a\sqrt{1 + m^2}.$$

Computing the sensitivity indices:

$$\mathcal{S}_m^J = \frac{m}{J}\frac{\partial J}{\partial m} = \frac{m}{a\sqrt{1+m^2}} \cdot \frac{am}{\sqrt{1+m^2}} = \frac{m^2}{1+m^2}. \tag{2.9}$$

$$\mathcal{S}_b^J = \frac{b}{J}\frac{\partial J}{\partial b} = 0, \tag{2.10}$$

since J does not depend on b.

The arc length is completely insensitive to the intercept. Geometrically, this is clear: shifting the line vertically does not change its length. Analytically, b does not appear in the formula for J, so its derivative is zero. A zero sensitivity index is not an error; it tells us that one POI has no local effect on the QOI, which is itself an informative finding.

Self-check 2.5. For the line $u(t) = 3t + 2$ on $[0, 4]$, compute $\mathcal{S}_m^J$ numerically. Verify: if m increases from 3 to 3.03 (a 1% increase), does J change by approximately $\mathcal{S}_m^J$%?

2.4.3 ▪ Area Under a Line

For the same line $u(t) = mt + b$ on $[0, a]$, the area is

$$J(m, b) = \int_0^a (mt + b)\, dt = \tfrac{1}{2}ma^2 + ba.$$

Both parameters now affect the area:

$$\mathcal{S}_m^J = \frac{m}{\frac{1}{2}ma^2 + ba} \cdot \tfrac{1}{2}a^2 = \frac{am}{am + 2b}, \tag{2.11}$$

$$\mathcal{S}_b^J = \frac{b}{\frac{1}{2}ma^2 + ba} \cdot a = \frac{2b}{am + 2b}. \tag{2.12}$$

Note that $\mathcal{S}_m^J + \mathcal{S}_b^J = 1$, so the two indices share a unit total for this particular linear functional. The reason is scaling. Doubling both m and b doubles J, so J is homogeneous of degree one in (m, b), and Euler's identity $m\,\partial J/\partial m + b\,\partial J/\partial b = J$ makes the two indices sum to 1. This is not a general property: sensitivity indices do not sum to any fixed value for arbitrary models or QOIs. Which parameter matters more depends on the values: $\mathcal{S}_m^J > \mathcal{S}_b^J$ when $am > 2b$, and the inequality reverses when $am < 2b$.

This example illustrates a general phenomenon: when two POIs both influence the same QOI, their relative importance depends on the nominal parameter values, not just on the model structure. SA must always be reported at specific nominal values.

2.4.4 ▪ Root of a Nonlinear Equation

Consider the nonlinear equation

$$f(u; \alpha) := \sin(\alpha u) - u = 0.$$

For $\alpha = 2$, the equation has three roots; for $\alpha > 1$, there is at least one nonzero root. We want to know how the positive root $u^*(\alpha)$ changes with α.

Differentiating $f(u; \alpha) = 0$ implicitly with respect to α:

$$u\cos(\alpha u) + \alpha\cos(\alpha u)\,\frac{\partial u}{\partial \alpha} - \frac{\partial u}{\partial \alpha} = 0.$$

Solving for $\partial u/\partial \alpha$:

$$\frac{\partial u}{\partial \alpha} = \frac{u\cos(\alpha u)}{1 - \alpha\cos(\alpha u)}.$$

At $\alpha = 2$, a numerical root solve (Newton's method, for instance) gives the positive root $u^* \approx 0.9477$, and then

$$\mathcal{S}_\alpha^{u^*} = \frac{\alpha}{u^*} \cdot \frac{u^*\cos(\alpha u^*)}{1 - \alpha\cos(\alpha u^*)} \approx -0.39.$$

A 1% increase in α decreases the positive root by approximately 0.39%. Once u^* is known, the index needs no perturbed solve: the derivative formula provides the estimate directly.

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2.4.5 ▪ A Linear Prediction Model (Statistical Connection)

Sensitivity indices apply equally to statistical models. Consider a linear prediction model for hospital readmission risk:

$$\hat{P} = \beta_0 + \beta_1 x_1 + \beta_2 x_2, \tag{2.13}$$

where x_1 = patient age (years) and x_2 = comorbidity count are the POIs, $\hat{P}$ (predicted readmission probability) is the QOI, and $\beta_0 = 0.05$, $\beta_1 = 0.002$, $\beta_2 = 0.01$ are estimated regression coefficients. At nominal values $x_1^* = 65$ years and $x_2^* = 3$ comorbidities, the predicted probability is $\hat{P}^* = 0.05 + 0.002(65) + 0.01(3) = 0.21$.

Applying the sensitivity index formula:

$$\mathcal{S}_{x_1}^{\hat{P}} = \frac{x_1^*}{\hat{P}^*} \frac{\partial \hat{P}}{\partial x_1} = \frac{65}{0.21} \times 0.002 \approx 0.619, \tag{2.14}$$

$$\mathcal{S}_{x_2}^{\hat{P}} = \frac{x_2^*}{\hat{P}^*} \frac{\partial \hat{P}}{\partial x_2} = \frac{3}{0.21} \times 0.01 \approx 0.143. \tag{2.15}$$

Age is the dominant driver: a 1% increase in age (roughly 0.65 years) increases predicted readmission risk by 0.619%. The index for the comorbidity count is harder to read. A 1% increase in a count of 3 is 0.03 of a condition, which no patient has. For a count-valued POI, read the raw coefficient instead: one more condition adds $\beta_2 = 0.01$ to $\hat{P}$.

Comparison with standardized regression coefficients. Statistical software reports standardized coefficients $\tilde{\beta}_i = \beta_i \, \sigma_{x_i} / \sigma_{\hat{P}}$, which normalize by standard deviations rather than by nominal values. If $\sigma_{x_1} = 15$ years, $\sigma_{x_2} = 1.5$, and $\sigma_{\hat{P}} \approx 0.04$, then $\tilde{\beta}_1 = 0.002 \times 15/0.04 = 0.75$ and $\tilde{\beta}_2 = 0.01 \times 1.5/0.04 = 0.375$: the same ranking (age dominates) but different magnitudes. The sensitivity index asks *"what if this patient's age changed by 1%?"*; the standardized coefficient asks *"what if age changed by one standard deviation across the patient population?"* Both are dimensionless measures of importance; they answer different questions.

2.5 ▪ The Sensitivity Index for the SIR Model

2.5.1 ▪ Model Derivation from First Principles

The SIR epidemic model [27, 1] divides a closed population of N individuals into three compartments: susceptible (S), infectious (I), and recovered (R), with $S(t) + I(t) + R(t) = N$ at all times. Figure 2.1 shows the flow diagram.

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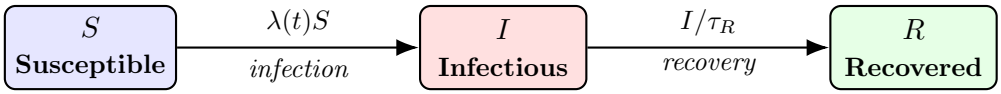


Figure 2.1. *SIR compartment flow diagram. The force of infection $\lambda(t) = c\beta I(t)/N$ is the per-capita rate at which any one susceptible individual becomes infected. Recovery occurs at rate $1/\tau_R$ per infectious individual per day.*

The force of infection. At any moment, each infectious individual makes c contacts per day.[2] Every contact with a susceptible individual transmits the infection with probability β. Of the c individuals contacted per day by one infectious person, a fraction S/N are susceptible. The rate at which one infectious individual generates new infections is therefore $c\beta S/N$ per day, and with I infectious individuals in the population the total rate of new infections is $c\beta SI/N$ per day. The *force of infection*

$$\lambda(t) = \frac{c\beta I(t)}{N} \tag{2.16}$$

is the per-capita infection rate at time t: the rate at which each susceptible individual acquires infection. Note that $\lambda(t)$ has units of day^{-1}: the product $\lambda(t) \cdot \Delta t$ is the probability of becoming infected during a short time interval Δt.

The ODE system. Translating the flow diagram into rates:

$$\frac{dS}{dt} = -\lambda(t)\, S \;=\; -\frac{c\beta SI}{N}, \tag{2.17}$$

$$\frac{dI}{dt} = \;\; \lambda(t)\, S - \frac{I}{\tau_R} \;=\; \frac{c\beta SI}{N} - \frac{I}{\tau_R}, \tag{2.18}$$

$$\frac{dR}{dt} = \frac{I}{\tau_R}. \tag{2.19}$$

[2]This chapter treats the contact rate as a single parameter c shared by all compartments, which makes $\mathcal{R}_0 = c\beta\tau_R$ a product of monomials with all sensitivity indices equal to $+1$. In heterogeneous-mixing models, such as the SIR_I framework developed in Hyman, Qu, and Xue [24], the contact rate is differentiated by compartment: susceptibles, infectious, and recovered individuals make c_S, c_I, and c_R contacts per day, respectively, with $\mathcal{R}_0 = c_I\beta/(\gamma_R+\nu_m)$. The equal-contact-rate case $c_S = c_I = c_R = c$, with no demography ($\nu_m = 0$), recovers the simplified model used here. That book also normalizes the state variables by the carrying capacity N^c, the population size at which births balance deaths in the absence of disease, and uses calligraphic $\mathcal{R}_0$ to distinguish the reproductive number from the recovered compartment R.

Each term has a biological origin. $-\lambda S$ in equation (2.17): susceptibles are lost at rate λ (the per-capita infection rate) times S (the number at risk). $+\lambda S$ in equation (2.18): these same individuals immediately enter the infectious compartment. $-I/\tau_R$ in equation (2.18): infectious individuals recover at rate $1/\tau_R$ per person per day, since the mean infectious period is τ_R days. $+I/\tau_R$ in equation (2.19): recovered individuals arrive at the same rate they depart from I.

Why $\mathcal{R}_0 = c\beta\tau_R$. In a fully susceptible population ($S \approx N$), one infectious individual generates new infections at rate $c\beta$ per day. The infectious period lasts on average τ_R days. The total number of secondary infections is therefore

$$\mathcal{R}_0 \;=\; \underbrace{c}_{\text{contacts/day}} \;\times\; \underbrace{\beta}_{\substack{\text{prob. infection}\\ \text{per contact}}} \;\times\; \underbrace{\tau_R}_{\text{infectious days}} \;=\; c\beta\tau_R. \tag{2.20}$$

Each factor is biologically independent: contacts per day depend on social behavior; transmission probability per contact depends on pathogen biology; and the infectious period depends on immune response. Because $\mathcal{R}_0$ is a product of three independent factors, varying any one of them by $x\%$ changes $\mathcal{R}_0$ by exactly $x\%$, a preview of why all three sensitivity indices equal 1. The ODE (2.18) gives the same threshold. At $S \approx N$ it reads $dI/dt = (c\beta - 1/\tau_R)\, I$, which is positive exactly when $c\beta\tau_R > 1$, that is, when $\mathcal{R}_0 > 1$. Table 2.1 collects the symbols, units, and nominal values used for this model throughout the book; the worked examples and the companion notebooks all start from these values.

2.5.2 ▪ The Basic Reproduction Number

The basic reproduction number

$$\mathcal{R}_0 = c\beta\tau_R$$

measures the average number of new infections produced by one infectious individual in an otherwise susceptible population (equation (2.20)). The rigorous definition of $\mathcal{R}_0$ via the next-generation operator is given in van den Driessche and Watmough [65] and Diekmann et al. [16].

In Chapter 4, we will compute time-dependent sensitivity indices for the full SIR model. Here we compute the static sensitivity indices for $\mathcal{R}_0$: a closed-form calculation that introduces all the key ideas in a familiar setting.

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Parameter	Symbol	Units	Nominal	Biological meaning
Contact rate	c	day^{-1}	5	Mean daily contacts per person
Transmission prob.	β	–	0.06	Probability of infection per contact with an infectious person
Infectious period	τ_R	days	7	Mean number of days a person remains infectious
Mean residence time	τ_m	days	10,000	Mean residence time $\tau_m = 1/\nu_m$; unused in this chapter's model, which has no demography. From Chapter 4 onward ν_m removes infectives and the threshold becomes $c\beta/(\gamma_R + \nu_m)$
Derived quantity $\mathcal{R}_0 = c\beta\tau_R$		–	$5 \times 0.06 \times 7 = 2.1$	Average secondary infections per case in a susceptible population

Table 2.1. *SIR model parameters, nominal values, and biological interpretation. Nominal values correspond to a moderately transmissible infection (influenza-like) in a well-mixed population. Typical ranges: $c \in [2, 20]$, $\beta \in [0.01, 0.5]$, $\tau_R \in [2, 14]$ days, giving $\mathcal{R}_0 \in [0.04, 140]$; the biologically plausible range for epidemic spread is $\mathcal{R}_0 > 1$.*

2.5.3 ▪ Reparameterizing the SIR Model

The recovery terms I/τ_R in equations (2.18)–(2.19) can also be written as $\gamma_R I$, with the recovery rate $\gamma_R = 1/\tau_R$. In that rate form, c, β, and γ_R appear directly in the model equations, which makes them a natural choice for POIs. However, we argued in Chapter 1 that POIs should be chosen to be interpretable in everyday terms.

The recovery rate γ_R (units: day^{-1}) is not directly interpretable. What a clinician understands and can measure is the *mean infectious period* $\tau_R = 1/\gamma_R$ (units: days): the average number of days a patient remains infectious. These two parameterizations are linked by:

$$\mathcal{S}^q_{\tau_R} = -\mathcal{S}^q_{\gamma_R} \qquad \text{for any QOI } q, \tag{2.21}$$

since $\partial q/\partial \tau_R = (\partial q/\partial \gamma_R)(\partial \gamma_R/\partial \tau_R) = -\gamma_R^2(\partial q/\partial \gamma_R)$, giving $\mathcal{S}^q_{\tau_R} = (\tau_R/q)(\partial q/\partial \tau_R) = -(\gamma_R/q)(\partial q/\partial \gamma_R) = -\mathcal{S}^q_{\gamma_R}$.

The magnitude is unchanged; the sign flips. Table 2.2 shows the effect of this reparameterization for $\mathcal{R}_0$.

POI	Interpretation	Form	$\mathcal{S}_p^{\mathcal{R}_0}$
Recovery rate γ_R (day^{-1})	–	rate	-1
Mean infectious period $\tau_R = 1/\gamma_R$ (days)	"average days infectious"	duration	$+1$
Contact rate c (day^{-1})	"contacts per day"	rate	$+1$
Transmission probability β (dimensionless)	"chance of infection per contact"	probability	$+1$

Table 2.2. *Sensitivity indices of $\mathcal{R}_0$ with respect to rate and duration parameterizations of the SIR model. The index for the mean infectious period $\tau_R = 1/\gamma_R$ has the same magnitude as the index for γ_R but the opposite sign. The duration form is easier to interpret: a longer infectious period increases transmission, and both the sign and the plain-language statement agree.*

2.5.4 ▪ Analytic Computation

With the interpretable POIs $\{\tau_R = 1/\gamma_R,\, c,\, \beta\}$, the sensitivity indices are:

$$\mathcal{S}_c^{\mathcal{R}_0} = \frac{c}{\mathcal{R}_0} \cdot \frac{\partial \mathcal{R}_0}{\partial c} = \frac{c}{c\beta/\gamma_R} \cdot \frac{\beta}{\gamma_R} = 1, \tag{2.22}$$

$$\mathcal{S}_\beta^{\mathcal{R}_0} = \frac{\beta}{\mathcal{R}_0} \cdot \frac{\partial \mathcal{R}_0}{\partial \beta} = \frac{\beta}{c\beta/\gamma_R} \cdot \frac{c}{\gamma_R} = 1, \tag{2.23}$$

$$\mathcal{S}_{\tau_R}^{\mathcal{R}_0} = -\mathcal{S}_{\gamma_R}^{\mathcal{R}_0} = -\frac{\gamma_R}{\mathcal{R}_0} \cdot \frac{\partial \mathcal{R}_0}{\partial \gamma_R} = -\frac{\gamma_R}{c\beta/\gamma_R} \cdot \left(\frac{-c\beta}{\gamma_R^2}\right) = +1. \tag{2.24}$$

All three POIs have sensitivity index $+1$ with the interpretable parameterization. In plain language: a 1% increase in contact rate, transmission probability, *or* mean infectious period each increases $\mathcal{R}_0$ by 1%. The three parameters are equally important for $\mathcal{R}_0$, and all three promote disease spread when increased, as every positive sign and every positive biological interpretation confirms.

Self-check 2.6. An intervention reduces the contact rate c by 20%. Using the sensitivity indices above, estimate the change in $\mathcal{R}_0$. If a second intervention reduces the mean infectious period τ_R by 15%, which intervention has the greater effect on $\mathcal{R}_0$?

2.6 ▪ Choosing Interpretable POIs

The SIR example illustrates a general principle that should guide every SA study before any calculation begins.

Choose each parameter of interest so that it can be discussed in ordinary language, not merely because it appears in the model equations.

Model equations are written in the language that is mathematically convenient, which is not always the language that is scientifically or opera-

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tionally natural. Rates (with units day^{-1}) are mathematically convenient in differential equations but not naturally interpretable; the corresponding mean times (with units days) are easier to discuss in a clinical or policy context.

The general rule for exit rates in compartmental models: if a parameter p is the per-capita rate at which individuals leave a state (units: time^{-1}), define the corresponding POI as $\tau = 1/p$ (the mean time spent in that state), compute $\mathcal{S}_\tau^q = -\mathcal{S}_p^q$, and report the result in terms of τ. The rule is for exit rates. The contact rate c counts contacts per day rather than departures from a state, is already interpretable, and stays a rate.

Two additional cases arise frequently:

Dimensionless probabilities. Transmission probabilities like β are already dimensionless and directly interpretable: "a 25% chance of transmission per contact" requires no conversion. The index's percent is still relative: a 1% increase in $\beta = 0.06$ takes it to 0.0606, not to 0.07.

Composite parameters. In the SIR model, c and β always appear together as the product $c\beta$ in the expression for the force of infection $\lambda = c\beta I/N$. Neither can be identified separately from incidence data alone; only the product $c\beta$ is identifiable (Section 3.5.1 defines the term). In this case, $c\beta$ is a more natural single POI than either factor individually when the parameters are estimated from incidence data. Keep c and β separate, as this chapter's self-checks and figures do, when an intervention acts on one of them: distancing on c, masks on β. A related issue arises whenever two POIs are not independently changeable, which leads to the topic of correlated POIs in Section 3.5, whose structural case this is.

A practical check: if you cannot explain what a 1% increase in a POI means in plain language to a colleague outside mathematics, in a quantity that colleague could observe or measure, reconsider the parameterization.

Even in highly cited published work, this distinction is sometimes overlooked. Sections 4.2 and 4.3 of Chitnis, Hyman & Cushing [15], a paper that has become a standard reference for SA in mathematical epidemiology, report sensitivity indices using rate parameters throughout. The SI values are mathematically correct, but a rate and its mean duration carry opposite signs. In the paper's Table 4, the index of $\mathcal{R}_0$ to the human recovery rate γ_h is negative (-0.43 at low transmission), while a longer infectious period $1/\gamma_h$ promotes spread. The paper reads this sign as intuitive, and for a rate it is; a reader who thinks in durations must invert it before the public health interpretation is clear. The present book recommends always asking, before reporting any SI: would a policy maker understand this sign immediately, or does it require translation?

One further misconception deserves attention here. A sensitivity index

near zero does *not* mean a parameter is globally unimportant. It means the model is approximately linear in that parameter near the nominal value, and the linear term happens to be small. If the model is nonlinear in that parameter, varying it over a wider range may reveal substantial effects that the SI misses entirely. Chapter 9 addresses this through global sensitivity analysis.

Self-check 2.7. A malaria model contains the parameter μ_v (mosquito mortality rate, units: day^{-1}). A colleague reports $\mathcal{S}_{\mu_v}^{\mathcal{R}_0} = -0.3$. Reparameterize in terms of mean mosquito lifespan $L = 1/\mu_v$ (units: days). What is $\mathcal{S}_{L}^{\mathcal{R}_0}$? Write a plain-language interpretation of each sensitivity index.

2.7 ▪ The Tornado Plot

When a model has several POIs, the collection of sensitivity indices becomes a list of numbers with signs and magnitudes. Reading such a list is manageable for three or four parameters; for ten or twenty, a visual display is essential.

The *tornado plot* is the standard display format for sensitivity indices. It is a horizontal bar chart with the following conventions:

- Bars extending to the *right* indicate positive sensitivity indices (POI and QOI move in the same direction).
- Bars extending to the *left* indicate negative sensitivity indices (opposite directions).
- Bars are sorted from largest to smallest by absolute magnitude, with the most influential parameter at the top.
- The length of each bar represents the magnitude of the index.

The sorted layout creates the "tornado" shape: wider at the top, narrower toward the bottom.

Figure 2.2 shows the tornado plot for $\mathcal{R}_0$ in the SIR model with POIs $\{c, \beta, \tau_R\}$. All three bars extend to the right (all indices are $+1$), indicating that each parameter promotes disease spread when increased. All bars are the same length, confirming that the three parameters are equally important for $\mathcal{R}_0$. *Unifying Idea 4* (a sensitivity index describes a model, not the world; Section 1.5). That equality is a fact about the algebra of $\mathcal{R}_0 = c\beta\tau_R$, not about epidemics. It says the three factors enter the product identically; it does not say a public-health program can move them equally, or that they cost the same to move.

Algorithm to produce a tornado plot for any set of sensitivity indices:

```
# TORNADO_PLOT: horizontal bar chart sorted by |SI|
# Input:  SI_values  - vector of sensitivity indices (positive or negati
```

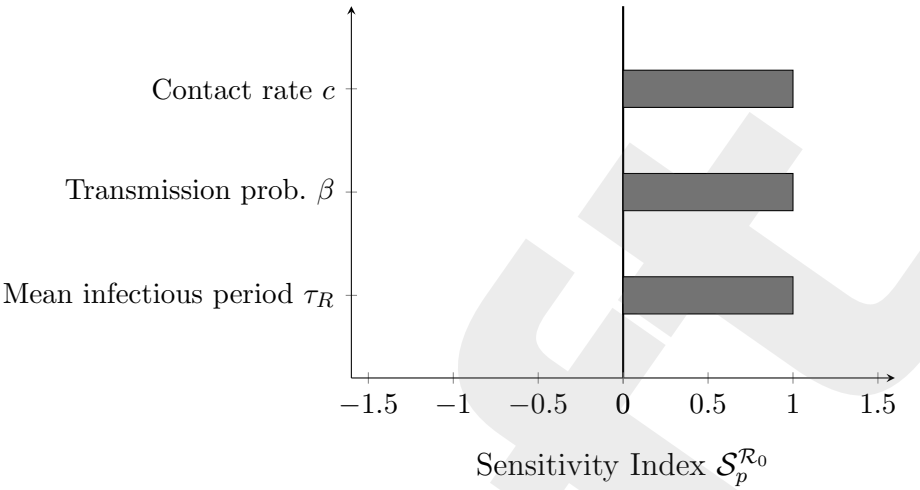


Figure 2.2. *Tornado plot for the sensitivity of $\mathcal{R}_0$ to the three* POIs *of the SIR model, parameterized by a contact rate, a probability and a duration, each interpretable as it stands. All three indices equal +1, so the bars are equal in length and all extend to the right. In models with more parameters and a wider range of index values, the sorted layout reveals immediately which parameters dominate.*

```
#         labels     - parameter names (strings)
#         QOI_name   - name of the quantity of interest

FUNCTION TORNADO_PLOT(SI_values, labels, QOI_name):
    # Sort parameters by absolute sensitivity, largest first
    order      = ARGSORT(ABS(SI_values), descending=True)
    SI_sorted  = SI_values[order]
    lab_sorted = labels[order]

    # Draw horizontal bars (positive → right, negative → left)
    BARH(SI_sorted, labels=lab_sorted)
    DRAW_VERTICAL_LINE(x=0)         # center reference line
    XLABEL("Sensitivity Index  SI_pˆ" + QOI_name)
    TITLE("Tornado Plot: Sensitivity of " + QOI_name)
END
```

Self-check 2.8. A model for antibiotic-resistance spread has sensitivity indices $\mathcal{S}^{\mathcal{R}_0}_{\tau_{\text{inf}}} = +1.4$, $\mathcal{S}^{\mathcal{R}_0}_{\beta} = +0.9$, $\mathcal{S}^{\mathcal{R}_0}_{\phi_{\text{hyg}}} = -0.6$ (with ϕ_{hyg} the fraction of patient contacts preceded by hand hygiene), $\mathcal{S}^{\mathcal{R}_0}_{c} = +0.5$.

(a) Sketch the tornado plot by hand. (b) In this model, at these nominal values, which parameter is most important? Least important? (c) If ϕ_{hyg} increases by 5%, what happens to $\mathcal{R}_0$?

2.8 ▪ A Preview: Second-Order Terms

The sensitivity index is built on a first-order approximation. For a QOI q that depends on a single POI p, the Taylor expansion gives

$$q(p+\delta p) = q(p) + \underbrace{\frac{\partial q}{\partial p}\,\delta p}_{\text{first-order}} + \underbrace{\frac{1}{2}\frac{\partial^2 q}{\partial p^2}\,(\delta p)^2}_{\text{second-order}} + \cdots$$

The sensitivity index is derived from the first-order term alone. When the second-order term is small relative to the first-order term (that is, when

$$\left|\frac{1}{2}\frac{\partial^2 q}{\partial p^2}\,\delta p\right| \ll \left|\frac{\partial q}{\partial p}\right|,$$

the sensitivity index provides an accurate approximation. When this condition fails, the linear approximation is poor and the SI overstates or understates the true sensitivity.

Chapter 3 develops practical methods for estimating the second-order term numerically and for assessing when it is large enough to matter. This is the first instance of Unifying Idea 3 from Chapter 1: local SA is a linearization, and linearizations have limits. The second-order term is the first correction beyond those limits.

For two POIs p_i and p_j that are varied together, whether or not they are correlated (a situation addressed in Chapter 3), the cross term

$$\frac{\partial^2 q}{\partial p_i\,\partial p_j}\,\delta p_i\,\delta p_j$$

also enters the expansion. This cross derivative quantifies the interaction between the two POIs: it measures how the sensitivity to p_i changes as p_j varies. Chapter 9 shows how Sobol interaction indices generalize this idea to the full range of parameter values.

2.9 ▪ Computational Laboratory: Sensitivity Indices for the SIR Model

This laboratory implements and visualizes the sensitivity indices developed in the chapter, using the SIR model as the working example. Complete the five parts in order; each builds on the previous.

Part 1: Analytic SI via the Symbolic Toolbox. Define $\mathcal{R}_0 = c\beta/\gamma_R$ using a symbolic differentiation tool. Compute $\mathcal{S}_c^{\mathcal{R}_0}$, $\mathcal{S}_\beta^{\mathcal{R}_0}$, $\mathcal{S}_{\gamma_R}^{\mathcal{R}_0}$ using the `diff` function and the derivative form of Definition 2.1. Simplify and confirm that all three equal ± 1 analytically.

Note: The code below requires the Symbolic Math Toolbox. If you do not have it, skip directly to Part 3 and verify the same results numerically. Parts 3–5 do not require the Symbolic Toolbox.

```
syms c beta gamma positive
R0   = c * beta / gamma;
SI_c = simplify( (c/R0) * diff(R0, c) )
SI_b = simplify( (beta/R0) * diff(R0, beta) )
SI_g = simplify( (gamma/R0) * diff(R0, gamma) )
```

Part 2: Reparameterize. Replace γ_R with the mean infectious period $\tau_R = 1/\gamma_R$. Rewrite $\mathcal{R}_0 = c\beta\tau_R$ symbolically. Compute $\mathcal{S}^{\mathcal{R}_0}_{\tau_R}$ and verify that it equals $-\mathcal{S}^{\mathcal{R}_0}_{\gamma_R}$ as predicted by equation (2.21).

Part 3: Numerical verification. Use the nominal values $c = 5$, $\beta = 0.06$, $\tau_R = 7$ days (so $\gamma_R = 1/7$). Compute $\mathcal{R}_0$ numerically. Estimate $\mathcal{S}^{\mathcal{R}_0}_{\tau_R}$ using a centered finite difference with $h = 10^{-5}\tau_R$. Compare with the analytic value of $+1$. How large is the relative error?

```
c = 5;  beta = 0.06;  tau = 7
R0    = c * beta * tau
h     = 1e-5 * tau                          # optimal step size (Chapter 3
R0_p  = c * beta * (tau + h)
R0_m  = c * beta * (tau - h)
SI_fd = (tau / R0) * (R0_p - R0_m) / (2*h)
PRINT "Analytic SI: 1.00000\n');
PRINT "FD estimate: %.5f\n', SI_fd);
```

Part 4: Tornado plots for both parameterizations. Using the `tornado_plot` function from Section 2.7, produce two tornado plots side by side: one with rate parameters $\{c, \beta, \gamma_R\}$ and one with interpretable parameters $\{c, \beta, \tau_R\}$. Observe how the sign of the γ_R/τ_R bar changes between the two plots and why.

MATLAB note: The `tornado_plot` function must be saved as a separate file named `tornado_plot.m` in your working directory (function files cannot be pasted into a script). Part 4 also uses `subplot` and cell array syntax for labels; the reference library (its address is in Section 1.7) provides `software/matlab/core/run_sir_example.m`, which demonstrates both if these are unfamiliar.

Part 5: Effect of nominal value. Compute $\mathcal{S}^{\mathcal{R}_0}_{c}$, $\mathcal{S}^{\mathcal{R}_0}_{\beta}$, $\mathcal{S}^{\mathcal{R}_0}_{\tau_R}$ at two different nominal transmission probabilities: $\beta_1 = 0.06$ and $\beta_2 = 0.6$. Do the SI values change? What does this tell you about the sensitivity of $\mathcal{R}_0$ to each parameter? (Recall that $\mathcal{R}_0 = c\beta\tau_R$ is a product of monomials;

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for such functions, the SI is constant regardless of the nominal value. This is a special property of multiplicative models that does not hold for more complex functions like those in Chapter 4.)

2.10 ▪ Exercises

Exercise 2.1 *(Bloom L2).* Explain the sign rule in your own words. Without performing any calculation, state the sign of the sensitivity index for each of the following, and explain your reasoning: (a) the sensitivity of lung capacity to cigarette consumption, treating capacity as the QOI and cigarettes per day as the POI; (b) the sensitivity of average crop yield to annual rainfall, near a region of the parameter space where crops are mildly drought-stressed; (c) the sensitivity of $\mathcal{R}_0$ to the mean infectious period τ_R.

Exercise 2.2 *(Bloom L2).* For the function $q = \sqrt{p}$ (defined for $p > 0$), compute $\mathcal{S}_p^q$ using all three equivalent forms of Definition 2.1. Verify that all three give the same result. Interpret the answer: if p doubles, by approximately what percentage does q change?

Exercise 2.3 *(Bloom L3).* Let $u(t) = \frac{1}{2}\lambda t^2$ be a parabola on $[0, b]$. The arc-length functional is $J = \int_0^b \sqrt{1 + (\lambda t)^2}\, dt$. Show that the sensitivity indices are

$$\mathcal{S}_b^J = \frac{2\mu_b}{\mu_b + (\operatorname{arcsinh}\mu_b)/\sqrt{1+\mu_b^2}}, \qquad \mathcal{S}_\lambda^J = \frac{\mu_b(1+\mu_b^2) - \sqrt{1+\mu_b^2}\,\operatorname{arcsinh}\mu_b}{\mu_b(1+\mu_b^2) + \sqrt{1+\mu_b^2}\,\operatorname{arcsinh}\mu_b},$$

where $\mu_b := \lambda b$ is dimensionless, and is unrelated to the demographic rate ν_m used elsewhere in this chapter. For $\mu_b = 1$, compute both indices numerically. Then check them with the derivative form (2.5): at $\lambda = b = 1$, compute $\partial J/\partial b$ and $\partial J/\partial \lambda$, and multiply them by b/J and λ/J.

Exercise 2.4 *(Bloom L3).* Let $u(t) = at^2 + bt + c$ and let $J_\pm = (-b \pm \sqrt{b^2 - 4ac})/(2a)$ denote the two real roots (assume $b^2 > 4ac$). Show that the sensitivity indices of the root J_+, the one taken with the plus sign, with respect to a, b, c can be written

$$\mathcal{S}_a^{J_+} = -\frac{aJ_+}{2aJ_+ + b}, \quad \mathcal{S}_b^{J_+} = -\frac{b}{2aJ_+ + b}, \quad \mathcal{S}_c^{J_+} = -\frac{c}{J_+(2aJ_+ + b)}.$$

For $a = 1$, $b = -3$, $c = 2$, compute all three indices. Interpret each sign.

Exercise 2.5 *(Bloom L3).* A SEIR model (Susceptible-Exposed-Infectious-Recovered) has reproduction number

$$\mathcal{R}_0 = \frac{c\beta\,\nu}{(\nu + \nu_m)(\gamma_R + \nu_m)},$$

where ν is the rate of progression from exposed to infectious, ν_m is the background mortality rate, and c, β, γ_R are as in the SIR model. Reparameterize using mean times: $\tau_E = 1/\nu$ (mean latent period), $\tau_I = 1/\gamma_R$ (mean infectious period), $\tau_m = 1/\nu_m$ (mean residence time). Compute $\mathcal{S}^{\mathcal{R}_0}_{\tau_E}$, $\mathcal{S}^{\mathcal{R}_0}_{\tau_I}$, $\mathcal{S}^{\mathcal{R}_0}_{\tau_m}$, $\mathcal{S}^{\mathcal{R}_0}_{c}$, $\mathcal{S}^{\mathcal{R}_0}_{\beta}$ and produce a tornado plot. Which parameter is most important?

Nominal values for the tornado plot: $c = 5$, $\beta = 0.06$, $\tau_E = 5.5$ days, $\tau_I = 7$ days, $\tau_m = 10{,}000$ days.

Exercise 2.6 *(Bloom L4).* A published paper reports $\mathcal{S}^{\mathcal{R}_0}_{\gamma_R} = -1.0$ for an SIR model, where γ_R is the recovery rate in units of day^{-1}. A second paper on the same model reports $\mathcal{S}^{\mathcal{R}_0}_{\tau_R} = +1.0$, where $\tau_R = 1/\gamma_R$ is the mean infectious period in days. A policy maker reading both papers is confused: one paper shows a negative index, the other shows a positive index for what appears to be the same parameter.

Write a one-paragraph explanation for the policy maker that (a) resolves the apparent contradiction, (b) explains which parameterization is more informative for policy, and (c) describes what each sensitivity index means in plain language.

Exercise 2.7 *(Bloom L4).* The following tornado plot data is given for a food-safety model with QOI J = mean number of illness cases per outbreak. The sensitivity indices are: $\mathcal{S}^{J}_{\tau_c} = +2.1$ (mean contamination duration, days), $\mathcal{S}^{J}_{\beta_f} = +1.8$ (food-to-person transmission probability), $\mathcal{S}^{J}_{\tau_s} = +1.3$ (mean hand-washing interval, days), $\mathcal{S}^{J}_{k_f} = +0.7$ (food contacts per day), $\mathcal{S}^{J}_{\nu_m} = -0.2$ (background clearance rate).

(a) Produce the tornado plot using the algorithm template from Section 2.7.

(b) Rank the five parameters by absolute importance in this model, at these nominal values.

(c) A public health program can achieve a 15% reduction in either τ_c or a 20% reduction in k_f. Which intervention reduces J more? By how much?

(d) The parameter ν_m has a small negative index. Write a one-sentence interpretation of what this means for disease control.

Exercise 2.8 *(Bloom L5).* You are advising the World Health Organization on a vaccine allocation strategy for a new respiratory pathogen. A compartmental model has been fitted to early outbreak data. The model has six

parameters; their sensitivity indices with respect to the QOI "total deaths at 6 months" are not yet known.

Design the SA study. Your design should specify:

(a) Which parameterization you will use (rates or durations) for each parameter, and why.

(b) What QOI or QOIs you would compute sensitivity indices for, and what additional QOI you might consider beyond the stated one.

(c) How you would present the results to a non-mathematical audience (what figure or table, and what the key message would be).

(d) One limitation of the local SA approach that the WHO advisors should be aware of in using these results.

Exercise 2.9 *(Bloom L4). Audit exercise.* A report on an SIR model states: "The sensitivity index of $\mathcal{R}_0$ with respect to the contact rate c is $+1$. The relationship between c and $\mathcal{R}_0$ is therefore linear, so doubling the contact rate doubles $\mathcal{R}_0$, and halving it halves $\mathcal{R}_0$." The numerical value quoted is correct.

(a) Identify the error in the interpretation. What does an index of $+1$ actually assert, and over what range?

(b) Construct a function of one variable whose sensitivity index equals $+1$ at the nominal point but which is not linear, and evaluate the error in the report's doubling claim for your function.

(c) The report's final sentence would be defensible for $\mathcal{R}_0 = c\beta\tau_R$ under one additional condition. State it.

(d) Rewrite the passage so that it is correct and no longer than the original.

Exercise 2.10 *(Bloom L6).* A modeling study reports the following: "Sensitivity analysis of $\mathcal{R}_0$ was performed with respect to all model parameters. The parameters are listed in Table 3 using the standard model notation. All sensitivity indices were positive, confirming that $\mathcal{R}_0$ is an increasing function of all model parameters."

Table 3 lists recovery rate γ_R, mortality rate ν_m, and waning immunity rate ρ alongside transmission and contact parameters.

Identify the error in the study's statement and explain specifically what is wrong, referring to the mathematical relationship between rates and their reciprocals. Restate the conclusion correctly.

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Notes

The three equivalent forms of the sensitivity index (limit, derivative, and logarithmic) appear in Arriola and Hyman [2], which also introduced the sign rule and the interpretable-parameterization convention adopted throughout this book. Earlier uses of normalized sensitivity in ecology and economics appear under the name *elasticity*; see, for example, the biological applications surveyed in Saltelli et al. [51].

The tornado diagram as a visualization tool for ranking sensitivity indices was popularized in risk analysis and decision science; see Saltelli et al. [56] for its use in the SA context.

Blower and Dowlatabadi [5] gave an early and influential application of sensitivity analysis to an HIV transmission model, demonstrating how tornado plots guide intervention priorities. Arriola and Hyman [3] apply the SI to $\mathcal{R}_0$ in an SEIR epidemic model.

For the near-zero-QOI regime and regularized sensitivity indices, see the discussion in Saltelli et al. [51], Chapter 3.

Answers to the self-checks

2.1. Stock 1: $P/E = 50/2.50 = 20$. Stock 2: $P/E = 45/3.00 = 15$. Stock 2 has a lower P/E ratio, meaning you pay less per dollar of earnings, a better value by this measure. The ratio provides a dimensionless comparison that is fair across the two stocks despite their different prices and earnings.

2.2. Absolute: $\delta I = 0.03 - 0.04 = -0.01$. Relative: $-0.01/0.04 = -0.25$, a 25% decrease. Regularized: I is a fraction of the population, and the state (S, I, R) sums to 1. Taking that nominal magnitude of the state as the reference scale, $u_{\text{ref}} = 1$, the first choice offered above, gives $-0.01/(1 + 0.04) \approx -0.0096$, close to absolute. Scaling by I alone instead, $u_{\text{ref}} = 0.04$, gives $-0.01/0.08 = -0.125$; the choice matters and should be stated. The reference scale is what makes the denominator dimensionally sound; it is not a bare 1 added to a fraction.

Since I is small relative to 1, relative change gives the most interpretable result: a 25% decrease in the infectious population is meaningful regardless of scale. For $I \ll 1$, absolute and regularized change are both small in magnitude and harder to interpret without the normalization that relative change provides.

2.3. *Ratio form:* $\delta A = \pi(r+\delta r)^2 - \pi r^2 = 2\pi r\,\delta r + \pi(\delta r)^2$, so $(\delta A/A)/(\delta r/r) = 2 + \delta r/r$, which tends to 2 as $\delta r \to 0$, so $\mathcal{S}_r^A = 2$.

Derivative form: $\partial A/\partial r = 2\pi r$, so $\mathcal{S}_r^A = (r/\pi r^2)(2\pi r) = 2$.

Logarithmic form: $\ln A = \ln \pi + 2 \ln r$, so $\partial \ln A/\partial \ln r = 2$.

All three give $\mathcal{S}_r^A = 2$: a 1% increase in radius produces a 2% increase in area. The circle's area is twice as sensitive to radius as to a parameter that appears linearly.

2.4. $\mathcal{S}_r^V = (r/V)(\partial V/\partial r) = (r/V) \cdot (2\pi r h/3) = 2$. $\mathcal{S}_h^V = 1$. Identical to the cylinder.

The reason: the sensitivity index depends on the *exponent* of each variable in the formula. Both $V_{\text{cone}} = \frac{1}{3}\pi r^2 h$ and $V_{\text{cylinder}} = \pi r^2 h$ have r to the power 2 and h

to the power 1. The constant $\frac{1}{3}$ cancels in the logarithmic form $\partial \ln V / \partial \ln r = 2$. This is the monomial rule of Section 2.3: for $q = k\, p_1^{a_1} p_2^{a_2} \cdots$, the sensitivity index with respect to p_i is the exponent a_i.

2.5. $\mathcal{S}_m^J = m^2/(1+m^2) = 9/(1+9) = 0.9$.
At $m = 3$: $J = 4\sqrt{10} \approx 12.649$. At $m = 3.03$: $J = 4\sqrt{1 + 3.03^2} = 4\sqrt{10.1809} \approx 12.763$. Change: $(12.763 - 12.649)/12.649 \approx 0.90\% \approx 0.9 \times 1\%$. ✓

2.6. First intervention: $\Delta\mathcal{R}_0/\mathcal{R}_0 \approx \mathcal{S}_c^{\mathcal{R}_0} \times (-20\%) = -20\%$. Second intervention: $\Delta\mathcal{R}_0/\mathcal{R}_0 \approx \mathcal{S}_{\tau_R}^{\mathcal{R}_0} \times (-15\%) = -15\%$. The first intervention has the greater effect. Both indices equal 1, so the percentage change in $\mathcal{R}_0$ equals the percentage change in the POI. The intervention that changes its POI by the larger percentage has the larger effect.

2.7. $\mathcal{S}_L^{\mathcal{R}_0} = -\mathcal{S}_{\mu_v}^{\mathcal{R}_0} = +0.3$.
Rate form: "A 10% increase in mosquito mortality rate decreases $\mathcal{R}_0$ by 3%." This requires mental inversion: higher mortality → shorter lifespan → less transmission.
Duration form: "A 10% increase in mean mosquito lifespan increases $\mathcal{R}_0$ by 3%." Direct: longer-lived mosquitoes transmit more malaria.

2.8. (a) Sorted order by |SI|: τ_{inf} (1.4, right), β (0.9, right), ϕ_{hyg} (0.6, *left*), c (0.5, right). (b) In this model, at these nominal values: most important τ_{inf}, least important c. Another model or another nominal point can rank them differently. (c) $\mathcal{R}_0$ decreases by approximately $0.6 \times 5\% = 3\%$. The negative sign and the biological interpretation agree: hand hygiene before more contacts blocks more transmission.

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Part II

Computing Sensitivity in Practice

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Chapter 3
Computing Sensitivity in Practice

A derivative is useful only if you can compute it honestly.

You have inherited a model from a collaborator. It is 500 lines of MATLAB code, produces reasonable epidemic projections, and has 11 parameters. Your collaborator has moved to another institution. The code runs correctly, but it uses conditional statements, lookup tables, and an internal adaptive ODE solver, none of which can be differentiated analytically. Your funding agency wants to know which parameters matter before committing to a three-year field measurement campaign.

How do you compute sensitivity indices for a model you cannot differentiate?

The answer is finite differences, and this chapter shows exactly how to do it, including how to avoid the mistakes that produce wrong results.

3.1 ▪ The Sensitivity Jacobian

Chapter 2 computed sensitivity indices one parameter at a time. For a model with n_p POIs and n_q QOIs, the complete local SA is captured by an $n_q \times n_p$ matrix; one entry per POI–QOI pair.

Definition 3.1 (Sensitivity Jacobian)**.** *Let $\vec{q} = (q_1, \ldots, q_{n_q})$ denote the vector of* QOI*s and $\vec{p} = (p_1, \ldots, p_{n_p})$ denote the vector of* POI*s. The* normalized sensitivity Jacobian *is the $n_q \times n_p$ matrix $\mathbf{S}$ with entries*

$$S_{ki} = \mathcal{S}_{p_i}^{q_k} = \frac{p_i}{q_k} \frac{\partial q_k}{\partial p_i}. \tag{3.1}$$

Each entry rescales the matching entry of the matrix of partial derivatives $\partial q_k / \partial p_i$, the Jacobian matrix of $\vec{q}$ with respect to $\vec{p}$ familiar from

calculus: row k is divided by q_k and column i is multiplied by p_i. Multiplying $\mathbf{S}$ by a vector of fractional changes in the POIs gives, to first order, the fractional change in every QOI.

For the SIR model with POIs $\{c, \beta, \tau_R, \nu_m\}$ (contact rate, transmission probability, mean infectious period, mortality rate) and QOIs $\{S(T), I(T), R(T)\}$ (the three compartment sizes at time T), the sensitivity Jacobian is a 3×4 matrix of twelve numbers. Each number answers one specific question: how much does compartment i change if POI j changes by 1%?

The tornado plots in Chapter 2 displayed single rows or columns of this matrix. The full matrix captures all POI–QOI relationships simultaneously.

Computing this matrix efficiently, in both time and accuracy, is the central computational challenge of sensitivity analysis.

A common initial reaction is to treat the sensitivity Jacobian as a collection of individual SI calculations, with the matrix label serving as convenient organization for numbers computed independently. It is more than that. The Jacobian has structure: its columns can be computed in parallel, many methods share a single baseline evaluation, and the adjoint method (Chapters 5 and 6) exploits the matrix structure to recover one complete *row* of the sensitivity Jacobian, all parameter sensitivities for one scalar QOI, from a single backward solve. For n_q QOIs, n_q backward solves after the one forward solve, $n_q + 1$ solves in all, recover all n_q rows. Understanding the Jacobian as a matrix, not just a table, is what makes efficient SA possible.

Self-check 3.1. A climate model has $n_p = 30$ POIs and $n_q = 8$ QOIs. How many individual sensitivity indices does the full SA require? If you had to compute each one separately, how many model evaluations would a naive brute-force approach require?

3.2 ▪ Finite Difference Formulas

A finite difference approximates the partial derivative $\partial q/\partial p_i$ by computing how the QOI changes when p_i alone is shifted by a small but finite amount h. Three approximations are in common use, differing in accuracy and cost.

3.2.1 ▪ Forward Difference

The simplest approximation uses one additional model evaluation:

$$\frac{\partial q}{\partial p_i} \approx \frac{q(p^* + h\vec{e}_i) - q(p^*)}{h}, \tag{3.2}$$

where $\vec{e}_i$ is the unit vector in the p_i direction and p^* is the nominal parameter vector. This is a first-order approximation: the error is $O(h)$, meaning that halving h approximately halves the error.

3.2.2 ▪ Centered Difference

Using one evaluation on each side of the nominal value gives a second-order approximation:

$$\frac{\partial q}{\partial p_i} \approx \frac{q(p^* + h\vec{e}_i) - q(p^* - h\vec{e}_i)}{2h}. \tag{3.3}$$

The error is $O(h^2)$: halving h reduces the error by a factor of four. For the same step size, centered differences are substantially more accurate than forward differences. The formula does not use $q(p^*)$, but the baseline run is still needed: it supplies the value q_k that normalizes each index in (3.1), which is why the centered count in Table 3.1 is $2n_p + 1$.

3.2.3 ▪ Fourth-Order Formula (Rarely Used in Practice)

When exceptionally high accuracy is required and model evaluations are inexpensive, one can eliminate the leading-order error term by combining four evaluations at two step sizes:

$$\frac{\partial q}{\partial p_i} \approx \frac{-q(p^* + 2h\vec{e}_i) + 8q(p^* + h\vec{e}_i) - 8q(p^* - h\vec{e}_i) + q(p^* - 2h\vec{e}_i)}{12h}. \tag{3.4}$$

The error is $O(h^4)$: halving h reduces the error by a factor of sixteen. This requires four evaluations per POI rather than two. In the applied SA literature this formula is uncommon; the centered difference (3.3) is the standard workhorse. The coefficients come from one step of Richardson extrapolation. Write $C(h)$ for the centered quotient (3.3), whose error is $(h^2/6)\, q''' + O(h^4)$ (Section 3.3). Then $[4C(h) - C(2h)]/3$ cancels the h^2 term and is exactly (3.4). Its small error is why the laboratory uses it as a reference value. Knowing which formula to use and what step size to choose are the practical skills taken up next.

3.2.4 ▪ Practical Workflow: Checking the Step with an Error Indicator

In practice, the centered difference (3.3) is the default choice. The forward difference (3.2) plays a secondary role not as a first approximation in its own right, but as a cheap error estimator: because it requires only the already-computed baseline $q(p^*)$ and the one-sided perturbed evaluation $q(p^* + h\vec{e}_i)$, no extra model runs are needed. The difference between the two approximations at the same h,

$$\text{estimated error} \approx \left| \frac{q(p^* + h\vec{e}_i) - q(p^*)}{h} - \frac{q(p^* + h\vec{e}_i) - q(p^* - h\vec{e}_i)}{2h} \right|, \tag{3.5}$$

provides a practical error indicator. Be clear about what it measures. The difference between the forward and centered quotients is dominated by the forward formula's $O(h)$ truncation term, since the centered one is already $O(h^2)$, so (3.5) estimates the error in the *forward* difference. It is a sound diagnostic for that, and a sound signal that h is in a sensible range, but it is not an error bar on the centered result and shrinking it to a tolerance does not bound the centered error. Use it to detect trouble, and set the step from the $\varepsilon_{\text{mach}}^{1/3}$ optimum in (3.9) (Section 3.3, below) rather than by driving this indicator down. Driving it down pulls h toward the much smaller forward-difference optimum (3.8), where rounding dominates the centered quotient. The centered optimum depends on higher-order derivatives that are rarely known, so in practice one evaluates it with a rough magnitude estimate and checks the indicator afterward.

The backward difference $[q(p^*) - q(p^* - h\vec{e}_i)]/h$ adds no independent check. The centered quotient is the average of the forward and backward quotients, so the backward quotient differs from the centered one by the same amount as the forward quotient, with the opposite sign. Comparing it with the centered result repeats (3.5).

3.2.5 ▪ Cost Comparison

Table 3.1 summarizes the cost of computing the complete sensitivity Jacobian by each method.

Method	Model evaluations	Error order	Best when
Forward difference	$n_p + 1$	$O(h)$	Error check only; not standalone
Centered difference	$2n_p + 1$	$O(h^2)$	Standard choice
Fourth-order FD	$4n_p + 1$	$O(h^4)$	Rare; cheap evaluations only
Analytic FSE (Chapter 4)	$n_p + 1$ solves	Solver tolerance	Differentiable model
Adjoint (Chapters 5–6)	2 solves	Solver tolerance	Many POIs, few QOIs

Table 3.1. *Cost of computing the complete sensitivity Jacobian by different methods, for a model with n_p* POI*s and n_q* QOI*s. Cost metric: finite-difference rows are counted in* model evaluations *(calls to the ODE solver or model function); FSE and adjoint rows are counted in* ODE solves*, with the baseline solve included (Section 1.2). For a system of n equations, one ODE solve $\approx n$ function evaluations per time step. The finite-difference counts assume a single baseline evaluation is reused across all* POI*s. The FSE cost follows the sequential convention of Table 4.1: the baseline solve of the n-equation model plus one augmented solve of $2n$ equations per* POI*, $n_p + 1$ solves in all. A simultaneous implementation replaces the n_p augmented solves by one solve of $n(1 + n_p)$ equations (Section 4.3). Since n_p of these solves are twice the baseline size, the FSE and finite-difference counts compare work only if the cost of a solve does not grow with its size. The adjoint cost assumes $n_q = 1$ scalar* QOI*; for n_q* QOI*s the adjoint requires $n_q + 1$ solves, the forward solve and one backward solve of size n per* QOI*. Error: the FSE and adjoint rows have no truncation error in h. Their equations give the derivative exactly, and their computed values are as accurate as the ODE solver's tolerances allow (Section 3.3).*

For the SIR model with $n_p = 4$ parameters and $n_q = 3$ state variables,

centered differences require $2 \times 4 + 1 = 9$ model evaluations. If each takes 1 second, the full SA takes 9 seconds.

For the malaria model of Chitnis, Hyman & Cushing [15] with $n_p = 17$ parameters and $n_q = 7$ state variables, centered differences require $2 \times 17 + 1 = 35$ evaluations. If each takes 1 minute: approximately 35 minutes. If each takes 1 hour: approximately 35 hours.

For a climate model with $n_p = 500$ parameters and model runs taking 6 hours each, centered differences require $2 \times 500 + 1 = 1001$ evaluations, at 6 hours each, that is approximately 250 days of computation. This is why Chapters 5 and 6 develop the adjoint method, which reduces the cost to 2 solves for one QOI, regardless of n_p.

Unifying Idea 1 (the sensitivity index is a normalized derivative; Section 1.5). Every entry of the sensitivity Jacobian is a normalized derivative: an instance of Unifying Idea 1 from Chapter 1. The finite-difference formulas in this chapter compute those derivatives numerically, for any model that can be evaluated, whether or not it can be differentiated analytically.

Self-check 3.2. For a model with $n_p = 20$ POIs and $n_q = 5$ QOIs, how many model evaluations does the centered-difference Jacobian require? (Remember: adding more QOIs is free: each pair of runs, one on each side of the nominal value, contributes an entire column of the Jacobian.) If each evaluation takes 45 seconds, how long does the complete SA take? At what value of n_p would centered differences take more than 24 hours, assuming each evaluation takes 2 minutes?

3.3 ▪ Step Size Selection: The U-Curve

The step size h controls a fundamental accuracy tradeoff. There is an optimal value: too large and the approximation misses the local curvature; too small and the computer's limited numerical precision destroys the result.

3.3.1 ▪ Two Sources of Error

Truncation error arises because finite differences are Taylor-series approximations. For the centered difference:

$$\frac{q(p+h) - q(p-h)}{2h} = q'(p) + \underbrace{\frac{h^2}{6}\, q'''(p) + \cdots}_{\text{truncation error}}. \tag{3.6}$$

This error decreases as $h \to 0$; on a log-log plot of error vs. h (with h increasing to the right, as in Figure 3.1) the truncation branch has slope $+2$.

Rounding error arises because the computed model output $\tilde{q}(p+h)$ differs from the true output $q(p+h)$ by a rounding error of order $\varepsilon_{\text{mach}}|q|$, where $\varepsilon_{\text{mach}} \approx 2.2 \times 10^{-16}$ in IEEE 754 double precision (Higham [21] treats floating-point rounding in full). When h is small, subtracting $\tilde{q}(p-h)$ from $\tilde{q}(p+h)$ amplifies this rounding error:

$$\text{rounding error in FD} \approx \frac{2\varepsilon_{\text{mach}}|q|}{2h} = \frac{\varepsilon_{\text{mach}}|q|}{h}. \tag{3.7}$$

This error *increases* as $h \to 0$; on the same log-log plot the rounding branch has slope -1. For the forward difference (3.2) the same reasoning gives a truncation error of about $(h/2)|q''(p)|$ and a rounding error of order $\varepsilon_{\text{mach}}|q|/h$; balancing the two gives the forward step in (3.8) below.

3.3.2 ▪ The U-Curve and the Optimal Step Size

The total error in the centered-difference approximation is the sum of truncation and rounding errors. As h decreases from a large value, truncation error falls while rounding error rises. The minimum total error occurs at the crossover, as measured in Figure 3.1.

Setting the two error contributions equal gives an approximate starting point for step-size selection:

$$h^*_{\text{forward}} \approx \sqrt{\varepsilon_{\text{mach}}} \cdot |p^*| \approx 1.5 \times 10^{-8} \cdot |p^*|, \tag{3.8}$$

$$h^*_{\text{centered}} \approx \varepsilon_{\text{mach}}^{1/3} \cdot \max\big(|p^*|,\, p_{\text{scale}}\big) \approx 6 \times 10^{-6} \cdot \max\big(|p^*|,\, p_{\text{scale}}\big). \tag{3.9}$$

For a parameter with nominal value $p^* = 0.3$, this suggests a starting centered-difference step of approximately $h \approx 1.8 \times 10^{-6}$.

The factor $\max(|p^*|, p_{\text{scale}})$ matters: scaling h by $|p^*|$ alone collapses to zero for a parameter whose nominal value is zero, so $p_{\text{scale}} > 0$ supplies a floor in the units of p.

These formulas should be understood as rough guides, not precise prescriptions. The balance being struck is between roundoff, of order $\varepsilon_{\text{mach}}|q|/h$, and truncation, of order $h^2|q'''|/6$; minimizing the sum gives $h^* \sim (\varepsilon_{\text{mach}}|q|/|q'''|)^{1/3}$. That ratio $|q|/|q'''|$ carries units of p^3, so its cube root is a length scale in parameter space, and the ratio is not a dimensionless quantity to be set to one. The estimates above replace that cube root with the only scale generally available in advance, the parameter's own magnitude. This assumes $|q'''| \approx |q|/|p^*|^3$, that is, that q varies on the scale of the parameter itself. Because $q'''(p)$ is unknown, the true optimal h remains model-dependent. Section 3.2 uses the error indicator (3.5) to check a step, not to choose one. The indicator answers a different question from the formulas and is not a substitute for them. Equation (3.9) tells you where to start; the indicator

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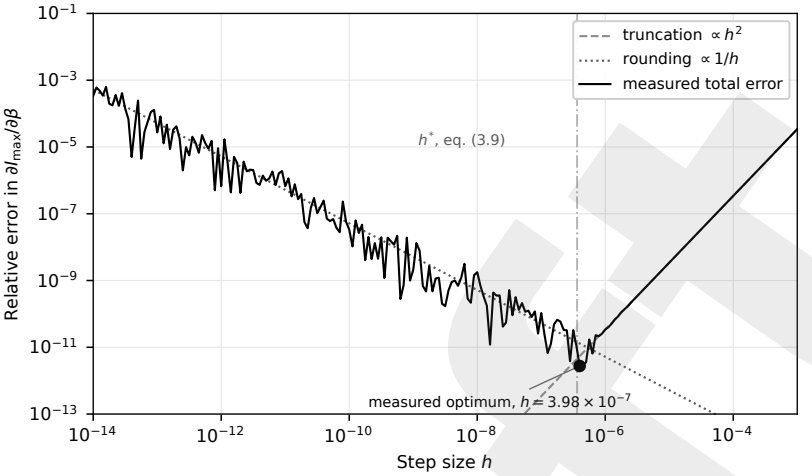


Figure 3.1. *The U-curve for centered differences in double precision, measured rather than sketched. The* QOI *is the closed-SIR peak prevalence* $I_{\max}(\beta) = N[1 - (1 + \ln r)/r]$ *with* $r = c\beta\tau_R$ *and* $N = 1000$*, whose derivative* $\partial I_{\max}/\partial\beta = 5888.391625$ *is known in closed form at* $c = 5$*,* $\tau_R = 7$*,* $\beta^* = 0.06$*, so the error can be measured. Truncation error (dashed) falls as* h^2*; rounding error (dotted) grows as* $1/h$*; the measured total (solid) is least for* h *of order* 10^{-7}*, where the relative error is of order* 10^{-12}*. Equation* (3.9) *predicts* $h^* = \varepsilon_{mach}^{1/3}|\beta^*| = 3.63 \times 10^{-7}$*, marked by the vertical line: the rule and the measurement agree in order of magnitude. The axis is the absolute step for* $\beta^* = 0.06$*; the rule* (3.9) *is relative,* $h^* \approx 6 \times 10^{-6}|\beta^*|$*. Right of the optimum the dashed truncation line lies under the measured total, and left of it the dotted rounding line lies inside the measured noise. Choosing* h *far below* h^* *is a common mistake and gives larger errors, not smaller ones; at* $h = 10^{-12}$ *the error is five orders of magnitude worse than at the optimum. Generated by* `generate_fig_ucurve.py`*.*

tells you whether the step you chose is in trouble, in either direction. Use the formula to pick h and the indicator to check it, rather than driving the indicator down as though it were an error bar on the centered result.

A common mistake in finite-difference SA is choosing h that is too small. Practitioners who set $h = 10^{-12}$ or $h = 10^{-15}$, reasoning that "smaller is more accurate," obtain results dominated by catastrophic cancellation (subtracting two nearly equal floating-point numbers, which destroys significant digits and amplifies rounding error). The U-curve explains why: below the optimal step size, rounding error grows as $1/h$ while truncation error continues to fall, so the total error increases rather than decreases. A single value of the error indicator (3.5) does not reliably detect this failure. Rounding can leave the forward and centered quotients about as close at $h = 10^{-12}$ as at the optimal step, and for some models they agree exactly. The reliable signal is how the indicator scales. In the reliable range it is the forward formula's truncation error, about $(h/2)|q''|$, so it grows tenfold when h grows tenfold. Compute it at h and at $10h$: if it does not scale this way, rounding

dominates and a larger step size is needed.

The U-curve also reveals a practical limitation for models implemented with adaptive ODE solvers. For a smooth analytic function the curve has a well-defined minimum, and in the reliable range the error indicator (3.5) shrinks in proportion to h. For a model that calls an ODE solver internally, the output is not mathematically smooth: the solver makes discrete step-size decisions that can cause the model output to change non-smoothly as a parameter varies by a small amount. In these cases the error indicator may not decrease monotonically, and the one-sided and centered approximations may disagree by much more than their expected relative gap of about $(h/2)|q''/q'|$. A relative gap of order 10^{-6} at steps near $10^{-6}|p^*|$ is normal for a smooth model and is not a sign of solver trouble. The remedy is to tighten the ODE solver's tolerances before changing h. An adaptive solver such as MATLAB's `ode45` or SciPy's `solve_ivp` takes two of them. The relative tolerance `RelTol` (`rtol` in SciPy) sets how many digits of each solution component the solver tries to keep. The absolute tolerance `AbsTol` (`atol`) takes over for a component near zero, where a relative test cannot be met. At each step the solver estimates the error that the step adds to a component y_i and shrinks the step until the estimate falls below roughly $\texttt{RelTol} \cdot |y_i| + \texttt{AbsTol}$. Both solvers default to $\texttt{RelTol} = 10^{-3}$ and $\texttt{AbsTol} = 10^{-6}$. The tolerances bound the error made in each step, not the error in the final solution. Errors from many steps accumulate and can grow along the trajectory, so a computed output can be less accurate than `RelTol` suggests. A practical check is to repeat the run with both tolerances ten times smaller: the digits that agree between the two runs give a working estimate of the accuracy. After tightening tolerances, pick h by (3.9) and check it with the indicator at two steps, as above.

ODE Solver Tolerance Rule

Before computing finite-difference sensitivities for a model that calls an ODE solver internally, tighten the solver tolerance to at least *ten times smaller* than the target step size h. As a practical starting point: set $\texttt{RelTol} = \texttt{AbsTol} = 10^{-8}$ whenever sensitivity analysis is the goal. If your target step size is $h \approx 10^{-5}|p^*|$ but the solver tolerance is 10^{-3}, the finite-difference quotient measures solver noise rather than true sensitivity. After tightening tolerances, pick h by (3.9) again and check it with the error indicator.

Self-check 3.3. For the SIR model with transmission probability $\beta^* = 0.06$, the theoretical formula (3.9) suggests a starting centered-difference step of $h \approx 3.6 \times 10^{-7}$. You then compute the centered difference and a one-sided

forward difference at this h and find they agree to four significant figures. (a) Is this step size acceptable? What does this agreement indicate? (b) Now suppose you repeat the check at $h = 10^{-12}$ and find the two approximations disagree by more than 50%. What does this signal, and what should you do?

3.4 ▪ Second-Order Terms

Chapter 2 previewed second-order terms as corrections to the first-order SI approximation. This section shows how to compute them.

3.4.1 ▪ Pure Second-Order Derivatives

The second derivative $\partial^2 q/\partial p_i^2$ measures the curvature of the QOI as a function of p_i. It can be estimated by reusing the three evaluations already computed for the centered first-order difference:

$$\frac{\partial^2 q}{\partial p_i^2} \approx \frac{q(p^* + h\vec{e}_i) - 2q(p^*) + q(p^* - h\vec{e}_i)}{h^2}. \tag{3.10}$$

No additional model evaluations are needed.

When the curvature is large, the first-order SI can overstate or understate the true change for larger perturbations, as Chapter 2 noted. It understates the change when the second-order term has the sign of the first-order term, as for $q = p^2$ under an increase in p. It overstates the change when the two terms have opposite signs, as for $q = \sqrt{p}$, whose index is $1/2$: a 10% rise in p raises q by 4.88%, not 5%. Specifically, if

$$\left| \frac{h}{2} \frac{\partial^2 q/\partial p_i^2}{\partial q/\partial p_i} \right| \ll 1,$$

the linear approximation is valid for perturbations of size h. If this ratio is not small, the sensitivity index provides only a local estimate, and Chapter 9's global SA methods are needed for a complete picture.

Unifying Idea 3 (local SA is a linearization; Section 1.5). The second-order terms computed in this section are the first quantitative correction beyond the linearization that defines every sensitivity index in this book. The first-order SI is valid when the model behaves approximately linearly near the nominal parameter values. The second-order term tells you how quickly that approximation degrades as perturbations grow. Chapter 9 provides the full global picture when the linearization is no longer adequate.

3.4.2 ▪ Mixed Second-Order Derivatives

The cross-derivative $\partial^2 q/\partial p_i\,\partial p_j$ measures how the sensitivity to p_i changes with p_j. For each pair (i, j), one additional model evaluation (at $p^* + h\vec{e}_i + h\vec{e}_j$) is required beyond the baseline and single-perturbation evaluations already performed for the first-order Jacobian:

$$\frac{\partial^2 q}{\partial p_i\,\partial p_j} \approx \frac{q(p^* + h\vec{e}_i + h\vec{e}_j) - q(p^* + h\vec{e}_i) - q(p^* + h\vec{e}_j) + q(p^*)}{h^2}. \tag{3.11}$$

For all $\binom{n_p}{2}$ pairs, this requires $\binom{n_p}{2}$ additional evaluations. For $n_p = 4$, that is 6 evaluations; for $n_p = 17$ (the malaria model), it is 136.

The cross-derivative $\partial^2 q/\partial p_i\,\partial p_j$ is large when the two POIs *interact*: varying them together is not the same as varying them separately. Chapter 9 shows that Sobol interaction indices S_{ij} are the global, variance-based analog of these local cross-derivatives.

Self-check 3.4. For the SIR model with $\mathcal{R}_0 = c\beta\tau_R$, compute $\partial^2\mathcal{R}_0/\partial c\,\partial\beta$ analytically. Is the cross-derivative large or small relative to the individual first-order derivatives $\partial\mathcal{R}_0/\partial c$ and $\partial\mathcal{R}_0/\partial\beta$? What does this imply about the interaction between c and β for $\mathcal{R}_0$?

3.5 ▪ Correlated POIs: The Most Common Mistake

> *A parameter can be uncertain without being important, and important without being uncertain. Correlation between parameters adds a third possibility: a parameter can appear important in isolation while its true effect is inseparable from another.*

Before computing any sensitivity indices, there is a question that must be asked: can these parameters actually be varied independently? In many models, the answer is no.

Every finite-difference SA computation implicitly assumes that each POI can be perturbed while all others are held exactly fixed. If the POIs are correlated, meaning that in the real world, changing one tends to change another, then this assumption is violated and the resulting sensitivity indices can be misleading. Section F.4 of Appendix F defines correlation and the coefficient ρ used below.

3.5.1 ▪ Diagnosing Correlation

Correlation among POIs arises from several sources.

Structural correlation occurs when parameters appear only as a product or ratio in the model equations. In the SIR model, the force of infection is $\lambda = c\beta I/N$. The parameters c and β appear only as the product $c\beta$. Observational data, incidence counts over time, can identify $c\beta$ as a unit but cannot separate the two factors. This kind is a property of the model's form rather than a co-movement: a policy can move c without moving β. It is grouped with the two kinds below because it needs the same care, and it differs from an interaction, which concerns how the effect of one POI depends on the value of another. Computing $\mathcal{S}_c^{\mathcal{R}_0}$ and $\mathcal{S}_\beta^{\mathcal{R}_0}$ separately is valid, and $\mathcal{S}_c^{\mathcal{R}_0}$ gives exactly the effect of changing c alone. What misleads is adding the two indices, ranking c and β as separate drivers, or estimating them separately from data that identify only the product $c\beta$. Reparameterizing to use $c\beta$ as a single POI is more natural in this case.

This observation is the starting point for *structural identifiability* analysis: the question of whether the parameters of a model can, in principle, be recovered from the observable outputs, regardless of measurement noise or sample size. A model is *structurally identifiable* in a parameter p if distinct values of p produce distinct output trajectories. The SIR model is not structurally identifiable in (c, β) separately, but is identifiable in the product $c\beta$. Sensitivity analysis reveals this: when two parameters have the same sensitivity index at every parameter value, not only at the nominal point (here $\mathcal{S}_c^q = \mathcal{S}_\beta^q$ for any QOI q), they are structurally correlated and the model cannot distinguish them. A sensitivity index of zero for a parameter is a related signal: the model output does not respond to that parameter at all, which may indicate that the parameter cannot be estimated from data. Chapter 9 returns to identifiability in the context of global SA, where variance-based methods provide computational tools for detecting structural correlation across the full parameter range. A full treatment of structural and practical identifiability is beyond the scope of this book; see Chapter 7 of Smith [60], on parameter identifiability and influence, and the references therein.

Empirical correlation occurs when parameter values are estimated from data that naturally links them. In the malaria model studied by Chitnis, Hyman & Cushing [15], the mosquito biting rate σ_v and the human biting saturation σ_h both appear in the force of infection. The paper's Appendix A computes σ_h from σ_v, the equilibrium population sizes, and field estimates of the number of bites on humans per mosquito per day, so the data link the two values. The paper then replaces them by the composite parameter $\zeta = \sigma_v\sigma_h/(\sigma_v N_v^* + \sigma_h N_h^*)$ and the ratio $\theta = \sigma_h/\sigma_v$, and it varies ζ with θ fixed, so that the two rates move together. Its stated reason is that the index of $\mathcal{R}_0$ to ζ, unlike the indices to σ_v and σ_h, does not depend on the

equilibrium population sizes (Section 4.2).

Constraint correlation occurs when a physical or biological requirement links parameters. In models where stage durations must sum to a total lifecycle, or where transition probabilities must sum to one, no parameter can change without forcing compensating changes in others.

3.5.2 ▪ The Impact of Ignoring Correlation

To see concretely what goes wrong, consider a model whose QOI is $q = p_1 + p_2$ where the two POIs satisfy $p_2 = 2p_1$ (a linear constraint). The standard sensitivity indices are:

$$\mathcal{S}^q_{p_1} = \frac{p_1}{q} \cdot 1 = \frac{p_1}{3p_1} = \frac{1}{3}, \qquad \mathcal{S}^q_{p_2} = \frac{p_2}{q} \cdot 1 = \frac{2p_1}{3p_1} = \frac{2}{3}.$$

These look reasonable: the second parameter is twice as important as the first. But in reality, p_1 and p_2 cannot be varied independently. Any change to p_1 forces a change to $p_2 = 2p_1$. The total effect of a 1% increase in p_1 is:

$$\Delta q = \frac{\partial q}{\partial p_1}\,\delta p_1 + \frac{\partial q}{\partial p_2}\,\delta p_2 = 1 \cdot (0.01p_1) + 1 \cdot (0.02p_1) = 0.03p_1.$$

The normalized total-derivative SI for p_1 is then $(p_1/q) \cdot (0.03p_1/0.01p_1) = (1/3) \cdot 3 = 1$, not $1/3$. The standard index understates the true sensitivity by a factor of three. This is equation (3.12) below with $\omega_{12} = \omega_{21} = 1$, since $p_2 = 2p_1$ makes the two proportional changes equal. The same equation gives p_2 a total index of $2/3 + 1/3 = 1$, so the two total indices sum to 2: total indices need not add to one, and here they no longer rank the two POIs, which move as one.

3.5.3 ▪ Four Practical Responses

Readers with a regression background will recognize correlated POIs *as the SA analog of multicollinearity.* In regression, when predictors are nearly collinear, the design matrix $X^\top X$ is nearly singular, coefficient estimates become unstable, and variance inflation factors (VIFs) grow large. In SA, when POIs are correlated, varying one parameter while holding others fixed may be physically impossible, as for constraint correlation. It may also be unlike any change the data show. The resulting sensitivity indices can be misleading for the same reason: the "separate" effects of the two parameters cannot be cleanly disentangled. The tools differ: VIF in regression versus reparameterization or total-derivative SI in sensitivity analysis, but the underlying statistical problem is identical: two predictors that move together cannot both be estimated or ranked in isolation.

Response 1: Diagnose first. Before any SA computation, examine the correlation structure of the POIs. Compute the Pearson correlation matrix from any available data on parameter values across studies or populations. Plot scatter matrices (a pairwise scatter matrix). A correlation well away from zero, say $|\rho_{ij}| > 0.5$, warrants a closer look; the cutoff is a rule of thumb, not a test. Data reveal only empirical correlation. Structural correlation shows in the model's form or in equal indices and calls for Response 2. Constraint correlation shows in the constraint itself, and the total derivative of Response 3 applies with the coupling the constraint fixes, as in the example above. Empirical correlation calls for Response 3 or 4.

Response 2: Reparameterize structurally correlated pairs. When two parameters appear only as a product or ratio, define that product or ratio as the natural POI. For the SIR model: use $z = c\beta$ as a single POI. Here $\mathcal{R}_0 = z\tau_R$ depends on c and β only through their product, so the composite sensitivity index is $\mathcal{S}_z^{\mathcal{R}_0} = 1$. It is not the sum $\mathcal{S}_c^{\mathcal{R}_0} + \mathcal{S}_\beta^{\mathcal{R}_0} = 2$. The sum measures a different perturbation: scaling c and β by the same fractional amount, which changes their product at twice that rate. Reparameterizing replaces two correlated POIs by one; it does not add their indices. (The distinction is developed in the exercises.)

Response 3: Total derivative for empirically correlated pairs. When empirical data establishes that a proportional change in p_i induces a proportional change in p_j with coefficient ω_{ij} (i.e., $\Delta p_j/p_j \approx \omega_{ij}\,\Delta p_i/p_i$), the total derivative SI is:

$$\mathcal{S}_{p_i}^{q}\big|_{\text{tot}} = \mathcal{S}_{p_i}^{q}\big|_{\text{std}} + \sum_{j\neq i} \mathcal{S}_{p_j}^{q}\,\omega_{ij}. \tag{3.12}$$

In plain language: ω_{ij} is the induced proportional change in p_j per unit proportional change in p_i. If a 1% increase in c tends to be accompanied by a 0.6% increase in β (from field data), then $\omega_{c\beta} = 0.6$. Estimate ω_{ij} from regression of observed parameter pairs, or from domain knowledge about shared drivers. A regression gives the associational slope (Section F.4); it stands in for the causal coefficient only when the caution below is met. Report both the standard and total-derivative SIs together with the coupling coefficients ω_{ij} so readers can assess the impact.

A caution on interpretation. The coefficient ω_{ij} should represent the *causally induced* change in p_j when p_i is actively perturbed, not merely the observed statistical correlation between them. If the correlation between c and β across field sites arises from a shared lurking variable (for example, both are higher in densely-populated urban settings) rather than from a direct biological mechanism, then a policy that reduces c through social distancing may not induce the correlated change in β. In that case the total-derivative SI overstates the intervention's effect. When the causal structure is uncertain, report both the standard and total-derivative SIs and state the assumption underlying the latter.

Response 4: Report transparently. When correlations exist but cannot be fully quantified, state explicitly in the SA report that the independence assumption was made, identify the correlated pairs, and flag which SI values are most likely affected.

Self-check 3.5. In the SIR model, the product $c\beta$ represents the effective transmission rate. Suppose a field study shows that c and β are positively coupled, with coupling coefficient $\omega_{c\beta} = 0.4$ (i.e., $\Delta\beta/\beta \approx 0.4\,\Delta c/c$), and suppose the coupling is causal: changing c induces the change in β. Use equation (3.12) to compute the total-derivative SI of $\mathcal{R}_0$ with respect to c at the nominal values. Compare with the standard SI $\mathcal{S}_c^{\mathcal{R}_0} = 1$.

3.6 ▪ Black-Box Models and the Pseudocode Template

When the model is inherited code, a compiled simulation, or a package you did not write, analytical differentiation is not possible. Finite differences are then the only option for local SA, and they work for *any* model that accepts a parameter vector and returns a QOI vector.

Students sometimes treat finite differences as an inferior approximation to be used only when analytic derivatives are unavailable. For black-box models, this framing is backwards: finite differences are not a fallback; they are the correct and complete tool. The accuracy they provide at the optimal step size is typically more than sufficient for SA decisions, and their applicability to models of arbitrary complexity makes them indispensable

in practice.

The following pseudocode computes the full normalized sensitivity Jacobian by centered differences, using the optimal step size from (3.9). Students should adapt this template for every black-box SA problem.

```
# From: software/matlab/core/sensitivity_jacobian.m
# (reference library; also reproduced in the companion notebooks)
# Works in MATLAB, Python, R, or any language - adapt syntax as needed

p_nom = [c=5, beta=0.06, tau=7, tau_m=10000]     # nominal parameter values

R0(p)  = p[c] * p[beta] * p[tau]              # model: R0 = c*beta*tau
S      = SENSITIVITY_JACOBIAN(R0, p_nom)      # returns SI vector
# S[j] = SI_{p_j}^{R0};  uses 2*n_p+1 = 9 model evaluations
```

When the QOI vanishes at the nominal point the normalized index is undefined, and the function returns `NaN` for that entry rather than substituting a regularized surrogate. The choice is deliberate: a reference scale u_{ref} has to come from the problem, and no template can supply one on the reader's behalf. Section 2.2.3 explains how to choose the scale when a regularized index is what you want. The `max` in the step-size formula prevents h from being identically zero when $p_{\mathrm{nom},i} = 0$.

To validate this template on any model, call it with the analytic sensitivity indices known from Chapter 2 and verify agreement. For the SIR model:

```
# Validation: SIR R0 = c*beta*tau - all SIs should equal 1 exactly
p_nom  = [c=5, beta=0.06, tau=7]
R0(p)  = p[c] * p[beta] * p[tau]
# step size, as implemented: h_j = h_scale * max(|p_nom[j]|, h_min/h_sca
h_scale = 6e-6;   h_min = 1e-10
S      = SENSITIVITY_JACOBIAN(R0, p_nom, h_scale, h_min)

PRINT "Sensitivity indices for R0:"
PRINT "  SI_c    =", S[c],    "  (exact: 1.0000)"
PRINT "  SI_beta =", S[beta], "  (exact: 1.0000)"
PRINT "  SI_tau  =", S[tau],  "  (exact: 1.0000)"
```

Self-check 3.6. The template above uses $h_{\min} = 10^{-10}$ as a floor for the step size. Why is this floor necessary, and what happens if a POI has nominal value exactly zero?

Adapting the template to a new model. To apply `sensitivity_jacobian` to any model, not just SIR, replace the function handle with your own model and update the nominal parameter vector:

```
# Example: pharmacokinetic two-compartment model
# p = [ka, ke, V]  (absorption rate, elimination rate, volume of distrib
# QOI = peak plasma concentration C_max

pk_model(p) = solve_pk(p)       # your function that returns C_max

p_nom = [ka=1.2, ke=0.3, V=5.0]
S = SENSITIVITY_JACOBIAN(pk_model, p_nom)
# S is a 3-element vector of SIs: [SI_ka, SI_ke, SI_V]
```

No other changes are needed: the step-size selection, the U-curve error indicator, and the normalization are all handled automatically for any differentiable model that can be called as a function. The only requirement is that `solve_pk(p)` accepts a parameter vector and returns a numeric scalar or vector of QOIs.

3.7 ▪ Computational Laboratory: SA of the SIR Model by Finite Differences

This laboratory puts the chapter's tools together and provides a complete reference implementation for black-box SA. Use the SIR ODE model with the four POIs $\{c, \beta, \tau_R, \nu_m\}$ and the three QOIs $\{S(T), I(T), R(T)\}$ at time $T = 60$ days.

Setup. Implement `SIR_model(p)` as a function that takes the parameter vector $p = [c, \beta, \tau_R, \nu_m]$ and returns $[S(60), I(60), R(60)]$ by integrating the SIR ODE. Use nominal values $c = 5$, $\beta = 0.06$, $\tau_R = 7$ days, $\nu_m = 0.0001$ day^{-1}, and initial conditions $S_0 = 999$, $I_0 = 1$, $R_0 = 0$.

Part 1: Compute the full sensitivity Jacobian. Call `SENSITIVITY_JACOBIAN(SIR_mo` `p_nom)` to produce the 3×4 sensitivity matrix. Display it as a labeled table with QOI names as row labels and POI names as column labels.

Part 2: Plot the U-curve. Fix all parameters at nominal values except β. Estimate $\partial I(60)/\partial\beta$ using centered differences for $h = 10^{-3}, 10^{-4}, \ldots, 10^{-13}$, or equivalently for relative steps h/β^* spanning the same range. Do not start at $h = 10^{-1}$: with $\beta^* = 0.06$ the left point of the centered difference would be $\beta - h = -0.04$, a negative transmission probability. Compute the absolute error using the fourth-order formula (3.4) as a reference value (computed at $h = 10^{-4}$ where it is accurate). Also plot the one-sided forward difference error at each h. Plot centered-difference error, forward-difference error, and

the discrepancy between the two (the practical error indicator (3.5)) vs. h on log-log axes. Identify the range of h where the error indicator is small and the centered approximation is reliable.

Part 3: Tornado plots. Using the Jacobian from Part 1, produce separate tornado plots for each QOI: the sensitivity of $S(60)$, $I(60)$, and $R(60)$ with respect to all four POIs. Interpret: which POI matters most for each QOI? Do the three plots give the same ranking?

Part 4: Correlation check. The contact rate c and transmission probability β appear only as the product $c\beta$ in the SIR force of infection. Confirm numerically that $\mathcal{S}_c^{I(60)} \approx \mathcal{S}_\beta^{I(60)}$ at the nominal values. Then reparameterize: define a new POI $\lambda_0 = c\beta$ and compute $\mathcal{S}_{\lambda_0}^{I(60)}$. Since $I(60)$ depends on c and β only through their product, the chain rule gives $\mathcal{S}_{\lambda_0}^{I(60)} = \mathcal{S}_c^{I(60)} = \mathcal{S}_\beta^{I(60)}$: all three are the same number, whatever that number turns out to be. Explain why the sum $\mathcal{S}_c^{I(60)} + \mathcal{S}_\beta^{I(60)}$ is then twice it, and why that is not a contradiction: the sum measures a directional perturbation that scales both c and β simultaneously, whereas $\mathcal{S}_{\lambda_0}$ treats λ_0 as a single free parameter. Report the value you compute; it is not 1.

Part 5: Correlation check from literature. The SIR force of infection is $\lambda = c\beta I/N$, so c and β appear only as the product $c\beta$ and are structurally correlated. Use the four-parameter SIR model; Part 4 has already confirmed that $\mathcal{S}_c^{I(60)} = \mathcal{S}_\beta^{I(60)}$, so start from that. Search the epidemiological literature for any reported correlations between contact rates c and transmission probabilities β across outbreak settings. Construct a 4×4 correlation matrix using whatever estimates you find. Equation (3.12) needs the coupling coefficient $\omega_{c\beta}$, not the correlation $\rho_{c\beta}$. On the relative scale the regression slope of $\Delta\beta/\beta$ on $\Delta c/c$ is $\rho_{c\beta}$ times the ratio of the coefficient of variation of β to that of c. It serves as $\omega_{c\beta}$ only if the change it describes is causal, as the caution in Section 3.5 requires. If you find no estimates, assume a moderate coupling $\omega_{c\beta} = 0.6$ and zero for all other pairs. Use equation (3.12) to compute the total-derivative SI of $I(60)$ with respect to c for this coupling. Compare with the standard SI and discuss the practical difference.

Extension. Using equation (3.10), compute the pure second-order sensitivities $\partial^2 I(60)/\partial p_i^2$ for each of the four POIs. For each parameter, assess whether the second-order term is large enough to matter for 10% perturbations. (Reuse evaluations from Part 1 to keep the additional cost minimal.)

3.8 ▪ Exercises

Exercise 3.1 *(Bloom L2).* Explain in your own words why making the step size h smaller does not always improve the accuracy of a finite-difference sensitivity estimate. Describe the two sources of error and explain why they

act in opposite directions as h changes.

Exercise 3.2 *(Bloom L3).* A model has $n_p = 15$ POIs and $n_q = 4$ QOIs.

(a) Fill in Table 3.1 for this model: how many model evaluations does each method require?

(b) If each evaluation takes 30 seconds, how long does the centered-difference Jacobian take?

(c) At what point (in terms of n_p and evaluation time) would the adjoint method of Chapters 5 and 6 be strictly necessary?

Exercise 3.3 *(Bloom L3).* For the function $q(p) = \sin(p^2)$ at $p^* = 1.3$, compute $\partial q/\partial p$ using centered differences for $h \in \{10^{-1}, 10^{-2}, \dots, 10^{-14}\}$. Compare each estimate to the exact value $q'(p) = 2p\cos(p^2)$. Plot error vs. h on log-log axes and identify the optimal h. Does it agree with the formula in equation (3.9)?

Exercise 3.4 *(Bloom L3).* For the SIR model $\mathcal{R}_0 = c\beta\tau_R$:

(a) Compute the pure second derivatives $\partial^2\mathcal{R}_0/\partial c^2$, $\partial^2\mathcal{R}_0/\partial\beta^2$, $\partial^2\mathcal{R}_0/\partial\tau_R^2$ analytically. What do you notice?

(b) Compute the cross derivatives $\partial^2\mathcal{R}_0/\partial c\,\partial\beta$, $\partial^2\mathcal{R}_0/\partial c\,\partial\tau_R$, $\partial^2\mathcal{R}_0/\partial\beta\,\partial\tau_R$ analytically.

(c) For a 10% perturbation to c and a simultaneous 10% perturbation to β, estimate the true change in $\mathcal{R}_0$ using both the first-order and the full second-order Taylor approximations. How large is the correction?

Exercise 3.5 *(Bloom L4).* In the SIR model, suppose field data suggests that the contact rate c and transmission probability β are positively correlated: across outbreak settings, when c increases by 10%, β tends to increase by 6%.

(a) Compute the standard sensitivity indices $\mathcal{S}_c^{\mathcal{R}_0}$ and $\mathcal{S}_\beta^{\mathcal{R}_0}$.

(b) Using equation (3.12), compute the total-derivative SI of $\mathcal{R}_0$ with respect to c, accounting for the correlation.

(c) Which is larger: the standard SI or the total-derivative SI? What does this mean for a policy maker who can reduce contact rates (through social distancing) but not transmission probability directly?

Exercise 3.6 *(Bloom L4).* A colleague provides you with the function `seir_model(p)` that implements a SEIR epidemic model with $n_p = 6$ parameters and returns the peak infection count as a single QOI. Each model evaluation takes approximately 0.2 seconds.

(a) How long will it take to compute the complete sensitivity Jacobian using centered differences?

(b) Apply the `sensitivity_jacobian` template to this model. What additional information do you need before interpreting the results? (Think about parameterization.)

(c) You find that two of the six parameters have $|\rho_{ij}| = 0.78$. What should you do before reporting the sensitivity indices?

Exercise 3.7 *(Bloom L5).* A government agency funds you to perform SA on a dengue fever transmission model with 9 parameters and 3 QOIs (peak infections, total deaths, days until outbreak peak). Each model evaluation takes 4 minutes.

Design the complete computational SA study. Your design should specify:

(a) Which finite-difference method you will use and why.

(b) The total computation time.

(c) How you will choose the step size for each parameter.

(d) What checks you will run before reporting any results.

(e) How you will present the results to a non-technical audience.

(f) One scenario in which the standard SA would give misleading results, and what you would do instead.

Exercise 3.8 *(Bloom L4).* *Audit exercise.* A colleague reports finite-difference sensitivity indices for a black-box model, computed with a forward difference at a relative step of $h = 10^{-12}$, and notes that this step was chosen "to make the truncation error negligible." The model is evaluated in double precision.

(a) The stated reasoning is correct as far as it goes. Explain why the conclusion does not follow, using the U-curve of Section 3.3.

(b) Estimate the order of magnitude of the total error at $h = 10^{-12}$ for a well-scaled model, and compare it with the error at the near-optimal step.

(c) The reported indices are finite, vary smoothly when plotted against the nominal values, and are internally consistent. Explain why none of those observations detects the problem, and state one diagnostic that would.

(d) Two of the reported parameters are known to be strongly correlated in the calibration data. State what the one-at-a-time indices mean under that condition and what they do not.

Exercise 3.9 *(Bloom L6).* A published sensitivity analysis reports: "All finite-difference computations used $h = 10^{-12}$ to ensure maximum precision. The code was run in double-precision floating-point arithmetic."

The authors report that for several parameters, the sensitivity indices are exactly zero to all reported decimal places.

Identify what is most likely wrong with these results and explain your reasoning quantitatively, using the U-curve analysis from Section 3.3. What should the authors have done, and how would you determine whether the zero SI values are genuine or numerical artifacts?

Notes

The analysis of finite-difference step-size selection, including the U-curve trade-off between truncation error ($O(h^2)$ for centered differences) and rounding error ($O(\varepsilon_{\text{mach}}/h)$), is a classical topic in numerical analysis. A careful treatment of the rounding-error side appears in Higham [21].

The Latin hypercube sampling design used in Chapter 9 was introduced by McKay, Beckman, and Conover [33]. Iman and Conover [25] established the rank-correlation variant that controls correlations among sampled parameters; it is one way to sample the correlated POIs of Section 3.5.

The condition-number diagnostics for identifying correlated POIs, and the connection to variance inflation factors in linear regression, are discussed in Saltelli et al. [51].

Answers to the self-checks

3.1. $30 \times 8 = 240$ sensitivity indices. Computing each index separately, with its own perturbed runs and one shared baseline run, is the naive approach: $2 \times 240 + 1 = 481$ model evaluations for centered differences (one perturbed run on each side of the

nominal value), or 240+1 = 241 for first-order (forward) differences (one perturbed run on one side; Section 3.2 defines both). A better approach perturbs one POI at a time and reads all eight QOIs from each run. It requires $30 + 1 = 31$ model evaluations for forward differences, or $2 \times 30 + 1 = 61$ for centered differences. The key: you do not need one evaluation per SI entry. For forward differences you need one evaluation per POI (plus one baseline), and then each evaluation contributes an entire column of the Jacobian. Centered differences need two per POI, one on each side.

3.2. $2 \times 20 + 1 = 41$ evaluations, regardless of $n_q = 5$; $41 \times 45 = 1{,}845$ seconds ≈ 31 minutes.

For 24 hours = 1,440 minutes with 2-minute evaluations: $2n_p + 1 \leq 720$ gives $n_p \leq 359$, so for $n_p \geq 360$, centered differences take more than 24 hours.

3.3. (a) Yes. Agreement to four significant figures shows that the forward difference's truncation error is small at this h. Rounding is small because h came from (3.9); a gap that grows tenfold at $10h$ would confirm it. For the centered formula, rounding and truncation errors are comparable at this h, by the design of (3.9). The agreement is a check on the step, not an error bar on the centered result (Section 3.2). The step size is acceptable.

(b) It signals that noise in the computed QOI, amplified by the $1/h$ in each quotient, swamps both approximations. Rounding error $\propto 1/h$, so at $h = 10^{-12}$ it is $3.6 \times 10^{-7}/10^{-12} = 3.6 \times 10^5$ times larger than at $h^* \approx 3.6 \times 10^{-7}$. For the closed-form peak of Figure 3.1 that still leaves a relative error near 10^{-6}, far below 50%, so machine rounding alone cannot open this gap. A gap this large points to a noisier output, such as a model whose ODE solver runs at a loose tolerance. The remedy is to tighten the solver tolerances, return h to the range suggested by the theoretical formula, and recheck the indicator at h and at $10h$. Moving to even larger h would re-introduce truncation error, so the reliable range lies in between.

3.4. $\partial^2 \mathcal{R}_0 / \partial c \, \partial \beta = \tau_R$ (positive, since $\mathcal{R}_0 = c\beta\tau_R$).

The first-order derivatives are $\partial \mathcal{R}_0/\partial c = \beta\tau_R$ and $\partial \mathcal{R}_0/\partial \beta = c\tau_R$.

Their sizes depend on the units of c and β, so compare the terms each contributes to a joint change. For relative changes $\delta c/c$ and $\delta\beta/\beta$ the exact change is $\Delta\mathcal{R}_0/\mathcal{R}_0 = \delta c/c + \delta\beta/\beta + (\delta c/c)(\delta\beta/\beta)$, and the last term comes from the mixed partial. It is the product of the two relative changes: small for small changes (0.01% against 2% for two 1% increases) but not for large ones (4% against 40% for two 20% increases). The mixed partial is nonzero, meaning c and β interact in their effect on $\mathcal{R}_0$: increasing both together increases $\mathcal{R}_0$ more than the sum of the individual increases. This is expected because $\mathcal{R}_0$ is multiplicative in c and β. For Sobol analysis, this interaction will produce a nonzero $S_{c\beta}$.

3.5. $\mathcal{S}_c^{\mathcal{R}_0} = 1$, $\mathcal{S}_\beta^{\mathcal{R}_0} = 1$. Here $\omega_{c\beta} = 0.4$ means $(\Delta\beta/\beta)/(\Delta c/c) = 0.4$, so equation (3.12) gives total SI $= 1 + 1 \times 0.4 = 1.4$. A 1% increase in c, with the 0.4% increase in β it induces, increases $\mathcal{R}_0$ by approximately 1.4%, not 1%. The total effect is 40% larger than the standard SI suggests. The result rests on the causal assumption: if c and β only share a driver, a policy that lowers c need not move β, and the standard SI of 1 describes its effect.

3.6. If $p_{\text{nom},i} = 0$, then $6 \times 10^{-6}|p_{\text{nom},i}| = 0$, giving $h = 0$ and a quotient 0/0, which MATLAB returns as `NaN` without an error. The floor $h_{\min} = 10^{-10}$ prevents this.

For a POI with nominal value zero, the step size is not scaled by the parameter magnitude, it is just the fixed small value 10^{-10}. The SI itself is then no help. The ratio form would divide by $p_i = 0$, and the derivative form (3.1) multiplies the derivative by $p_i = 0$, so it gives 0 whatever the derivative is. The reference routine `sensitivity_jacobian.m` therefore stops with an error when a nominal value is zero, rather than returning an index. If a POI is exactly zero at its nominal value, the investigator should either reparameterize or report the absolute (unnormalized) derivative rather than the normalized SI.

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Part III

Analytic Forward Sensitivity Analysis

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Chapter 4
Analytic Forward Sensitivity Analysis

Differentiate the model, then watch which parameters matter, and when.

Chapter 3 showed how to compute sensitivity indices for any model by finite differences. For the malaria model with 17 POIs and 7 state variables, the centered-difference Jacobian requires $2 \times 17 + 1 = 35$ model evaluations (one baseline solve plus two perturbed solves per parameter, each giving all 7 state components simultaneously), about 35 minutes if each run takes a minute, or 35 hours if each takes an hour.

There is a faster and more accurate alternative, provided the model can be differentiated analytically: differentiate the governing equations directly. The result is a new system of equations, the *forward sensitivity equations*, whose solution is the exact derivative. For an equilibrium output the cost is one linear solve per POI. For the malaria model's endemic equilibrium, that is 17 linear solves of a 7×7 system after the one run that finds the equilibrium. Each solve takes milliseconds (Section 4.2). For an output that depends on time, each POI adds one linear ODE solve instead (Section 4.4).

The catch is that the model must be differentiable (Section 7.4 treats models with a switch) and the analyst must be patient enough to derive the sensitivity equations. This chapter develops that derivation systematically, shows that it always produces a *linear* system no matter how nonlinear the original model is, and ends by quantifying the remaining computational bottleneck that motivates Chapters 5 and 6.

4.1 ▪ The FSE Derivation Pattern

The forward sensitivity equation (FSE) for any model is obtained by a single operation: differentiate the governing equation with respect to a POI and apply the chain rule. The result is always a linear equation in the sensitivity variables, regardless of whether the original model is linear or nonlinear.

4.1.1 ▪ The Pattern in Three Steps

Step 1. Write the governing equation in the form $\mathcal{F}(u;p) = 0$ or $\mathcal{F}(u;p) = f$, where u is the solution and p is the POI. For an ODE $\dot{u} = F(u;p)$, this means $\mathcal{F}(u;p) = \dot{u} - F(u;p) = 0$.

Step 2. Differentiate both sides with respect to p. Apply the chain rule to every term that depends on u. Collect terms. The result is an equation for $\partial u/\partial p$.

Step 3. Recognize that the equation from Step 2 is *linear in* $\partial u/\partial p$, even if $\mathcal{F}$ is nonlinear in u.

The linearity in Step 3 is not a coincidence. By the chain rule, every term of the differentiated equation contains $\partial u/\partial p$ at most once, to the first power, times a factor that does not depend on it. So differentiating a nonlinear equation in u produces a linear equation in $\partial u/\partial p$; the proof of Theorem 4.1 spells this out. This structural fact is what makes analytic SA computationally tractable.

Misconception to confront. A common reaction to the FSE is: "Deriving and solving these equations must be at least as hard as solving the original model." This is wrong in two ways. First, the FSE is always *linear* in the sensitivity variable, even when the original model is highly nonlinear; a linear system is categorically easier to solve than a nonlinear one. Second, the FSE for an ODE uses the *same* coefficient matrix F_u as the original model. An implicit (stiff) solver already forms F_u as it steps, and the FSE can reuse that work; an explicit solver never forms it, so you supply F_u yourself. With either kind of solver the additional cost per parameter is one linear ODE solve, not one full nonlinear solve. The FSE is not a second model; it is a linear companion to the first.

4.1.2 ▪ Application to Three Model Types

The same three-step pattern applies to algebraic systems, difference equations, and ordinary differential equations. Sections 4.2–4.3 work through each type in detail. Here we display the pattern side by side.

Model type	Governing equation	Forward sensitivity equation
Algebraic	$f(u;p)=0$	$f_u \frac{\partial u}{\partial p} = -f_p$
Linear system	$A\vec{u}=\vec{b}$	$A \frac{\partial \vec{u}}{\partial p} = \frac{\partial \vec{b}}{\partial p} - \frac{\partial A}{\partial p}\vec{u}$
ODE IVP	$\dot{u}=F(u;p)$	$\frac{d}{dt}\left[\frac{\partial u}{\partial p}\right] = F_u \frac{\partial u}{\partial p} + F_p$
OΔE (difference eq.)	$u_{n+1}=G(u_n;p)$	$\frac{\partial u_{n+1}}{\partial p} = G_u \frac{\partial u_n}{\partial p} + G_p$

In each case, the coefficient multiplying the unknown sensitivity (f_u, A, F_u, G_u) is the Jacobian of the model with respect to the solution u, evaluated at the nominal solution: at the single equilibrium point for the algebraic rows, and at each point of the nominal trajectory, $u(t)$ or u_n, for the ODE and OΔE rows. The sensitivity variable ($\partial u/\partial p$) always appears linearly. In the algebraic, linear-system and OΔE rows the right-hand side contains only known quantities; in the OΔE row $\partial u_n/\partial p$ is known from the previous step. In the ODE row the right-hand side contains the unknown itself, multiplied by F_u.

Unifying Idea 1 (the sensitivity index is a normalized derivative; Section 1.5). Every entry of the sensitivity Jacobian that this chapter computes is a normalized derivative: an instance of Unifying Idea 1 from Chapter 1. The FSE computes the raw derivative $\partial u/\partial p$ without the $O(h^2)$ truncation error of the finite-difference approach; normalizing it by p/u adds only arithmetic. For an ODE model, the solver's tolerances set the accuracy of the computed values. The FSE is not a different kind of SA result; it is the same sensitivity index, obtained by a more accurate and often faster computational route.

Self-check 4.1. For the logistic ODE $\dot{u} = ru(1-u/K)$, write down the FSE for $\partial u/\partial r$ and for $\partial u/\partial K$. Identify the coefficient F_u in each case.

4.2 ▪ Algebraic Systems and Equilibria

For static problems, equations whose solutions do not change in time, the FSE is a linear system that can be solved by standard linear algebra.

4.2.1 ▪ A Single Nonlinear Equation

For the scalar equation $f(u;p)=0$, differentiating with respect to p gives

$$f_u \frac{\partial u}{\partial p} + f_p = 0,$$

so $\partial u/\partial p = -f_p/f_u$ provided $f_u \neq 0$. This is exactly the implicit function theorem. When $f_u = 0$, the derivative is undefined, which corresponds to a bifurcation point, a situation studied in Chapter 8.

4.2.2 ▪ Linear Systems: The FSE for $A\vec{u} = \vec{b}$

For the system $A\vec{u} = \vec{b}$, differentiating with respect to any scalar parameter $p \in \{a_{ij}, b_i\}$ gives:

$$A\,\frac{\partial \vec{u}}{\partial p} = \frac{\partial \vec{b}}{\partial p} - \frac{\partial A}{\partial p}\,\vec{u}. \tag{4.1}$$

The right-hand side is a known vector once the forward solution $\vec{u}$ is known. Solving equation (4.1) requires one additional linear solve with the same coefficient matrix A for each parameter p. Since the LU factorization of A is computed only once, the cost for m parameters is one LU factorization plus m back-substitutions.

Important: Do not solve (4.1) by computing A^{-1}. Explicit matrix inversion is numerically unstable and computationally wasteful. In MATLAB, use the backslash operator (`A\b`), which solves the system by LU factorization without forming the inverse.

4.2.3 ▪ The Competitive Lotka–Volterra Model

Consider the two-species competitive Lotka–Volterra model [36]:

$$\frac{du_1}{dt} = u_1(1 - u_1 - a u_2), \tag{4.2}$$

$$\frac{du_2}{dt} = b u_2(1 - u_2 - c u_1), \tag{4.3}$$

with parameters a, b, $c > 0$ satisfying $0 < a, c < 1$. The negative cross-terms $-au_2$ and $-cu_1$ represent interspecific competition: each species inhibits the other's growth. The condition $0 < a, c < 1$ ensures that neither species is a sufficiently strong competitor to exclude the other, so a stable coexistence equilibrium exists. The coexistence equilibrium $(\bar{u}_1, \bar{u}_2)$ is

$$\bar{u}_1 = \frac{1 - a}{1 - ac}, \qquad \bar{u}_2 = \frac{1 - c}{1 - ac}. \tag{4.4}$$

To find how this equilibrium depends on a and c, we differentiate the equilibrium equations

$$f_1(\bar{u}_1, \bar{u}_2; a, c) := \bar{u}_1(1 - \bar{u}_1 - a\bar{u}_2) = 0,$$
$$f_2(\bar{u}_1, \bar{u}_2; a, c) := b\bar{u}_2(1 - \bar{u}_2 - c\bar{u}_1) = 0,$$

with respect to a to obtain the FSE

$$D_{\vec{u}}[\vec{f}]\,\frac{\partial \vec{u}}{\partial a} = -\frac{\partial \vec{f}}{\partial a}, \tag{4.5}$$

where $D_{\vec{u}}[\vec{f}]$ is the Jacobian matrix of $\vec{f}$ with respect to $\vec{u}$, evaluated at the equilibrium:

$$D_{\vec{u}}[\vec{f}]\big|_{\text{eq}} = \begin{pmatrix} 1 - 2\bar{u}_1 - a\bar{u}_2 & -a\bar{u}_1 \\ -bc\bar{u}_2 & b(1 - 2\bar{u}_2 - c\bar{u}_1) \end{pmatrix}, \tag{4.6}$$

and

$$\frac{\partial \vec{f}}{\partial a} = \begin{pmatrix} -\bar{u}_1\bar{u}_2 \\ 0 \end{pmatrix}.$$

Solving the 2×2 linear system (4.5) gives $\partial\bar{u}_1/\partial a$ and $\partial\bar{u}_2/\partial a$ exactly. The sensitivity indices follow from Definition 2.1.

A key observation: although the equilibrium equations (4.2)–(4.3) are nonlinear, the FSE (4.5) is a *linear system* in the sensitivity variables. This is always true for equilibrium problems, as the following result shows.

Theorem 4.1 (Linearity of the FSE)**.** *Let $\vec{f}(\vec{u};\vec{p}) = \vec{0}$ be a system of nonlinear equations with $\vec{f}$ differentiable in $\vec{u}$ and $\vec{p}$. The forward sensitivity equation for $\partial\vec{u}/\partial p_j$ is the linear system*

$$D_{\vec{u}}[\vec{f}]\,\frac{\partial \vec{u}}{\partial p_j} = -\frac{\partial \vec{f}}{\partial p_j},$$

provided the Jacobian $D_{\vec{u}}[\vec{f}]$ is nonsingular.

Proof. Differentiating $\vec{f}(\vec{u}(\vec{p});\vec{p}) = \vec{0}$ with respect to p_j and applying the chain rule gives

$$D_{\vec{u}}[\vec{f}]\,\frac{\partial \vec{u}}{\partial p_j} + \frac{\partial \vec{f}}{\partial p_j} = \vec{0},$$

where $D_{\vec{u}}[\vec{f}]$ denotes the Jacobian matrix of $\vec{f}$ with respect to $\vec{u}$, evaluated at the nominal solution $(\vec{u}(\vec{p});\vec{p})$. Rearranging yields

$$D_{\vec{u}}[\vec{f}]\,\frac{\partial \vec{u}}{\partial p_j} = -\frac{\partial \vec{f}}{\partial p_j}.$$

The right-hand side $-\partial\vec{f}/\partial p_j$ is a known vector once $\vec{u}(\vec{p})$ is available. Linearity in $\partial\vec{u}/\partial p_j$ follows from the fact that differentiation is itself a linear operation: the quantity $\partial\vec{u}/\partial p_j$ appears in the left-hand side only through the matrix-vector product $D_{\vec{u}}[\vec{f}]\,(\partial\vec{u}/\partial p_j)$. No powers, products, or compositions of $\partial\vec{u}/\partial p_j$ appear, regardless of the nonlinearity of $\vec{f}$ in $\vec{u}$. Nonsingularity of $D_{\vec{u}}[\vec{f}]$ guarantees a unique solution. □

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This theorem is one of the most practically useful facts in SA. Regardless of how complex or nonlinear the model is, its FSE at an equilibrium is always a linear system, and linear systems can be solved efficiently.

Self-check 4.2. For the competitive Lotka–Volterra model with $a = 0.4$, $b = 1$, $c = 0.3$: (a) Compute the coexistence equilibrium using equation (4.4). (b) Assemble the Jacobian (4.6) at this equilibrium. (c) Solve the FSE for $\partial\bar{u}_1/\partial a$ and $\partial\bar{u}_2/\partial a$. (d) Compare with the analytic derivatives of (4.4).

4.2.4 ▪ The Car Rental Problem: Sensitivity of an OΔE Fixed Point

Consider the car rental Markov chain: two airports A and B with transition probabilities $\alpha = \Pr(A \to A)$ and $\beta = \Pr(B \to B)$. The discrete-time model is

$$A_{n+1} = \alpha A_n + (1-\beta)B_n, \tag{4.7}$$

$$B_{n+1} = (1-\alpha)A_n + \beta B_n. \tag{4.8}$$

The fixed point (A^*, B^*) gives the long-run fractions of cars at each airport, so it is normalized by $A^* + B^* = 1$:

$$A^* = \frac{1-\beta}{2-\alpha-\beta}, \qquad B^* = \frac{1-\alpha}{2-\alpha-\beta}.$$

Write M for the transition matrix of (4.7)–(4.8). The fixed point solves $(M-I)\vec{x} = \vec{0}$ with $\vec{x} = (A^*, B^*)$, and $M-I$ is singular because each column of M sums to one. Differentiating gives $(M-I)\,\partial\vec{x}/\partial\alpha = -(\partial M/\partial\alpha)\,\vec{x}$, and the normalization adds $\partial A^*/\partial\alpha + \partial B^*/\partial\alpha = 0$. Replacing one row of $M-I$ by this condition gives a system of the form (4.1) with a nonsingular matrix in the role of A. The result is:

$$\mathcal{S}_\alpha^{A^*} = \frac{\alpha}{2-\alpha-\beta}, \qquad \mathcal{S}_\alpha^{B^*} = -\frac{\alpha(1-\beta)}{(1-\alpha)(2-\alpha-\beta)}. \tag{4.9}$$

The pair for β follows by exchanging the roles of the two airports, and comparing the two pairs tells management which transition probability has the greater effect on the long-run car distribution. At nominal values $\alpha = 0.7$ and $\beta = 0.3$, the formula gives $\mathcal{S}_\alpha^{A^*} = 0.7/(2-0.7-0.3) = 0.70$: a 1% increase in the probability of keeping a car at airport A produces a 0.70% increase in the long-run fraction of cars at A.

4.3 ▪ Forward Sensitivity for ODE Systems

For a dynamical model governed by an ODE, the sensitivity $\partial u/\partial p$ changes over time as the solution evolves. The FSE is itself an ODE that must be solved alongside the original model.

4.3.1 ▪ Derivation of the ODE FSE

Consider the scalar, autonomous ODE with parameter p:

$$\frac{du}{dt} = F(u;p), \qquad u(0) = u_0(p). \tag{4.10}$$

Define the sensitivity variable $w(t) = \partial u(t)/\partial p$. Differentiating (4.10) with respect to p and exchanging the order of $\partial/\partial t$ and $\partial/\partial p$ (allowed when F is continuously differentiable in u and p, the standard condition under which the solution depends differentiably on the parameter):

$$\frac{d}{dt}\left[\frac{\partial u}{\partial p}\right] = \frac{\partial F}{\partial u}\,\frac{\partial u}{\partial p} + \frac{\partial F}{\partial p}. \tag{4.11}$$

In terms of w:

$$\dot{w} = F_u(u;p)\,w + F_p(u;p), \qquad w(0) = \frac{du_0}{dp}. \tag{4.12}$$

The coefficient $F_u(u(t);p)$ depends on the forward solution $u(t)$, so the FSE (4.12) must be solved simultaneously with the original ODE (4.10). Together they form an *augmented system* of two ODEs.

For the vector ODE $\dot{\vec{u}} = \vec{F}(\vec{u};\vec{p})$ with n state variables and m parameters:

$$\frac{d}{dt}\left[\frac{\partial \vec{u}}{\partial p_j}\right] = J_F(\vec{u};\vec{p})\,\frac{\partial \vec{u}}{\partial p_j} + \frac{\partial \vec{F}}{\partial p_j}, \tag{4.13}$$

where $J_F = \partial\vec{F}/\partial\vec{u}$ is the $n \times n$ Jacobian of $\vec{F}$ with respect to $\vec{u}$. One equation (4.13) is needed per parameter p_j, each an n-dimensional ODE system. For m parameters: the augmented system has $n + mn = n(1+m)$ equations.

Theorem 4.1 extends to the dynamical case: each FSE is *linear in the sensitivity variable* $\partial\vec{u}/\partial p_j$, even though $\vec{F}$ is nonlinear in $\vec{u}$. The coefficient $J_F(\vec{u}(t);\vec{p})$ is time-varying, but the structure is linear. If the trajectory settles at an equilibrium at which every eigenvalue of J_F has negative real part, the sensitivity settles too: its time derivative tends to zero, and the limit of (4.13) is the linear system of Theorem 4.1. The ODE FSE of this section then ends at the equilibrium FSE of Section 4.2.

Self-check 4.3. Write down the augmented ODE system (original ODE + FSE) for the logistic equation $\dot{u} = ru(1-u/K)$ with respect to both parameters r and K simultaneously. How many total ODEs does the augmented system have?

4.3.2 ▪ Time-Dependent Sensitivity Indices

For dynamical models, the sensitivity index is time-dependent:

$$\mathcal{S}_p^u(t) = \frac{p}{u(t)}\, w(t) = \frac{p}{u(t)}\, \frac{\partial u(t)}{\partial p}. \tag{4.14}$$

Where $u(t)$ is zero or near zero, as $R(t)$ is at $t = 0$, Remark 2.2 applies. The relative importance of POIs can change over time. A parameter that strongly influences the early phase of an epidemic may become less important as the system approaches equilibrium.

4.3.3 ▪ The SIR Model: Full FSE Treatment

Chapter 2 introduced the SIR model without vital dynamics (Equations (2.17)–(2.19)). For the FSE treatment we extend it to include demography (births and deaths), adding a fourth POI $\tau_m = 1/\nu_m$ (mean residence time). This extension does not change the analysis for c, β, and τ_R; it adds the sensitivity with respect to τ_m and illustrates how the FSE scales to larger parameter sets. The chapter keeps c and β as two POIs, although Section 3.5 recommends merging them into $c\beta$, so that the tie between their indices can be seen. A study would carry $c\beta$ alone and save three equations.

The SIR model with demography is:

$$\frac{dS}{dt} = \nu_m N - \frac{c\beta SI}{N} - \nu_m S, \tag{4.15}$$

$$\frac{dI}{dt} = \frac{c\beta SI}{N} - (\gamma_R + \nu_m) I, \tag{4.16}$$

$$\frac{dR}{dt} = \gamma_R I - \nu_m R. \tag{4.17}$$

With interpretable POIs $\vec{p} = (c, \beta, \tau_R, \tau_m)$ where $\tau_R = 1/\gamma_R$ (mean infectious period, days) and $\tau_m = 1/\nu_m$ (mean residence time, days), the augmented system for a single POI p_j adds 3 sensitivity ODEs:

$$\frac{dw_{Sj}}{dt} = J_{11} w_{Sj} + J_{12} w_{Ij} + J_{13} w_{Rj} + \frac{\partial F_S}{\partial p_j}, \tag{4.18}$$

$$\frac{dw_{Ij}}{dt} = J_{21} w_{Sj} + J_{22} w_{Ij} + J_{23} w_{Rj} + \frac{\partial F_I}{\partial p_j}, \tag{4.19}$$

$$\frac{dw_{Rj}}{dt} = J_{31} w_{Sj} + J_{32} w_{Ij} + J_{33} w_{Rj} + \frac{\partial F_R}{\partial p_j}, \tag{4.20}$$

where J_{ij} are the entries of the Jacobian J_F of (F_S, F_I, F_R) with respect to (S, I, R); Chapter 6 writes J_F out in full in (6.20). The equations contain the

rates γ_R and ν_m, while the POIs are durations, so each forcing term takes one more chain-rule step: $\partial F/\partial \tau_R = -\gamma_R^2\, \partial F/\partial \gamma_R$, as in the derivation of (2.21), and likewise $\partial F/\partial \tau_m = -\nu_m^2\, \partial F/\partial \nu_m$. For $m = 4$ POIs, the full augmented system has $3 + 4 \times 3 = 15$ ODEs.

The initial conditions for the FSE are zero: $w_{Sj}(0) = w_{Ij}(0) = w_{Rj}(0) = 0$ (assuming the initial conditions do not depend on the parameters).

Figure 4.1 shows the time-dependent SI for $I(t)$ with respect to c, β, τ_R, and τ_m. Through the growth phase the contact rate c and transmission probability β dominate, and they peak near day 29 at 6.799. The mean infectious period τ_R overtakes them in absolute value only in a narrow window that opens at the epidemic peak, day 44, and closes at day 51, when what remains to be settled is the timing of the turnover rather than its size. After day 51 transmission dominates again, and by day 90 its index has grown to -8.6 against -0.7 for τ_R: a larger contact rate burns through the susceptible pool sooner, so it leaves fewer infectious people late, and the index that was strongly positive during growth becomes strongly negative during decline. Two lessons follow, and neither is visible from a single number. The ranking changes, and it changes twice rather than once. The sign changes too, at day 48 for c and β: the same parameter that drives the epidemic up is the one that has already exhausted it.

4.4 ▪ The Proliferation Problem and Its Consequences

> *Before improving the model, find out which knob actually moves the machine.*

The FSE has no finite-difference truncation error and is efficient for a single parameter. For m parameters the cost scales linearly, but there are two ways to arrange it and they are not equally cheap. The *simultaneous* form integrates one system of dimension $n(1 + m)$, carrying every sensitivity block alongside the state; that is what §4.3 constructs. The *sequential* form runs m separate solves of dimension $2n$, one per parameter. Both give identical sensitivities, but the simultaneous form is cheaper because the state trajectory and its step-size selection are computed once rather than m times. For the four-parameter SIR model of this chapter the simultaneous form needed about 1900 right-hand-side evaluations against about 6700 for the sequential, a factor of roughly 3.5 (SciPy's default Runge–Kutta solver at relative tolerance 10^{-10} and absolute tolerance 10^{-12}; the sequential count is the four $2n$-equation solves, with no separate baseline). Table 4.1 gives the sequential cost for representative models.

Counting convention. Throughout this section, "solve counts" refer to the

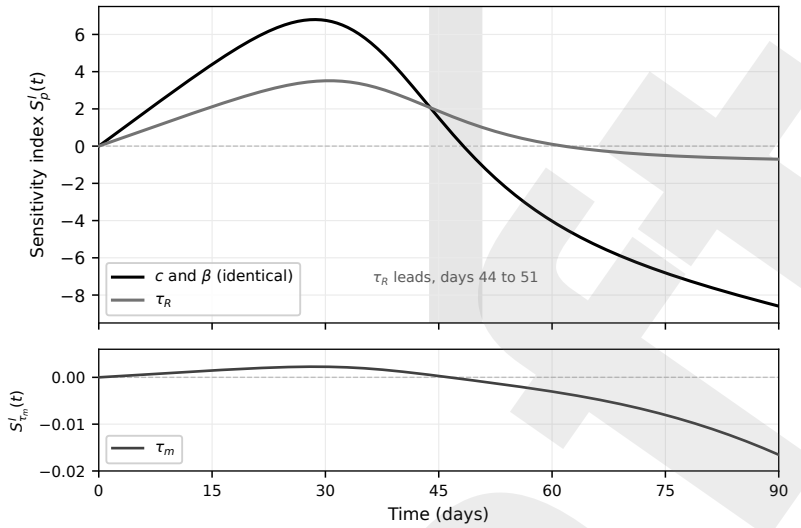


Figure 4.1. *Computed time-dependent sensitivity indices $\mathcal{S}_p^I(t)$ for the SIR model at the nominal parameters of this chapter ($c = 5$, $\beta = 0.06$, $\tau_R = 7$, $\tau_m = 10^4$, $N = 1000$, with $S(0) = 999$, $I(0) = 1$, $R(0) = 0$), computed by centered differences in each* POI *(relative step 10^{-6}), which agree with the FSE to the digits quoted. The indices for c and β are equal to machine precision, differing by about 10^{-15}, and are drawn as a single curve: the two* POI*s enter only through the product $c\beta$, so no computation can separate them. Both peak at* 6.799 *near day 29 and cross zero at day 48. The mean infectious period τ_R peaks later, at* 3.513 *near day 31, and crosses zero at day 62. It leads in absolute value only in the shaded window, which opens at the epidemic peak on day 44 and closes on day 51; before and after it, transmission dominates. The mean residence time τ_m is three orders of magnitude smaller and so is given its own panel with its own zero line; it changes sign at day 46 rather than staying negative throughout. Generated by* `generate_fig_sir_timevarySI.py`*, which reproduces these values byte for byte.*

total number of ODE solves, including the one baseline forward solve (of the original n-equation system) required before any sensitivity computation begins. In the sequential arrangement the FSE adds m augmented solves on top of this baseline, for a total of $m+1$; the simultaneous arrangement adds one solve of a larger system instead. The counts below use the sequential convention, which is the conservative one. Each $2n$-equation solve carries the state, so in that arrangement the baseline is not strictly needed; the book counts it anyway, as it counts the forward solve of every method (Section 1.2).

Figure 4.2 shows what those solves buy: the normalized indices for the SIR model vary over the epidemic, so a single end-of-run number would hide the ranking changing with time. For the SIR model, 5 solves is extremely fast in practice. For the climate model, 201 solves at 6 hours each is just

Model	n states	m POIs	Augmented size	FSE cost
SIR	3	4	$3 + 12 = 15$	5 ODE solves
SEIR (adds Exposed compartment)	4	6	$4 + 24 = 28$	7 ODE solves
Malaria model	7	17	$7 + 119 = 126$	18 ODE solves
Climate model	500	200	$500 + 100{,}000$	201 ODE solves

Table 4.1. *Augmented system size and FSE cost for representative models. The FSE cost column follows the sequential $m + 1$ convention: m augmented solves plus the one baseline forward solve. The costs are for an output that depends on time; an equilibrium output needs one model run and then one linear solve per* POI *(Section 4.2). A simultaneous implementation replaces these by a single solve of dimension $n(1+m)$ and is typically several times cheaper. For the climate model, 201 solves at 6 hours each is just over 50 days of computation. The count prices each augmented solve as one model run, though a larger system costs more per step, so those 50 days are a lower bound. This is the cost that motivates the adjoint method in Chapter 6.*

over 50 days. This is the computational wall that motivates the adjoint approach.

Unifying Idea 2 (the adjoint is the transpose; Section 1.5). Table 4.1 reveals the bottleneck that Chapters 5 and 6 resolve. The sensitivity Jacobian can be computed by finite differences, by the analytic FSE or by the adjoint, and the answer is the same matrix of normalized derivatives. The adjoint (Unifying Idea 2 of Chapter 1, developed in Chapter 5) reduces the cost from $m + 1$ solves to 2 solves for one QOI by exploiting the transposed structure of the problem. The cost reduction is proportional to m; for $m = 200$, the adjoint is 100 times faster than the FSE.

Unifying Idea 3 (local SA is a linearization; Section 1.5). Every sensitivity counted in Table 4.1 is a linearization. The FSE equations give the exact derivative of a nonlinear model, which is a linear approximation to the model's response. That approximation is valid when parameters are near their nominal values. The proliferation problem scales with m; the validity limitation is independent of m. Chapter 8 addresses the second problem; Chapters 5 and 6 address the first.

Self-check 4.4. For the malaria model with $n = 7$ state variables and $m = 17$ POIs, and one response functional, a scalar QOI (say, the infected-host fraction $i_h(T)$ at a final time T): (a) How many ODE solves does the FSE approach require? (b) If each ODE solve takes 2 minutes, how long does the complete FSA take? (c) How many ODE solves will the adjoint method in Chapter 6 require?

4.5 ▪ Choosing a Method

With three computational methods now available (finite differences, analytic FSE, and the adjoint previewed above), the right choice depends on

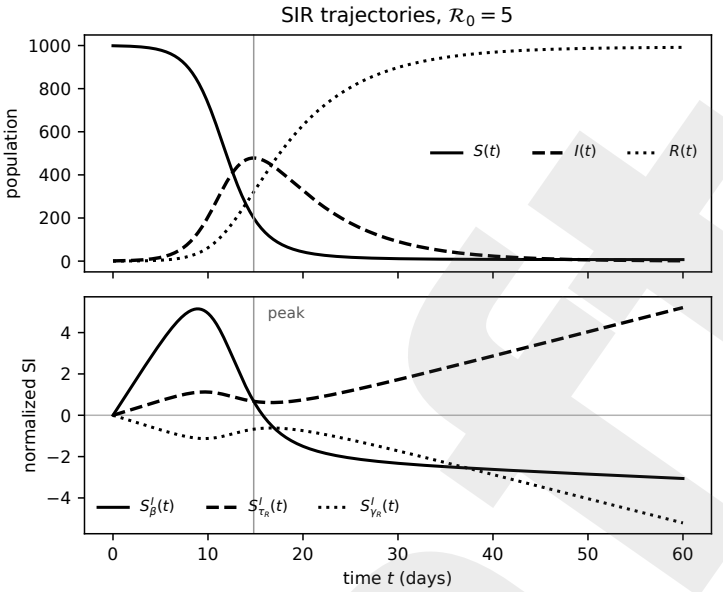


Figure 4.2. *Forward sensitivity analysis of the SIR model via the FSE. The figure integrates the SIR model without demography at $\mathcal{R}_0 = c\beta\tau_R = 5$, reached by raising $c\beta$ to 5/7 per day with $\tau_R = 7$ days; in the legend, β stands for that rate. Top: compartment trajectories. Bottom: time-dependent normalized sensitivity indices $S^I_\beta(t)$, $S^I_{\tau_R}(t)$, and $S^I_{\gamma_R}(t)$. Before the epidemic peak, transmission drives $I(t)$ upward ($S^I_\beta, S^I_{\tau_R} > 0$); after the peak, recovery dominates, and the plotted index $S^I_{\tau_R}$ keeps growing, reaching $+5.21$ at $t = 60$ while S^I_β has turned negative. Stated in the recovery rate the same fact reads $S^I_{\gamma_R} = -5.21 < 0$ with magnitude increasing, since $\gamma_R = 1/\tau_R$ flips the sign (§2.5); the figure plots the duration, not the rate. By $t = 60$, $I(t)$ has fallen below 1% of its peak, so Remark 2.2 applies to those late values. The sensitivity indices are not constant in time. The figure uses $\mathcal{R}_0 = 5$ rather than the chapter's nominal $\mathcal{R}_0 = c\beta\tau_R = 2.1$ so that the post-peak regime is legible: at the nominal value the peak falls near day 44, as Figure 4.1 shows, leaving too little of a 60-day window after it to see the indices exchange rank. The two values do not behave alike for τ_R. At the nominal value $S^I_{\tau_R}$ falls after the peak and turns negative near day 62 (Figure 4.1). At $\mathcal{R}_0 = 5$ it stays positive and keeps growing.*

the model and the problem. Chapter 7 collects the full comparison once automatic differentiation is also in hand; Chapter 9 extends it to the global methods.

For this course, the default progression is:

1. Start with finite differences (Chapter 3) to get SA results quickly, with no derivation needed.
2. Switch to analytic FSE (this chapter) when the model is differentiable and accuracy matters.
3. Use the adjoint (Chapters 5–6) when m is large compared with the number of QOIs (rule (5.5)).
4. Check for correlated POIs before reporting any results (Chapter 3).
5. Consider global SA (Chapter 9) when parameter uncertainty is large.

A common assumption is that finite differences are adequate for any SA study once the step size is chosen correctly. For black-box models, this is true: FD is the only option and produces reliable results at the optimal step size. For differentiable models, the analytic FSE is simultaneously more accurate (no $O(h^2)$ truncation error) and often faster (no repeated model evaluations at perturbed parameter values). In its sequential form the FSE needs the same $m + 1$ solves as forward differences, so its saving in time comes from the simultaneous form (Section 4.4); its gain in accuracy holds in both. The two methods give the same numbers when FD is implemented correctly; the FSE gives those same numbers without truncation error and without the step-size selection problem. For an ODE model its computed values still carry the solver's error, set by `RelTol` and `AbsTol`, so set them tightly, as for finite differences (`RelTol = AbsTol =` 10^{-8}, Section 3.3).

4.6 ▪ Computational Laboratory: Analytic FSA of the SIR Model

This laboratory implements the augmented ODE system for the SIR model and compares the analytic FSE approach with the finite-difference method from Chapter 3.

Setup. Use the SIR model (4.15)–(4.17) with POIs $\vec{p} = (c, \beta, \tau_R, \tau_m)$ (interpretable parameterization), nominal values $c = 5$, $\beta = 0.06$, $\tau_R =$ 7 days, $\tau_m =$ 10,000 days, and initial conditions $S_0 = 999$, $I_0 = 1$, $R_0 = 0$, $N = 1000$.

Part 1: Implement the augmented ODE system. Implement a function `sir_augmented(t, y, c, beta, tau_R, tau_m)` that returns the derivatives for the full 15-equation augmented system: 3 original state equations plus $4 \times 3 = 12$ FSE equations. The FSE equations require the Jacobian of (F_S, F_I, F_R) with respect to (S, I, R).

State vector layout. The 15-component state vector `y` is organized as follows:

Index	Meaning
1	$S(t)$
2	$I(t)$
3	$R(t)$
4–6	$(\partial S/\partial c,\ \partial I/\partial c,\ \partial R/\partial c)$
7–9	$(\partial S/\partial \beta,\ \partial I/\partial \beta,\ \partial R/\partial \beta)$
10–12	$(\partial S/\partial \tau_R,\ \partial I/\partial \tau_R,\ \partial R/\partial \tau_R)$
13–15	$(\partial S/\partial \tau_m,\ \partial I/\partial \tau_m,\ \partial R/\partial \tau_m)$

The initial condition is `y0 = [S0; I0; 0; zeros(12,1)]`. The sensitivity variables all start at zero because the initial conditions do not depend on the

parameters. The reference library provides `software/matlab/fse/sir_augmented.m` for this system, to check against after completing your own implementation.

Part 2: Solve and extract sensitivities. Solve the augmented system from $t = 0$ to $t = 90$ days using an ODE solver. Extract the time-dependent sensitivity indices $\mathcal{S}_c^I(t)$, $\mathcal{S}_\beta^I(t)$, $\mathcal{S}_{\tau_R}^I(t)$, $\mathcal{S}_{\tau_m}^I(t)$.

Part 3: Plot time-dependent SIs. Plot all four $\mathcal{S}_{p_j}^I(t)$ curves on a single axis, alongside the epidemic curve $I(t)$. Identify: (a) at $t = 10$ days, which parameter matters most? (b) at $t = 50$ days, has the ranking changed? (c) which parameter has the largest absolute SI at the epidemic peak?

Part 4: Compare with finite differences. At $t = 30$ days and $t = 70$ days, compare the FSE-based SI values against the centered-difference estimates from Chapter 3. Are they equal to 4 significant figures? Report any discrepancies and explain their likely source.

Part 5: Cost comparison. Time the two approaches: the augmented ODE (all 4 sensitivities at once) versus centered differences, which call `SIR_model` $2 \times 4 + 1 = 9$ times (one baseline plus a pair per POI). Which is faster? Which is more accurate? At what value of m (number of POIs) would you expect the two approaches to take approximately equal time?

4.7 ▪ Exercises

Exercise 4.1 *(Bloom L2).* State the three-step FSE derivation pattern in your own words, without using any symbols. Then explain, in one sentence, why the FSE is always linear in the sensitivity variable even when the original equation is nonlinear.

Exercise 4.2 *(Bloom L3).* For the nonlinear equation $g(u; p) := u^3 - 3pu + 2 = 0$:

(a) Write the FSE for $\partial u/\partial p$.

(b) At $p = 1$, find all real roots u numerically.

(c) Compute $\mathcal{S}_p^u$ at each root.

(d) At $p = 1$, for which root is the sensitivity largest in magnitude? Verify your answer at the simple root near $u = -2$ using a centered finite difference with optimal h.

(e) Explain why the same centered difference cannot be used at the positive root. For $p < 1$ the two positive roots are complex, so no nearby real positive branch exists on the left of $p = 1$. Show instead, by one-sided continuation for $p > 1$, that the two positive roots satisfy

$u_\pm \approx 1 \pm \sqrt{p-1}$, and explain why this square-root scaling makes $\partial u/\partial p$ unbounded as $p \downarrow 1$.

Exercise 4.3 *(Bloom L3).* For the logistic ODE $\dot{u} = ru(1-u/K)$, $u(0) = u_0$, with $r = 1$, $K = 10$ and $u_0 = 1$:

(a) Write the FSE for $w_r = \partial u/\partial r$.

(b) Verify that the FSE is linear in w_r by identifying the coefficient (which depends on $u(t)$) and the forcing term.

(c) Implement the augmented system (u, w_r, w_K) and solve from $t = 0$ to $t = 20$.

(d) Plot $\mathcal{S}_r^u(t)$ and $\mathcal{S}_K^u(t)$. Interpret: early in the growth phase vs. near the carrying capacity.

Exercise 4.4 *(Bloom L3).* For the competitive Lotka–Volterra model (4.2)–(4.3) with $a = 0.5$, $b = 2$, $c = 0.4$:

(a) Compute the coexistence equilibrium.

(b) Assemble and solve the FSE for $\partial\vec{u}/\partial c$.

(c) Compute the normalized sensitivity indices $\mathcal{S}_c^{\bar{u}_1}$ and $\mathcal{S}_c^{\bar{u}_2}$.

(d) Using equation (4.4), compute the same indices analytically. Verify agreement.

Exercise 4.5 *(Bloom L4).* The SEIRS model (Susceptible-Exposed-Infectious-Recovered-Susceptible) has 4 state variables and 7 parameters.

(a) How many ODEs are in the augmented FSE system for all 7 parameters?

(b) If each ODE solve takes 3 minutes, how long does the full FSA take using the FSE approach? The adjoint approach (Chapter 6)?

(c) Suppose a modeler decides to reduce computation time by computing sensitivities only for the 3 parameters they expect to be most important, based on biological intuition. Identify one scenario where this strategy would lead to a misleading conclusion.

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Exercise 4.6 *(Bloom L4).* Using your MATLAB implementation from the laboratory:

(a) At three time points, $t = 5$ (early epidemic), $t = t_{\text{peak}}$ (peak infections), and $t = 80$ (late); report the tornado plot of SIs for $I(t)$.

(b) Does the parameter ranking change across the three time points?

(c) A public health officer says: "We should target the parameter with the largest SI at the epidemic peak, since that is when the most people are affected." Critique this statement using your results.

Exercise 4.7 *(Bloom L5).* A colleague is modeling cholera transmission with a SEIR-type model having 8 parameters and 4 state variables. She asks for advice on the SA methodology. Design a complete FSA study, specifying:

(a) Whether to use finite differences or analytic FSE, and why.

(b) How you would validate the implementation.

(c) What QOIs you would compute sensitivity indices for, and at what time points.

(d) How you would present the results to identify the two most critical intervention targets.

(e) One situation in which the FSE results might be misleading, and what additional analysis you would recommend.

Exercise 4.8 *(Bloom L6).* A published paper reports: "We performed sensitivity analysis of the SEIR model using the forward sensitivity equations. All sensitivity indices were computed at $t = 365$ days, giving a single set of values for the entire epidemic."

The model simulates a disease whose outbreak dynamics play out over approximately 60 days, but the model runs for a year.

Identify at least two methodological issues with this approach. For each issue, state what the authors should have done instead and explain what information they likely missed.

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Notes

The forward sensitivity equation framework, differentiating the governing equation with respect to parameters to obtain a linear system for the sensitivities, is the continuous analog of forward-mode algorithmic differentiation (Chapter 7). Early systematic treatments appear in the Russian literature on sensitivity theory; an accessible modern account is given by Cacuci [10].

The SIR model with demography, adding birth and death rates ν_m to the basic Kermack–McKendrick model, is treated in detail by Anderson and May [1], Chapter 3. Van den Driessche and Watmough [65] give a criterion for the existence and stability of endemic equilibria for $\mathcal{R}_0$ near one. The SEIR extension, adding an exposed compartment with mean latent period τ_E, is analyzed in Murray [36].

The proliferation problem (cost growing linearly in m) motivates the adjoint approach of Chapters 5 and 6. Cacuci [9] compares the two formalisms: the forward one suits few parameter changes and many responses, and the adjoint one is more economical for many parameters and few responses.

Answers to the self-checks

4.1. $F(u; r, K) = ru(1 - u/K)$. Computing partial derivatives: $F_u = r(1 - 2u/K)$, $F_r = u(1 - u/K)$, $F_K = ru^2/K^2$.

FSE for $w_r = \partial u/\partial r$: $\dot{w}_r = r(1 - 2u/K)\, w_r + u(1 - u/K)$.

FSE for $w_K = \partial u/\partial K$: $\dot{w}_K = r(1 - 2u/K)\, w_K + ru^2/K^2$. Both start from $w_r(0) = w_K(0) = 0$ when $u(0)$ does not depend on r or K.

Both FSEs have the same coefficient $F_u = r(1 - 2u/K)$ multiplying the unknown sensitivity; only the right-hand-side forcing term differs. The coefficient depends on the forward solution $u(t)$, so the FSE must be solved simultaneously with the original ODE.

4.2. (a) $\bar{u}_1 = (1-0.4)/(1-0.12) = 0.6/0.88 \approx 0.6818$; $\bar{u}_2 = (1-0.3)/0.88 = 0.7/0.88 \approx 0.7955$.

(b) $D_u f = \begin{pmatrix} 1 - 2(0.6818) - 0.4(0.7955) & -0.4(0.6818) \\ -1(0.3)(0.7955) & 1 - 2(0.7955) - 0.3(0.6818) \end{pmatrix} \approx \begin{pmatrix} -0.6818 & -0.2727 \\ -0.2386 & -0.7955 \end{pmatrix}$

(c) Since $f_1 = u_1(1 - u_1 - a u_2)$ gives $\partial f_1/\partial a = -u_1 u_2$ and f_2 does not involve a, the right-hand side is $-\partial \vec{f}/\partial a = (0.6818 \times 0.7955,\, 0)^T \approx (0.5424, 0)^T$. Solving the 2×2 system $D_u f\, \partial \bar{\vec{u}}/\partial a = -\partial \vec{f}/\partial a$ without rounding:

$$\frac{\partial \bar{u}_1}{\partial a} = -0.9039256198, \qquad \frac{\partial \bar{u}_2}{\partial a} = +0.2711776860.$$

(d) Differentiating (4.4) directly,

$$\frac{\partial \bar{u}_1}{\partial a} = -\frac{1-c}{(1-ac)^2} = -\frac{0.7}{0.88^2} = -0.9039256198, \qquad \frac{\partial \bar{u}_2}{\partial a} = \frac{c(1-c)}{(1-ac)^2} = \frac{0.21}{0.88^2} = +0.2711776860$$

The two agree to machine precision, as they must: the FSE is the derivative of the equilibrium condition, so it cannot disagree with the derivative of the equilibrium.

Carrying the arithmetic unrounded is what makes the agreement visible; rounding to three figures would hide whether the match is exact or merely close.

4.3. The augmented system consists of: Original: $\dot{u} = ru(1 - u/K)$. FSE for $w_r = \partial u/\partial r$: $\dot{w}_r = r(1 - 2u/K)w_r + u(1 - u/K)$. FSE for $w_K = \partial u/\partial K$: $\dot{w}_K = r(1 - 2u/K)w_K + ru^2/K^2$. Initial values: $u(0) = u_0$ and, by (4.12), $w_r(0) = w_K(0) = 0$ when u_0 does not depend on r or K.

Total: 3 ODEs. In general, for n state variables and m parameters, the augmented system has $n + mn = n(1 + m)$ equations.

4.4. (a) 17 FSE solves (one per parameter), plus 1 forward solve = 18 total. (b) $18 \times 2 = 36$ minutes. (c) For one QOI the adjoint requires exactly 2 solves (1 forward + 1 adjoint), regardless of m. For 17 parameters, the adjoint is 9 times faster than the FSE.

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Part IV

The Adjoint Approach

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Chapter 5
The Adjoint in Linear Algebra

The adjoint is the same sensitivity question, asked from the other end.

Chapter 4 showed that analytic forward sensitivity analysis for a model with $m = 50$ parameters requires 50 linear solves, one per parameter, after the nominal solve for $\vec{u}$: 51 in all. Each solve uses the same coefficient matrix A, so we pay once for the LU factorization and 51 times for back-substitution. That is still 51 solves. An LU factorization of an $n \times n$ matrix costs about $2n^3/3$ arithmetic operations and each pair of triangular solves about $2n^2$, so for large n the shared factorization dominates, and a count of solves measures time only when each solve is a fresh factorization or an ODE solve.

Now suppose we care about a single response functional, not the entire solution $\vec{u}$. The quantity of interest is $J = \vec{c}^T\vec{u}$: the pollution level at one monitoring station, the total disease burden, the profit at equilibrium. We want $\partial J/\partial p$ for all 50 parameters p.

Is there a way to compute all 50 derivatives simultaneously, in the cost of the nominal solve and a *single* further linear solve?

Yes. The answer is the adjoint. The adjoint approach to sensitivity analysis was developed for nonlinear systems by Cacuci [9] and has since become an active area of research in design optimization for aeronautics [17].

5.1 ▪ A Preview: The Same Answer, Two Solves Instead of Fifty-One

Before the general theory, work through a small example by hand to see exactly what the adjoint does and why it is efficient.

The Setup

Consider the 2×2 linear system

$$A\vec{u} = \vec{b}(p_1, p_2), \qquad A = \begin{pmatrix} 3 & 1 \\ 1 & 2 \end{pmatrix}, \qquad \vec{b} = \begin{pmatrix} p_1 \\ p_2 \end{pmatrix}, \tag{5.1}$$

where p_1 and p_2 are two parameters of interest. The scalar response functional is $J = u_1 + u_2 = \vec{c}^T\vec{u}$ with $\vec{c} = (1,1)^T$. We want $\partial J/\partial p_1$ and $\partial J/\partial p_2$.

The Direct (Forward) Approach: Three Solves

Differentiating $A\vec{u} = \vec{b}$ with respect to p_1: $A\,(\partial\vec{u}/\partial p_1) = \partial\vec{b}/\partial p_1 = (1,0)^T$. Differentiating with respect to p_2: $A\,(\partial\vec{u}/\partial p_2) = \partial\vec{b}/\partial p_2 = (0,1)^T$. Beyond the nominal solve for $\vec{u}$, that is two linear solves (one per parameter) with the same matrix A. For a 2×2 system by hand we write out A^{-1}; in code, solve by LU factorization and never form the inverse (Section 4.2):

$$A^{-1} = \tfrac{1}{5}\begin{pmatrix} 2 & -1 \\ -1 & 3 \end{pmatrix}, \quad \text{so} \quad \frac{\partial\vec{u}}{\partial p_1} = \tfrac{1}{5}\begin{pmatrix} 2 \\ -1 \end{pmatrix}, \qquad \frac{\partial\vec{u}}{\partial p_2} = \tfrac{1}{5}\begin{pmatrix} -1 \\ 3 \end{pmatrix}.$$

The derivatives of $J = \vec{c}^T\vec{u}$ are:

$$\frac{\partial J}{\partial p_1} = \vec{c}^T\frac{\partial\vec{u}}{\partial p_1} = (1,1)\cdot\tfrac{1}{5}(2,-1)^T = \tfrac{1}{5}, \qquad \frac{\partial J}{\partial p_2} = \vec{c}^T\frac{\partial\vec{u}}{\partial p_2} = (1,1)\cdot\tfrac{1}{5}(-1,3)^T = \tfrac{2}{5}.$$

Two sensitivity solves give two derivatives; with the nominal solve, that is three solves. For 50 parameters: 51 solves.

The Adjoint Approach: Two Solves

Instead of solving for $\partial\vec{u}/\partial p_j$ separately for each j, introduce the *adjoint vector* $\vec{v}$ satisfying

$$A^T\vec{v} = \vec{c}. \tag{5.2}$$

This is one linear solve with the transposed matrix, the second solve after the nominal one. Since A is symmetric here, $A^T = A$, so $\vec{v} = A^{-1}\vec{c}$:

$$\vec{v} = A^{-1}\vec{c} = \tfrac{1}{5}\begin{pmatrix} 2 & -1 \\ -1 & 3 \end{pmatrix}\begin{pmatrix} 1 \\ 1 \end{pmatrix} = \tfrac{1}{5}\begin{pmatrix} 1 \\ 2 \end{pmatrix}.$$

What does $\vec{v}$ represent? The adjoint vector is indexed by the *equations* of the system, not by its states. Its ith entry measures how much J moves when the ith right-hand side b_i moves, so $\vec{v} = \partial J/\partial\vec{b}$. The state side of the

question is already settled and is not interesting here: $\partial J/\partial \vec{u} = \vec{c} = (1,1)^T$, so u_1 and u_2 contribute to $J = u_1 + u_2$ equally, by construction. What the adjoint adds is the translation from a perturbation of the *data* to a change in the response, after the coupling imposed by A has been accounted for.

In the 2×2 example, $\vec{v} = \frac{1}{5}(1,2)^T$ says that adding a unit to b_2 moves J *twice as much* as adding a unit to b_1. That asymmetry comes entirely from A, since $\vec{c}$ weights the two states equally. This is what makes the adjoint more than a convenient shortcut: $\vec{v}$ is the sensitivity of the response to the forcing, which is the quantity a modeler can actually act on, and no rearrangement of $\vec{c}$ alone would produce it.

Now observe: for any parameter p_j,

$$\frac{\partial J}{\partial p_j} = \vec{c}^T \frac{\partial \vec{u}}{\partial p_j} = (A^T \vec{v})^T \frac{\partial \vec{u}}{\partial p_j} = \vec{v}^T \underbrace{A \frac{\partial \vec{u}}{\partial p_j}}_{= \partial \vec{b}/\partial p_j} = \vec{v}^T \frac{\partial \vec{b}}{\partial p_j}.$$

This is the key step: $A\,(\partial \vec{u}/\partial p_j) = \partial \vec{b}/\partial p_j$ is the FSE right-hand side, a known vector. No additional solve is needed, just a dot product.

Checking against our direct answers:

$$\frac{\partial J}{\partial p_1} = \vec{v}^T \frac{\partial \vec{b}}{\partial p_1} = \tfrac{1}{5}(1,2)\cdot(1,0)^T = \tfrac{1}{5} \checkmark \qquad \frac{\partial J}{\partial p_2} = \vec{v}^T \frac{\partial \vec{b}}{\partial p_2} = \tfrac{1}{5}(1,2)\cdot(0,1)^T = \tfrac{2}{5} \checkmark$$

One adjoint solve for $\vec{v}$, then one dot product per parameter. For 50 parameters: still one adjoint solve, then 50 dot products, against one solve per parameter for the FSE. Counting the nominal solve on both sides, as §5.4 does, that is 51 solves against 2, a factor of about 25. Counting only the sensitivity solves gives 50 against 1; the two conventions differ by roughly a factor of two, which is why the chapter fixes the first and keeps to it.

Self-check 5.1. Add a third parameter p_3 to the system above by setting $\vec{b} = (p_1 + p_3,\ p_2)^T$. Without solving any new linear systems, use the adjoint vector $\vec{v}$ already computed to find $\partial J/\partial p_3$.

The pattern above is the entire idea of the adjoint, stripped to its essentials. Section 5.2 states it precisely.

5.2 ▪ The Strategy, Stated Explicitly

Section 5.1 showed that all m sensitivities can be recovered without solving m separate systems. This section shows why the trick works.

The adjoint is not a new kind of problem. It is a deliberate trick: introduce an auxiliary vector, choose it to cancel the unknown sensitivity

variable, and read off the result. The three steps below carry out the trick in symbols.

Step 1. We want $\partial J/\partial p_j$ for all parameters p_j. Since $J = \vec{c}^{\,T}\vec{u}$, this is $\partial J/\partial p_j = \vec{c}^{\,T}(\partial\vec{u}/\partial p_j)$. But $\partial\vec{u}/\partial p_j$ requires a separate FSE solve for every j (50 solves for 50 parameters, after the nominal solve).

Step 2. Introduce an auxiliary vector $\vec{v}$ and observe that for any vector $\vec{v}$:

$$\vec{c}^{\,T}\frac{\partial\vec{u}}{\partial p_j} = \vec{v}^{\,T}\left[A\frac{\partial\vec{u}}{\partial p_j}\right] + \left(\vec{c}^{\,T} - \vec{v}^{\,T}A\right)\frac{\partial\vec{u}}{\partial p_j}.$$

The first term on the right is $\vec{v}^{\,T}\vec{r}_j$, where $\vec{r}_j = A(\partial\vec{u}/\partial p_j)$ is the FSE right-hand side. By (4.1), $\vec{r}_j = \partial\vec{b}/\partial p_j - (\partial A/\partial p_j)\vec{u}$, a known vector once $\vec{u}$ is known. The second term vanishes if we choose $\vec{v}$ so that $\vec{v}^{\,T}A = \vec{c}^{\,T}$.

Step 3. Choose $\vec{v}$ to satisfy the **adjoint equation**:

$$A^T\vec{v} = \vec{c}. \tag{5.3}$$

Then $\vec{c}^{\,T} - \vec{v}^{\,T}A = \vec{0}$ and the sensitivity formula reduces to:

$$\frac{\partial J}{\partial p_j} = \vec{v}^{\,T}\vec{r}_j \qquad \text{for all } j = 1, \ldots, m. \tag{5.4}$$

One solve for $\vec{v}$ (equation (5.3)) recovers all m sensitivities through formula (5.4). The trick works because the adjoint equation does not depend on j: $\vec{v}$ is determined by $\vec{c}$ and A alone. If the response vector $\vec{c}$ itself depends on p_j, the product rule adds one term, $(\partial\vec{c}/\partial p_j)^T\vec{u}$, to (5.4); it needs no further solve.

The adjoint is not a separate, unrelated problem. It is constructed entirely from the coefficient matrix A (via transposition) and the response vector $\vec{c}$ of the forward problem. Everything about the adjoint is determined by the forward problem.

Connection to Linear Programming Duality

Readers who have seen linear programming will recognize a structural parallel. The primal LP $\max\{\vec{c}^{\top}\vec{x} : A\vec{x} = \vec{b},\ \vec{x} \geq 0\}$ has a dual problem $\min\{\vec{b}^{\top}\vec{y} : A^{\top}\vec{y} \geq \vec{c}\}$, whose constraint involves the transposed matrix $A^{\top}$. The LP dual variable $\vec{y}$ measures the sensitivity of the optimal objective value to perturbations in the right-hand side $\vec{b}$, which is what the adjoint variable $\vec{v}$ does here: it measures the sensitivity of $J = \vec{c}^{\top}\vec{u}$ to perturbations in $\vec{b}$ (the parameter forcing). The adjoint equation $A^{\top}\vec{v} = \vec{c}$ is a direct generalization of the LP dual constraint. For ODE models the connection runs through Lagrange multipliers: Chapter 6 derives the adjoint variable as one (Section 6.3), and Appendix E.4 identifies the adjoint ODE with the costate equation of optimal control.

Self-check 5.2. For the system $A\vec{u} = \vec{b}$ with $A = \begin{pmatrix} 2 & 1 \\ 0 & 3 \end{pmatrix}$ and response functional $J = u_1$ (so $\vec{c} = (1, 0)^T$): write down the adjoint equation (5.3) and solve for $\vec{v}$. What does the sign of v_2 say about how J responds to b_2?

5.3 ▪ The Commutative Diagram

The three-step strategy and the 2×2 preview both express the same mathematical fact: the FSE route and the adjoint route reach the same sensitivity, at different cost. Figure 5.1 shows this as a diagram, read it as a summary of the algebra already developed, not as an introduction to the idea.

The diagram commutes: both paths from the forward problem to the sensitivity give identical results. This can be verified directly from the algebra in Section 5.2.

Theorem 5.1 (Forward–Adjoint Equivalence)**.** *Let $A\vec{u} = \vec{b}$ with A nonsingular (so that $\vec{u}$, $\partial\vec{u}/\partial p_j$ and $\vec{v}$ below exist and are unique), $J = \vec{c}^{T}\vec{u}$, and FSE $A\,\partial\vec{u}/\partial p_j = \vec{r}_j$. Let $\vec{v}$ solve the adjoint equation $A^T\vec{v} = \vec{c}$. Then*

$$\frac{\partial J}{\partial p_j} = \vec{c}^{T}\frac{\partial \vec{u}}{\partial p_j} = \vec{v}^{T}\vec{r}_j$$

for every parameter p_j.

Proof. $\vec{c}^{T}(\partial\vec{u}/\partial p_j) = (A^T\vec{v})^T(\partial\vec{u}/\partial p_j) = \vec{v}^{T}A(\partial\vec{u}/\partial p_j) = \vec{v}^{T}\vec{r}_j$. The second equality uses $(A^T\vec{v})^T = \vec{v}^{T}(A^T)^T = \vec{v}^{T}A$, which follows from the rule $(BC)^T = C^TB^T$ with $B = A^T$, $C = \vec{v}$. With $\vec{w} = \partial\vec{u}/\partial p_j$, the proof is the identity $\langle A\vec{w}, \vec{v}\rangle = \langle \vec{w}, A^T\vec{v}\rangle$ that defines the transpose (Section A.3). □

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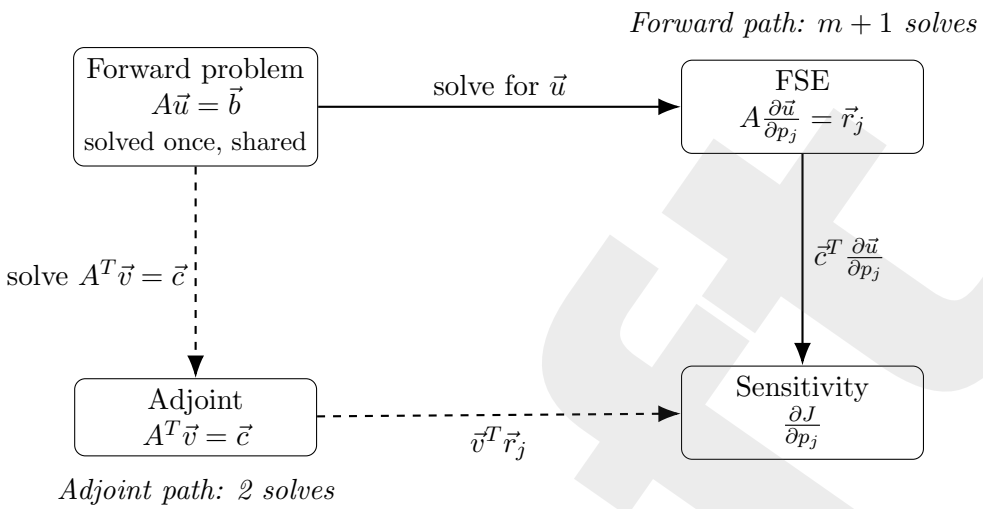


Figure 5.1. *The commutative diagram for sensitivity analysis. The forward path (solid arrows) computes $\partial J/\partial p_j$ by solving the FSE for each parameter, $m+1$ solves total. The adjoint path (dashed arrows) shares the forward solve for $\vec{u}$, which $\vec{r}_j$ generally needs, solves the adjoint equation once, then evaluates the sensitivity formula for each parameter; 2 solves total. Both paths deliver the same result.*

The adjoint gives *exactly* the same sensitivity values as the FSE. It is not an approximation. Students sometimes expect that a method requiring fewer solves must sacrifice accuracy; here it does not.

Self-check 5.3. Using Theorem 5.1, state in words why the adjoint equation does not need to be re-solved when a new parameter p_k is added to the sensitivity study.

5.4 ▪ Cost Comparison and the Decision Rule

5.4.1 ▪ When to Use the Adjoint

Both the FSE and the adjoint produce the same sensitivity Jacobian. Their costs differ dramatically when m and r are unequal, where m is the number of POIs and r is the number of response functionals.

For the FSE: one solve per POI, plus the nominal solve for $\vec{u}$ itself $\rightarrow$ $m+1$ solves total.

For the adjoint: one solve per response functional, plus that same nominal solve $\rightarrow$ $r+1$ solves total. Counting the nominal solve on both sides is what makes the comparison fair, and it is the convention used in Chapter 4: for the malaria output $i_h(T)$, seventeen parameters cost eighteen ODE solves by the FSE and two by the adjoint. The nominal solve is needed whenever $\vec{r}_j$ contains $\vec{u}$, as it does when A depends on p_j or the model is nonlinear,

and it supplies the value of J that normalizes each index; for an equilibrium model it is the nonlinear computation of the equilibrium, not a linear solve.

(5.5)

$$\boxed{m > r \implies \text{use adjoint}; \qquad r > m \implies \text{use FSE}; \qquad m = r \implies \text{either}.}$$

In most SA applications, there are many POIs (m large) and a small number of QOIs of primary interest (r small), so the adjoint is preferred. The canonical situation is $r = 1$: one scalar response functional (peak infection count, total burden, profit at equilibrium) and many parameters.

5.4.2 ▪ Cost Comparison Plot

Figure 5.2 shows the number of linear solves required by each method as a function of the number of parameters.

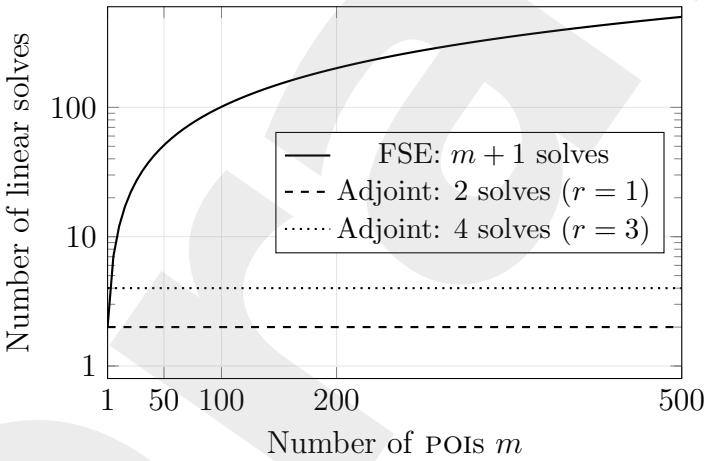


Figure 5.2. *Cost comparison for FSE vs. adjoint sensitivity analysis, measured in number of linear solves. For $r = 1$ response functional (dashed), the adjoint always requires exactly 2 solves regardless of m. For $r = 3$ (dotted), the adjoint requires 4 solves. The FSE requires $m + 1$ solves. The crossover occurs at $m = r$; for $m > r$, the adjoint wins.*

For the competitive population model ($m = 3$ parameters, $r = 2$ response functionals $\{\bar{u}_1, \bar{u}_2\}$): FSE requires 4 solves; adjoint requires 3. The advantage is modest here. For the malaria model ($m = 17$, $r = 1$): FSE requires 18 solves; adjoint requires 2, the factor of 9 quoted in Chapter 4. Those counts are ODE solves for the output $i_h(T)$. For the endemic equilibrium the counts are the same, but only the first solve, which finds the equilibrium, is nonlinear; each of the others is a linear solve. For a climate model ($m = 200$, $r = 2$): FSE requires 201 solves; adjoint requires 3: a factor of 67 improvement.

Self-check 5.4. A model has $m = 30$ POIs and $r = 5$ response functionals. Each linear solve takes 2 minutes. How long does the FSE approach take? How long does the adjoint approach take? At what value of r would the two approaches have equal cost?

5.5 ▪ The Competitive Population: Full Adjoint Treatment

We now apply the adjoint to the competitive population equilibrium from Chapter 4, comparing the forward and adjoint approaches side by side. The forward problem is the equilibrium condition $\vec{f}(\vec{u}) = \vec{0}$, which is nonlinear; its solution is (4.4). Only the FSE (4.5) is linear, with matrix $A = D_{\vec{u}}[\vec{f}]$ as in equation (4.6). The comparison makes the cost advantage concrete.

5.5.1 ▪ FSE Approach: Three Parameters, Four Solves

For parameters a, b, c, the FSE requires solving $A(\partial\vec{u}/\partial p_j) = -\partial\vec{f}/\partial p_j$ for $j = a, b, c$, three linear solves with the same matrix A, one LU factorization and three back-substitutions. With the nominal solve for $\vec{u}$, the FSE costs four solves.

5.5.2 ▪ Adjoint Approach: Three Solves for Two Response Functionals

Now suppose we want $\partial J_1/\partial p_j$ and $\partial J_2/\partial p_j$ for all three parameters, where $J_1 = \bar{u}_1$ and $J_2 = \bar{u}_2$. The response vectors are $\vec{c}_1 = (1, 0)^T$ and $\vec{c}_2 = (0, 1)^T$.

The two adjoint equations are:

$$A^T\vec{v}_1 = \vec{c}_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \qquad A^T\vec{v}_2 = \vec{c}_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}. \tag{5.6}$$

Two adjoint solves (one per response functional) give $\vec{v}_1$ and $\vec{v}_2$. With the same nominal solve for $\vec{u}$, the adjoint costs three solves.

A note on correlated QOIs. $J_1 = \bar{u}_1$ and $J_2 = \bar{u}_2$ share one equilibrium $\vec{u}$, so each parameter moves both. The adjoint needs no special treatment for this: it solves $A^T\vec{v}_k = \vec{c}_k$ for each response functional separately.

The sensitivities follow from formula (5.4):

$$\frac{\partial J_k}{\partial p_j} = \vec{v}_k^T\vec{r}_j, \qquad \vec{r}_j = -\frac{\partial\vec{f}}{\partial p_j}, \tag{5.7}$$

where $\vec{r}_j = -\partial\vec{f}/\partial p_j$ is the equilibrium form of the FSE right-hand side, from differentiating $\vec{f}(\vec{u}; \vec{p}) = \vec{0}$ (Theorem 4.1). This gives all $2 \times 3 = 6$ sensitivities from two adjoint solves rather than three FSE solves: three solves in all against four, with the nominal solve counted on both sides.

5.5.3 ▪ Verification

Theorem 5.1 guarantees the two approaches give identical results. At the nominal values $a = 0.4$, $b = 1$, $c = 0.3$ of Self-check 4.2, whose answer prints $\vec{u}$, the matrix A and the derivatives with respect to a:

	$\mathcal{S}_a^{\bar{u}_1}$	$\mathcal{S}_b^{\bar{u}_1}$	$\mathcal{S}_c^{\bar{u}_1}$	$\mathcal{S}_a^{\bar{u}_2}$	$\mathcal{S}_b^{\bar{u}_2}$	$\mathcal{S}_c^{\bar{u}_2}$
FSE (4 solves)	-0.530	0	0.136	$+0.136$	0	-0.292
Adjoint (3 solves)	-0.530	0	0.136	$+0.136$	0	-0.292

The rows are identical. The adjoint is not an approximation. The row labels count every solve, the nominal solve included.

A structural observation: because $\vec{c}_1$ and $\vec{c}_2$ are the unit vectors $\vec{e}_1$ and $\vec{e}_2$, the two adjoint solutions $\vec{v}_1, \vec{v}_2$ are the columns of $(A^T)^{-1}$, which equals $(A^{-1})^T$. Collecting all adjoint solutions into a matrix therefore gives the transpose of the inverse. This connection is intellectually interesting but not computationally useful: we never form A^{-1} explicitly, we extract the product $\vec{v}_k^T \vec{r}_j$ for each required sensitivity. With r responses that costs r solves, far cheaper than the full inverse when $r \ll n$; here $r = n = 2$, so the two adjoint solves cost as much as the inverse.

5.6 ▪ Reverse Mode and Backpropagation

The adjoint strategy of the preceding sections applies to linear systems. This section extends it to arbitrary differentiable computations via the computation graph, revealing the connection to algorithmic differentiation and, through it, to the training of neural networks.

5.6.1 ▪ The Computation Graph

Every differentiable computation can be represented as a directed acyclic graph (its arrows form no loops, so each value is computed only from values before it) in which each node holds an intermediate value and each edge represents an elementary operation (addition, multiplication, division, elementary function).

For $\mathcal{R}_0 = c\beta/\gamma_R$, the computation graph is:

$$c,\ \beta \xrightarrow{\times} v_1 = c\beta, \qquad v_1,\ \gamma_R \xrightarrow{\div} v_2 = v_1/\gamma_R = \mathcal{R}_0. \tag{5.8}$$

Each arrow takes every value on its left as an input: the product node has inputs c and β, and the quotient node has inputs v_1 and γ_R. Reading left to right is the forward evaluation: given c, β, γ_R, compute $\mathcal{R}_0$.

5.6.2 ▪ Forward Mode: One Pass Per Input

In *forward mode* differentiation, we propagate derivatives $\dot{v}_i = \partial v_i/\partial p$ through the graph for one input p at a time.

For $p = c$: set $\dot{c} = 1$, $\dot{\beta} = 0$, $\dot{\gamma_R} = 0$. Propagate forward:

$$\dot{v}_1 = \dot{c}\beta + c\dot{\beta} = \beta, \qquad \dot{R}_0 = \dot{v}_1/\gamma_R - v_1\dot{\gamma_R}/\gamma_R^2 = \beta/\gamma_R.$$

So $\partial\mathcal{R}_0/\partial c = \beta/\gamma_R$. Repeat with $\dot{\beta} = 1$ to get $\partial\mathcal{R}_0/\partial\beta$. Repeat with $\dot{\gamma_R} = 1$ to get $\partial\mathcal{R}_0/\partial\gamma_R$. Total: 3 forward passes for 3 parameters.

5.6.3 ▪ Reverse Mode: Two Passes for All Inputs

In *reverse mode* differentiation, we propagate the adjoint variable $\bar{v}_i = \partial J/\partial v_i$ backward through the graph.

Set $\bar{R}_0 = \partial J/\partial\mathcal{R}_0 = 1$ (we are computing $\partial\mathcal{R}_0/\partial(\cdot)$, so $J = \mathcal{R}_0$). Propagate backward:

$$\bar{v}_1 = \bar{R}_0 \cdot \frac{\partial\mathcal{R}_0}{\partial v_1} = 1 \cdot (1/\gamma_R) = 1/\gamma_R, \tag{5.9}$$

$$\bar{c} = \bar{v}_1 \cdot \frac{\partial v_1}{\partial c} = (1/\gamma_R) \cdot \beta = \beta/\gamma_R, \tag{5.10}$$

$$\bar{\beta} = \bar{v}_1 \cdot \frac{\partial v_1}{\partial \beta} = (1/\gamma_R) \cdot c = c/\gamma_R, \tag{5.11}$$

$$\bar{\gamma_R} = \bar{R}_0 \cdot \frac{\partial\mathcal{R}_0}{\partial\gamma_R} = 1 \cdot (-v_1/\gamma_R^2) = -c\beta/\gamma_R^2. \tag{5.12}$$

After the forward evaluation, one backward pass gives all three derivatives simultaneously:

$$\frac{\partial\mathcal{R}_0}{\partial c} = \bar{c} = \frac{\beta}{\gamma_R}, \quad \frac{\partial\mathcal{R}_0}{\partial\beta} = \bar{\beta} = \frac{c}{\gamma_R}, \quad \frac{\partial\mathcal{R}_0}{\partial\gamma_R} = \bar{\gamma_R} = -\frac{c\beta}{\gamma_R^2}.$$

These match the analytic results. Reverse mode costs two passes regardless of the number of inputs: the forward evaluation that stores the node values and one backward pass. Each pass costs time proportional to the graph size. In this graph each value feeds one later node. When a value feeds two, as ν and ν_m do in Exercise 5.5, its adjoint is the sum of the two backward contributions; Figure 1.3 in Chapter 1 shows this for x_2.

Table 5.1 summarizes the comparison.

5.6.4 ▪ The Backpropagation Connection

Note for readers without machine learning background: this subsection connects the adjoint to neural network training. If backpropagation is unfamiliar, skip ahead to Unifying Idea 2 below and continue from there; the

	Forward mode	Reverse mode
Propagates	$\dot{v}_i = \partial v_i/\partial p$ (one p)	$\bar{v}_i = \partial J/\partial v_i$ (one J)
Direction	Input $\to$ output	Output $\to$ input
Cost	1 pass per input	$1 + r$ passes
Preferred when	$m \leq r$ (few inputs)	$m > r$ (few outputs)
Matrix analog	FSE: $A\,\partial\vec{u}/\partial p_j = \vec{r}_j$	Adjoint: $A^T\vec{v} = \vec{c}$

Table 5.1. *Forward mode and reverse mode differentiation: a side-by-side comparison. The matrix analogs confirm that the two modes are the computation-graph generalizations of the FSE and adjoint approaches from Sections 5.2–5.4.*

adjoint ODE does not depend on the neural network material. Return here after completing Chapter 6 if you want the connection.

If you have encountered training a neural network, you have already used the adjoint under a different name.

A neural network is a composition of parameterized functions. Training minimizes a loss function $J(\theta)$ over parameters θ using gradient descent, which requires $\partial J/\partial\theta_i$ for every parameter θ_i. The loss function is a scalar ($r = 1$) and modern networks have millions of parameters ($m \gg 1$), so reverse mode is overwhelmingly preferred. Backpropagation is exactly the reverse-mode computation described above, applied to the computation graph of the network.

To map notation: the adjoint variable $\lambda_i = \partial J/\partial v_i$ of later chapters records how much J changes per unit change at node i, the role played by $\bar{v}_i$ in reverse mode.

The adjoint equation for the ODE case, derived in Chapter 6, is the continuous limit of this discrete backward accumulation. The backward ODE for $\vec{\lambda}(t)$ is the functional-analytic version of the backward pass through the computation graph.

Unifying Idea 2 (the adjoint is the transpose; Section 1.5). The statement holds in whatever space you are working in.

In this chapter: the adjoint equation is $A^T\vec{v} = \vec{c}$, the transposed coefficient matrix applied to the adjoint vector.

In the computation graph: reverse mode is the chain rule applied in reverse order, accumulating $\bar{v}_i = \partial J/\partial v_i$ backward through the graph.

In Chapter 6: the adjoint will be an ODE running backward in time, with a terminal condition at $t = T$. The integration-by-parts identity that produces this ODE is the functional-analytic analog of the matrix transposition that produced A^T here.

One idea. Three settings. One cost advantage: when $m \gg r$, the adjoint/reverse-mode is the preferred computational strategy.

Self-check 5.5. Use the reverse-mode computation of equations (5.9)–(5.12) to compute $\mathcal{S}_c^{\mathcal{R}_0}$, $\mathcal{S}_\beta^{\mathcal{R}_0}$, and $\mathcal{S}_{\gamma_R}^{\mathcal{R}_0}$. Express each as a pure number and verify using the Chapter 2 results.

5.7 ▪ Bridge to Chapter 6

Students sometimes expect the adjoint for differential equations to require advanced functional analysis (Hilbert spaces, adjoint operators, and formal operator theory). Appendix A provides the L^2 inner product and orthogonality needed for that formalism; Appendix B develops the full abstract operator framework that makes the connection rigorous. The preceding sections show why this expectation is misplaced: the core idea is the matrix transpose. Every formal construct in Chapter 6 is the natural extension of the matrix algebra developed here. Reading Chapter 6 with this chapter in mind, the abstract notation should feel like familiar territory.

Unifying Idea 1 (the sensitivity index is a normalized derivative; Section 1.5). Every sensitivity value computed in Section 5.5, via the adjoint, is a normalized derivative, an instance of Unifying Idea 1 from Chapter 1. The adjoint is a computational strategy; the quantity it produces is still the SI defined in Chapter 2. The two commutative paths in Figure 5.1 produce the same sensitivity index by different routes.

Chapter 5 has established the adjoint for linear systems. Chapter 6 extends these results to ordinary differential equations.

The translation is direct:

Chapter 5 (linear algebra)	Chapter 6 (dynamical systems)
Coefficient matrix A	Differential operator $d/dt - J_F(t)$
Transpose A^T	Adjoint operator $-(d/dt + J_F(t)^T)$
Adjoint equation $A^T\vec{v} = \vec{c}$	Adjoint ODE $-\dot{\vec{\lambda}} = J_F^T\vec{\lambda} + g_u$
Terminal condition: none	Terminal condition: $\vec{\lambda}(T) = h_u^T$
Sensitivity formula $\vec{v}^T\vec{r}_j$	Sensitivity formula $\int_0^T \vec{\lambda}^T F_{p_j}\, dt$
Cost: 2 solves for any m ($r = 1$)	Cost: 2 ODE solves for any m ($r = 1$)

The integration-by-parts identity in Chapter 6 plays the role of the algebraic transposition here. The terminal condition on $\vec{\lambda}(T)$ is the time-dependent analog of the right-hand side $\vec{c}$ in $A^T\vec{v} = \vec{c}$. The cost advantage

is identical: two solves, the forward solve and one adjoint solve, recover all m sensitivities of one response functional regardless of m.

5.8 ▪ Computational Laboratory: Forward vs. Adjoint

This laboratory compares the FSE and adjoint computations on the competitive population model, then implements reverse-mode differentiation for $\mathcal{R}_0$.

Part 1: Competitive population via FSE. At nominal values $a = 0.4$, $b = 1$, $c = 0.3$ (here c is the competition coefficient of (4.3), not the contact rate), compute the equilibrium using equation (4.4). Assemble the Jacobian A and compute the LU factorization using your language's LU routine. Solve the three FSEs ($\partial\vec{u}/\partial a$, $\partial\vec{u}/\partial b$, $\partial\vec{u}/\partial c$) using `U\(L\r)` for each. Record the time.

Part 2: Competitive population via adjoint. Solve the two adjoint equations $A^T\vec{v}_1 = \vec{c}_1$ and $A^T\vec{v}_2 = \vec{c}_2$ using the same LU factorization (with transposed triangular solves). Compute all 6 sensitivities via $\vec{v}_k^T\vec{r}_j$. Record the time and verify that all 6 values agree with Part 1 to machine precision ($\varepsilon_{\text{mach}} \approx 2.2 \times 10^{-16}$, Section 3.3).

Note on the transposed solve. Many LU routines give $A = P^TLU$, where the permutation matrix P records the row swaps (pivoting) that the routine makes to keep the elimination stable; then $A^T = U^TL^TP$. Solving $A^T\vec{v}_1 = \vec{c}_1$ therefore proceeds in three steps: $U^T\vec{w}_1 = \vec{c}_1$, then $L^T\vec{w}_2 = \vec{w}_1$, then $\vec{v}_1 = P^T\vec{w}_2$. In languages with matrix-transpose syntax, this is written as `P' * (L' \ (U' \ c1))`, reading right to left: first solve `U'\c1`, then `L'\`, then multiply by `P'` at the end. The script names the response vectors `c1` and `c2` because `c` already holds the scalar competition coefficient.

```
%% Assemble Jacobian at equilibrium (see Chapter 4)
A = jac_symp(ubar, a, b, c);
[L, U, P] = lu(A);

%% FSE: 3 solves
for j = 1:3
    rj = rhs_symp(ubar, j);          %% FSE right-hand side
    dwdp(:,j) = U \ (L \ (P * rj));
end

%% Adjoint: 2 solves
c1 = [1;0]; c2 = [0;1];
v1 = P' * (L' \ (U' \ c1));   %% solve A^T v1 = c1
```

```
v2 = P' * (L' \ (U' \ c2));   %% solve A^T v2 = c2
for j = 1:3
    rj = rhs_symp(ubar, j);
    dJdp(1,j) = v1' * rj;     %% dJ1/dp_j
    dJdp(2,j) = v2' * rj;     %% dJ2/dp_j
end
```

Part 3: Timing as a function of system size. For random systems of size $n = 5, 10, 50, 100$ with $m = 20$ parameters and $r = 1$ response functional, time both approaches. Plot computation time vs. n on log-log axes. Does the ratio of times approach the solve-count ratio $(m+1)/(r+1)$, here 21 against 2? That ratio assumes every solve costs the same, as it does when each solve factors A afresh. With one shared LU factorization, as in Parts 1 and 2, explain why the measured ratio is smaller.

Part 4: Reverse-mode $\mathcal{R}_0$ by hand. Implement the reverse-mode computation (5.9)–(5.12) for $\mathcal{R}_0 = c\beta/\gamma_R$. Verify all three sensitivities against a symbolic computation tool, as in Part 1 of the Chapter 2 laboratory. Count the number of arithmetic operations (each addition, subtraction, multiplication and division once) in forward mode vs. reverse mode for this graph.

Part 5: Verify that adjoint = 50 FSE solves. Set up a random 10×10 system with $m = 50$ parameters and $r = 1$. Compute $\partial J/\partial p_j$ for all j by: (a) the forward path, the nominal solve and 50 FSE solves, 51 in all; (b) the adjoint path, the nominal solve and 1 adjoint solve, 2 in all. Verify that the maximum absolute difference is less than 10^{-12}.

5.9 ▪ Exercises

Exercise 5.1 *(Bloom L2).* State the three-step adjoint strategy from Section 5.2 in your own words, using no symbols. Then explain: why does the adjoint equation $A^T\vec{v} = \vec{c}$ not depend on which parameter p_j we are differentiating with respect to? Why does this independence make the adjoint efficient?

Exercise 5.2 *(Bloom L3).* For the linear system

$$A\vec{u} = \vec{b}, \qquad A = \begin{pmatrix} 3 & 1 \\ 1 & 2 \end{pmatrix}, \quad \vec{b} = \begin{pmatrix} b_1(p) \\ b_2(p) \end{pmatrix} = \begin{pmatrix} p \\ 2p \end{pmatrix},$$

with response functional $J = u_1 + u_2$ (so $\vec{c} = (1,1)^T$):

(a) Solve the forward problem to find $\vec{u}(p)$ at $p = 1$.

(b) Write and solve the FSE for $\partial \vec{u}/\partial p$.

(c) Write and solve the adjoint equation.

(d) Use formula (5.4) to compute $\partial J/\partial p$.

(e) Verify by differentiating $J(p) = u_1(p) + u_2(p)$ directly.

Exercise 5.3 *(Bloom L3).* For each scenario below, state whether you would use the FSE or the adjoint, and calculate the number of linear solves each approach requires.

(a) $m = 5$ parameters, $r = 20$ response functionals.

(b) $m = 100$ parameters, $r = 3$ response functionals.

(c) $m = 8$ parameters, $r = 8$ response functionals.

(d) $m = 1{,}000$ parameters, $r = 1$ response functional. Each solve takes 0.1 seconds. How much faster is the preferred method?

Exercise 5.4 *(Bloom L4).* For the competitive population equilibrium with parameters $a = 0.5$, $b = 2$, $c = 0.4$:

(a) Compute all 6 sensitivities ($\mathcal{S}_a^{\bar{u}_1}$, $\mathcal{S}_b^{\bar{u}_1}$, $\mathcal{S}_c^{\bar{u}_1}$, $\mathcal{S}_a^{\bar{u}_2}$, $\mathcal{S}_b^{\bar{u}_2}$, $\mathcal{S}_c^{\bar{u}_2}$) using the FSE approach (4 solves, counting the nominal solve).

(b) Compute the same 6 sensitivities using the adjoint approach (3 solves).

(c) Verify that the two methods agree.

(d) Which approach requires fewer solves? By how much? At what values of m and r would the FSE be preferred?

Exercise 5.5 *(Bloom L4).* The SEIR basic reproduction number is $\mathcal{R}_0 = c\beta\nu/[(\nu+\nu_m)(\gamma_R+\nu_m)]$ with parameters c, β, ν (exposure rate), γ_R (recovery rate), ν_m (mortality rate).

(a) Draw the computation graph for $\mathcal{R}_0$ (intermediate nodes and edges).

(b) Implement the forward-mode computation to find $\partial \mathcal{R}_0/\partial \nu$. Count the number of arithmetic operations.

(c) Implement the reverse-mode computation to find all five derivatives $\partial \mathcal{R}_0/\partial c$, $\partial \mathcal{R}_0/\partial \beta$, $\partial \mathcal{R}_0/\partial \nu$, $\partial \mathcal{R}_0/\partial \gamma_R$, $\partial \mathcal{R}_0/\partial \nu_m$ in one backward pass after the forward evaluation.

(d) Compute the five normalized sensitivity indices and produce a tornado plot using interpretable POIs (mean times, not rates), at $\tau_E = 5.5$ days, $\tau_I = 7$ days and $\tau_m = 10{,}000$ days.

Exercise 5.6 *(Bloom L5).* A steady-state water-quality model of a river system has $m = 40$ parameters (pollution source rates, degradation rates, flow speeds, mixing coefficients) and $r = 2$ response functionals: (1) pollutant concentration at a downstream monitoring station, and (2) the steady pollutant load entering a protected estuary.

Design a complete SA study using the adjoint method. Specify:

(a) The form of the adjoint equation for this model.

(b) How many adjoint solves are required and why.

(c) How you would validate the implementation.

(d) How you would use the sensitivity results to advise regulators on which pollution sources to prioritize.

(e) One situation where the adjoint results might be misleading, and what you would do to detect and correct it.

Exercise 5.7 *(Bloom L6).* A colleague states: "The adjoint always gives the same answer as forward SA, just faster. So if we already have the FSE code working, there is no reason to implement the adjoint."

Evaluate this claim carefully.

(a) Is the first sentence (same answer, just faster) correct? Give a precise mathematical justification.

(b) Is the second sentence (no reason to implement the adjoint) correct? Describe at least two situations where switching from FSE to adjoint would be strongly advisable.

(c) For the FSE to be preferred over the adjoint, what relationship between m and r must hold? Give a specific numerical example where FSE is strictly better.

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Notes

The adjoint of a linear operator on a Hilbert space is the abstract generalization of the matrix transpose; a rigorous treatment appears in Appendix B, which builds on the inner products of Appendix A, and in standard functional analysis texts. Horn and Johnson [23] give a comprehensive treatment of matrix adjoints and their use in sensitivity analysis of linear systems. The sensitivity Jacobian (Definition 3.1) is the same object that Smith [60] uses in Chapter 7 of that book to drive parameter identifiability analysis and parameter selection; a reader who completes Chapter 6 will find that chapter of Smith immediately accessible.

Reverse-mode algorithmic differentiation on a computation graph, which is the computational realization of the adjoint idea, is a generalization of the backpropagation algorithm of neural network training, which became widely known through the work of Rumelhart, Hinton, and Williams; see the historical account in Baydin et al. [4].

The cost argument, that the forward solve and one adjoint solve recover all m sensitivities for one response functional, against $m+1$ solves for forward SA, appears in Giles and Pierce [17]; Cacuci [9] states the same economy for nonlinear systems with many parameters and few responses. Giles and Pierce give an accessible exposition of the adjoint idea in the context of aeronautical design optimization.

Answers to the self-checks

5.1. $\partial\vec{b}/\partial p_3 = (1,0)^T$. $\partial J/\partial p_3 = \vec{v}^T(1,0)^T = \frac{1}{5}(1,2)\cdot(1,0)^T = \frac{1}{5}$. No new solve needed. Adding parameters costs only one dot product each once $\vec{v}$ is known. This is the computational power of the adjoint.

5.2. $A^T\vec{v} = \vec{c}$ becomes $\begin{pmatrix} 2 & 0 \\ 1 & 3 \end{pmatrix}\vec{v} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$. Solving: $v_1 = 1/2$, $v_2 = -1/6$. This single solve (one forward substitution, since A^T is lower triangular) gives us the sensitivity of $J = u_1$ to any number of parameters, via formula (5.4). Since $\vec{v} = \partial J/\partial\vec{b}$, $v_2 = -1/6$ says that adding a unit to b_2 lowers $J = u_1$ by $1/6$. The second equation gives $u_2 = b_2/3$, and the first, $2u_1 + u_2 = b_1$, then pushes u_1 down by half that amount. Here A is not symmetric, so the transpose matters: solving $A\vec{v} = \vec{c}$ instead would give $\vec{v} = (1/2, 0)^T$ and miss the effect of b_2.

5.3. The adjoint equation $A^T\vec{v} = \vec{c}$ depends only on A (the system coefficient matrix, at the nominal parameters) and $\vec{c}$ (the response vector). Adding a parameter changes neither: A may depend on p_k, but its nominal value is the same matrix. Adding parameter p_k changes only the FSE right-hand side $\vec{r}_k = \partial\vec{b}/\partial p_k - (\partial A/\partial p_k)\vec{u}$, which is a known vector once the forward solution $\vec{u}$ is computed. The sensitivity $\partial J/\partial p_k = \vec{v}^T\vec{r}_k$ is then obtained by a single dot product, no additional linear solve.

5.4. FSE: $31 \times 2 = 62$ minutes, counting the baseline solve. Adjoint: $6 \times 2 = 12$ minutes, likewise. Savings: 50 minutes. Leaving the shared baseline solve out of both counts

gives 60 and 10 minutes, the same saving; the totals follow the convention used above. Equal cost when $r = m = 30$.

5.5. $\partial \mathcal{R}_0/\partial c = \beta/\gamma_R$. $\mathcal{S}_c^{\mathcal{R}_0} = (c/\mathcal{R}_0)(\partial \mathcal{R}_0/\partial c) = (c/(c\beta/\gamma_R)) \cdot (\beta/\gamma_R) = 1$.
$\partial \mathcal{R}_0/\partial \beta = c/\gamma_R$. $\mathcal{S}_\beta^{\mathcal{R}_0} = (\beta/(c\beta/\gamma_R))(c/\gamma_R) = 1$.
$\partial \mathcal{R}_0/\partial \gamma_R = -c\beta/\gamma_R^2$. $\mathcal{S}_{\gamma_R}^{\mathcal{R}_0} = (\gamma_R/(c\beta/\gamma_R))(-c\beta/\gamma_R^2) = -1$.
All three match the Chapter 2 results. The reverse-mode computation recovers the full sensitivity Jacobian in two passes: the forward evaluation and one backward pass.

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Chapter 6
The Adjoint for Dynamical Systems

To find what shaped an outcome, trace its influence backward through time.

The forward model tells you where you end up.
The adjoint tells you which moments mattered.

Climate models routinely have 500 or more tunable parameters. Forward sensitivity analysis using the FSE approach requires one ODE solve per parameter plus the baseline: 501 solves on a modern workstation, each taking 6 hours. Total: 3,006 hours, or about 125 days.

The adjoint approach requires two ODE solves for a single scalar QOI: one forward solve to generate the trajectory, one backward solve for the adjoint. Total cost: approximately 12 hours, plus storage and interpolation overhead for the forward solution.

The mathematics behind this 250-to-1 speedup is a direct extension of the matrix adjoint from Chapter 5. In that chapter, the key operation was transposing the coefficient matrix: $A \to A^T$. Here, the key operation is transposing the differential operator, which turns a forward ODE into a backward ODE. This chapter derives that operation step by step, explains in plain language why the backward direction arises, and implements the full calculation for the SIR epidemic model. The notation is one step heavier than in Chapter 5, because integration by parts and continuous-time functionals replace matrix transposes. The underlying idea is the same transposition argument in a new space. The adjoint sensitivity equations for ODE systems, including their software implementation, are treated in depth by Cao et al. [13]; a recent comparison of forward and adjoint methods on biological models is given by Mester et al. [34].

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6.1 ▪ The Timeline Figure

Figure 6.1 shows the most important fact of this chapter before any equations appear: the adjoint ODE runs backward in time, opposite to the forward ODE.

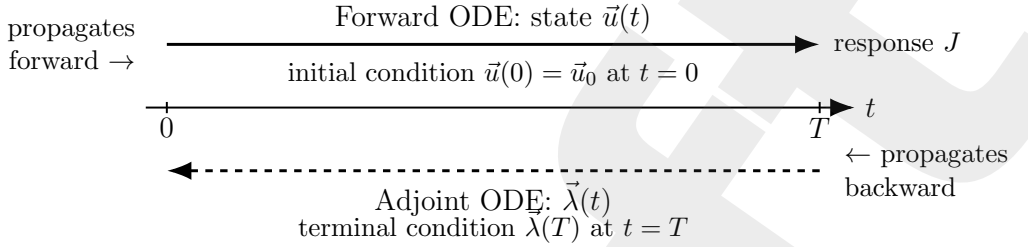


Figure 6.1. *The timeline for forward and adjoint sensitivity analysis. The forward ODE propagates the model state $\vec{u}(t)$ from the initial condition at $t = 0$ toward the response J at $t = T$. The adjoint ODE propagates the sensitivity weight $\vec{\lambda}(t)$ backward from a terminal condition at $t = T$, in the direction opposite to the forward problem. The two ODEs are not independent: the adjoint coefficient depends on the forward solution $\vec{u}(t)$ at every time step.*

The forward ODE begins at $t = 0$ and moves right, building up the state $\vec{u}(t)$ until it reaches the response J at time T. The adjoint ODE begins at $t = T$ and moves left, distributing the response backward through time.

This opposite direction is not a numerical trick. It is a mathematical necessity, explained in Section 6.2.

6.2 ▪ Why Backward? Intuition Before Derivation

The adjoint ODE runs backward because the question it answers is naturally backward: how much did each moment in time contribute to the final value of J?

Think of the epidemic. A perturbation applied to the infectious compartment at $t = 0$ has the entire simulation period to propagate: it changes the dynamics at $t = 1$, $t = 2$, and all the way to T. Its influence on J is large. A perturbation applied at $t = T - \epsilon$ has almost no time to propagate before the simulation ends. Its influence on J is negligible.

So the "weight" we should assign to perturbations decreases as we move from left to right along the time axis, and increases as we move from right to left. The adjoint variable $\lambda(t)$ is this weight: $\lambda(t)$ measures how sensitive the response J is to a perturbation of the state at time t.

At $t = T$, the weight is determined by the response functional (how does J change if the final state changes?). At earlier times, $\lambda(t)$ is computed by propagating this weight backward, accounting for how perturbations at time t have time to amplify before reaching T.

For a burden functional such as $J = \int_0^T I\,dt$ in the SIR model, the adjoint is largest near $t = 0$ and decays toward T: an early infection has the whole remaining epidemic in which to generate descendants, a late one does not. That shape is common but it is not a general rule. It depends jointly on the Jacobian, the running cost, and the terminal condition, and the scalar decay example later in this chapter reverses it: there $\lambda(t) = e^{p(t-T)}$ is *smallest* at $t = 0$ and largest at $t = T$. Read the shape off the dynamics rather than assuming it. The adjoint ODE starts at $t = T$, the only time at which we know λ directly, and integrates backward to find $\lambda(t)$ at all earlier times.

This must be understood before the derivation. The backward direction of the adjoint is not a consequence of the algebra; the algebra is a consequence of this backward structure.

The adjoint and forward problems are not independent: the coefficient in the adjoint ODE at every time t involves $\vec{u}(t)$, the forward solution. The adjoint can only be solved after the forward problem is solved and $\vec{u}(t)$ is stored at every time point needed.

Self-check 6.1. Suppose the response functional is $J = S(T)$ (the susceptible population at the final time) for the SIR model with $c = 5$, $\beta = 0.06$, $\tau_R = 7$, $\tau_m = 10{,}000$, $N = 1000$ and one initial infective. This epidemic is slow: it peaks near $t = 44$ and is still running at the end of a 90-day simulation. At which time ($t = 5$, $t = 44$, or $t = 80$) would you expect $\lambda_I(t)$ to be largest in magnitude, and what is its sign? Explain without any calculation.

6.3 ▪ The Lagrange Multiplier Derivation

6.3.1 ▪ A Warm-Up: Exponential Decay

Before the five-step derivation, work through the simplest possible case to see exactly what the adjoint ODE looks like and how it gives the answer. The warm-up uses three formulas derived below: the adjoint ODE (6.12), its terminal condition (6.13) and the sensitivity integral (6.15). It only substitutes into them.

Let $\dot{u} = -pu$ with $u(0) = 1$ and $J = u(T)$. The forward solution is $u(t) = e^{-pt}$, so $J = e^{-pT}$ and $dJ/dp = -Te^{-pT}$ (by direct differentiation; the "correct" answer).

The adjoint approach finds the same result differently. For $F(u; p) = -pu$ and $g = 0$, $h(u) = u$:

- Adjoint ODE: $-\dot{\lambda} = F_u\,\lambda = -p\lambda$, so $\dot{\lambda} = p\lambda$.
- Terminal condition: $\lambda(T) = +h'(u(T)) = +1$.
- Solution (running backward from T): $\lambda(t) = +e^{p(t-T)}$.

- Sensitivity formula: $dJ/dp = +\int_0^T \lambda(t)\, F_p\, dt$. Here $F_p = -u$ and $\lambda(t) = e^{p(t-T)}$ (backward from $\lambda(T) = +1$): $dJ/dp = +\int_0^T e^{p(t-T)}(-u(t))\, dt = -\int_0^T e^{p(t-T)} u_0 e^{-pt}\, dt = -u_0 e^{-pT} T = -T e^{-pT}$. ✓

The adjoint gave the same derivative using two ODE solves (one forward, one backward) for this single scalar QOI, with no dependence on m (how many parameters there are). If p were joined by 49 other parameters, the cost would still be two ODE solves plus 50 integrals, one per parameter.

Now we derive this procedure in full generality.

6.3.2 ▪ The Five-Step Derivation

The derivation proceeds through five numbered steps, each announced before it is executed. The scalar ODE case is treated fully; the vector case follows by replacing scalars with vectors and matrices.

6.3.3 ▪ Setup

Consider the initial value problem

$$\dot{u} = F(u; p), \qquad u(0) = u_0, \tag{6.1}$$

with a response functional

$$J = \int_0^T g(u)\, dt + h(u(T)). \tag{6.2}$$

Here g is the running cost (an integrand depending on the solution u but *not* directly on the parameter p) and h is a terminal cost (depending only on the final state). We also assume the initial condition u_0 is independent of p, so $\partial u_0/\partial p = 0$; this holds for all examples in this chapter. This covers the main cases in practice: total infectious person-time $J = \int I\, dt$, final state $J = u(T)$, and weighted combinations. When g depends explicitly on p, an additional term $\int_0^T g_p\, dt$ appears in the sensitivity formula; see Cao et al. [13] for the fully general treatment. We want dJ/dp without solving the FSE.

The strategy is identical to Chapter 5: introduce an auxiliary variable, choose it to cancel the unknown sensitivity $\partial u/\partial p$, and read off the result from what remains. In Chapter 5 the auxiliary variable was a vector $\vec{v}$ satisfying $A^T\vec{v} = \vec{c}$; here it is a function $\lambda(t)$ satisfying a backward ODE.

Step 1: Introduce the Lagrange multiplier.

We introduce an auxiliary function $\lambda(t)$, called the adjoint variable or Lagrange multiplier, and observe that, because $u(t)$ satisfies the ODE (6.1):

$$\int_0^T \lambda(t)\big[F(u;p) - \dot{u}\big]\,dt = 0.$$

This is the ODE constraint written as a vanishing integral. Define the augmented functional

$$\tilde{J} = J + \int_0^T \lambda(t)\big[F(u;p) - \dot{u}\big]\,dt. \tag{6.3}$$

Since the bracketed term is zero, $\tilde{J} = J$. The strategy, carried over from Chapter 5, is to choose $\lambda(t)$ so that $\partial u/\partial p$ disappears when we differentiate $\tilde{J}$ with respect to p.

Physical analogy. Think of a building inspector who keeps a ledger beside the building. The ledger is $\lambda(t)$. It is not the strain itself, and it does not change the building; it records how much each moment in time contributes to the structure's overall stress J. Adding the constraint integral adds an entry that is identically zero as long as the beams obey their equilibrium equations: it costs nothing, but it opens the door to measuring sensitivity.

Step 2: Differentiate with respect to p.
Denote $w = \partial u/\partial p$. Differentiating (6.3):

$$\frac{d\tilde{J}}{dp} = \int_0^T g_u\, w\,dt + h'(u(T))\,w(T) + \int_0^T \lambda\big[F_u\, w + F_p - \dot{w}\big]\,dt. \tag{6.4}$$

Differentiating λ itself would add $\int_0^T (\partial\lambda/\partial p)\big[F - \dot{u}\big]\,dt$, which is zero because the bracket vanishes for every p. The unknown $w = \partial u/\partial p$ appears four times: in $g_u w$, in $h'(u(T))\,w(T)$, in $F_u w$ and, as $\dot{w}$, in the last integral. The next step eliminates it by choosing λ appropriately.

Physical analogy. Differentiating with respect to p is like asking: if we nudge a design parameter, such as the stiffness of one beam, how does the overall stress J respond? The sensitivity $w = \partial u/\partial p$ represents the ripple that nudge sends through the entire structure over time. It appears everywhere because the system is coupled. The ledger $\lambda(t)$ will be chosen so that those ripples never have to be computed.

Step 3: Integrate by parts.
The term $\int_0^T \lambda\dot{w}\,dt$ contains $\dot{w}$, and no single choice of λ can cancel both w and $\dot{w}$ at every time. So we move the time derivative from w to λ by

integrating by parts. This is the key step, the ODE analog of the matrix transposition $v^T A = (A^T v)^T$ in Chapter 5.

The standard integration-by-parts identity, written out step by step, gives

$$\begin{aligned}\int_0^T \lambda\, \dot{w}\, dt &= \big[\lambda\, w\big]_0^T - \int_0^T \dot{\lambda}\, w\, dt \\ &= \lambda(T)\, w(T) - \lambda(0)\, w(0) - \int_0^T \dot{\lambda}\, w\, dt. \end{aligned} \tag{6.5}$$

The first line is the identity itself; the second evaluates the bracket at the endpoints, exposing two boundary terms that will behave very differently.

Substituting (6.5) into (6.4) and moving the integral $\int_0^T \lambda\, F_p\, dt$ to the left:

$$\begin{aligned}\frac{d\tilde{J}}{dp} - \int_0^T \lambda\, F_p\, dt &= \int_0^T g_u\, w\, dt + h'(u(T))\, w(T) \\ &\quad - \lambda(T)\, w(T) + \lambda(0)\, w(0) + \int_0^T \dot{\lambda}\, w\, dt + \int_0^T \lambda\, F_u\, w\, dt. \end{aligned} \tag{6.6}$$

Collecting the w-terms and reading off the condition for them to vanish:

$$\frac{d\tilde{J}}{dp} - \int_0^T \lambda\, F_p\, dt = \underbrace{\big[h'(u(T)) - \lambda(T)\big] w(T) + \lambda(0)\, w(0) + \int_0^T \big[g_u + \dot{\lambda} + \lambda F_u\big] w\, dt}_{\text{terms involving } w = \partial u/\partial p}. \tag{6.7}$$

Reading the two boundary terms. The two boundary terms in (6.7) require separate treatment, and each is handled by a different mechanism.

The $t = 0$ term vanishes automatically. The Setup requires $u(0) = u_0$ to be independent of p, so

$$w(0) = \frac{du(0)}{dp} = \frac{du_0}{dp} = 0, \qquad \text{hence} \qquad \lambda(0)\, w(0) = 0. \tag{6.8}$$

No condition on λ at $t = 0$ is required; the initial time is free.

The $t = T$ term is the opposite situation: it does not vanish on its own. The $-\lambda(T)\, w(T)$ produced by integration by parts meets the $h'(u(T))\, w(T)$ already generated by differentiating $h(u(T))$ in Step 2, and the two combine into a single coefficient of the unknown $w(T)$,

$$\big[h'(u(T)) - \lambda(T)\big]\, w(T). \tag{6.9}$$

This pairing is the second payoff of IBP: it manufactures a term whose coefficient we control, namely $\lambda(T)$, directly alongside a term whose coefficient we do not. Setting

$$\lambda(T) = +h'(u(T)) \tag{6.10}$$

zeroes the bracket and eliminates $w(T)$ from the sensitivity formula in one stroke. This is the terminal condition that Step 4 will formally adopt.

Remark 6.1. This is the moment where the terminal condition is not arbitrary; it is chosen precisely to eliminate the unknown $w(T)$ from the sensitivity formula.

Integration by parts is not merely an algebraic trick. It is exactly the operation that moves the time derivative from w to λ, transforming the FSE into the adjoint ODE, just as matrix transposition moved the linear operator from u to v in Chapter 5.

Physical analogy. Think of two runners passing a baton around a track. Forward mode carries $\dot{w}$; the baton is held by the sensitivity variable. Integration by parts hands the baton to λ: now $\dot{\lambda}$ carries the time derivative instead, and w appears without any derivative. This exchange is what makes it possible to choose λ to cancel w in the next step; without it, we could not get rid of $\dot{w}$ by specifying an ODE for λ alone.

Step 4: Choose λ to eliminate w.

For the sensitivity formula (6.7) to be free of w, the coefficient of w must vanish everywhere. This requires two conditions on λ; a third term vanishes by the Setup.

First, the integrand coefficient of $w(t)$ must vanish for all $t \in (0, T)$:

$$g_u + \dot{\lambda} + \lambda F_u = 0, \quad \text{i.e.,} \quad \dot{\lambda} = -F_u\,\lambda - g_u. \tag{6.11}$$

Multiplied by -1, this is the standard form used in the rest of the book:

$$\boxed{-\dot{\lambda} = F_u\,\lambda + g_u.} \tag{6.12}$$

Second, the coefficient of $w(T)$ must vanish:

$$h'(u(T)) - \lambda(T) = 0, \quad \text{i.e.,} \quad \lambda(T) = +h'(u(T)). \tag{6.13}$$

Third, the term $\lambda(0)w(0)$ must vanish. Since $u(0) = u_0$ does not depend on p, we have $w(0) = \partial u_0/\partial p = 0$, so this term vanishes automatically.

Terminal Condition: Two Cases That Cover Smooth, Fixed-Horizon QOIs

The terminal condition $\vec{\lambda}(T) = +\nabla h(\vec{u}(T))$ covers every smooth QOI evaluated at a fixed horizon, through two cases that combine additively:

(1) *Terminal cost* $J = h(\vec{u}(T))$: set $\vec{\lambda}(T) = +\nabla h(\vec{u}(T))$. For $J = S(T)$ (final susceptibles), $\lambda_S(T) = +1$, all others zero. A point evaluation $J = \vec{c}^{\top}\vec{u}(T)$ is this case with $h(\vec{u}) = \vec{c}^{\top}\vec{u}$, giving $\vec{\lambda}(T) = +\vec{c}$; it is not a separate rule.

(2) *Running cost* $J = \int_0^T g(\vec{u})\,dt$ with no terminal term ($h = 0$): set $\vec{\lambda}(T) = \vec{0}$. This is the case for $J = \int_0^T I\,dt$ (total burden).

The two combine: if $J = h(\vec{u}(T)) + \int_0^T g\,dt$, add the terminal conditions from (1) and (2). *Memorize these two cases.* They do not cover a QOI defined by an optimization over time, such as the peak prevalence $I_{\max}$ used in the Chapter 9 laboratory, or the time at which it occurs. Such a QOI is not a smooth function of the state at a fixed T, and Chapter 8 treats what goes wrong when it is differentiated as though it were.

The terminal condition (6.13) does not arise because we are running the ODE backward. It arises because the response functional involves the final state $h(u(T))$, and differentiating this with respect to p produces a boundary term at $t = T$. This is a fundamental distinction: the terminal condition is imposed by the response functional, not by the direction of integration.

Physical analogy. Choosing λ to kill every w-term does not change how the structure responds to a nudge; it arranges the ledger so that the ripple w never has to be measured. The terminal condition $\lambda(T) = +h'(u(T))$ is the inspector's final reading: at the end of the day the ledger must close, recording exactly how sensitive the last measured outcome is to the final state. Once $\lambda(t)$ is chosen, w has nowhere to appear in the sensitivity formula; it has been designed out.

Step 5: Read off the sensitivity formula.

With $\lambda(t)$ satisfying (6.12)–(6.13), all terms involving w in (6.7) vanish. The boxed formula below is the general result, stated without assuming anything about the initial condition; the assumption made at the start of the derivation, that u_0 does not depend on p, is what removes its second

term:

$$\boxed{\frac{dJ}{dp} = + \int_0^T \lambda(t)\, F_p(u(t); p)\, dt + \lambda(0)\, \frac{du_0}{dp}.} \tag{6.14}$$

When the initial condition does not depend on p, the second term vanishes. The sensitivity with respect to any parameter p_j is then:

$$\frac{dJ}{dp_j} = + \int_0^T \lambda(t)\, \frac{\partial F}{\partial p_j}\, dt. \tag{6.15}$$

After the forward solve, one backward ODE solve for $\lambda(t)$ gives all m sensitivities via m scalar integrals: two solves in all, exactly the cost advantage we sought.

Physical analogy. The inspector's record $\lambda(t)$ is now complete: one backward sweep through time, consulting the forward solution $u(t)$ at each moment, accumulates everything needed. Reading off dJ/dp_j is then as simple as computing a running total, one scalar integral per parameter, with no additional solves. The backward sweep is the price paid once; the m integrals are the dividend collected once for every parameter in the model.

6.3.4 ▪ The Vector Case

For the vector system $\dot{\vec{u}} = \vec{F}(\vec{u}; \vec{p})$ with $\vec{u} \in \mathbb{R}^n$ and functional $J = \int_0^T g(\vec{u})\, dt + h(\vec{u}(T))$, where, as before, g does not depend directly on $\vec{p}$, the derivation is identical with scalars replaced by vectors and F_u replaced by the Jacobian $J_F = \partial \vec{F}/\partial \vec{u}$. The adjoint ODE and terminal condition become:

$$-\dot{\vec{\lambda}} = J_F^T\, \vec{\lambda} + \nabla_u g^T, \tag{6.16}$$

$$\vec{\lambda}(T) = +\nabla_u h^T \big|_{t=T}. \tag{6.17}$$

Linearity of the adjoint ODE. Just as Theorem 4.1 showed that the FSE is always linear in the forward sensitivity variable $\partial \vec{u}/\partial p_j$, the adjoint ODE is always linear in $\vec{\lambda}(t)$, even when the forward ODE is nonlinear in $\vec{u}(t)$. The coefficient $J_F^T(\vec{u}(t))$ is a matrix that depends on the forward solution, but not on $\vec{\lambda}$ itself. This linearity is the ODE counterpart of the algebraic linearity from Chapter 5, and it is why the adjoint is tractable: regardless of how complex the forward dynamics are, the adjoint ODE is a linear ODE that can be solved efficiently.

Sign convention. We write the adjoint in backward-time form as $-\dot{\vec{\lambda}} = \cdots$ because that is what the derivation gives: the boxed equation (6.12) follows

from collecting the terms in (6.11), and it fixes the sign before any solver is chosen. Software must then match the equation, not the other way round. In particular, integrating with a solver over $[T, 0]$ traverses time downward while solving that same equation; the direction of travel is not a sign change, and an implementation that flips the sign because it runs backward has introduced a second error that happens to cancel the first. The sensitivity formula (6.18) then needs no further sign change when reading off dJ/dp_j from the integral.

The sensitivity formula becomes:

$$\frac{dJ}{dp_j} = + \int_0^T \vec{\lambda}^T \frac{\partial \vec{F}}{\partial p_j}\, dt + \vec{\lambda}(0)^T \frac{\partial \vec{u}_0}{\partial p_j}. \tag{6.18}$$

Unifying Idea 2 (the adjoint is the transpose; Section 1.5). The derivation is now complete, and this is the idea in its full generality. The adjoint in Chapter 5 was $A^T\vec{v} = \vec{c}$: the transposed coefficient matrix A^T applied to $\vec{v}$. The adjoint here is $-\dot{\vec{\lambda}} = J_F^T\vec{\lambda} + \nabla_u g^T$: the transposed Jacobian J_F^T applied to $\vec{\lambda}$, with the sign reversal arising from the integration by parts (6.5). The integration by parts is the exact functional-analytic analog of the matrix transposition $A \to A^T$ from Chapter 5. Appendix A.3 makes this analogy precise as integration by parts, Appendix B.2 states the Lagrange identity and derives the sensitivity formula (6.18) rigorously from it, and Appendix B.4 sets the objects of Chapters 5 and 6 side by side. The terminal condition (6.17) is the time-dependent analog of the right-hand side $\vec{c}$ in $A^T\vec{v} = \vec{c}$. This is one idea.

Self-check 6.2. For the scalar ODE $\dot{u} = -pu$, $u(0) = u_0$, take the running cost $J = \int_0^T u\, dt$ ($g(u) = u$, $h = 0$) in place of the warm-up's $J = u(T)$: (a) Write the adjoint ODE and terminal condition using equations (6.12) and (6.13). (b) Solve the adjoint ODE analytically. (c) Use formula (6.15) to compute dJ/dp. (d) Verify by solving the forward ODE analytically and differentiating.

6.4 ▪ The SIR Model: Full Adjoint Treatment

We apply the derivation to the SIR model, writing all adjoint equations explicitly and implementing the full calculation in the laboratory (Section 6.6).

6.4.1 ▪ Setup and Response Functional

The SIR model (4.15)–(4.17) with interpretable POIs $\vec{p} = (c, \beta, \tau_R, \tau_m)$:

$$\begin{aligned} F_S &= \nu_m N - \frac{c\beta SI}{N} - \nu_m S, \\ F_I &= \frac{c\beta SI}{N} - (\gamma_R + \nu_m) I, \\ F_R &= \gamma_R I - \nu_m R, \end{aligned}$$

where $\gamma_R = 1/\tau_R$ and $\nu_m = 1/\tau_m$. Here N is a fixed parameter. Births balance deaths, so $S + I + R$ stays equal to N along every solution, and the Jacobian below does not differentiate through N. That choice leaves every dJ/dp_j unchanged but does affect $\vec{\lambda}$ (Self-check 6.3). The response functional is

$$J = \int_0^T I(t)\, dt, \tag{6.19}$$

the total number of infectious person-days: a measure of the total disease burden. Here $g(\vec{u}) = I$, $h = 0$.

6.4.2 ▪ The Jacobian

The Jacobian $J_F = \partial(F_S, F_I, F_R)/\partial(S, I, R)$ evaluated at the forward solution $(S(t), I(t), R(t))$:

$$J_F = \begin{pmatrix} -\dfrac{c\beta I}{N} - \nu_m & -\dfrac{c\beta S}{N} & 0 \\ \dfrac{c\beta I}{N} & \dfrac{c\beta S}{N} - \gamma_R - \nu_m & 0 \\ 0 & \gamma_R & -\nu_m \end{pmatrix}. \tag{6.20}$$

6.4.3 ▪ Adjoint ODEs

With $g_S = 0$, $g_I = 1$, $g_R = 0$ (since $g = I$), the adjoint system $-\dot{\vec{\lambda}} = J_F^T \vec{\lambda} + \nabla g^T$ written out fully is:

$$-\dot{\lambda}_S = \left(-\frac{c\beta I}{N} - \nu_m \right) \lambda_S + \frac{c\beta I}{N}\, \lambda_I, \tag{6.21}$$

$$-\dot{\lambda}_I = -\frac{c\beta S}{N}\, \lambda_S + \left(\frac{c\beta S}{N} - \gamma_R - \nu_m \right) \lambda_I + \gamma_R \lambda_R + 1, \tag{6.22}$$

$$-\dot{\lambda}_R = -\nu_m \lambda_R. \tag{6.23}$$

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Each adjoint equation uses one *column* of J_F, because the matrix is transposed: the equation for λ_I collects the second column of (6.20), $(-c\beta S/N,\ c\beta S/N - \gamma_R - \nu_m,\ \gamma_R)$. Terminal conditions (since $h = 0$):

$$\lambda_S(T) = \lambda_I(T) = \lambda_R(T) = 0. \tag{6.24}$$

These three ODEs are integrated backward from $t = T$ to $t = 0$, using the stored forward solution $(S(t), I(t), R(t))$ to evaluate the time-varying coefficients. For a terminal cost such as $J = I(T)$, the running-cost term $+1$ in (6.22) drops out ($g = 0$), and case (1) of the box above gives $\vec{\lambda}(T) = (0, 1, 0)^T$ instead.

6.4.4 ▪ Sensitivity Formulas

For each POI, the sensitivity is the integral (6.18) evaluated using the stored $\vec{\lambda}(t)$:

$$\frac{dJ}{d\beta} = + \int_0^T \frac{cSI}{N}(\lambda_I - \lambda_S)\, dt, \tag{6.25}$$

$$\frac{dJ}{dc} = + \int_0^T \frac{\beta SI}{N}(\lambda_I - \lambda_S)\, dt, \tag{6.26}$$

$$\frac{dJ}{d\tau_R} = + \int_0^T \left[\frac{\partial F_I}{\partial \tau_R}\lambda_I + \frac{\partial F_R}{\partial \tau_R}\lambda_R\right] dt = + \int_0^T \frac{I}{\tau_R^2}(\lambda_I - \lambda_R)\, dt, \tag{6.27}$$

$$\frac{dJ}{d\tau_m} = + \int_0^T \left[\frac{\partial F_S}{\partial \tau_m}\lambda_S + \frac{\partial F_I}{\partial \tau_m}\lambda_I + \frac{\partial F_R}{\partial \tau_m}\lambda_R\right] dt. \tag{6.28}$$

Normalized sensitivity indices follow from Definition 2.1: $\mathcal{S}^J_{p_j} = (p_j/J)(dJ/dp_j)$.

Multiple QOIs and correlated outputs. When the study involves $r > 1$ response functionals, for example both $J_1 = \int_0^T I\, dt$ (total burden) and $J_2 = S(T)$ (final susceptibles), the adjoint is solved once per QOI, giving r backward solutions $\vec{\lambda}^{(1)}(t), \ldots, \vec{\lambda}^{(r)}(t)$. Both J_1 and J_2 depend on the same forward trajectory $\vec{u}(t)$, so a parameter change moves both. As for the equilibrium in Chapter 5, the adjoint needs no special treatment for this.

6.4.5 ▪ The Adjoint Variable and Its Interpretation

Figure 6.2 shows $I(t)$ and $\lambda_I(t)$ computed for nominal parameter values over a 90-day simulation.

The adjoint variable $\lambda_I(t)$ has a concrete epidemiological interpretation: it measures the marginal benefit of reducing the infectious population at

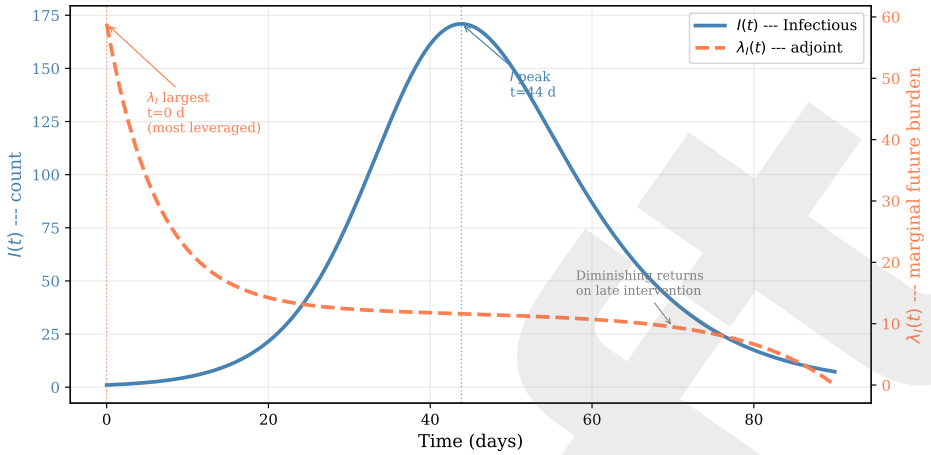


Figure 6.2. *Computed $I(t)$ (solid, left axis) and the adjoint variable $\lambda_I(t)$ (dashed, right axis) for the SIR model at nominal parameter values ($c = 5$, $\beta = 0.06$, $\tau_R = 7$, $N = 1000$, with $S(0) = 999$, $I(0) = 1$, $R(0) = 0$) over a 90-day simulation, obtained from a forward solve followed by a backward adjoint solve. The curves are drawn without the demographic turnover term, which at $\tau_m = 10^4$ days moves either curve by less than 2% of its peak. The right axis is in days, the units of $\lambda_I = \partial J/\partial I$ for $J = \int I\,dt$. The adjoint variable is largest near $t = 0$, when a perturbation to the infectious compartment has the most time to affect the total burden $J = \int I\,dt$, and decays toward zero as $t \to T$. This gives $\lambda_I(t)$ a concrete interpretation as the marginal value of reducing the infectious population at time t: interventions applied when λ_I is large avert the most infectious person-time.*

time t. When λ_I is large (early epidemic), reducing I by one person-day saves many future infections. When λ_I is small (late epidemic), most transmission has already occurred, so the marginal benefit is low.

Put concretely: $\lambda_I(t)$ measures the additional *infectious person-time* accumulated over the rest of the window from having one extra infectious individual present at time t, propagated through all secondary, tertiary, and downstream infections, under the linearized dynamics about the nominal trajectory. This is a local, first-order statement: it is exact for infinitesimal perturbations and remains a good approximation for small finite perturbations, but does not account for nonlinear feedback as the perturbation grows large. To connect this to the epidemic dynamics: the force of infection $c\beta I(t)/N$ (equation (2.16)) gives the instantaneous probability per day that any one susceptible individual becomes infected. The adjoint variable $\lambda_I(t)$ carries this one step further; it measures the cumulative downstream burden on $J = \int_0^T I\,dt$ attributable to one infectious individual present at time t. Early in the epidemic, when the susceptible pool is large and the force of infection $c\beta I(t)/N$ is growing, one extra infection generates many secondary cases; late in the epidemic, when susceptibles are depleted, the same extra infection propagates little further transmission. Quantitatively:

one extra infectious person at time t increases the force of infection by $c\beta/N$ per susceptible per day and initiates a cascade of secondary, tertiary, and subsequent infections whose total contribution to J is $\lambda_I(t)$. Note the units. Since $J = \int_0^T I\,dt$ has dimensions of person-time, $\lambda_I = \partial J/\partial I$ has dimensions of *time*, not of a case count. Reading it as "cases averted" requires a separate incidence model, since cumulative prevalence and case count are different quantities. With that caveat, λ_I still ranks *when* to act: early action stops a cascade, late action only a trickle. Turning that ranking into a statement about cost-effectiveness requires a control and a cost model; the relevant object is then $\delta J = \int_0^T \vec{\lambda}(t)^\top F_p(t)\,\delta p(t)\,dt$ for a control $p(t)$, a parameter that varies in time (Appendix E.4). It weights $\vec{\lambda}$ by how the control actually enters the dynamics.

This interpretation is available from the adjoint computation alone, with no additional work.

The terminal condition $\vec{\lambda}(T) = \vec{0}$ is not the same as starting the forward ODE from initial condition $\vec{0}$ and running it in reverse. The two problems have different structures: the forward ODE is driven by $F(\vec{u};\vec{p})$, while the adjoint ODE is driven by $J_F^T\vec{\lambda}$ with a forcing term from g. The backward direction comes from the integration by parts, not from reversing time. The distinction matters for implementation: you run the ODE solver with `tspan = [T, 0]`, not with a negated vector field.

Unifying Idea 3 (local SA is a linearization; Section 1.5). The adjoint variable $\lambda_I(t)$ illustrates this idea. The adjoint variable displays the local sensitivity of J throughout the simulation period. This continuous record of sensitivity is the local SA analog of what global SA asks globally: how does J respond to perturbations at each moment in time? Chapter 9 will ask the same question for perturbations of the parameters across their full uncertainty ranges rather than perturbations of the state at specific times. The adjoint provides the linearization; global SA provides the full nonlinear picture.

Unifying Idea 1 (the sensitivity index is a normalized derivative; Section 1.5). The sensitivity indices $\mathcal{S}_{p_i}^J$ computed in this chapter via the adjoint are exactly the normalized derivatives of Unifying Idea 1 from Chapter 1: the ratio of relative changes in J to relative changes in each POI p_i. The adjoint is simply a dramatically more efficient algorithm for computing those same derivatives when the model has many POIs and few QOIs.

Self-check 6.3. The adjoint variable $\lambda_R(t)$ for the recovered compartment satisfies $-\dot{\lambda}_R = -\nu_m\lambda_R$ with $\lambda_R(T) = 0$. (a) Solve this ODE analytically (backward from T). (b) What does the solution tell you about the sensitivity of J to perturbations of the recovered compartment?

6.5 ▪ Discrete vs. Continuous Adjoint

The adjoint ODE (6.16) is the *continuous adjoint*: it is derived by differentiating the continuous ODE and then discretizing for numerical integration. An alternative is the *discrete adjoint*: discretize the ODE first and then adjoint the discrete system. For linear ODEs with fixed step sizes, the two approaches give identical results. For nonlinear ODEs with adaptive step-size control, they can differ.

In practice: if your ODE solver uses fixed step sizes, either adjoint is acceptable. If it uses adaptive step sizes (as most library solvers do by default), the continuous adjoint requires storing the forward solution densely enough to interpolate accurately, while the discrete adjoint requires differentiating through the step-size controller. For the SA computations in this book, the continuous adjoint with dense output storage is the more practical approach; set the solver tolerances by the ODE Solver Tolerance Rule of Section 3.3, and see Section 7.4 for adaptive step-size solvers. Sirkes and Tziperman [58] compare the continuous and discrete adjoints for geophysical models solved with fixed time steps.

Neural ODEs. In 2018, Chen et al. [14] demonstrated that the adjoint calculation derived in this chapter can be used to train certain deep neural networks. The network defines $\vec{F}(\vec{u};\theta)$ for parameters θ; the loss function plays the role of J; and the training algorithm integrates the adjoint ODE backward to compute $\partial J/\partial\theta_i$ for all network parameters. In machine learning, "backpropagation through time" names the same backward pass for recurrent neural networks, which apply one set of parameters again at every time step; it is the discrete version of this backward integration. The continuous limit is the adjoint ODE derived here.

Beyond ODEs: A Pointer to the Wider Adjoint Universe

The adjoint ODE in this chapter is the simplest instance of a broader mathematical structure that scales to much larger problems. When the governing model is a partial differential equation (PDE) rather than an ODE, such as a groundwater transport model, an atmospheric circulation model or a structural mechanics simulation, the adjoint becomes a PDE solved backward in both time and space, with boundary conditions imposed at the final time in the same way that the terminal condition $\vec{\lambda}(T) = +\nabla_u h^T$ is imposed here. Cacuci [9] derived this adjoint formalism for nonlinear functional equations in 1981; Giles and Pierce [17] review its use in aerodynamic design optimization. One example they report minimizes the drag of a business jet whose wing surface is described by 90 shape parameters.

In data assimilation, which fits a dynamical model to time-series obser-

vations, the adjoint provides the gradient of the data-mismatch functional with respect to all model parameters simultaneously. This is the mathematical engine behind the 4D-Var algorithm (four-dimensional variational assimilation, the standard method in operational numerical weather prediction) where the model has millions of state variables and the assimilation window spans several hours. The MATLAB adjoint in this chapter is the direct analog of all these applications: the mathematical structure (introduce a Lagrange multiplier function, differentiate, integrate by parts, choose the multiplier to cancel the unknown sensitivity) is identical in every setting; only the function space changes. A systematic graduate-level development of this adjoint formalism for general nonlinear functional equations, including PDEs and integral equations, is in Cacuci [10].

6.6 ▪ Computational Laboratory: The SIR Adjoint in Practice

This laboratory implements the full forward-adjoint computation for the SIR model and verifies the result against the FSE approach from Chapter 4.

Setup. Use the SIR model with POIs $\vec{p} = (c, \beta, \tau_R, \tau_m)$, nominal values $c = 5$, $\beta = 0.06$, $\tau_R = 7$, $\tau_m = 10{,}000$, initial conditions $S_0 = 999$, $I_0 = 1$, $R_0 = 0$, $N = 1000$, and $T = 90$ days. The response functional is $J = \int_0^{90} I(t)\, dt$.

Part 1: Solve the forward SIR with dense output. Solve the SIR system using an adaptive ODE solver with option `'Refine', 8` or dense interpolation, storing the solution on a fine uniform grid with spacing $\Delta t = 0.1$ days. Store (S, I, R) at every grid point; this array is needed for the adjoint coefficient evaluation.

Interpolation (required before the backward solve). Before integrating the adjoint backward, build interpolants of the forward solution so the adjoint ODE can evaluate $S(t)$, $I(t)$, and $R(t)$ at any time during the backward pass:

```
# Build smooth interpolants from forward solution grid
# (use cubic/pchip or similar smooth interpolant)
S_interp = INTERPOLANT(t_fwd, U[:, S_col])
I_interp = INTERPOLANT(t_fwd, U[:, I_col])
R_interp = INTERPOLANT(t_fwd, U[:, R_col])
# These allow evaluation at any t in [0, T] during the backward adjoint
```

Pass these interpolants into the adjoint function. Skipping this step is the most common cause of errors in the adjoint lab.

Part 2: Solve the adjoint system backward. Implement the adjoint ODEs (6.21)–(6.23). Note that these depend on the forward solution $(S(t), I(t), R(t))$, which must be interpolated from the stored grid.

```
# Adjoint ODE for SIR model - integrate BACKWARD from T to 0
# PLUS convention: -d(lambda)/dt = J_F^T * lambda + grad_u(g)
# g = I(t), so grad_u(g) = (0, 1, 0)

FUNCTION sir_adjoint(t, lam, params):
    N = 1000;  gamma_R = 1/tau_R;  nu_m = 1/tau_m
    Sv = S_interp(t);  Iv = I_interp(t);  Rv = R_interp(t)

    iS = 0;  iI = 1;  iR = 2          %% index positions
    lS = lam[iS];  lI = lam[iI];  lR = lam[iR]
    lam_dot = [0, 0, 0]

    # d(lS)/dt  (no running cost term for S)
    lam_dot[iS] = -( (-c*beta*Iv/N - nu_m)*lS + (c*beta*Iv/N)*lI )

    # d(lI)/dt  (+1 from grad_u(g) = 1, PLUS convention)
    lam_dot[iI] = -( (-c*beta*Sv/N)*lS + (c*beta*Sv/N - gamma_R - nu_m)*
                    + gamma_R*lR + 1 )

    # d(lR)/dt  (no running cost term for R)
    lam_dot[iR] = -( -nu_m*lR )

    RETURN lam_dot
END

# Solve: lam(T) = (0,0,0),  tspan = [T, 0]  (backward)
LAM = SOLVE_ODE(sir_adjoint, tspan=[T, 0], y0=[0,0,0])
```

Solve from $t = T$ to $t = 0$ with terminal conditions $\vec{\lambda}(T) = (0, 0, 0)$.

Part 3: Plot $I(t)$ and $\lambda_I(t)$. On a single axis, plot the epidemic curve $I(t)$ (left scale) and the adjoint variable $\lambda_I(t)$ (right scale). Annotate the epidemic peak. Confirm that λ_I is largest near $t = 0$ and smallest near $t = T$.

Part 4: Compute sensitivities and produce a tornado plot. Evaluate the integrals (6.25)–(6.28) numerically using the trapezoidal rule. Normalize to obtain $\mathcal{S}^J_{p_j}$ for all four POIs. Produce a tornado plot and identify which parameter has the greatest influence on the total disease burden.

Part 5: Verify against Chapter 4 FSE results. From the Chapter 4 computational lab, retrieve the SI values computed by the FSE at a single time point. Compare with the adjoint-based SIs for $J = \int I\,dt$. Note: the two computations answer *different* questions (the FSE computes $\mathcal{S}_{p_j}^{I(t)}$ at a fixed t, while the adjoint computes $\mathcal{S}_{p_j}^{\int I\,dt}$), so the numerical values will differ. Instead, verify that $\mathcal{S}_\beta^J = \mathcal{S}_c^J$ to at least 4 significant figures (the two indices are equal by the product structure of $c\beta$ in the SIR model, but their common value depends on the parameters and is not generally 1) and discuss whether the ratio holds exactly.

Part 6: Timing comparison. Time the adjoint computation (Parts 1–4) against the FSE approach from Chapter 4 (solve the 15-equation augmented system once; it carries all four sensitivities). At what number of parameters m would you expect the adjoint to be faster? Does the timing crossover occur near $m = r = 1$, as the solve counts predict ($m+1$ sequential FSE solves against $r + 1$ adjoint solves)?

6.7 ▪ Exercises

Exercise 6.1 *(Bloom L3).* For the scalar ODE $\dot{u} = ru(1 - u/K)$ (logistic) with $J = \int_0^T u\,dt$ (total population over time):

(a) Write the adjoint ODE and terminal condition.

(b) Show that the adjoint ODE is linear in $\lambda(t)$ even though the forward ODE is nonlinear in $u(t)$.

(c) Explain why the adjoint ODE depends on the forward solution $u(t)$.

(d) In MATLAB, solve the forward and adjoint ODEs simultaneously (forward from 0 to T; store; then backward from T to 0) for $r = 0.5$, $K = 100$, $u_0 = 10$, $T = 20$. Plot $u(t)$ and $\lambda(t)$.

Exercise 6.2 *(Bloom L3).* State in your own words what integration by parts accomplishes in Step 3 of the Lagrange multiplier derivation, and why this is analogous to the matrix transposition $A \to A^T$ in Chapter 5. Then verify equation (6.5) directly by differentiating the product λw and integrating over $[0, T]$.

Exercise 6.3 *(Bloom L4).* Using your MATLAB implementation from the laboratory:

(a) Compute $\mathcal{S}_\beta^J$, $\mathcal{S}_c^J$, $\mathcal{S}_{\tau_R}^J$, $\mathcal{S}_{\tau_m}^J$ via the adjoint for $J = \int_0^{90} I\,dt$.

(b) Verify that $\mathcal{S}_c^J = \mathcal{S}_\beta^J$ to at least 6 significant figures. Explain this result using the structure of the SIR model.

(c) If you could implement one intervention that reduces a single parameter by 20%, which parameter would you target to most reduce the total disease burden? Justify using the tornado plot.

(d) Compare the computation time with the FSE approach from Chapter 4.

Exercise 6.4 *(Bloom L4).* Fill in the computation cost table for the SIR model with $r = 1$ response functional ($J = \int I\, dt$) and $n = 3$ state variables:

m (number of POIs)	FSE cost (ODE solves)	Adjoint cost (ODE solves)
4		
10		
50		
500		

If each ODE solve takes 30 seconds, at what value of m does the adjoint become more than 10 times faster than the FSE?

Exercise 6.5 *(Bloom L4).* Explain in plain language, without any equations, why:

(a) The adjoint ODE must be integrated backward from $t = T$ rather than forward from $t = 0$.

(b) The adjoint variable $\lambda_I(t)$ for the SIR model and response $J = \int I\, dt$ is larger near $t = 0$ than near $t = T$.

(c) A public health agency that wants to minimize total disease burden should focus interventions on the early phase of the epidemic, when λ_I is large. Use $\lambda_I(t)$ to quantify this recommendation.

(d) The terminal condition $\vec{\lambda}(T) = \vec{0}$ is not the same as running the forward ODE backward with initial condition $\vec{0}$.

Exercise 6.6 *(Bloom L5).* Extend the SIR adjoint treatment to the SEIR model:

$$\begin{aligned}\dot{S} &= \nu_m N - c\beta SI/N - \nu_m S,\\ \dot{E} &= c\beta SI/N - (\nu + \nu_m)E,\\ \dot{I} &= \nu E - (\gamma_R + \nu_m)I,\\ \dot{R} &= \gamma_R I - \nu_m R,\end{aligned}$$

with POIs $\vec{p} = (c, \beta, \tau_E, \tau_I, \tau_m)$ (interpretable parameterization: $\nu = 1/\tau_E$, $\gamma_R = 1/\tau_I$, $\nu_m = 1/\tau_m$) and response functional $J = \int_0^T I(t)\,dt$.

Design the complete adjoint SA study, specifying:

(a) The 4×4 Jacobian J_F at a generic point (S, E, I, R).

(b) All four adjoint ODEs and their terminal conditions.

(c) The sensitivity formulas dJ/dp_j for all five POIs.

(d) The MATLAB implementation strategy (how you would store the forward solution and integrate the adjoint backward).

(e) What additional cost the SEIR adjoint requires compared to the SIR adjoint (in terms of ODE solves and storage).

Exercise 6.7 *(Bloom L6).* A climate modeler reports: "We implemented the adjoint to compute sensitivities for our model with 200 parameters, but the results disagree with finite-difference estimates by approximately 15% for some parameters. The adjoint must have a bug."

Identify three distinct causes that could produce this discrepancy without any coding error. For each cause:

(a) State the cause precisely.

(b) Describe a diagnostic test that would confirm or rule it out.

(c) Describe the correction if the cause is confirmed.

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Notes

The adjoint equation for systems of ordinary differential equations, running backward from $t = T$ with a terminal condition determined by the response functional, was established in the nonlinear setting by Cacuci [9]. The formulation adopted in this chapter follows Cao, Li, and Petzold [13], who treat the adjoint system for differential-algebraic equations, with ODEs as the index-0 case, and implement it in the solver DASPKADJOINT. That solver handles the memory challenge of backward integration by two-level checkpointing of the forward trajectory (Section 7.4).

The Pontryagin maximum principle, which gave the first systematic treatment of adjoint variables for optimal control problems in continuous time, appears in Pontryagin et al. [44]; the 4D-Var algorithm in numerical weather prediction, which is essentially the adjoint of the atmospheric model applied to data assimilation, is surveyed by Navon [38].

Sirkes and Tziperman [58] discuss the distinction between the *continuous adjoint* (derived from the continuous ODE and then discretized) and the *discrete adjoint* (the exact adjoint of the discrete numerical scheme). For adaptive-step-size solvers these can differ; the continuous adjoint adopted in this chapter is the standard choice in mathematical epidemiology.

Smith [60] develops the adjoint sensitivity analysis procedure (ASAP) for general parameter-dependent dynamical systems in Chapter 8 of that book, on local sensitivity analysis, presenting both perturbation and variational derivations within a functional-analytic framework. The first edition of Smith [59] developed the forward (FSAP) and adjoint (ASAP) sensitivity analysis procedures in functional-analytic detail, including supporting appendix material. The second edition references this material as supplemental and focuses its Chapter 8 on a more compact treatment grounded in adjoint methods, parameter subset selection, and complex-step approximation. The chapters of the present book complement Smith's treatment by developing the matrix, computation-graph, and ODE adjoint settings as instances of one organizing idea, at a level designed to be the reader's first careful encounter with the adjoint method.

Answers to the self-checks

6.1. $\lambda_I(t)$ measures how much $J = S(T)$ changes per extra infective at time t. It is negative at all three times: an extra infective causes further infections, and each one removes a susceptible before T. How long the perturbation has to propagate does not by itself fix the size. In a basic SIR epidemic that burns out well before T, one extra case causes about the same number of further infections whenever it arrives, and the three values come out nearly equal. Here the epidemic is still running at $T = 90$. At $t = 5$ an extra infective adds about half again to an

infectious pool of about two, so it moves the whole epidemic earlier, and an earlier epidemic has removed more susceptibles by day 90. At $t = 44$ the extra case adds only its own chain of infections, and at $t = 80$ that chain has 10 days to run. So $|\lambda_I|$ should be largest at $t = 5$ and smallest at $t = 80$. The computed values agree: $\lambda_I(5) \approx -1.63$, $\lambda_I(44) \approx -0.63$ and $\lambda_I(80) \approx -0.37$.

6.2. (a) $F = -pu$, so $F_u = -p$, and $g_u = 1$. Adjoint ODE: $-\dot\lambda = -p\lambda + 1$, i.e., $\dot\lambda = p\lambda - 1$. Terminal condition: $\lambda(T) = +h'(u(T)) = 0$.

(b) The general solution is $\lambda(t) = 1/p + Ce^{pt}$, and $\lambda(T) = 0$ gives $C = -e^{-pT}/p$. Solving from $t = T$ back to $t = 0$: $\lambda(t) = \big(1 - e^{p(t-T)}\big)/p$, which is positive for $t < T$.

(c) $F_p = -u$ and $u(t) = u_0 e^{-pt}$. Sensitivity formula: $dJ/dp = -\int_0^T \lambda(t)u(t)\,dt = -\dfrac{u_0}{p}\int_0^T \big(e^{-pt} - e^{-pT}\big)\,dt = -\dfrac{u_0}{p}\left[\dfrac{1 - e^{-pT}}{p} - Te^{-pT}\right]$.

(d) $J = \int_0^T u_0 e^{-pt}\,dt = u_0(1 - e^{-pT})/p$, so $dJ/dp = u_0 Te^{-pT}/p - u_0(1 - e^{-pT})/p^2$, the same as (c). ✓

6.3. (a) $-\dot\lambda_R = -\nu_m \lambda_R$ gives $\dot\lambda_R = \nu_m \lambda_R$ with $\lambda_R(T) = 0$. Solution: $\lambda_R(t) = \lambda_R(T)\, e^{\nu_m (t-T)} = 0$ for all $t \in [0, T]$. (The equation is linear with no forcing term, and $\lambda_R \equiv 0$ satisfies it and the terminal condition; by uniqueness it is the only solution.) This uses N held at its constant value, as in the Jacobian (6.20). If $N = S + I + R$ were differentiated as a state, F_S and F_I would depend on R through N, and λ_R would no longer vanish.

(b) $\lambda_R(t) = 0$ means that perturbations to $R(t)$ have no effect on $J = \int I\,dt$. This makes biological sense: recovered individuals do not contribute to future transmission (in the basic SIR model without waning immunity), so changing $R(t)$ does not change the integral of I.

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Chapter 7
Adjoint Sensitivity in Practice

Reverse the calculation, and one output can speak about many inputs.

You need the sensitivities of a complex disease model of $n = 30$ coupled ODEs with $m = 30$ parameters. The model is your own code, built from smooth operations and integrated with a fixed step, so the code is differentiable. You know from Chapter 6 that the adjoint requires only 2 ODE solves, against the 31 of a *sequential* FSE, one augmented solve per parameter plus the baseline. A *simultaneous* FSE is instead a single solve of dimension $n(1+m)$: cheaper than 31 separate solves, but still growing with m, whereas the adjoint cost does not.

The adjoint system has one equation per state, so the work of deriving it grows with n, not with m. Deriving it by hand for 30 coupled nonlinear ODEs would take weeks and would almost certainly introduce errors in the derivation.

Is there a way to obtain the adjoint computation without deriving the adjoint equations by hand? Yes, algorithmic differentiation does this automatically [37]. And there is a way to use it from MATLAB today, without writing a single AD routine from scratch.

7.1 ▪ The Annuity Function: A Worked Comparison

Before examining the computational tools, we work through a concrete example from finance where all three methods, forward mode, reverse mode, and symbolic differentiation, can be compared exactly. Finance is a natural first case: AD is used there to compute the sensitivities of prices to their inputs (the "Greeks" of option pricing).

7.1.1 ▪ The Problem

An annuity pays a fixed amount P per period for n periods at interest rate r. A payment due k periods from now is worth $P(1+r)^{-k}$ today, because that sum, invested at rate r, grows to P in k periods. Summing this geometric series over $k = 1, \dots, n$ gives the present value

$$A(P, r, n) = P\,\frac{1 - (1+r)^{-n}}{r}. \tag{7.1}$$

We treat n as a real number, so that $\partial A/\partial n$ is defined, and we want all three sensitivity indices: $\mathcal{S}_P^A$, $\mathcal{S}_r^A$, $\mathcal{S}_n^A$.

7.1.2 ▪ Forward Mode: Three Passes

In forward mode we differentiate the computation one parameter at a time, propagating $\dot v_k = \partial v_k/\partial p$ through intermediate values. We trace the computation of A through the following steps:

$$v_1 = (1+r)^{-n}, \quad v_2 = 1 - v_1, \quad v_3 = v_2/r, \quad A = P v_3.$$

For $p = P$: set $\dot P = 1$, $\dot r = 0$, $\dot n = 0$.

$$\dot v_1 = 0, \quad \dot v_2 = 0, \quad \dot v_3 = 0, \quad \dot A = \dot P\, v_3 + P\, \dot v_3 = v_3.$$

So $\partial A/\partial P = v_3 = (1 - (1+r)^{-n})/r$.

For $p = r$: set $\dot r = 1$, $\dot P = 0$, $\dot n = 0$.

$$\dot v_1 = -n(1+r)^{-n-1}, \quad \dot v_2 = n(1+r)^{-n-1}, \quad \dot v_3 = \frac{\dot v_2\, r - v_2}{r^2}, \quad \dot A = P\, \dot v_3.$$

For $p = n$: set $\dot n = 1$, $\dot P = 0$, $\dot r = 0$.

$$\dot v_1 = -\ln(1+r)\,(1+r)^{-n}, \quad \dot v_2 = \ln(1+r)\,(1+r)^{-n}, \quad \dot v_3 = \dot v_2/r, \quad \dot A = P\, \dot v_3.$$

AD never forms these expressions. At a given (P, r, n) each line is evaluated to a number as soon as its inputs are known, and only the number is kept; we print the formulas so that each line can be checked. Three passes through the computation graph give three derivatives. Each pass repeats the evaluation of A and adds its own tangent operations, so the total cost grows in proportion to the number of parameters. The operation counts below make this exact.

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7.1.3 ▪ Reverse Mode: Two Passes

The mechanism is the one shown in Figure 1.3, where the same computation graph is traversed in both directions: forward mode carries a tangent from one input to the output, and reverse mode carries an adjoint from the output back to every input at once. What follows is that picture applied to the annuity. Figure 7.1 draws the annuity's graph in both directions.

In reverse mode, we compute all three derivatives in one backward pass. Set $\bar{A} = 1$. Work backward through the computation:

$$\begin{aligned}
\bar{P} &= \bar{A}\, v_3 = v_3, \\
\bar{v}_3 &= \bar{A}\, P = P, \\
\bar{v}_2 &= \bar{v}_3 / r, \\
\bar{r} &\mathrel{+}= \bar{v}_3 \cdot (-v_2/r^2), \\
\bar{v}_1 &= -\bar{v}_2, \\
\bar{r} &\mathrel{+}= \bar{v}_1 \cdot \frac{\partial v_1}{\partial r} = \bar{v}_1 \cdot \big(-n(1+r)^{-n-1}\big), \\
\bar{n} &= \bar{v}_1 \cdot \frac{\partial v_1}{\partial n} = \bar{v}_1 \cdot \big(-\ln(1+r)(1+r)^{-n}\big).
\end{aligned}$$

Two points of discipline are visible here. The sweep must visit each node only after every node that consumes it, so $\bar{v}_1$ is assigned before it is used. And r feeds two downstream nodes, v_1 and v_3, so $\bar{r}$ *accumulates*: the two contributions are added, not assigned. Writing $=$ in place of $+=$ silently drops one path and is among the most common errors in hand-written reverse sweeps.

One backward pass gives $\partial A/\partial P$, $\partial A/\partial r$, and $\partial A/\partial n$ simultaneously. Total cost: two passes, one forward pass that records every intermediate value and one backward pass that reads them back to accumulate derivatives, independent of how many parameters m there are. The record kept by the forward pass is called the *tape*: it holds each operation and its intermediate value, here v_1, v_2, v_3 and A. For the annuity the two passes cost $21/6 = 3.5$ times one function evaluation, as counted below.

7.1.4 ▪ Symbolic Differentiation

As a third comparison, we differentiate the formula (7.1) by hand (or by a computer algebra system). Define $v = (1+r)^{-n}$. Then $A = P(1-v)/r$, giving:

$$\frac{\partial A}{\partial P} = \frac{1-v}{r}, \quad \frac{\partial A}{\partial r} = P\left(\frac{nv/(1+r) - (1-v)/r}{r}\right), \qquad \frac{\partial A}{\partial n} = P\,\frac{\ln(1+r)\,v}{r}.$$

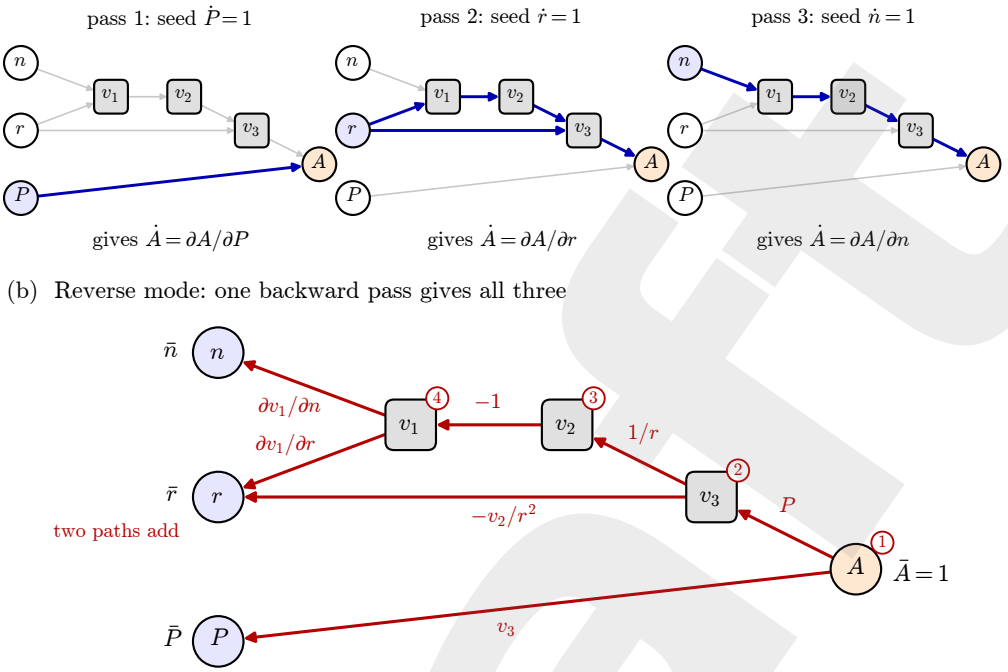


Figure 7.1. *Computation graph of the annuity function* (7.1)*: $A = Pv_3$, with $v_1 = (1+r)^{-n}$, $v_2 = 1 - v_1$ and $v_3 = v_2/r$. (a) Forward mode seeds one input at a time. Each pass carries a tangent along the blue edges, and the tangent on the gray edges is zero. Three inputs cost three passes. (b) Reverse mode seeds $\bar{A} = 1$ and makes one backward sweep. Each edge carries its local partial derivative. The circled numbers give the order of visits: each node is visited only after every node that consumes it. Two paths reach r, through v_1 and through v_3, so $\bar{r}$ adds their two contributions. The sweep returns $\bar{P}$, $\bar{r}$ and $\bar{n}$, the three partial derivatives of A, in one backward pass.*

Symbolic differentiation produces exact closed-form expressions that can be evaluated at any (P, r, n) with a single function evaluation; no passes through the computation graph are required at query time. The cost is paid upfront in the derivation; the payoff is that evaluation is cheap thereafter.

7.1.5 ▪ Exact Operation Counts

For the annuity function we can count arithmetic operations exactly for the procedures printed above, though not for a machine. Real implementations fuse operations, cache intermediates, and manage a tape differently, so these numbers are meant to expose the scaling in m, not to predict a runtime. We count each addition, subtraction, multiplication, division, power, logarithm and negation as one operation. A stored intermediate such as $1 + r$, $-n$ or v_1 is reused at no cost. A unit seed or a zero tangent is not computed,

and $\bar{r}$ starts at zero, so its first contribution is assigned rather than added. The function A requires: 1 addition, 1 negation, 1 power, 1 subtraction, 1 division, 1 multiplication = 6 operations.

Forward mode repeats these 6 primal operations in each pass and adds that pass's tangent operations. Counted line by line, they are 0 for $p = P$ (the seed gives $\dot{A} = v_3$ directly), 3+1+4+1 = 9 for $p = r$, and 3+1+1+1 = 6 for $p = n$. For $m = 3$ parameters: $3 \times 6 + 0 + 9 + 6 = 33$ operations.

Reverse mode records the 6 primal operations once. Its seven sweep lines cost $0 + 0 + 1 + 4 + 1 + 5 + 4 = 15$ operations, so reverse mode costs $6 + 15 = 21$ operations regardless of m.

The ratio: forward mode costs $33/6 = 5.5$ evaluations of A here, close to $2m$, and each added parameter adds a pass. Reverse mode costs $21/6 = 3.5$ evaluations, whatever m is. For the annuity with $m = 3$, forward mode uses $33/21 \approx 1.6$ times as many operations as reverse mode, a small gain with only three parameters. For a climate model with $m = 500$, forward mode needs 500 passes against the 2 of reverse mode, so the gap grows in proportion to m. For a neural network with $m = 10^7$ parameters and one scalar loss function, reverse mode still requires only two passes, one forward and one backward. These are the same two passes as for the annuity, while forward mode would require 10^7 separate passes. This is why reverse-mode AD made modern deep learning computationally feasible.

Symbolic differentiation evaluates A and the three formulas above in 15 operations, sharing $1 + r$, v and $(1 - v)/r$. Centered finite differences make $2m + 1 = 7$ evaluations, or $7 \times 6 = 42$ operations before the differences are formed.

Method	Passes	Operations, $m = 3$	Scales as
Forward mode	$m = 3$	33	$O(m)$
Reverse mode	2	21	$O(1)$ in m
Symbolic diff	1 (at query time)	15	$O(m)$
Finite diff	$2m + 1 = 7$ evals	42 (evals only)	$O(m)$, approx

Table 7.1. *Operation counts for the annuity* (7.1) *with* $m = 3$ *parameters, under the counting convention stated above.*

At nominal values $P = 1000$, $r = 0.05$, $n = 20$, all methods give: $\mathcal{S}_P^A = 1$, $\mathcal{S}_r^A \approx -0.4240$, $\mathcal{S}_n^A \approx 0.5902$. (The derivation of the analytic formulas is Exercise 7.1.)

Self-check 7.1. In the reverse sweep above, $\bar{r}$ collects two contributions, one through v_3 and one through v_1. Convert each into its share of $\mathcal{S}_r^A = (r/A)\,\bar{r}$, evaluate both at $P = 1000$, $r = 0.05$, $n = 20$, and check that they sum to

-0.4240. Then explain why $\mathcal{S}_P^A = 1$ needs no sweep at all.

The same adjoint computation applies directly to epidemic models. Replace "payment rate" P with "infection rate" $c\beta SI/N$, replace "present value of payments" A with "total epidemic burden" $J = \int_0^T I\,dt$, and replace the interest rate r with the recovery rate $\gamma_R = 1/\tau_R$. The annuity's terminal condition ($\lambda(T) = 0$ because no payments arrive after the policy expires) becomes the adjoint terminal condition for $J = \int_0^T I\,dt$ (no additional burden after the observation window closes). The backward solve is the same computation in different clothing: one backward solve, after the forward solve, recovers sensitivities with respect to every epidemic parameter at once.

7.2 ▪ Algorithmic Differentiation

Algorithmic differentiation (AD), also called automatic differentiation, is the chain rule applied systematically to every arithmetic operation in a computer program. It is neither symbolic differentiation (which manipulates formulas) nor numerical differentiation (which approximates derivatives). It is exact to machine precision and applies to any differentiable code.

Students sometimes expect AD to produce a formula for the derivative, like a computer algebra system would. It does not. AD produces the numerical value of the derivative at a specific input, by accumulating the chain rule through the sequence of arithmetic operations the program performs.

Similarly, reverse-mode AD is not the same as numerical differentiation. A centered-difference estimate of $\partial A/\partial r$ has error $O(h^2)$ from the truncation of the Taylor series; reverse-mode AD gives the exact derivative (to floating-point precision) with no step-size selection required.

7.2.1 ▪ Forward Mode

Forward-mode AD attaches a tangent value $\dot{v}_k = \partial v_k/\partial p$ to each intermediate variable v_k, for one chosen parameter p. At each arithmetic operation, the tangent is updated using the chain rule:

$$v_k = f(v_i, v_j) \implies \dot{v}_k = \frac{\partial f}{\partial v_i}\dot{v}_i + \frac{\partial f}{\partial v_j}\dot{v}_j.$$

At the end of the computation, $\dot{v}_{\text{output}}$ contains $\partial J/\partial p$. Repeat for each parameter: m passes total. Forward-mode AD is the computation-graph form of the FSE, the pairing drawn in Figure 1.3 and Table 5.1.

7.2.2 ▪ Reverse Mode

Reverse-mode AD attaches an adjoint value $\bar{v}_k = \partial J/\partial v_k$ to each intermediate variable. In the backward pass, these are accumulated using the chain rule in reverse:

$$v_k = f(v_i, v_j) \implies \bar{v}_i \mathrel{+}= \frac{\partial f}{\partial v_i}\bar{v}_k, \quad \bar{v}_j \mathrel{+}= \frac{\partial f}{\partial v_j}\bar{v}_k.$$

The forward pass computes and stores all intermediate values. The backward pass accumulates all $\partial J/\partial p_i$ simultaneously. Total cost: one forward pass + one backward pass, regardless of m.

Reverse-mode AD is the discrete-computation-graph version of the adjoint ODE derived in Chapter 6. The adjoint ODE propagates $\vec{\lambda}(t)$ backward through a continuous system; reverse-mode AD propagates $\bar{v}_k$ backward through a discrete computation graph. The underlying mathematical operation is identical: transposition of the derivative operator. On a computation graph the transpose is concrete. Collect the local partial derivatives in a matrix G with $G_{ki} = \partial f_k/\partial v_i$. Forward mode sums along row k, $\dot{v}_k = \sum_i G_{ki}\dot{v}_i$; reverse mode sums down column i, $\bar{v}_i = \sum_k G_{ki}\bar{v}_k$, which is multiplication by G^T. Each reversed arrow of Figure 7.1(b) carries the local partial derivative G_{ki} of the forward step it reverses, so the sweep multiplies by G^T.

Unifying Idea 2 (the adjoint is the transpose; Section 1.5). Reverse-mode AD implements this idea automatically: it constructs the transpose through the backward pass without requiring the analyst to derive the adjoint equations by hand. For a model written as differentiable code, this eliminates the main practical barrier to adjoint SA: instead of deriving and implementing the adjoint ODE, the analyst uses an AD library that differentiates the code directly. The adjoint equation is implicit in the reverse accumulation; the analyst never writes it explicitly.

7.2.3 ▪ Available Tools

Earlier AD tools such as ADIFOR, DAFOR and TAMC are now outdated or unmaintained. Table 7.2 lists the current ecosystem. For a comprehensive survey of AD methods, see Baydin et al. [4] and Margossian [31].

Self-check 7.2. Explain in one sentence each why: (a) AD is not symbolic differentiation; (b) reverse-mode AD is not the same as centered finite differences; (c) forward-mode AD requires m passes while reverse-mode requires 2.

Language	Tool	Mode(s)	Notes
MATLAB	Symbolic Math Toolbox	Symbolic	Exact; limited scalability
MATLAB	Deep Learning Toolbox (`dlarray` + `dlgradient`)	Reverse	Reverse-mode AD for differentiable code; designed for deep-learning workflows
Python	JAX	Forward and reverse	NumPy-compatible; JIT compilation
Python	PyTorch `autograd`	Reverse	Widely used in ML
Julia	ForwardDiff.jl	Forward	High performance
Julia	Zygote.jl	Reverse	Source-to-source; flexible
C/Fortran	ADOL-C	Forward and reverse	Mature; still active
C/Fortran	Tapenade	Forward and reverse	Source transformation

Table 7.2. *AD tools in wide use by language at the time of writing. The first row is not an AD tool: the Symbolic Math Toolbox differentiates formulas, not code, and is listed for comparison. Every tool whose mode includes reverse provides reverse-mode AD for arbitrary differentiable code; ForwardDiff.jl is forward mode only. JIT (just-in-time) compilation turns code into machine code when it first runs; source-to-source tools such as Zygote.jl and Tapenade write derivative code from the model's own code. For large-scale scientific computing, JAX (Python) or ADOL-C (C/Fortran) are the most commonly used options.*

7.3 ▪ The Master Decision Table

With the local methods now in hand, Table 7.3 consolidates the decision rules developed across Chapters 3–7; its last row points ahead to the global methods of Chapter 9.

There is no single universally correct method. The right choice depends on the model, the available tools, and the question being asked. A useful heuristic: start with finite differences (Chapter 3) to get SA results quickly, then switch to the adjoint (Chapter 6 or AD) if the computation time is unacceptable or if m is large.

One principle applies universally: whatever method you use, always sanity-check at least two sensitivity indices against an independent method before trusting the full result. For finite differences, verify one analytic SI if

Model type	m vs. r	Method	Accuracy	Cost
Black-box	Any	FD (Chapter 3)	$O(h^2)$	$2m$+1 evals
Analytic, simple	$m \leq r$	FSE (Chapter 4)	Solver tol.	m+1 solves
Analytic, diff. code	$m > r$	AD, reverse	Rounding	2 passes
Analytic, manual	$m > r$	Adjoint (Chapter 6)	Solver tol.	2 solves
Correlated POIs	Any	Reparameterize	–	Depends
Near bifurcation	Any	FSE + caution	Solver tol.*	m+1 solves
Large uncertainty	Any	Global SA (Chapter 9)	Sampling error	$N(m+2)$ evals

Table 7.3. *Master decision table for SA method selection; see Pianosi et al. [43] for a broader SA survey. The primary driver is whether the model is black-box or differentiable, and whether $m > r$ or $m \leq r$. The first four rows choose a method by how the model is available: "analytic" means its equations are in hand, "simple" and "manual" mean the sensitivity or adjoint equations are derived by hand, and "diff. code" means the model is coded in differentiable operations. The last three rows name situations that change that choice; when one applies, follow its row. No row covers a model with a switch (an* `if` *statement,* `min` *or* `max`*): Section 7.4 shows why AD fails there, and an adjoint without a jump condition fails in the same way; finite differences then work only if the switching time is located accurately. (*Near a bifurcation, the FSE still has a unique solution, but an equilibrium's index can grow without bound, and any linear prediction fails for changes that cross the threshold; use the result with caution and consult Chapter 8.)* ***Accuracy:*** *the FSE and adjoint equations give the derivative exactly, and their computed values carry the ODE solver's error, set by* `RelTol` *and* `AbsTol` *(Section 3.3). For an algebraic model, rounding is the only error. Reverse-mode AD differentiates the code to rounding error; when the code integrates an ODE, that derivative carries the integrator's error in the same way.* ***Cost note:*** *"2 passes" (reverse AD) and "2 solves" (analytic adjoint) assume $r = 1$ scalar* QOI*. For r* QOI*s they become $r + 1$, the forward solve or recording pass and one backward solve or pass per* QOI*, against the FSE's $m + 1$; for $m > r$ the adjoint is cheaper. "Evals" are runs of the n-equation model, while m of the FSE's $m+1$ solves have dimension $2n$ (Table 4.1), so the two counts compare work only if the cost of a solve does not grow with its size.* ***AD vs analytic adjoint:*** *use reverse-mode AD when the model is implemented in differentiable code and memory allows storing the computation tape. Use the analytic adjoint (Chapter 6) when memory is constrained or the numerical integrator (e.g., any adaptive-step ODE solver) is not itself differentiable. A nonlinear ODE or PDE model is no obstacle in itself: integrated by a fixed-step scheme written in differentiable code, it suits reverse-mode AD (Section 7.4).*

available. For the adjoint, verify two values against finite differences. This cross-check catches the majority of implementation errors.

Finite differences and the adjoint do not compete. FD is the right tool for black-box models; the adjoint is the right tool for models whose code is available and differentiable, with many parameters. Where both apply, each checks the other, as in the cross-check above. But each tool has failure modes that practitioners encounter repeatedly.

7.4 ▪ Practical Guidance and Common Pitfalls

7.4.1 ▪ Discretize-Then-Adjoint vs. Adjoint-Then-Discretize

When computing adjoint sensitivities for ODE systems, two strategies are available. In the *adjoint-then-discretize* approach (used throughout Chapter 6), the adjoint ODE is derived analytically from the continuous ODE and then solved numerically. In the *discretize-then-adjoint* approach, the ODE is first discretized into a finite-difference recurrence, and the adjoint is then derived for that discrete system: it is the reverse sweep of Section 7.1 carried back through the solver's time steps, with the stored values of every step as the tape.

For linear ODEs with fixed step sizes, both approaches give identical results. For nonlinear ODEs with adaptive solvers, they can differ: the adjoint-then-discretize approach may give sensitivities that disagree with finite-difference verification because the continuous adjoint assumes a smooth forward trajectory that the adaptive solver does not produce. The discretize-then-adjoint approach is always consistent with finite differences but requires differentiating through the step-size control logic, which is non-trivial.

For the applications in this book, the adjoint-then-discretize approach with a fixed-step-size forward solver is the most practical choice. Sirkes and Tziperman [58] discuss the implications for geophysical models, where the adjoint of the discretized equations can show strongly nonphysical behavior.

7.4.2 ▪ Memory Requirements for the Adjoint

The adjoint ODE requires the forward solution $\vec{u}(t)$ stored at every time point where the adjoint ODE coefficients are evaluated. For a system with $n = 100$ state variables solved over 10,000 time steps in double precision, the stored forward solution requires $100 \times 10{,}000 \times 8$ bytes $= 8$ MB, modest. For a climate model with $n = 10^6$ state variables and 10^6 time steps, the requirement is 8×10^{12} bytes $= 8$ TB; infeasible to store in memory.

For large systems, two strategies address this: *Checkpointing* stores the forward solution only at a subset of time points and recomputes intermediate values as needed during the backward pass. *Adjoint-of-checkpoint* schemes (Griewank & Walther [18]) achieve optimal trade-offs between memory and recomputation cost. For the models in this course, dense storage is always adequate.

7.4.3 ▪ Adaptive Step-Size Solvers

As noted in Chapter 3, adaptive ODE solvers introduce non-smoothness in the model output as a function of parameters: the step-size controller makes discrete decisions that change discontinuously with the parameter value. The continuous adjoint ODE from Chapter 6 assumes a smooth forward solution; when adaptive solvers introduce discontinuities, the continuous adjoint may give inaccurate results.

The safest approach: use fixed-step-size integration (e.g., `ode4` in your preferred language, a 4th-order Runge–Kutta) for the forward solve when computing adjoint sensitivities. This eliminates the non-smoothness and makes the continuous and discrete adjoints equivalent. Alternatively, use tight tolerances (`RelTol` $\leq 10^{-8}$) with adaptive solvers to minimize the non-smoothness.

Self-check 7.3. You compute adjoint sensitivities for the SIR model using an adaptive ODE solver for the forward solve, then verify against centered finite differences. The adjoint gives $\mathcal{S}_\beta^J = 1.04$ while centered FD gives 0.87. Name two possible causes of this discrepancy, and describe the diagnostic steps you would take to identify which cause applies.

7.4.4 ▪ Discontinuities Break AD

AD requires the model to be differentiable at the nominal parameter values. Common sources of non-differentiability in practice include `if` statements whose conditions depend on parameters,[3] `min` and `max` functions, piecewise-defined model components such as quarantine thresholds or saturation functions with hard cutoffs, and look-up tables interpolated without smoothing. When these are present, AD will silently compute the derivative of one branch, which may be zero or undefined at the branch point. The diagnostic: compare AD results against centered finite differences. If they disagree significantly for small h, suspect a discontinuity. Replacing piecewise functions with smooth approximations (sigmoid functions, smooth max/min) restores differentiability.

Self-check 7.4. A disease model contains the piecewise function: $\beta(I) = \beta_0$ if $I < 0.1N$, else $\beta_0/2$. This represents reduced contact when hospitalizations are high. (a) Is this function differentiable with respect to I every-

[3] *If-statements are where smooth calculus goes to lose its temper.* When an `if` condition switches branches as a parameter is perturbed, a nominal evaluation and a perturbed one follow different computational paths, and the resulting "derivative" is the derivative of whichever branch happened to be active, which may be zero or undefined at the transition.

where? (b) If you run reverse-mode AD through this function at $I = 0.09N$, what derivative will AD report? Is this reliable? (c) Propose a smooth approximation that would make the model fully differentiable.

Unifying Idea 1 (the sensitivity index is a normalized derivative; Section 1.5). Every method in this chapter (forward mode, reverse mode, symbolic differentiation, finite differences, the FSE, the adjoint ODE) computes the same object: Unifying Idea 1, the sensitivity index defined in Chapter 2. The methods differ in computational cost, accuracy, and applicability to different model types. They all target the same number, $\mathcal{S}^J_{p_j} = (p_j/J)(dJ/dp_j)$, and differ in how closely they reach it.

7.5 ▪ Computational Laboratory: Four Methods, One Model

This laboratory computes the same sensitivity, $\partial J/\partial\beta$ for the SIR model with $J = \int_0^{90} I\,dt$, using all four methods, and compares their accuracy, speed, and ease of implementation.

Part 1: Finite differences (Chapter 3 method). Use the `sensitivity_jacobia` template from Chapter 3 with the SIR model. Record the estimated $\mathcal{S}^J_\beta$ and the computation time.

Part 2: Analytic FSE (Chapter 4 method). Solve the 6-equation augmented system from Chapter 4 for the β parameter alone (3 SIR state equations plus 3 sensitivity equations $\partial\dot{S}/\partial\beta$, $\partial\dot{I}/\partial\beta$, $\partial\dot{R}/\partial\beta$). Extract $\partial I(t)/\partial\beta$ at all time points and integrate numerically. Record the result and time.

Part 3: Adjoint ODE (Chapter 6 method). Solve the adjoint system from Chapter 6. Use formula (6.18) to compute $dJ/d\beta$. Record the result and time. *Reading the adjoint.* The adjoint variable $\lambda_I(t)$ you compute in Part 3 carries a concrete epidemiological interpretation: it measures the additional infectious person-time that would follow from one extra infectious person introduced at time t, counting all downstream secondary and subsequent transmission through the end of the observation window. It is not a count of additional cases. As Chapter 6 notes, $\lambda_I = \partial J/\partial I$ has dimensions of time when J is measured in person-time, and converting it to cases averted needs a separate incidence model. As in Chapter 6, that reading holds under the linearization about the nominal trajectory: it is exact for infinitesimal perturbations and degrades once a perturbation is large enough for nonlinear feedback to matter. Biologically, this means $\lambda_I(t)$ is large early in an epidemic (when a single infection can trigger many further cases) and small near the end (when susceptibles are depleted and further spread is limited). This interpretation, derived directly from the adjoint

computation, is developed fully in Chapter 6 (§ 6.4).

Part 4: Symbolic sanity check (not part of the timed comparison). Compute $\mathcal{S}_\beta^{\mathcal{R}_0}$ for $\mathcal{R}_0 = c\beta/\gamma_R$ using a symbolic or AD tool of your choice (e.g., SymPy, Symbolics.jl, MATLAB Symbolic Toolbox; Part 1 of the Chapter 2 laboratory shows the MATLAB calls). This sanity check verifies that your AD setup returns the analytically known result $\mathcal{S}_\beta^{\mathcal{R}_0} = 1$ (by the monomial rule), confirming your pipeline is working before applying it to the time-dependent integral $J = \int_0^T I\,dt$. Note: symbolic tools compute sensitivities of closed-form expressions, not of ODE solutions over time, so this is a *calibration* step, not a fourth method for computing $dJ/d\beta$.

```
# Symbolic differentiation of R0 = c*beta/gamma_R
# Use your preferred symbolic tool

DECLARE_SYMBOLIC  c, beta, gamma_R  (positive)
R0    = c * beta / gamma_R
SI_b  = (beta / R0) * DIFF(R0, with_respect_to=beta)
       # Expected result: SI_b = 1

# For time-dependent sensitivity via reverse-mode AD:
p_nom = [c=5, beta=0.06, tau_R=7, tau_m=10000]
p_ad  = AD_VAR(p_nom)                    # register as differentiable
J     = SIR_integral(p_ad)               # model must use AD-compatible o
dJdp  = GRAD(J, p_ad)                   # reverse pass
PRINT "AD sensitivity for beta:", dJdp[beta]
```

Part 5: Comparison table. Fill in the following table from your measurements:

Method	$\mathcal{S}_\beta^J$	Rel. error	Time (s)	Lines of new code
Finite differences				
Analytic FSE				
Adjoint ODE				
Automatic differentiation (AD)				

Discuss: which method is easiest to implement? Most accurate? Fastest? Which would you use in practice for this model?

Extension: introducing a discontinuity. Modify the SIR model to include a treatment threshold: if $I(t) > 0.1N$, reduce τ_R by 20% (faster treatment when infection is high). Re-run all four methods. Which methods give consistent results? Which methods silently produce wrong answers? How does the U-curve (Chapter 3) change for finite differences near β?

Connection to optimization. The adjoint computed in this chapter is the same object as the Lagrange multiplier in constrained optimization and the costate variable in optimal control. Appendix E develops this connection through LP duality and Pontryagin's minimum principle (the continuous-time optimality condition that generalizes the Euler–Lagrange equations).

7.6 ▪ Exercises

Exercise 7.1 *(Bloom L3).* For the annuity function (7.1) at nominal values $P = 1000$, $r = 0.05$, $n = 20$:

(a) Compute A numerically.

(b) Compute all three sensitivities $\mathcal{S}_P^A$, $\mathcal{S}_r^A$, $\mathcal{S}_n^A$ analytically by differentiating (7.1).

(c) Implement forward mode by hand for $p = r$ (as in Section 7.1) and verify your answer from (b).

(d) Implement reverse mode for all three parameters in one backward pass and verify all three answers.

(e) Count the number of arithmetic operations in each approach.

Exercise 7.2 *(Bloom L4).* For each of the following SA problems, select the most appropriate method from Table 7.3 and justify in two sentences:

(a) A climate model with 800 parameters written in Fortran. Each model run takes 10 hours. You want sensitivities for 2 response functionals.

(b) A pharmacokinetic model with 5 parameters and analytic closed-form solutions. You want sensitivities for 8 drug concentration time points.

(c) A traffic simulation with 40 parameters implemented as a compiled C++ executable. Parameter values can be set via a configuration file. Each run takes 3 minutes.

(d) A 10-parameter epidemic model written in your preferred language with an adaptive ODE solver. You want one response functional. The model contains an `if` statement that triggers a quarantine intervention when $I > 0.05N$.

Exercise 7.3 *(Bloom L4).* Using automatic differentiation (AD):

(a) Wrap the function $\mathcal{R}_0(c, \beta, \tau_R) = c\beta\tau_R$ as a differentiable computation and use reverse-mode AD to compute all three partial derivatives at $c = 5$, $\beta = 0.06$, $\tau_R = 7$.

(b) Compare with the analytic derivatives.

(c) Wrap the SIR equilibrium calculation (solve $\vec{f}(\vec{u}) = 0$ numerically) and compute $\partial \bar{I}/\partial\beta$ at the endemic equilibrium. Compare with the FSE result from Chapter 4.

(d) What happens when you try to apply reverse-mode AD through an adaptive-step ODE solver? Explain why, and what the remedy is.

Exercise 7.4 *(Bloom L5).* Design the complete SA workflow for a vector-borne disease transmission model with the following properties: $n = 8$ state variables (human and mosquito compartments), $m = 14$ parameters (mortality rates, biting rates, transmission probabilities, mean durations), $r = 2$ response functionals (basic reproduction number and endemic human infection prevalence), each ODE solve taking approximately 2 minutes.

Specify:

(a) Which SA method you would use and why.

(b) The expected total computation time.

(c) How you would validate the implementation.

(d) How you would handle the fact that two parameters (σ_v and σ_h, the mosquito and human biting rates) are structurally correlated.

(e) What output format you would use to present results to a non-mathematical public health audience.

Exercise 7.5 *(Bloom L5). Audit exercise.* A manuscript under your review reports sensitivities of a scalar QOI with respect to 600 POIs in a differentiable ODE model. The authors state that they used a central finite-difference scheme "for robustness, since adjoint methods can be numerically delicate," and report a total run time of several days on a cluster.

(a) Using the master decision table of Section 7.3, state which method the problem structure indicates and estimate the cost ratio between it and the scheme the authors used.

(b) The authors' stated reason is not baseless. Give the circumstance in which it would be the right call, and state what evidence the manuscript would need to present for you to accept it here.

(c) The reported indices may well be correct. Explain why a referee should still ask for a change, and distinguish that request from a claim that the results are wrong.

(d) Draft the two sentences you would write in your referee report.

Exercise 7.6 *(Bloom L6).* A research group uses reverse-mode AD to compute adjoint sensitivities for a mosquito population model with adaptive-step-size ODE integration. They report that some sensitivity indices agree with finite differences but others disagree by 10–20%, with no obvious pattern. The discrepancies are largest for parameters that appear in the mortality rate expressions.

(a) Identify the most likely technical cause of the discrepancy.

(b) Describe two diagnostic tests that would confirm or rule out this cause.

(c) Propose two remedies and explain the trade-offs between them.

(d) At what point would you conclude that the disagreement indicates a genuine SA result (the linearization is poor for these parameters) rather than a numerical artifact?

Notes

Forward-mode algorithmic differentiation is credited to Wengert (1964), and reverse mode as it is used today to Speelpenning (1980); Linnainmaa (1970) is often cited as the first published description of reverse mode [4]. The definitive reference is Griewank and Walther [18], which covers operator overloading, source transformation, checkpointing, and the relationship between AD and the adjoint method. Naumann [37] treats the combinatorial and software aspects of AD in depth.

The connection between reverse-mode AD and backpropagation in deep learning is made precise by Baydin et al. [4], who give a unified treatment and a historical account of the parallel development of these ideas in scientific computing and machine learning. Margossian [31] surveys reverse-mode AD implementations from the perspective of statistical computing, including Stan. The neural ODE framework of Chen et al. [14] applies the

continuous adjoint of Chapter 6 to learn the right-hand side of the ODE from data, connecting Chapters 6 and 7.

Mester, Landeros, Rackauckas, and Lange [34] explain the mathematical basis of adjoint, complex-perturbation and forward-mode sensitivity analysis and compare them on biological models.

Answers to the self-checks

7.1. Through v_3: $\bar{r}\mathrel{+}=\bar{v}_3(-v_2/r^2)=-Pv_2/r^2=-A/r$, whose share is $(r/A)(-A/r)=-1$ at any nominal values. Through v_1: $\bar{v}_1=-\bar{v}_2=-P/r$, so $\bar{r}\mathrel{+}=(-P/r)\big(-n(1+r)^{-n-1}\big)=Pnv_1/\big(r(1+r)\big)$, whose share is $rnv_1/\big((1+r)(1-v_1)\big)$. With $v_1=1.05^{-20}=0.3769$ it is 0.5760, and the sum is $-1+0.5760=-0.4240$, as printed.

$\mathcal{S}_P^A$ needs no sweep because A is linear in P: $\partial A/\partial P=(1-(1+r)^{-n})/r=A/P$, so $\mathcal{S}_P^A=(P/A)(A/P)=1$ for any r and n, the monomial rule of Chapter 2. A 1% increase in the payment produces exactly a 1% increase in present value.

7.2. (a) Symbolic differentiation manipulates formulas algebraically; AD applies the chain rule numerically to the sequence of operations in the code, producing a number rather than a formula.

(b) Centered finite differences approximate the derivative with error $O(h^2)$ via $(f(p+h)-f(p-h))/(2h)$; reverse-mode AD computes the exact derivative (to floating-point precision) by accumulating the chain rule backward through the computation graph.

(c) Forward mode propagates the tangent $\dot{v}_k=\partial v_k/\partial p$ for one parameter p per pass, requiring one pass per parameter. Reverse mode propagates the adjoint $\bar{v}_k=\partial J/\partial v_k$ for one response J per backward pass, giving all m derivatives simultaneously. The backward pass needs the values that one forward pass recorded, so reverse mode takes two passes in all.

7.3. Possible causes: (1) An error in the adjoint code itself, such as a wrong sign or a missing term. The finite-difference cross-check exists to catch exactly this, so it comes first. (2) The adaptive step-size control makes the computed output non-smooth in β, which corrupts the continuous adjoint and the difference quotient alike. (3) The finite-difference step size h was chosen poorly (e.g., too small, causing catastrophic cancellation, or too large, causing truncation error). Diagnostics: (a) Run the adjoint code on a case with a known answer, such as the warm-up of Section 6.3, and check each adjoint equation against the Jacobian term by term. (b) Recompute both the adjoint and the finite differences with the fixed-step `ode4`: if the discrepancy disappears, the solver is the culprit. (c) Plot the U-curve for the FD estimate to verify the step size is optimal; if the FD value changes significantly with h, the FD is unreliable. Always use finite differences as a sanity check for any adjoint implementation. The gap here, 0.17 on 0.87, is about 20%, far above the 1% at which a discrepancy warrants investigation.

7.4. (a) No. The function has a jump discontinuity at $I=0.1N$, so $\partial\beta/\partial I$ is undefined there. (b) At $I=0.09N$ the condition $I<0.1N$ holds, so the function takes the β_0 branch and AD differentiates that branch, which is zero. That zero is the true derivative of $\beta(I)$ at this point. It is unreliable inside an ODE model for another reason: as the epidemic grows, the trajectory crosses $I=0.1N$, and the crossing

time depends on the parameters. AD follows one branch at a time and misses the jump at the crossing, so the derivative of an output such as $\int_0^T I\,dt$ can be badly wrong even though every local derivative of $\beta(I)$ is right. (c) Replace the step with a sigmoid: $\beta(I) = \beta_0\big(1 - 0.5\,\sigma(k(I/N - 0.1))\big)$, where $\sigma(x) = 1/(1 + e^{-x})$ and $k > 0$ controls sharpness. As $k \to \infty$ this recovers the discontinuous original. The step is spread over a band about $4/k$ wide in I/N, across which σ runs from 0.12 to 0.88. With $k = 100$ the band is $0.08N$ to $0.12N$, and at $I = 0.09N$ the smoothed value is $0.87\beta_0$, not β_0. With $k = 1000$ the band is $0.098N$ to $0.102N$, and at $0.09N$ the smoothed value is β_0 to within 10^{-4}. Choose k so that the band is narrow compared with the distance from the operating point to the threshold. The price is steepness: inside the band $|\partial\beta/\partial I|$ reaches $k\beta_0/(8N)$, which makes the ODE stiff for large k: inside the band the feedback of I on $\beta(I)$ acts on a time scale far shorter than the epidemic's, so an explicit solver must take very small steps to stay stable.

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Part V

Limitations and Extensions

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Chapter 8

Caveat Emptor, Limitations of Local Sensitivity Analysis

Local sensitivity is a linear truth, and nature is rarely linear for long.

In 2019, Saltelli et al. [49] reviewed the sensitivity analyses in 280 highly cited papers drawn from 19 subject areas. One third or more of those analyses moved one factor at a time, a class that includes derivatives computed at a single nominal parameter value. Such methods explore thin corridors of the input space, so they cannot answer a global question (how does output uncertainty relate to input uncertainty across the full parameter range?). A derivative can be computed correctly and still answer a different question than the one the authors claimed to answer.

The deeper problem is not computational but epistemological: sensitivity analysis is still often defined by method rather than purpose [47]. Practitioners reach for the method from their own disciplinary tradition rather than the one best suited to their question. A local method applied to answer a global question will produce results that are internally correct but wrong for the stated purpose. Before choosing an SA method, two questions must be answered. First, what is the goal? It may be factor prioritization (which inputs most reduce output variance?), factor fixing (which inputs can be frozen without changing the output?), or factor mapping (which inputs drive the output above a threshold?). Second, do the method's assumptions match the model's behavior?

Assuming local sensitivity holds globally is the triumph of mathematical hope over nonlinear reality.

This chapter asks the question that should be asked before any SA is begun: is local SA the right tool for this question? *Unifying Idea 4* (a

sensitivity index describes a model, not the world; Section 1.5). A second question travels with it. When an index misbehaves near a threshold, the reader has to decide whether the system is genuinely poised there or whether the model has been pushed outside the range in which it represents anything. The mathematics cannot settle that; only knowledge of the system can.

8.1 ▪ Method-Question Mismatch

Local sensitivity analysis is built on a derivative. A derivative is a local object: it measures the behavior of a function at one point, and it is only meaningful in the vicinity of that point. This makes local SA powerful for some questions and completely wrong for others.

Local SA can answer: *which parameter most affects the model near the nominal parameter values?*

Local SA cannot answer: *which parameter most affects the model across its full range of uncertainty?* That question requires the parameter to actually vary across its range, and a derivative does not do that.

Local SA cannot answer: *how does the model behave at parameter values far from the nominal?* The derivative approximates linear behavior; real models are often highly nonlinear.

The principle is simple and unforgiving. The method must match the question. Using local SA to answer a global question is not merely suboptimal; it can produce a ranking of parameter importance that is entirely wrong. A parameter that appears unimportant locally may dominate the output variance when varied across its full uncertainty range. A parameter that appears important locally may have negligible effect at realistic perturbation sizes if the model is highly nonlinear in that parameter.

A small sensitivity index does not mean a parameter can be ignored globally. It means only that a small relative change in the parameter predicts a small relative change in the output. Three things make it small: a small slope, a nominal parameter value near zero, or a large nominal output q^*. Section 8.5 shows a specific example where the local ranking is the opposite of the global ranking.

Self-check 8.1. A published SA study reports: "All sensitivity indices are less than 0.05, confirming that the model is robust to parameter uncertainty." The study used nominal parameter values from the literature, with no information on parameter uncertainty ranges. Identify two ways in which the study's conclusion goes beyond what local SA can actually establish.

8.2 ▪ Bifurcations and SA Failure

Local SA requires the model output to be differentiable with respect to the parameters at the nominal values. When this fails, local SA fails completely. The most important way it fails is at a bifurcation.

A bifurcation is a parameter value at which the qualitative behavior of the model changes: two equilibria merge into one, a stable fixed point becomes unstable, or an epidemic threshold is crossed [63, 29]. At a fold the derivative $\partial u/\partial p$ is infinite, and at a transcritical point the normalized index diverges. Local SA breaks down entirely. Why it breaks down depends on what is differentiated, and Section 8.4.1 sorts the cases. For an equilibrium at a fold or transcritical point, the Jacobian is singular at the bifurcation itself, so the FSE $D_{\vec{u}}[\vec{f}\,](\partial\vec{u}/\partial p) = \vec{r}$ may have no solution or infinitely many (the Fredholm alternative, Appendix B.3). Near it the Jacobian is only nearly singular, and the FSE has a unique solution, which can be ill-conditioned. The time-dependent FSE, a linear initial-value problem, always has a unique solution. For models with periodic limit cycles, sensitivity near a bifurcation of the periodic orbit involves Floquet multipliers; this is treated in Appendix D.

8.2.1 ▪ The Quadratic Example

Consider the scalar equation $f(u;p) = u^2 + p = 0$. For $p < 0$, there are two real roots: $u = \pm\sqrt{-p}$. At $p = 0$, the two roots merge. For $p > 0$, there are no real roots.

The FSE for the sensitivity of the positive root $u^* = \sqrt{-p}$ is:

$$\frac{\partial u^*}{\partial p} = \frac{-\frac{\partial f}{\partial p}}{\frac{\partial f}{\partial u}} = \frac{-1}{2u^*} = \frac{-1}{2\sqrt{-p}}.$$

The normalized SI is:

$$\mathcal{S}_p^{u^*} = \frac{p}{u^*}\frac{\partial u^*}{\partial p} = \frac{p}{\sqrt{-p}} \cdot \frac{-1}{2\sqrt{-p}} = \frac{1}{2}.$$

This is constant (independent of p) for all $p < 0$; the positive root is always in exact proportion $1/2$ to any fractional change in p.

But as $p \to 0^-$:

$$\frac{\partial u^*}{\partial p} = \frac{-1}{2\sqrt{-p}} \to -\infty.$$

The derivative diverges at the bifurcation. Near $p = 0$, the model is infinitely sensitive to the parameter p: any small positive perturbation to p destroys the equilibrium entirely.

Unifying Idea 3 (local SA is a linearization; Section 1.5). This is the idea in its most dramatic form. The sensitivity index is the derivative-based linearization of the model response. At a fold such as this one, the function $u^*(p)$ has infinite curvature, its graph has a vertical tangent, and no linear approximation is valid. The linearization does not just become inaccurate at the fold; it becomes undefined.
Unifying Idea 1 (the sensitivity index is a normalized derivative; Section 1.5). The index is a normalized derivative, so it can fail through either factor. At a fold the raw derivative $\partial u^*/\partial p$ becomes infinite. The normalized index then diverges as well whenever the fold sits at a nonzero parameter value, as in Exercise 8.1, whose fold is at $p = 1$. The quadratic above is the exception: its fold sits at $p = 0$, so the factor p in the normalization cancels the divergence and the index stays $1/2$. At a transcritical bifurcation the raw derivative stays finite, but the equilibrium it is normalized against approaches zero, so the normalized index diverges. Section 8.4.1 sorts these cases by what is differentiated.
Unifying Idea 2 (the adjoint is the transpose; Section 1.5). The adjoint route inherits the same difficulty, by a mechanism that depends on what is being differentiated. For the sensitivity of an *equilibrium*, the adjoint system is algebraic and inherits the singularity directly: the transpose of a singular matrix is also singular, so at a fold or transcritical point the adjoint system has no unique solution. For a time-dependent functional over a *finite* horizon the adjoint ODE remains solvable, but the near-zero eigenvalue of J_F leaves the adjoint with a slowly decaying mode, so $\vec{\lambda}$ can grow large as the horizon lengthens. Forward and adjoint approaches fail together, for the same underlying reason.

Self-check 8.2. For $f(u;p) = u^2 + p$ with $p = -4$: (a) Find the two roots $u^*_\pm$. (b) Compute $\mathcal{S}_p^{u^*_+}$ analytically. Compare with the derivative $\partial u^*_+/\partial p$. (c) Estimate $\partial u^*_+/\partial p$ numerically using a centered difference with $h = 0.1, 0.01, 0.001$. At what value of p near zero does the finite-difference estimate become unreliable?

8.2.2 ▪ The Epidemic Threshold

The quadratic example above illustrates a fold bifurcation (root collision): two equilibria merge and annihilate as a parameter crosses zero. The epidemic threshold is a structurally different bifurcation, a *transcritical* bifurcation in which two equilibria exchange stability rather than colliding. Both defeat local SA, but for reasons worth separating. At a fold, the *raw derivative* of the equilibrium diverges. At a transcritical bifurcation the raw derivative stays finite; what diverges is the *normalized* index, because the

quantity it is normalized against is itself going to zero. Either way, the linearization stops being a fair description of the model near the threshold.

The SIS (*Susceptible–Infectious–Susceptible*) model gives the cleanest version of this behavior, because its endemic equilibrium has a closed form. Infectious individuals return directly to the susceptible class rather than gaining permanent immunity [65]. In a closed population of size N:

$$\dot{S} = -\frac{\beta S I}{N} + \gamma_R I, \qquad \dot{I} = \frac{\beta S I}{N} - \gamma_R I, \tag{8.1}$$

with $S + I = N$ constant. Writing $i = I/N$ and substituting $S = N(1 - i)$ gives the scalar equation $\dot{i} = i\big(\beta(1 - i) - \gamma_R\big)$, whose equilibria are the disease-free state $i^* = 0$ and, for $\mathcal{R}_0 > 1$,

$$i^* = 1 - \frac{\gamma_R}{\beta} = 1 - \frac{1}{\mathcal{R}_0}, \qquad \mathcal{R}_0 = \frac{\beta}{\gamma_R}. \tag{8.2}$$

The disease-free state is stable for $\mathcal{R}_0 < 1$ and the endemic equilibrium is stable for $\mathcal{R}_0 > 1$. The two exchange stability at $\mathcal{R}_0 = 1$: this is the transcritical bifurcation.

The sensitivity of the endemic level to $\mathcal{R}_0$ is

$$\frac{\partial i^*}{\partial \mathcal{R}_0} = \frac{1}{\mathcal{R}_0^2}, \qquad \mathcal{S}^{i^*}_{\mathcal{R}_0} = \frac{\mathcal{R}_0}{i^*} \cdot \frac{1}{\mathcal{R}_0^2} = \frac{1}{\mathcal{R}_0(1 - 1/\mathcal{R}_0)} = \frac{1}{\mathcal{R}_0 - 1}.$$

The raw derivative is well behaved at the threshold: it tends to 1 as $\mathcal{R}_0 \to 1^+$. The normalized index is not. It diverges, $\mathcal{S}^{i^*}_{\mathcal{R}_0} \to +\infty$, precisely because i^* is vanishing. Figure 8.1 shows both the equilibrium branch and the diverging sensitivity index.

SA of the endemic equilibrium for $\mathcal{R}_0$ values close to 1 must be interpreted with extreme caution. A small change in any parameter that affects $\mathcal{R}_0$ could eliminate the endemic equilibrium entirely or cause it to surge. Local SA correctly identifies this extreme sensitivity but cannot predict the qualitative change.

For $\mathcal{R}_0 \leq 1$: SA of the endemic equilibrium is undefined; the equilibrium does not exist. Computing SA under the assumption that i^* exists when $\mathcal{R}_0 < 1$ is a mathematical error. Depending on how the equilibrium is formulated and solved, a numerical routine may return a negative or complex value without warning.

Sensitivity to the individual parameters. Differentiating (8.2) with respect to β and γ_R:

$$\frac{\partial i^*}{\partial \beta} = \frac{\gamma_R}{\beta^2}, \qquad \frac{\partial i^*}{\partial \gamma_R} = -\frac{1}{\beta}.$$

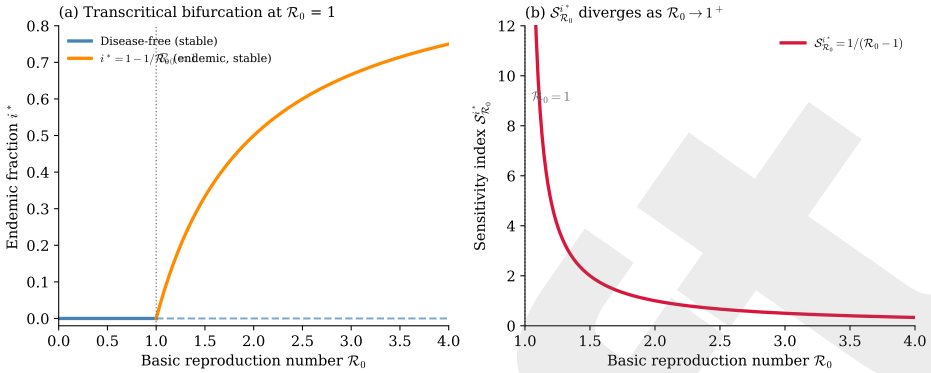


Figure 8.1. *Bifurcation of the SIS endemic equilibrium at $\mathcal{R}_0 = 1$. (a) The infectious fraction $i^* = \max(0, 1 - 1/\mathcal{R}_0)$: zero (disease-free state, stable) for $\mathcal{R}_0 < 1$, rising continuously for $\mathcal{R}_0 > 1$ (endemic equilibrium, stable). The dashed line continues the disease-free state for $\mathcal{R}_0 > 1$, where it is unstable. The kink at $\mathcal{R}_0 = 1$ marks the transcritical bifurcation. (b) The sensitivity index $\mathcal{S}^{i^*}_{\mathcal{R}_0} = 1/(\mathcal{R}_0 - 1)$ diverges as $\mathcal{R}_0 \to 1^+$. Any parameter that shifts $\mathcal{R}_0$ near the epidemic threshold produces an arbitrarily large proportional change in the endemic level. Local SA correctly captures this infinite sensitivity but cannot predict the qualitative shift from disease-free to endemic behavior; that requires knowing that a bifurcation exists.*

The normalized sensitivity indices are:

$$\mathcal{S}^{i^*}_{\beta} = \frac{\partial i^*}{\partial \beta} \cdot \frac{\beta}{i^*} = \frac{\gamma_R/\beta^2 \cdot \beta}{1 - 1/\mathcal{R}_0} = \frac{1/\mathcal{R}_0}{(\mathcal{R}_0 - 1)/\mathcal{R}_0} = \frac{1}{\mathcal{R}_0 - 1}, \tag{8.3}$$

$$\mathcal{S}^{i^*}_{\gamma_R} = \frac{\partial i^*}{\partial \gamma_R} \cdot \frac{\gamma_R}{i^*} = \frac{-1/\beta \cdot \gamma_R}{1 - 1/\mathcal{R}_0} = -\frac{1}{\mathcal{R}_0 - 1}. \tag{8.4}$$

The chain rule gives both results at once from the index with respect to $\mathcal{R}_0$. Since $\mathcal{R}_0 = \beta/\gamma_R$, $\mathcal{S}^{\mathcal{R}_0}_{\beta} = 1$ and $\mathcal{S}^{\mathcal{R}_0}_{\gamma_R} = -1$, so $\mathcal{S}^{i^*}_{\beta} = \mathcal{S}^{i^*}_{\mathcal{R}_0}\mathcal{S}^{\mathcal{R}_0}_{\beta} = 1/(\mathcal{R}_0 - 1)$ and $\mathcal{S}^{i^*}_{\gamma_R} = -1/(\mathcal{R}_0 - 1)$. Both indices diverge as $\mathcal{R}_0 \to 1^+$, confirming that threshold sensitivity failure is a structural property of the bifurcation, not a feature specific to SIR.

A structural constraint. Notice that $\mathcal{S}^{i^*}_{\beta} + \mathcal{S}^{i^*}_{\gamma_R} = 0$ for all $\mathcal{R}_0 > 1$. This is not a coincidence. Since $i^* = 1 - \gamma_R/\beta$ depends on β and γ_R only through their ratio, rescaling both parameters by the same factor leaves i^* unchanged: i^* is homogeneous of degree 0 in (β, γ_R). Euler's theorem for homogeneous functions then gives $\beta\, \partial i^*/\partial\beta + \gamma_R\, \partial i^*/\partial\gamma_R = 0$, which is exactly $\mathcal{S}^{i^*}_{\beta} + \mathcal{S}^{i^*}_{\gamma_R} = 0$. In biological terms: if transmission and recovery rates both double, the endemic level is unchanged, so their sensitivities must cancel. This is a case where the sum-to-zero property from Chapter 2 arises from a structural symmetry in the model rather than a normalization identity.

Numerical illustration. Table 8.1 shows how the sensitivity indices grow as $\mathcal{R}_0$ approaches 1 from above, using nominal parameters $\beta = 0.3$ day^{-1}, $\gamma_R = 0.1$ day^{-1} ($\mathcal{R}_0 = 3$) as a baseline and reducing β to move $\mathcal{R}_0$ toward threshold.

$\mathcal{R}_0$	i^*	$\mathcal{S}_\beta^{i^*} = -\mathcal{S}_{\gamma_R}^{i^*}$	Interpretation
3.0	0.667	+0.5	Low: standard operating range
1.5	0.333	+2.0	Elevated: 1% change ⇒ 2% change in i^*
1.2	0.167	+5.0	High: treat as warning
1.1	0.091	+10.0	Very high: ranking unreliable
1.01	0.010	+100.	Near-threshold: SA has broken down

Table 8.1. *Sensitivity index $\mathcal{S}_\beta^{i^*} = 1/(\mathcal{R}_0 - 1)$ for the SIS endemic equilibrium, with $\mathcal{S}_{\gamma_R}^{i^*} = -\mathcal{S}_\beta^{i^*}$. As $\mathcal{R}_0 \to 1^+$, both indices diverge and local SA loses quantitative meaning. The practical threshold for caution is $|\mathcal{S}^{i^*}| > 10$, i.e. $\mathcal{R}_0 < 1.1$.*

The same conclusion for the SIR model with demography. The SIR model of Chapter 4, Equations (4.15)–(4.17), also has a transcritical bifurcation at $\mathcal{R}_0 = 1$. Its endemic level is

$$i^* = \frac{\nu_m}{\gamma_R + \nu_m}\left(1 - \frac{1}{\mathcal{R}_0}\right), \tag{8.5}$$

which carries a prefactor the SIS model does not. That prefactor is small whenever the mean residence time $\tau_m = 1/\nu_m$ is long compared with the infectious period, so the endemic level itself is small and is approached only over demographic timescales. It has no effect on the index with respect to $\mathcal{R}_0$. Any constant prefactor cancels in the normalization,

$$\mathcal{S}_{\mathcal{R}_0}^{i^*} = \frac{\mathcal{R}_0}{k(1 - 1/\mathcal{R}_0)} \cdot \frac{k}{\mathcal{R}_0^2} = \frac{1}{\mathcal{R}_0 - 1} \qquad \text{for any } k > 0,$$

and since $\mathcal{R}_0 = c\beta/(\gamma_R + \nu_m)$ is proportional to c and to β, the chain rule gives $\mathcal{S}_\beta^{i^*} = \mathcal{S}_c^{i^*} = 1/(\mathcal{R}_0 - 1)$ as well. These indices transfer unchanged to the model used throughout the book. This scale-invariance is worth noting in its own right: a normalized sensitivity index is blind to the overall size of the response it describes, which is exactly what makes it comparable across parameters with different units, and exactly what makes it fragile when the response approaches zero. The recovery-rate index does not transfer, because the prefactor itself depends on γ_R:

$$\mathcal{S}_{\gamma_R}^{i^*} = -\frac{\gamma_R}{\gamma_R + \nu_m} \cdot \frac{\mathcal{R}_0}{\mathcal{R}_0 - 1},$$

which is -1.91 at the book's nominal values, where the SIS form (8.4) would give -0.91. The structural constraint survives over three rates. With $\mathcal{R}_0 = c\beta/(\gamma_R + \nu_m)$ the endemic level is $i^* = \nu_m/(\gamma_R + \nu_m) - \nu_m/(c\beta)$, which is homogeneous of degree 0 in (β, γ_R, ν_m), so Euler's theorem gives $\mathcal{S}^{i^*}_{\beta} + \mathcal{S}^{i^*}_{\gamma_R} + \mathcal{S}^{i^*}_{\nu_m} = 0$. Both threshold indices still diverge as $\mathcal{R}_0 \to 1^+$.

Operational rule: when to stop reporting SI as a ranking. A sensitivity index is meaningful as a ranking tool only when the linearization is a fair local approximation. Two practical rules of thumb (these are heuristic thresholds, not mathematical guarantees; the appropriate cutoff depends on the model and the application):

1. *Magnitude check.* If $|\mathcal{S}^q_p| > 10$ for a normalized index, treat the result as a qualitative warning ("this parameter region is highly sensitive") rather than a quantitative ranking. An SI of 100 does not mean the parameter matters $10\times$ more than one with SI 10; it may mean the model is near a threshold.
2. *Condition-number check.* Compute $\kappa(J_F)$ (the condition number of the model Jacobian at the operating point). If $\kappa(J_F) > 10^4$, the sensitivity computation is ill-conditioned: small errors in the model evaluation may be amplified by $\kappa(J_F)$ in the sensitivity values. Always report $\kappa(J_F)$ alongside SI values computed near a bifurcation or near parameter degeneracy. The check needs a Jacobian with more than one state and no conserved total (§8.3 defines κ). The scalar SIS equation has a 1×1 Jacobian, so $\kappa = 1$ at every $\mathcal{R}_0$, and in the two-equation form (8.1) the conserved total $S + I = N$ makes J_F singular at every state. For a one-state model the threshold shows in the size of the Jacobian instead, $|J_F|/\gamma_R = \mathcal{R}_0 - 1 = 1/\mathcal{S}^{i^*}_{\beta}$, so the magnitude check already catches it. For the SIR model with demography (laboratory Part 4) the check works: $\kappa_2(J_F)$ at the endemic equilibrium is about 4×10^3 at $\mathcal{R}_0 = 3$, passes 10^4 near $\mathcal{R}_0 = 1.35$, and reaches about 3×10^5 at $\mathcal{R}_0 = 1.01$.

Takeaway. The divergence of local SA near $\mathcal{R}_0 = 1$ is a property of the transcritical bifurcation structure, not of the specific model. A compartmental epidemic model with a generic forward transcritical bifurcation at $\mathcal{R}_0 = 1$ will exhibit the same behavior. (In a model with a *backward bifurcation* the endemic branch bends back below $\mathcal{R}_0 = 1$, so for some $\mathcal{R}_0 < 1$ a stable endemic equilibrium coexists with the stable disease-free state: the bistability of Section 8.4. Such models, and models with non-smooth incidence, fall outside this rule because their threshold is not simply $\mathcal{R}_0 = 1$; for bistable models Section 8.4 calls for a separate SA at each stable equilibrium.) The magnitude check ($|\mathcal{S}^q| > 10$) applies equally to SIS, SEIR, and

any other model of this type. The condition-number check applies where rule 2 says it does, and Section 8.4.1 sorts which check suits which quantity.

8.3 ▪ Ill-Conditioning

For linear algebraic models, trouble near a singularity takes a different form. When the coefficient matrix A in $A\vec{u} = \vec{b}$ is nearly singular, small changes in the right-hand side $\vec{b}$ (or in the matrix entries) can produce large changes in the solution. Those large sensitivities are genuine: the exact derivatives $\partial\vec{u}/\partial\vec{b}$ are the entries of A^{-1}, which grow as A nears singularity. What ill-conditioning adds is a separate cost: the computed solution, and every index derived from it, can lose accuracy.

8.3.1 ▪ The Condition Number

The *condition number* $\kappa(A) = \|A\|\|A^{-1}\|$, where the matrix norm $\|A\| = \max_{\vec{x}\neq 0} \|A\vec{x}\|/\|\vec{x}\|$ is the largest factor by which A stretches a vector (see Appendix C.2 for the SVD-based definition $\kappa_2 = \sigma_1/\sigma_n$), measures how much the solution $\vec{u}$ can change relative to a change in the data. For the 2-norm,

$$\kappa_2(A) = \frac{\sigma_{\max}(A)}{\sigma_{\min}(A)}, \tag{8.6}$$

where $\sigma_{\max}$ and $\sigma_{\min}$ are the largest and smallest singular values of A. For a symmetric matrix the singular values are the absolute values of the eigenvalues, so κ_2 is the ratio of the largest to the smallest $|\lambda|$; Appendix C.2 gives the general route. A matrix with $\kappa_2(A) = 1$ is perfectly conditioned (orthogonal, or a nonzero scalar multiple of an orthogonal matrix); a matrix approaching singularity has $\sigma_{\min} \to 0$ and $\kappa_2(A) \to \infty$.

Remark 8.1. Readers from statistics will recognize the condition number as the SA analog of the *variance inflation factor* (VIF). In regression, a large VIF warns that the design matrix $X^\top X$ is nearly singular and that coefficient estimates are unreliable; in SA, a large $\kappa(A)$ issues the same warning about the sensitivity Jacobian. Both diagnose a nearly degenerate linear system in which "separate" effects cannot be cleanly estimated.

The key bound: if the data changes by a relative amount ϵ, the solution changes by at most $\kappa(A)\,\epsilon$:

$$\frac{\|\delta\vec{u}\|}{\|\vec{u}\|} \leq \kappa(A)\,\frac{\|\delta\vec{b}\|}{\|\vec{b}\|}.$$

When $\kappa(A) \gg 1$, even tiny changes in the data produce large changes in the solution.

8.3.2 ▪ SA and Ill-Conditioning

For SA, ill-conditioning raises two questions that are easy to conflate. A colleague who reports "the sensitivity of u_1 to b_1 is extremely large" is reporting two things at once: (a) a genuine model property: the exact sensitivities are entries of A^{-1}, which are large when A is nearly singular, so the output truly varies strongly with this parameter; and (b) a warning about accuracy: the computed index, like the computed solution, may have lost digits.

The two hold together and are independent, and $\kappa(A)$ answers (b). If $\kappa(A) \approx 10^6$, then numerical errors of size $\varepsilon_{\text{mach}} \approx 10^{-16}$ can contaminate the solution at the level of 10^{-10}: 6 digits of the solution may be meaningless. A small product $\kappa(A)\,\varepsilon_{\text{mach}}$ therefore clears a large computed index. A large $\kappa(A)$ cannot mark one as an artifact, because it also means a large exact sensitivity in some direction. Nor can a small residual $\|\vec{b} - A\hat{\vec{u}}\|$: a backward-stable solve leaves a small residual at any $\kappa(A)$. To check the digits themselves, recompute in higher precision or compare with an analytic value (§8.4.1).

Ill-conditioning is itself an SA result: it warns the investigator that the computed solution, and any sensitivity derived from it, is fragile to small errors in the data or the model entries. It is worth separating two ideas that are easy to conflate. Conditioning is a property of the forward map: how much the solution of $A\vec{u} = \vec{b}$ moves when the data move. Identifiability is a property of the inverse problem: whether the parameters can be recovered from the observations at all. Ill-conditioning of A often accompanies poor identifiability and is a useful warning sign, but the two are not the same statement, and a large $\kappa(A)$ can also reflect a genuinely large but well-posed response. This is not a failure of SA; it is one of the most important things SA can reveal.

Self-check 8.3. The 2×2 matrix $A(\epsilon) = \begin{pmatrix} 1 & 1 \\ 1 & 1+\epsilon \end{pmatrix}$ with $\epsilon > 0$: (a) Compute $\det(A(\epsilon))$ and show $A \to$ singular as $\epsilon \to 0$. (b) Compute the condition number $\kappa_2(A)$ for $\epsilon = 1, 0.1, 0.01$. (c) For $\vec{b} = (2, 2+\epsilon)^T$, solve $A\vec{u} = \vec{b}$ and compute $\mathcal{S}_\epsilon^{u_1}$. What happens to this SI as $\epsilon \to 0$?

8.4 ▪ Separatrices and Trajectory Sensitivity

Dynamical systems often have regions of qualitatively different behavior separated by a *separatrix*: a boundary in state space that trajectories cannot cross. SA computed on one side of a separatrix describes behavior qualitatively different from SA on the other side.

The epidemic threshold is a related but distinct phenomenon, and the distinction is worth keeping. The condition $S(0) > \gamma_R N/(c\beta)$ (equivalently $\mathcal{R}_0 > 1$) is a threshold in parameter and initial-condition space, not a separatrix in state space: it decides whether an outbreak grows, but it is not a barrier that trajectories are forbidden to cross. Both kinds of boundary defeat local SA, for different reasons.

Consider the epidemic peak $I_{\max}$ as β varies across the threshold. Below threshold the infectious compartment never grows, so $I_{\max} = I(0)$. Above threshold the peak rises from that floor, and expanding about $\mathcal{R}_0 = 1$ gives

$$I_{\max} - I(0) \;\approx\; \tfrac{1}{2} N\,(\mathcal{R}_0 - 1)^2.$$

So $I_{\max}$ does not jump at the threshold. It is continuous there, and its first derivative is continuous as well and equal to zero. That is the trap, and it is worse than a jump would be: $\mathcal{S}_\beta^{I_{\max}} = 0$ exactly at the threshold, so local SA reports that β does not matter at the precise parameter value where β decides whether there is an epidemic at all. The response is quadratic, and a first-order method is blind to it. This is the misconception that opened this chapter, in its sharpest form.

The practical implication: if any parameter perturbation of interest crosses a qualitative threshold (endemic vs. disease-free, outbreak vs. no outbreak, stable vs. oscillatory), local SA cannot be used to predict the response. The model must be analyzed separately in each qualitative regime, and the boundary itself requires special treatment.

A related limitation applies to models with multiple stable attractors. When a model can settle into qualitatively different long-run states depending on initial conditions, the sensitivity index computed near one attractor says nothing about behavior near another. Extensions of the SIR model that exhibit bistability (where both a disease-free and an endemic equilibrium are stable for the same parameter values) require separate SA studies for each attractor. Local SA at one equilibrium cannot warn the analyst that another equilibrium exists.

8.4.1 ▪ What Is Differentiated Decides the Failure

Sections 8.2 to 8.4 have shown local SA failing near a threshold in several ways. What goes wrong depends on what is being differentiated, and so does the remedy. We sort the cases by that quantity and give for each what to compute instead.

(1) *An equilibrium at a fold*, $u^*(p)$ (§8.2; Exercise 8.1). F_u is singular at the fold, so the raw derivative diverges, and past the fold the equilibrium does not exist. The normalized index diverges too unless the

fold sits at $p = 0$, as in the quadratic. *Instead:* trace the branch by continuation (solve at a sequence of nearby parameter values, starting each solve from the previous root) and draw the bifurcation diagram (§8.7, Part 1).

(2) *An equilibrium at a transcritical threshold*, $i^*(\mathcal{R}_0)$ (§8.2.2). The raw derivative stays finite, because $\partial\vec{F}/\partial p$ vanishes as J_F becomes singular. The normalized index diverges as $i^* \to 0$, and a relative change in β of about $1/\mathcal{S}_\beta^{i^*}$ removes the equilibrium. *Instead:* draw the bifurcation diagram (§8.7, Part 2), and report $\mathcal{R}_0 - 1$ with the change in each parameter that reaches $\mathcal{R}_0 = 1$.

(3) *The solution of $A\vec{u} = \vec{b}$ as A nears singularity* (§8.3). Nothing fails in the model: the large indices are entries of A^{-1}, the genuine response. The computation can lose about $\log_{10}\kappa(A)$ digits. *Instead:* read $\kappa(A)$ as that bound, and check the digits in higher precision or against an analytic value; a small residual cannot check them.

(4) *A quantity on a finite-time trajectory*, such as $I(T)$ (§8.7, Part 3). The forward sensitivity equations are a linear initial-value problem and always have a solution, so the index stays finite across $\mathcal{R}_0 = 1$. What fails is the linear prediction for finite perturbations that carry the trajectory across a threshold or a separatrix (§8.4). *Instead:* rerun the model at the perturbed values, or sample across the threshold with the methods of Chapter 9.

(5) *A threshold quantity*, such as which equilibrium is stable, or $I_{\max}$ (§8.4). The stable equilibrium switches at $\mathcal{R}_0 = 1$, so that output is not differentiable there and is flat on either side. $I_{\max}$ is flat at the threshold, so $\mathcal{S}_\beta^{I_{\max}} = 0$. No local index can flag the switch. *Instead:* use factor mapping: sample the plausible ranges (§9.2) and report the fraction with $\mathcal{R}_0 > 1$.

(6) *A Floquet multiplier* μ_k of a periodic orbit (Appendix D). At a period-doubling, and as an orbit shrinks to a Hopf point, μ_k typically reaches the unit circle at a finite rate, so its derivative stays finite; at a fold of cycles it diverges. *Instead:* follow μ_k with (D.3) to locate where $|\mu_k| = 1$, and run the model on both sides of that value.

A large index is therefore not by itself a malfunction, and a small or zero index is no reassurance. Every case ends with the same step: find where the threshold lies, and compare that distance with the plausible range of each parameter.

DRAFT

8.5 ▪ When Global SA Is Needed

Local SA studies the neighborhood. Global SA studies the country.

Local SA fails to answer the right question in three specific situations. When any of these conditions holds, Chapter 9's global SA methods are required. The practical decision framework for choosing between local and global SA is developed at length by Saltelli et al. [56, 54].

To reiterate the misconception that opened this chapter: a sensitivity index near zero does not mean a parameter is globally unimportant. It means that the linearization at the nominal value is nearly flat, that the nominal value p^* sits near zero, or that the nominal output q^* is large. The last two make the factor p^*/q^* in the index small. The example in §8.5.1 is the second case. What happens away from the nominal, across the full uncertainty range of the parameter, requires global SA to determine.

8.5.1 ▪ Large Parameter Uncertainty

A parameter known only to within a factor of 2 or 3 has a plausible range that runs from a half or a third of its nominal value up to twice or three times it. Such ranges are common in epidemiology, ecology, and environmental modeling, and the linearization underlying local SA may be very poor across them.

As a concrete example, consider a model where the QOI depends on a parameter p through $q(p) = e^p$. At nominal value $p^* = 0$: $\mathcal{S}_p^q = p^*(dq/dp)/q = 0 \cdot 1/1 = 0$. The normalized index is zero. But if the uncertainty range is $[-2, 2]$, the QOI varies from $e^{-2} \approx 0.14$ to $e^2 \approx 7.4$: a factor of about 55. Note where the information is lost. The raw derivative is perfectly informative here, $dq/dp = 1$ at $p^* = 0$; it is the normalization by a nominal value of zero that drives the index to zero. This is a reference-point artifact, not a statement that the model is insensitive to p, and it is a standing hazard whenever a parameter's nominal value sits at or near the origin. The relative-change and regularized measures of Section 2.2.3 exist for exactly this situation.

8.5.2 ▪ Highly Nonlinear Models

Two models with identical local SIs for a parameter p may behave very differently when p is perturbed by 20%: one model may be nearly linear (the SI is a good approximation), while the other may be highly curved (the SI may understate or overstate the true change).

The sigma-normalized sensitivity index (Section 9.1) provides the minimal bridge: it weights the local SI by the parameter's standard deviation,

giving a measure that at least accounts for the relative magnitudes of parameter uncertainties. It does not see curvature, since it is still a linearization. A direct test for nonlinearity compares the linear prediction with a model run at the 20% change itself.

8.5.3 ▪ Correlated Parameters

Local SA under the independence assumption gives misleading results when parameters are correlated (Chapter 3). Global SA methods in Chapter 9 include specific tools for handling this case. Two meanings need separating. Parameters linked by the model's form, such as c and β entering only as $c\beta$, are reparameterized (Section 3.5; Table 7.3). Inputs whose distributions are dependent (Section F.4) need the tools of Section 9.7.

8.5.4 ▪ The Portfolio Example

Consider a model $Y = s_1Q_1 + s_2Q_2$ with $s_1 = 2$, $s_2 = 1$ and parameter standard deviations $\sigma_1 = 1$, $\sigma_2 = 3$.

The model slopes are $\partial Y/\partial Q_1 = s_1 = 2$ and $\partial Y/\partial Q_2 = s_2 = 1$, so on the unnormalized derivative alone parameter 1 appears more important. (The normalized index $\mathcal{S}^Y_{Q_i} = s_iQ_i/Y$ would additionally depend on the nominal Q_i, which this example leaves unspecified; the comparison here is between slopes and σ-weighted contributions.)

The sigma-normalized index: $S^\sigma_i = (s_i\sigma_i)/\sigma_Y$, where $\sigma_Y = \sqrt{4+9} = \sqrt{13}$, because the variances of independent inputs add (Section F.2 of Appendix F). So $S^\sigma_1 = 2/\sqrt{13} \approx 0.55$ and $S^\sigma_2 = 3/\sqrt{13} \approx 0.83$.

Parameter 2 dominates the output variance, even though it has a smaller slope. The shares of the variance are the squares of the indices, not the indices themselves: $4/13 \approx 0.31$ and $9/13 \approx 0.69$, which sum to one (Section 9.4). The ranking by slope is inverted because parameter 2 has much larger uncertainty. Chapter 9's Sobol indices provide the rigorous global version of this sigma-normalized assessment, and Section 9.1 opens with this same example.

Remark 8.2. For a linear model $Y = \sum_i s_iQ_i$ with independent inputs, the sigma-normalized index $S^\sigma_i = (s_i\sigma_i)/\sigma_Y$ is identical to the *standardized regression coefficient* reported by most statistical software, the "standardized beta" obtained by regressing Y on the Q_i's after standardizing all variables to zero mean and unit variance. The sigma-normalized sensitivity index and the standardized coefficient are the same quantity, derived from the same model and evaluated at the same nominal point; the difference is notation and disciplinary convention. The plain relative index $\mathcal{S}^Y_p$ of Chapter 2 is a different quantity, and carries no variance weighting.

Self-check 8.4. For the SIR model, suppose β has uncertainty $\sigma_\beta = 0.02$ and τ_R has uncertainty $\sigma_{\tau_R} = 3$ days, at nominal values $\beta^* = 0.06$ and $\tau_R^* = 7$ days.

(a) From Chapter 2, $\mathcal{S}_\beta^{\mathcal{R}_0} = \mathcal{S}_{\tau_R}^{\mathcal{R}_0} = 1$. Which parameter appears more important from local SA alone?

(b) Compute σ_β/β^* and σ_{τ_R}/τ_R^* (the relative uncertainties). Which parameter has larger relative uncertainty?

(c) Compute the sigma-normalized indices for $\mathcal{R}_0$ with respect to β and τ_R. Which parameter drives more variance in $\mathcal{R}_0$?

8.6 ▪ When SA Guides Data Collection

Sections 8.2 to 8.5 have identified when SA can mislead. This section completes the practical argument the book has been building since Chapter 1: how to use SA *correctly* to make decisions about field measurement.

The decision chain. A sensitivity study produces a ranked list of parameters. Translating that list into a field-measurement plan requires three additional steps that SA alone cannot provide:

1. *Identifiability:* Can the high-sensitivity parameter actually be estimated from the available data, or is it structurally confounded with another parameter? Section 3.5 showed that parameters with identical sensitivity indices are a strong warning sign of non-identifiability: if two parameters always appear together in the model equations, their individual effects on any QOI are indistinguishable, and field data cannot separate them. This does not prove non-identifiability in general (equal SIs for one QOI do not preclude different SIs for another), but it is a reliable indicator worth investigating before designing field studies. For the SIR model, c and β always appear as the product $c\beta$, so field studies should estimate $c\beta$ directly (e.g., from secondary attack rates) rather than c and β separately.

2. *Measurability:* Is the high-sensitivity parameter actually measurable with available instruments and budget? The mean infectious period τ_R appears in the top tier of every SIR sensitivity analysis, and it is measurable: clinicians record the duration from symptom onset to recovery. The residence time τ_m has negligible SI for short-horizon models and is measurable anyway from demographic records. A parameter with large SI but no feasible measurement pathway cannot be pinned down by data. Its *influence* should be reduced instead, by redesigning the intervention so that the outcome depends less on it.

3. *Actionability:* Can the high-sensitivity parameter be reduced by a feasible public health intervention? SA of the SIR model with $J = \max_t I(t)$ consistently identifies c (contact rate) and β (transmission probability) as the most influential parameters. Contact rates can be reduced by social distancing, school closures, and remote-work policies. Transmission probability per contact can be reduced by mask requirements, hand-hygiene campaigns, and vaccination. These two routes correspond to different interventions with different costs, compliance rates, and social acceptability. SA identifies both as equally valuable targets; the choice between them requires policy judgment beyond the model.

The two-budget problem. A common practical scenario: a government agency has funding to (1) measure exactly k_{meas} parameters precisely in the field, and (2) implement exactly k_{int} interventions. The correct SA-guided procedure is:

1. Rank all parameters by how much of the uncertainty in peak I each one carries. Use the sigma-normalized index $S^{\sigma}_{p_j}$ when the linearization holds across the plausible ranges, and otherwise Morris screening (a rapid method for ranking parameters with many fewer model evaluations than full global SA, developed in Section 9.6) or Sobol indices. A ranking by the local index $|\mathcal{S}^{\text{peak } I}_{p_j}|$ alone ignores the uncertainty ranges, as the portfolio example shows, and near the threshold it is empty: at $\mathcal{R}_0 = 1$ every peak index is zero (§8.4.1).
2. From the top-ranked parameters, apply the identifiability filter: remove any parameter that is structurally confounded with a higher-ranked parameter.
3. Apply the measurability filter: remove any parameter for which no feasible measurement protocol exists within the budget.
4. The remaining top k_{meas} parameters are the measurement priorities.
5. From the ranked list, apply the actionability filter separately: the top k_{int} actionable parameters are the intervention priorities. These sets may overlap.

What the model cannot tell you. The complete picture requires one more piece: parameter inference itself, the step that actually quantifies how much a given measurement budget reduces parameter uncertainty. SA tells you *which* parameters matter most; parameter estimation tells you *how precisely* you can learn them from data; parameter selection and influence analysis tell you *how to design the measurement study* to learn those parameters as efficiently as possible.

Beyond inference lies the deeper question of *model discrepancy*: what to do when the assumed model form is itself wrong, which is the natural

complement to the SA failure modes catalogued in this chapter. SA tells you when a method gives the wrong answer for your model; model discrepancy theory tells you when your model gives the wrong answer for the world.

These disciplines, SA, parameter estimation, optimal experimental design, and model discrepancy, form a complete pipeline for model-based decision making. This book has equipped you for the first stage.

If a parameter is influential and poorly known, that is not a nuisance. That is your research agenda.

8.7 ▪ Computational Laboratory: Bifurcation and Ill-Conditioning

Part 1: The quadratic bifurcation. For $f(u;p) = u^2 + p$, compute the positive root $u^* = \sqrt{-p}$ and the SI $\mathcal{S}_p^{u^*} = 1/2$ analytically. Plot u^* vs. p for $p \in [-4, -0.01]$. Estimate $\partial u^*/\partial p$ by centered differences at $p = -4, -1, -0.1, -0.01, -0.001$. Compare the finite-difference estimates with the analytic value. At what value of p does the finite-difference estimate become unreliable?

Part 2: SIS bifurcation diagram using MATCONT. Download and install MATCONT (`https://matcont.sourceforge.io`). Define the SIS equilibrium equation $\dot{i} = i(\beta(1-i) - \gamma_R) = 0$ as a MATCONT system. Use MATCONT's continuation capability to trace the endemic equilibrium $i^* = 1 - 1/\mathcal{R}_0$ as $\mathcal{R}_0$ increases from 0.5 to 3.0. MATCONT will automatically detect the transcritical bifurcation at $\mathcal{R}_0 = 1$ and flag it. Plot the bifurcation diagram (equilibrium value vs. $\mathcal{R}_0$). If MATCONT is unavailable, trace the equilibrium manually: for $\mathcal{R}_0 \le 1$, $i^* = 0$ (disease-free); for $\mathcal{R}_0 > 1$, $i^* = 1 - 1/\mathcal{R}_0$. (For the SIR model with demography the branch carries the extra factor $\nu_m/(\gamma_R + \nu_m)$ of Equation (8.5); the shape of the diagram, and the index with respect to $\mathcal{R}_0$, are unchanged.)

Part 3: FSA comparison across $\mathcal{R}_0 = 1$. Using the full SIR ODE system, compute $\mathcal{S}_\beta^{I(T)}$ at $T = 90$ days for $\mathcal{R}_0 = 0.8, 1.0, 1.2, 2.0$. At each value, use centered finite differences. Tabulate the results. At $\mathcal{R}_0 = 1$: does the SI blow up, return NaN, or behave normally? Try $h = 0.01, 0.001, 0.0001$ to see what happens.

Part 4: Condition number of the SIR Jacobian as a function of $\mathcal{R}_0$. For the SIR endemic equilibrium, the Jacobian J_F at (S^*, I^*, R^*) depends on $\mathcal{R}_0$. Compute $\kappa_2(J_F)$ using your language's condition-number routine for $\mathcal{R}_0 \in \{1.01, 1.05, 1.1, 1.5, 2.0, 3.0\}$. Plot $\kappa_2(J_F)$ vs. $\mathcal{R}_0$. Observe the spike as $\mathcal{R}_0 \to 1^+$: the Jacobian becomes nearly singular at the bifurcation. What does this imply about SA of the SIR endemic equilibrium for parameter values that place $\mathcal{R}_0$ close to 1?

Part 5: Sigma-normalized SIs for the SIR model. Using the SIR model with uncertainty ranges $\sigma_c = 1$ (contacts/day), $\sigma_\beta = 0.02$, $\sigma_{\tau_R} = 3$ days, $\sigma_{\tau_m} = 500$ days (all at 1σ), compute the sigma-normalized indices for $\mathcal{R}_0$. Compare the ranking with the local SI ranking from Chapter 2. Which parameter most affects $\mathcal{R}_0$ given its realistic uncertainty range?

8.8 ▪ Exercises

Exercise 8.1 *(Bloom L3).* For the equation $g(u; p) = u^3 - 3pu + 2 = 0$:

(a) Find all real roots at $p = 1$. Which root changes most rapidly with p?

(b) Compute $\mathcal{S}_p^{u^*}$ for each real root at $p = 1$ using the implicit function theorem.

(c) Plot the roots as functions of p for $p \in [0.8, 1.2]$. At what value of p do two roots merge?

(d) At the bifurcation value of p, what does the FSE predict? Verify numerically by *one-sided* continuation from $p > 1$. A centered difference through $p = 1$ is not available on the positive branch: for $p < 1$ those two roots are complex, so the branch does not exist on that side.

Exercise 8.2 *(Bloom L3).* Using the SIR model with $c = 5$, $\tau_R = 7$ days, $\tau_m = 10{,}000$ days:

(a) Compute $\mathcal{S}_\beta^{I(T)}$ for $\beta = 0.04, 0.06, 0.08, 0.12$ at $T = 90$ days using centered finite differences. Note: with demography the threshold is $\mathcal{R}_0 = c\beta/(\gamma_R + \nu_m)$. At $\tau_m = 10{,}000$ days this differs from $c\beta\tau_R$ by 0.07%, so to the precision quoted it takes values $1.4, 2.1, 2.8, 4.2$.

(b) Now recompute for $\beta = 0.02$ ($\mathcal{R}_0 = 0.7$). Compare with the results for $\mathcal{R}_0 > 1$. Explain any qualitative differences.

(c) If you were to add a bifurcation analysis tool (e.g., MATCONT), what would you expect the bifurcation diagram to look like for the endemic equilibrium of this SIR model as β varies?

Exercise 8.3 *(Bloom L4).* For the 3×3 linear system $A\vec{u} = \vec{b}$ where:

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9+\epsilon \end{pmatrix}, \quad \vec{b} = A\vec{u}_{\text{exact}}, \quad \vec{u}_{\text{exact}} = (1, 1, 1)^T.$$

(a) For $\epsilon = 1, 0.1, 0.01, 0.001, 0.0001$, compute $\kappa(A)$ and $\|\hat{\vec{u}} - \vec{u}_{\text{exact}}\|$ where $\hat{\vec{u}} = A^{-1}\vec{b}$ (via your linear system solver).

(b) Plot $\|\hat{\vec{u}} - \vec{u}_{\text{exact}}\|$ vs. $\kappa(A)$ on log-log axes. What is the approximate slope?

(c) What does this experiment tell you about interpreting large SI values for nearly singular models?

Exercise 8.4 *(Bloom L6).* You are asked to perform SA on an epidemic model whose best-fit parameter values give $\mathcal{R}_0 = 1.04$, very close to the epidemic threshold. Write a one-page advisory report for a public health agency explaining:

(a) What local SA can and cannot tell you at this parameter value.

(b) What specific numerical problems to expect when computing sensitivity indices near $\mathcal{R}_0 = 1$.

(c) What additional analyses you would recommend before advising on interventions.

(d) How you would explain the concept of a bifurcation and its implications to a non-mathematical policy maker.

Exercise 8.5 *(Bloom L4).* *Audit exercise.* A policy brief reports local sensitivity indices for an endemic equilibrium of an SIR model at a parameter set for which $\mathcal{R}_0 = 1.02$, and concludes that "the equilibrium prevalence is extremely sensitive to the recovery rate, so recovery is the highest-priority intervention target." The indices are large in magnitude and were computed correctly.

(a) Explain why indices of large magnitude are expected here, and why that magnitude is not by itself evidence of an intervention priority.

(b) State what happens to the quantity being differentiated as $\mathcal{R}_0 \to 1^+$, and what that implies for the validity of the linearization.

(c) The brief's conclusion also assumes something about the recovery rate that is outside every index reported. Identify it.

(d) Recommend the analysis you would run instead, and state what question it would answer that the local indices cannot.

Exercise 8.6 *(Bloom L6).* A published modeling paper reports: "Sensitivity analysis of the endemic equilibrium of our malaria model found that all indices are less than 0.1, indicating the model is robust to parameter uncertainty. The basic reproduction number at fitted parameter values is $\mathcal{R}_0 = 1.15$."

Identify at least three specific methodological concerns with this analysis and its conclusions. For each concern, state: (a) what the problem is; (b) what evidence the authors would need to provide to address it; and (c) what alternative analysis would give a more complete picture.

Notes

The methodological critique of sensitivity analysis practice that motivates this chapter is developed systematically by Saltelli et al. [49], who reviewed the SA in 280 highly cited papers and found that many fail to explore the space of the input factors: one third or more of the analyses moved one factor at a time.

The sigma-normalized sensitivity index S_i^{σ}, which bridges local SA (Chapter 2) and global SA (Chapter 9), is discussed in Saltelli et al. [53], Section 1.2.4; its coincidence with the standardized regression coefficient for a linear model is shown in Section 1.2.5 of the same book.

The structural identifiability problem for ODE models asks whether the parameter vector can in principle be recovered from perfect observations of the state. Pianosi et al. [43] (Section 2.4) note that sensitivity analysis is closely related to identifiability analysis, which asks whether the parameters of a model can be estimated from the available inputs and outputs. Mester et al. [34] compare the differential methods that compute the sensitivities themselves: forward, adjoint, and complex-perturbation sensitivity analysis.

Smith [60] develops the steps that follow this chapter at the graduate level: frequentist and Bayesian inference in Chapters 11 and 12 of that book; parameter identifiability and influence in its Chapter 7, and parameter subset selection in its chapter on local sensitivity analysis (Chapter 8); and model discrepancy in its Chapter 14.

Answers to the self-checks

8.1. (1) "Robust to parameter uncertainty": local SA says nothing about parameter uncertainty. It describes behavior near the nominal value. If the uncertainty range for some parameters is large (e.g., ±50%), the model could be highly sensitive to that parameter globally even though the local SI is small.

(2) The conclusion depends implicitly on the nominal parameter values chosen. If the nominal values are changed, the local SI values change too (for nonlinear

models). A study with different nominal values might find very different rankings. This sensitivity of local SA to the choice of nominal values is itself a limitation that the study does not address.

8.2. (a) $u_{\pm}^* = \pm\sqrt{4} = \pm 2$. (b) $\partial u_+^*/\partial p = -1/(2\sqrt{-p}) = -1/4$. $\mathcal{S}_p^{u_+^*} = (p/u_+^*)(-1/(2\sqrt{-p})) = (-4/2)(-1/4) = 1/2$.

(c) Numerically at $p = -4$: all three step sizes give ≈ -0.250 ✓. As $p \to 0^-$, the function $u_+^* = \sqrt{-p}$ has a square-root singularity; finite differences become inaccurate when the curvature term dominates the step-size error, which occurs when $h \gtrsim |p|$. Near $p = 0$, even very small h is not small compared to $|p|$, so finite differences fail.

8.3. (a) $\det = \epsilon \to 0$ as $\epsilon \to 0$. Singular at $\epsilon = 0$. (b) A is symmetric, so its singular values are the absolute values of its eigenvalues, $\lambda_\pm = \left(2 + \epsilon \pm \sqrt{4 + \epsilon^2}\right)/2$, both positive, and $\kappa_2(A) = \lambda_+/\lambda_-$. For $\epsilon = 1$: $\kappa_2 \approx 6.85$. For $\epsilon = 0.1$: $\kappa_2 \approx 42.1$. For $\epsilon = 0.01$: $\kappa_2 \approx 402$. As $\epsilon \to 0$: $\kappa \to \infty$.

(c) $A^{-1} = \epsilon^{-1}\begin{pmatrix} 1+\epsilon & -1 \\ -1 & 1 \end{pmatrix}$. $\vec{u} = A^{-1}\vec{b} = (1,1)^T$ for all $\epsilon > 0$. $\mathcal{S}_\epsilon^{u_1}$: differentiate $u_1 = (2(1+\epsilon) - (2+\epsilon))/\epsilon = \epsilon/\epsilon = 1$ with respect to ϵ, but the simplification $u_1 = 1$ holds exactly, so $\mathcal{S}_\epsilon^{u_1} = 0$. The sensitivity vanishes!

This illustrates a subtlety: a large condition number does not make every index large. Here $\mathcal{S}_\epsilon^{u_1} = 0$ because $\vec{b}$ moves with A, while $\partial u_1/\partial b_2 = -1/\epsilon = -100$ at $\epsilon = 0.01$: the exact sensitivity is large in some directions only. What the condition number bounds is the accuracy of the *computed* $\vec{u}$: at $\epsilon = 0.01$ about $\log_{10}\kappa_2$ digits, between two and three, may be lost.

8.4. (a) Both have SI = 1, so local SA alone says they are equally important.

(b) $\sigma_\beta/\beta^* = 0.02/0.06 \approx 0.33$ (33%); $\sigma_{\tau_R}/\tau_R^* = 3/7 \approx 0.43$ (43%). The mean infectious period has larger relative uncertainty.

(c) Apply the delta method (F.6) first to get $\sigma_{\mathcal{R}_0}$: $\partial\mathcal{R}_0/\partial\beta = c\tau_R = 5 \times 7 = 35$; $\partial\mathcal{R}_0/\partial\tau_R = c\beta = 5 \times 0.06 = 0.3$. $\sigma_{\mathcal{R}_0}^2 \approx (35 \times 0.02)^2 + (0.3 \times 3)^2 = 0.49 + 0.81 = 1.30$, so $\sigma_{\mathcal{R}_0} \approx 1.14$. Using $S_i^\sigma = (\partial\mathcal{R}_0/\partial p_i)\,\sigma_i/\sigma_{\mathcal{R}_0}$ (Equation (9.1)): $S_\beta^\sigma = 35 \times 0.02/1.14 \approx 0.61$; $S_{\tau_R}^\sigma = 0.3 \times 3/1.14 \approx 0.79$. The mean infectious period drives more output variance despite equal local SIs, because τ_R has larger relative uncertainty. The ratio $S_{\tau_R}^\sigma/S_\beta^\sigma \approx 1.3$ matches the ratio of $\sigma_{\tau_R}\,\partial\mathcal{R}_0/\partial\tau_R$ to $\sigma_\beta\,\partial\mathcal{R}_0/\partial\beta = 0.90$ to 0.70. Chapter 9 will quantify this more precisely with Sobol indices.

Note. A common shortcut computes $\mathcal{S}_p^{\mathcal{R}_0} \cdot (\sigma_p/p^*)$ (the local SI times the relative uncertainty) as a quick approximation to S_p^σ. This shortcut equals the full formula only when $\sigma_{\mathcal{R}_0} = \mathcal{R}_0^*$, that is, when the output's coefficient of variation is exactly 1, which is a large uncertainty rather than a small one. The exact relation is

$$S_p^\sigma = \mathcal{S}_p^Y \frac{\mathrm{CV}_p}{\mathrm{CV}_Y}, \qquad \mathrm{CV}_p = \sigma_p/p^*, \quad \mathrm{CV}_Y = \sigma_Y/Y^*,$$

so the shortcut is exact precisely when $\mathrm{CV}_Y = 1$ and is otherwise wrong by the factor $1/\mathrm{CV}_Y$ in either direction. For accurate sigma-normalized indices, always use the delta-method σ_Y in the denominator.

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Chapter 9
Global Sensitivity Analysis and Sobol Indices

The point is not to remove uncertainty.
The point is to find out which uncertainty is worth worrying about.

Consider two epidemiological models. Both give $\mathcal{S}_\beta^{\mathcal{R}_0} = +1$ at the same nominal parameter values. In Model A, $\mathcal{R}_0$ is nearly linear in β across the full biologically plausible range $[0.01, 0.15]$. In Model B, $\mathcal{R}_0$ is highly nonlinear: it grows in proportion to β up to the nominal value and then levels off, flat across the upper end of the range.

For Model A, the local SI correctly identifies β as an important parameter. For Model B, local SA gives the same SI value of $+1$; but it is now describing only the slope at the nominal point. A 20% increase in β from the nominal produces very different effects in the two models, and no derivative computed at the nominal value can see this.

Chapter 8 showed when local SA fails. This chapter provides the tools to complement and extend it; global SA answers questions that local SA cannot, while local SA remains the starting point for understanding which parameters matter and why. The tools rest on expectation, variance, conditional expectation and correlation, and Appendix F supplies them for a reader who has not met them.

Before choosing a global SA method, two questions must be answered. *What is the goal?* Three distinct purposes call for different methods [56, 53]: *factor prioritization* (which inputs most reduce output variance if their uncertainty were eliminated?), *factor fixing* (which inputs can be held constant without materially changing the output?), and *factor mapping* (which input combinations drive the output above a critical threshold?). *Do the method's assumptions match the model?* First-order Sobol indices answer the factor-prioritization question; total-order indices answer the factor-fixing question. Using either one to answer a question outside its scope produces results

that are numerically correct but scientifically misleading, the central lesson of Chapter 8. *Unifying Idea 4* (a sensitivity index describes a model, not the world; Section 1.5). All three purposes are stated about inputs to a model. Factor fixing licenses holding a parameter constant *in the model*; it is not a finding that the quantity is constant in the world, and the variance being decomposed is the variance the assumed input distributions put there.

9.1 ▪ Why Local SA Is Not Enough: The Sigma-Normalized Bridge

Local SA studies the neighborhood. Global SA studies the country.

The two can give very different answers about which parameters matter most, as the following example shows.

9.1.1 ▪ The Portfolio Example

Return to the model of Section 8.5.4, $Y = s_1 Q_1 + s_2 Q_2$ where $s_1 = 2$, $s_2 = 1$ and the parameters have standard deviations $\sigma_1 = 1$, $\sigma_2 = 3$.

Local SA gives the raw derivatives $\frac{\partial Y}{\partial Q_1} = 2 > \frac{\partial Y}{\partial Q_2} = 1$: parameter 1 appears more important. (These are raw slopes, not the normalized indices of Chapter 2, which would need nominal values Q_1^*, Q_2^* and Y^*.) But the parameters have very different uncertainty ranges, and Q_2 varies three times as much as Q_1.

The output variance is:

$$\mathrm{Var}(Y) = s_1^2\sigma_1^2 + s_2^2\sigma_2^2 = 4 + 9 = 13 \quad \Longrightarrow \quad \sigma_Y = \sqrt{13}.$$

The *sigma-normalized sensitivity index*

$$S_i^\sigma = \frac{\sigma_i}{\sigma_Y}\frac{\partial Y}{\partial Q_i} \tag{9.1}$$

accounts for both the derivative and the parameter uncertainty:

$$S_1^\sigma = \frac{1}{\sqrt{13}} \cdot 2 \approx 0.55, \qquad S_2^\sigma = \frac{3}{\sqrt{13}} \cdot 1 \approx 0.83.$$

Parameter 2 dominates, opposite to the local SA ranking. The local ranking was misleading because it ignored the factor-of-3 difference in parameter uncertainty.

Like the normalized SI in Chapter 2, S_i^σ is dimensionless: σ_i carries the units of Q_i and σ_Y carries the units of Y, so $(\sigma_i/\sigma_Y)\,\partial Y/\partial Q_i$ is a pure number. This means S_1^σ and S_2^σ can be compared directly even when the two

parameters are measured in completely different units, the same reason the tornado plot in Chapter 2 sorted by $|\mathcal{S}_p^q|$ rather than by the raw derivative. *Unifying Idea 1* (the sensitivity index is a normalized derivative; Section 1.5). The sigma-normalized index is still the sensitivity index from Chapter 2, rescaled by the ratio of relative uncertainties: with $\mathrm{CV}_i = \sigma_i/Q_i^*$ and $\mathrm{CV}_Y = \sigma_Y/Y^*$, $S_i^\sigma = \mathcal{S}_{Q_i}^Y \mathrm{CV}_i/\mathrm{CV}_Y$. The parameter's relative uncertainty raises the index and the output's lowers it. Global SA builds on local SA; it does not replace it. The local SI from Chapter 2 reappears inside the variance-based formulas of this chapter whenever the response is close to linear over the input range, which is exactly the delta-method setting. The connection is real but not universal: Morris, PRCC, and Borgonovo's moment-independent measures are built from other quantities and contain no local index.

9.1.2 ▪ Two SIR Models

For the SIR model, suppose two outbreak settings give the same $\mathcal{S}_\beta^{\mathcal{R}_0} = 1$. In Setting A, β is tightly constrained from contact-tracing data ($\sigma_\beta = 0.005$), while in Setting B, β must be estimated from seroprevalence data ($\sigma_\beta = 0.03$). Local SA says β is equally important in both settings. In Setting B the contribution of β to the uncertainty in $\mathcal{R}_0$, measured by $|\partial\mathcal{R}_0/\partial\beta|\,\sigma_\beta$, is six times larger. The sigma-normalized index rises by less than six, because $\sigma_{\mathcal{R}_0}$ in the denominator rises with σ_β as well; with the nominal values of the self-check below and its standard deviations $\sigma_c = 0.5$ and $\sigma_{\tau_R} = 2$ for the other two parameters, the factor is about 3.2. These uncertainties propagate to $\mathcal{R}_0$ through the standard delta-method formula (F.6) of Appendix F.

Self-check 9.1. For the SIR model with $c^* = 5$, $\beta^* = 0.06$ and $\tau_R^* = 7$, $\mathcal{R}_0 = c\beta\tau_R$ has $\mathcal{R}_0^* = 2.1$ and local SIs $\mathcal{S}_c^{\mathcal{R}_0} = \mathcal{S}_\beta^{\mathcal{R}_0} = \mathcal{S}_{\tau_R}^{\mathcal{R}_0} = 1$. The parameter standard deviations are $\sigma_c = 0.5$, $\sigma_\beta = 0.01$ and $\sigma_{\tau_R} = 2$. First compute $\sigma_{\mathcal{R}_0}$ from the delta method $\mathrm{Var}(\mathcal{R}_0) \approx \sum_i (\partial\mathcal{R}_0/\partial p_i)^2\sigma_i^2$, then rank the three parameters by their sigma-normalized index S_i^σ.

9.1.3 ▪ Time-Varying Sensitivity

The analyses above collapse the SIR model's time-dependent output into a single scalar QOI: $\mathcal{R}_0$, or peak infections, or final size. This is often appropriate, but it discards information. For ODE models, sensitivity indices can be computed as functions of time: $S_{p_i}^{q_j}(t)$ tells which parameters matter most at each moment in the epidemic trajectory.

In the SIR model the answer changes substantially over the course of the outbreak. Near the start, the trajectory is exponentially growing and β (the

transmission rate) dominates: small changes in β compound rapidly. Near the peak, γ_R (the recovery rate) becomes comparably important because the balance between new infections and recoveries determines when the peak occurs. Toward the end, both rates shape the final size, but initial-condition parameters that seemed irrelevant during growth can re-emerge as influential. A global SA computed against only the peak-infections QOI would miss the period when γ_R most needs careful estimation.

Time-varying global SA works by running the Saltelli algorithm (Section 9.5) against the model output at each recorded time step $t_1, t_2, \ldots, t_T$. For ODE models this costs no additional model evaluations: a single run of the SIR system produces the full trajectory $\vec{u}(t)$ at every time step, so the $N(m+2)$ evaluations of the Saltelli construction automatically yield model output at all T recorded times. The sensitivity indices $\hat{S}_i(t_k)$ and $\hat{S}_{T_i}(t_k)$ are then computed by applying the Jansen estimator to the outputs at each time step t_k separately; an arithmetic operation with negligible cost compared to the ODE solves. Wu et al. [66] apply this approach to epidemic models and show that parameter importance rankings shift substantially across the trajectory, with direct implications for which parameters to prioritize for data collection at each stage of an outbreak.

9.2 ▪ Sampling Methods

All global SA methods require evaluating the model at many parameter combinations. Choosing those combinations efficiently is the first computational task.

9.2.1 ▪ Random Sampling

A note on notation before the methods begin. Through the modeling chapters N has denoted the total population, $S + I + R = N$. From here to the end of the chapter it denotes the *sample size*, the number of parameter vectors drawn, which is the standard usage in the sensitivity literature and the N of the Saltelli cost $N(m + 2)$. No population N appears in this chapter, so the two never meet on a page.

The simplest approach: draw N independent samples from the joint parameter distribution (often assumed uniform or independent log-normal over plausible ranges). For N large, random sampling provides good coverage on average, but can leave regions of parameter space unsampled by chance.

9.2.2 ▪ Latin Hypercube Sampling

Latin Hypercube Sampling (LHS) divides each parameter's range into N equal strata and ensures that exactly one sample falls in each stratum for every parameter [33]. This eliminates the clustering that random sampling can produce and improves coverage with smaller N.

Definition 9.1 (LHS Sample)**.** *An LHS of size N for m parameters with ranges $[l_j, u_j]$, $j = 1, \ldots, m$, is an $N \times m$ matrix X such that each column contains exactly one value from each of the N equal-width strata $[l_j + (k-1)(u_j - l_j)/N,\ l_j + k(u_j - l_j)/N]$ for $k = 1, \ldots, N$.*

The standard LHS algorithm permutes the strata independently for each parameter and adds uniform jitter within each stratum:

```
# LHS sample: N points in m dimensions (see repository for full implemen
set_random_seed(42)
X = LHS_SAMPLE(N, m, lower=lb, upper=ub)  # N-by-m sample matrix
# Each column j has exactly one sample in each of the N strata of [lb[j]
```

For the same N, LHS provides substantially better coverage than random sampling, particularly for estimating marginal effects of individual parameters; see Helton and Davis [19] for convergence analysis and uncertainty propagation results. For Sobol index estimation (Section 9.5), LHS is not the default; this chapter fills the Saltelli matrices from a scrambled Sobol' sequence (§9.2.3).

Self-check 9.2. For $N = 4$ samples and $m = 1$ parameter on $[0, 1]$: (a) Draw a random sample of size 4. Is it possible that all 4 points fall in $[0, 0.5]$? Is this possible with LHS? (b) Write out explicitly what an LHS of size 4 looks like for one parameter.

9.2.3 ▪ Low-Discrepancy Sampling and Scrambled Sobol' Sequences

The Monte Carlo convergence rate $O(1/\sqrt{N})$ for the Jansen estimators is a consequence of drawing the rows of matrices A and B independently and uniformly at random. Independence is a liability rather than an asset: independent draws cluster and leave gaps, so many of the $N(m+2)$ model evaluations are wasted on redundant regions of parameter space.

The key insight is that the estimators require only that each row of A (and B) be *marginally uniform*, that is, each component Q_i is uniformly distributed over its range. Independence among rows is not required. Replacing independent uniform draws with a *low-discrepancy sequence* that

is marginally uniform but deliberately spread across $[0,1]^m$ gives the same unbiased estimators with substantially lower variance.

Discrepancy. The *discrepancy* of a point set measures how uniformly points cover the unit hypercube. Low-discrepancy sequences (LDS) achieve discrepancy $O((\log N)^m/N)$, compared to the typical $O(1/\sqrt{N})$ of independent uniform samples. For smooth integrands with $m \leq 10$, this translates directly into faster convergence of the Sobol estimators.

Scrambled Sobol' sequences. The current best practice is the *scrambled Sobol' sequence* [40]. A Sobol' sequence fills $[0,1]^m$ one stratum at a time; scrambling randomizes the digits of each coordinate in a way that keeps this stratification. Each point is then marginally uniform, the joint distribution covers the space uniformly, and the sequence is *randomized*; the estimator is unbiased and its variance is estimable by repeating the scramble. Owen [39] proved that scrambled nets achieve variance $O((\log N)^{m-1}/N^3)$ for smooth integrands, a superlinear improvement over the $O(1/N)$ of Latin hypercube sampling.

Independence is not the critical property. Independence guarantees that the standard confidence interval formulas apply. Scrambled Sobol' sequences are not independent: pairs of points carry small correlations, slightly negative on average. That negative dependence is the source of the variance reduction described below. They also satisfy a central limit theorem. Confidence intervals are constructed by repeating the scrambled estimation with 30 or more independent scrambles. Each scramble is a full design, so R scrambles cost R times the $N(m+2)$ runs of one estimate. The practical consequence: for a given budget N, scrambled Sobol' sampling often substantially reduces the confidence interval width on $\hat{S}_i$ and $\hat{S}_{T_i}$ compared to independent uniform sampling, in smooth, low-dimensional problems frequently by a factor of two or more (about seven in the worked example of §9.5), at no additional model evaluations. The *identically distributed* property, each sample point is uniform over its range, is what the unbiasedness of the estimators relies on; statistical independence among the sample points is not required, and for a low-discrepancy design it is precisely what one gives up in exchange for the more even coverage that lowers estimator variance. Figure 9.1 illustrates the coverage difference.

Practical guidance. SALib [20] uses scrambled Sobol' sequences by default since version 1.4. In MATLAB, `sobolset` in the Statistics and Machine Learning Toolbox provides the same construction. For $m \leq 20$ and $N \leq 10{,}000$, scrambled Sobol' sampling is usually the better choice, giving lower variance than IID or LHS for integrands with enough smoothness. The advantage narrows as m grows or when the integrand is rough, and for a genuinely discontinuous response there is no guarantee of improvement.

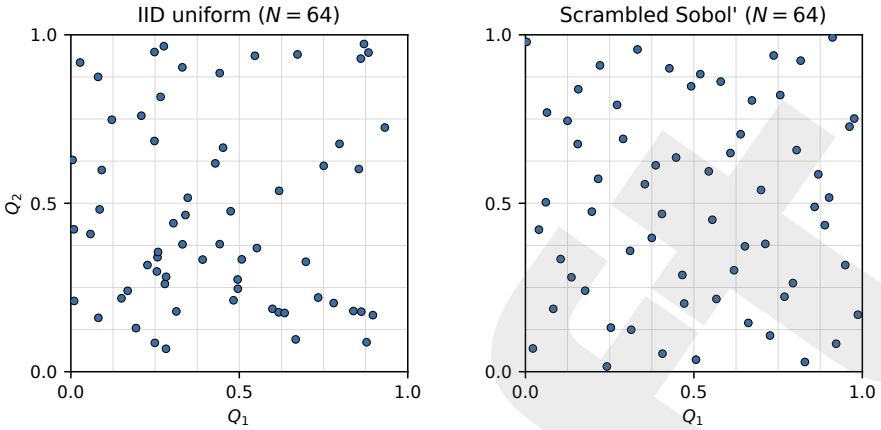


Figure 9.1. *Comparison of $N = 64$ points in two dimensions. Left: independent identically distributed (IID) uniform samples show visible clusters and empty regions that waste model evaluations. Right: scrambled Sobol' sequence [40] achieves near-uniform coverage. With $N = 64 = 2^6$ points in two dimensions the sequence is a $(0, 6, 2)$-net in base 2, and its scrambling (SciPy's linear matrix scramble with a digital shift) preserves that property, so every one of the 8×8 cells contains exactly one point. This is a guarantee rather than an average: in the left panel 24 of the 64 cells are empty and one holds five points. Both strategies produce marginally uniform samples, so both yield unbiased Sobol estimators; the LDS strategy produces much lower estimator variance for the same computational budget N.*

For a comprehensive treatment of quasi-Monte Carlo methods, including higher-dimensional constructions and their error analysis, see Owen [41].

9.3 ▪ Regression-Based SA: LHS/PRCC

The Partial Rank Correlation Coefficient (PRCC) is the dominant global SA method in mathematical epidemiology. It is computationally cheap, handles monotone (but not necessarily linear) relationships, and the Marino et al. [32] MATLAB implementation is widely used and freely available; it builds on the rank-correlation approach of Iman and Conover [25] and the epidemiological SA tradition of Blower and Dowlatabadi [5].

Remark 9.2. The PRCC is a *nonparametric partial correlation coefficient*. Step 2 of the calculation below fits a linear regression on rank-transformed data, and Step 3 extracts the partial correlation coefficient for each predictor. Readers from statistics will recognize this as a rank-based analog of the partial correlation coefficient from multiple regression. It is not the partial regression coefficient, the slope that Step 2 fits; on ranks that slope is the standardized rank regression coefficient, a different number. PRCC for p_j measures the linear relationship between $\mathrm{rank}(p_j)$ and $\mathrm{rank}(Y)$ after removing the linear effects of all other rank-transformed parameters. The rank transformation replaces the linearity assumption with a monotonic-

ity assumption, extending applicability to nonlinear but order-preserving responses.

9.3.1 ▪ The PRCC Calculation

Given N LHS samples of the m POIs and the corresponding N model outputs:

Step 1. Rank-transform all inputs and the output separately. Replace each value with its rank among the N observations.

Step 2. Fit a linear regression of the rank-transformed output on all rank-transformed inputs.

Step 3. For each parameter p_j, compute the partial correlation coefficient between the rank-transformed p_j and the rank-transformed output, controlling for all other parameters. This is the PRCC for p_j.

Step 4. Compute a t-test p-value for each PRCC to assess statistical significance.

The PRCC ranges from -1 to $+1$. Its sign has the same interpretation as the SI sign rule from Chapter 2: positive means POI and QOI increase together; negative means they move in opposite directions.

9.3.2 ▪ When PRCC is Appropriate

PRCC requires monotone relationships: the QOI must consistently increase (or consistently decrease) as each POI increases, across the full range of parameter values. It does not require linearity. For most compartmental epidemic models, the relationships are approximately monotone, which is why PRCC dominates the field.

Before applying PRCC, verify the monotonicity assumption by producing a scatter matrix: plot the model output against each POI for the full LHS sample using a pairwise scatter matrix. A roughly monotone cloud (upward or downward sloping, without reversals) supports the use of PRCC. A hump-shaped or U-shaped cloud signals non-monotonicity, and Sobol indices should be used instead.

PRCC and Sobol indices are not equivalent. PRCC assumes monotone relationships; Sobol indices make no such assumption. When a relationship is nonmonotone (a QOI first increases then decreases as a parameter increases), PRCC may give a near-zero coefficient even for a highly important parameter, while the Sobol total index will correctly identify it as important. For the models in this course, both methods typically agree, but verifying agreement is a useful consistency check.

For the SIR model, $\mathcal{R}_0 = c\beta/\gamma_R$ is monotone increasing in β and monotone decreasing in γ_R, which is why PRCC performs reliably on this par-

ticular model. For output quantities that exhibit threshold or bifurcation behavior, such as the time to peak infections in models with multiple competing timescales and parameter-dependent qualitative changes, monotonicity can fail even when the individual input-output relationships look locally linear. In such cases PRCC will underestimate the importance of parameters that drive threshold-crossing, while the Sobol total index captures the full nonlinear contribution [57].

9.4 ▪ Sobol Decomposition and Variance-Based SA

Variance-based SA is the most rigorous framework for global SA: it quantifies exactly how much of the total output variance is attributable to each POI or group of POIs [61].

9.4.1 ▪ The Intuition: Conditional Expectation

Suppose parameter Q_1 is fixed at q_1 while all other parameters vary across their ranges. The expected output $\mathbb{E}[Y \mid Q_1 = q_1]$, the average of Y over the other inputs with Q_1 held at q_1 (Section F.3), is some function of q_1. If Q_1 is important, this conditional expectation varies substantially as q_1 changes. The variance of this conditional expectation, $\mathrm{Var}_{Q_1}[\mathbb{E}_{Q_{-1}}(Y \mid Q_1)]$, measures how much of Y's variability is explained by Q_1 alone. Normalizing by the total variance gives the first-order Sobol index.

9.4.2 ▪ The ANOVA Decomposition

Why does a unique decomposition exist? Because statistical independence allows us to cleanly separate what each parameter contributes alone from what it contributes jointly. When the Q_i are independent, projecting f onto the subspace of functions of Q_i alone gives a well-defined component $f_i(Q_i)$, and these projections do not interfere with each other. The condition $\int_0^1 f_i(Q_i)\, dQ_i = 0$ is the mathematical expression of this separation. It is an orthogonality: under the inner product $\langle f, g\rangle = \int fg$ of Appendix A.4, it says that f_i is orthogonal to the constant function 1, that is, f_i has zero mean. Distinct components are orthogonal in the same sense, so they contribute zero covariance. When parameters are correlated, this orthogonality breaks down, and the decomposition loses its uniqueness (Section 9.7 returns to this point). With independence, the decomposition is both unique and additive in variance.

For independent POIs $\vec{Q} = (Q_1, \ldots, Q_m)$ uniformly distributed on $[0,1]^m$, any square-integrable model $Y = f(\vec{Q})$ admits the unique *ANOVA decom-*

position:

$$f(\vec{Q}) = f_0 + \sum_i f_i(Q_i) + \sum_{i<j} f_{ij}(Q_i, Q_j) + \cdots + f_{12\cdots m}(\vec{Q}), \tag{9.2}$$

where $f_0 = \mathbb{E}[Y]$, $f_i(Q_i) = \mathbb{E}[Y \mid Q_i] - f_0$, and the higher-order terms capture interactions. Uniqueness follows from the orthogonality conditions $\int_0^1 f_u(\vec{Q}_u)\, dQ_i = 0$ for each $i \in u$; see Sobol [61] and Section 4.3 of Saltelli et al. [53], which states the result. The terms are orthogonal, so their variances add:

$$D = \mathrm{Var}(Y) = \sum_i D_i + \sum_{i<j} D_{ij} + \cdots + D_{12\cdots m}, \tag{9.3}$$

where $D_i = \mathrm{Var}[\mathbb{E}(Y \mid Q_i)]$ and $D_{ij} = \mathrm{Var}[\mathbb{E}(Y \mid Q_i, Q_j)] - D_i - D_j$.

Worked example: a model with an interaction term. Consider $f(Q_1, Q_2) = Q_1 + Q_2 + Q_1 Q_2$ with $Q_1, Q_2 \sim \mathcal{U}(0,1)$ independent. Compute the ANOVA components directly from the definitions:

$$\begin{aligned}
f_0 &= \mathbb{E}[Y] = \tfrac{1}{2} + \tfrac{1}{2} + \tfrac{1}{4} = \tfrac{5}{4},\\
f_1(Q_1) &= \mathbb{E}[Y \mid Q_1] - f_0 = \big(Q_1 + \tfrac{1}{2} + \tfrac{Q_1}{2}\big) - \tfrac{5}{4} = \tfrac{3Q_1}{2} - \tfrac{3}{4} = \tfrac{3}{2}\big(Q_1 - \tfrac{1}{2}\big),\\
f_2(Q_2) &= \mathbb{E}[Y \mid Q_2] - f_0 = \tfrac{3}{2}\big(Q_2 - \tfrac{1}{2}\big) \quad \text{(by symmetry)},\\
f_{12}(Q_1, Q_2) &= f - f_0 - f_1 - f_2 = Q_1 Q_2 - \tfrac{Q_1}{2} - \tfrac{Q_2}{2} + \tfrac{1}{4} = \big(Q_1 - \tfrac{1}{2}\big)\big(Q_2 - \tfrac{1}{2}\big).
\end{aligned}$$

Verify orthogonality: $\int_0^1 f_{12}\, dQ_1 = \big(\int_0^1 (Q_1 - \tfrac{1}{2})\, dQ_1\big)(Q_2 - \tfrac{1}{2}) = 0$. ✓

The variances are $D_1 = D_2 = \big(\tfrac{3}{2}\big)^2 \mathrm{Var}(Q_1) = \tfrac{9}{4} \cdot \tfrac{1}{12} = \tfrac{3}{16}$, since $\mathrm{Var}(Q_1) = 1/12$ (Section F.2), and $D_{12} = \mathrm{Var}(f_{12}) = \mathrm{Var}(Q_1 - \tfrac{1}{2})\,\mathrm{Var}(Q_2 - \tfrac{1}{2}) = \tfrac{1}{144}$, since the variance of a product of independent factors with mean zero is the product of their variances. Total variance $D = \tfrac{3}{16} + \tfrac{3}{16} + \tfrac{1}{144} = \tfrac{27}{144} + \tfrac{27}{144} + \tfrac{1}{144} = \tfrac{55}{144}$. The first-order indices $S_1 = S_2 = \tfrac{3/16}{55/144} = \tfrac{27}{55} \approx 0.491$, and the interaction index $S_{12} = \tfrac{1/144}{55/144} = \tfrac{1}{55} \approx 0.018$. Note $S_1 + S_2 + S_{12} = 1$. For the additive model $f = Q_1 + Q_2$ (no interaction), $S_1 = S_2 = \tfrac{1}{2}$ and $S_{12} = 0$: the indices sum to 1 with nothing left over for interactions. This is why $\sum S_i < 1$ signals the presence of interactions.

9.4.3 ▪ Sobol Indices

Definition 9.3 (Sobol Indices)**.** *The* first-order Sobol index *for parameter i is:*

$$S_i = \frac{D_i}{D} = \frac{\mathrm{Var}[\mathbb{E}(Y \mid Q_i)]}{\mathrm{Var}(Y)}. \tag{9.4}$$

The total Sobol index *for parameter i [22] is:*

$$S_{T_i} = 1 - \frac{\mathrm{Var}[\mathbb{E}(Y \mid \vec{Q}_{-i})]}{\mathrm{Var}(Y)} = \frac{\mathbb{E}[\mathrm{Var}(Y \mid \vec{Q}_{-i})]}{\mathrm{Var}(Y)}, \tag{9.5}$$

where $\vec{Q}_{-i}$ denotes all parameters except Q_i; the two forms agree by the law of total variance (F.7)*, applied with $\vec{Q}_{-i}$ in place of Q_i.*

The first-order index measures the *direct* effect of Q_i; the total index measures the direct effect plus all interactions with other parameters. When $S_{T_i} = S_i$ for all i, the model is additively separable (no interactions). When $S_{T_i} > S_i$, parameter i interacts with other parameters.

Remark 9.4 (ANOVA and Sobol indices). Readers from statistics will recognize $S_i = \mathrm{Var}[\mathbb{E}(Y \mid Q_i)]/\mathrm{Var}(Y)$ as the quantity known as $\boldsymbol{\eta^2}$ (eta-squared) in classical ANOVA, the proportion of total response variance explained by one factor acting alone. The two frameworks share the same mathematical skeleton. In classical ANOVA, the total sum of squares decomposes as $SS_{\text{total}} = SS_1 + SS_2 + SS_{12} + SS_{\text{residual}}$, where each SS is computed from an observed dataset and SS_{residual} captures unexplained noise. In the Sobol framework, the total variance decomposes as $D = D_1 + D_2 + D_{12}$ with no residual term, because the model is deterministic and all variance is explained by the inputs. Three differences matter in practice: (1) classical ANOVA requires categorical factors or discretized inputs; Sobol indices work with continuous parameters; (2) classical ANOVA estimates variance components from sample data; Sobol indices are population quantities estimated by Monte Carlo; (3) classical ANOVA requires balance and orthogonality across observed data; Sobol indices require independence of the input distributions. Despite these differences, for a linear model with independent inputs, the two frameworks coincide exactly: $S_i = \eta^2$. The Sobol framework is the natural generalization of ANOVA to nonlinear deterministic models with continuous inputs.

Unifying Idea 1 (the sensitivity index is a normalized derivative; Section 1.5). The first-order Sobol index $S_i = D_i/D$ uses the same ratio structure as the sensitivity index from Chapter 2; it measures one variance as a fraction of another. The connection is explicit: for a linear model $Y = \sum_i a_i Q_i$ with independent $Q_i \sim \mathcal{U}[l_i, u_i]$, the first-order Sobol index equals the square of the sigma-normalized index: $S_i = (S_i^\sigma)^2$. Global SA builds on local SA. For a linear model S_i and S_i^σ give the same ranking by magnitude but different values. The raw local slopes can give a different ranking, as the portfolio of Section 9.1 shows.

Unifying Idea 3 (local SA is a linearization; Section 1.5). The Sobol decomposition resolves the tension that this idea exposed in Chapter 8. Local

SA is a linearization, valid near the nominal values. Sobol indices make no linearity assumption: they measure the actual variance contribution across the full parameter range. When the linearization is poor (Model B from the chapter opening), local and global SA give different answers, and Sobol indices give the correct answer.

In the language of Appendix A, the ANOVA decomposition (9.2) is the orthogonal decomposition of f in the Hilbert space $L^2([0,1]^m)$ with respect to the inner product $\langle f, g\rangle = \int_{[0,1]^m} f(\vec{q})\, g(\vec{q})\, d\vec{q}$. The orthogonality explains both why the variances add in (9.3) and why the method requires independent POIs: independence is needed for the inner product to factorize and the projections to be orthogonal.

Self-check 9.3. For $Y = c_1 Q_1 + c_2 Q_2$ with $Q_i \sim \mathcal{U}(0,1)$ independent: (a) Compute $\mathbb{E}[Y \mid Q_1]$ and $\mathrm{Var}[\mathbb{E}(Y \mid Q_1)]$. (b) Compute $\mathrm{Var}(Y)$ using the independence of Q_1 and Q_2. (c) Show that $S_1 = c_1^2/(c_1^2 + c_2^2)$ and $S_2 = c_2^2/(c_1^2 + c_2^2)$. (d) Verify $S_1 + S_2 = 1$ (no interaction for additive models).

9.5 ▪ Computing Sobol Indices: The Saltelli Algorithm

A naive Monte Carlo estimate of S_i fixes $Q_i = q_i$ at each of N values and averages the model over N samples of the other parameters at each. That costs $O(N^2)$ evaluations per index, or $O(mN^2)$ for all m first-order indices. The insight of Saltelli [48] is that all first-order and total Sobol indices can be estimated with $N(m+2)$ evaluations using two carefully constructed sample matrices. The design and estimators for the total sensitivity index are developed in Saltelli et al. [50].

9.5.1 ▪ The Saltelli Construction

Table 9.1 fixes the notation used below. This chapter fills A and B, each $N \times m$, from one scrambled Sobol' sequence (§9.2.3) in $2m$ dimensions, with N a power of two. The first m coordinates of each point form a row of A, and the last m form the same row of B. In Python, `scipy.stats.qmc.Sobol` with `scramble=True` applies a linear matrix scramble followed by a digital shift. Confidence intervals come from repeating the whole estimation under independent scrambles (§9.2.3); the bootstrap described below is the fallback, valid only when the rows are IID. Define the m matrices $A_B^{(i)}$: matrix A with its i-th column replaced by the i-th column of B. The model is evaluated at all rows of A, B, and all m matrices $A_B^{(i)}$: a total of $N(m+2)$ evaluations. Figure 9.2 shows the structure for $m = 3$ columns.

Matrix	Contents
A	Base sample matrix ($N \times m$): each row follows the joint input distribution (by default, the first m coordinates of a scrambled Sobol' point)
B	Second sample matrix ($N \times m$): each row follows the same input distribution, independent of the same row of A (by default, the last m coordinates of the same point)
$A_B^{(i)}$	Matrix A with its i-th column replaced by the i-th column of B; all other columns come from A

Table 9.1. *Saltelli matrix notation. The key idea: $A_B^{(i)}$ and A share the values of all parameters* except *Q_i, which is drawn from B. This "pick-freeze" design isolates the variance contribution of Q_i from all other parameters.*

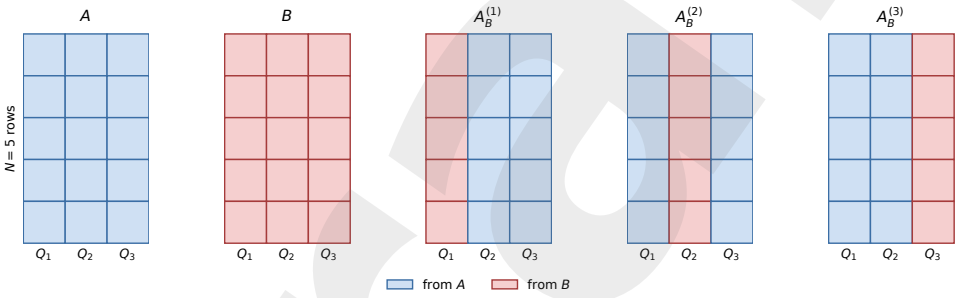


Figure 9.2. *Saltelli pick-freeze matrix structure for $m = 3$ parameters (columns) and $N = 5$ sample rows. Blue cells carry values from the base matrix A; red cells carry values from the matrix B. $A_B^{(i)}$ shares every column with A* except *column i, which is "frozen" from B. This design ensures that $A_B^{(i)}$ and A differ* only *in parameter Q_i, so the pair carries everything Q_i contributes, alone and through all its interactions: it estimates $D\,S_{T_i}$ (see (9.7)). The total cost is $N(m+2)$ model evaluations: N for A, N for B, and N for each of the m hybrid matrices.*

9.5.2 ▪ The Jansen Estimators

For the first-order index, use the Jansen [26] estimator:

$$\hat{S}_i = 1 - \frac{\frac{1}{2N}\sum_{j=1}^{N}[f(B_j) - f(A_{B,j}^{(i)})]^2}{\hat{D}}, \tag{9.6}$$

where $\hat{D}$ is the variance of the $2N$ values $f(A_j)$ and $f(B_j)$ pooled: $\hat{D} = \frac{1}{2N}\sum_j [f(A_j)^2 + f(B_j)^2] - \left(\frac{1}{2N}\sum_j [f(A_j) + f(B_j)]\right)^2$.

For the total index:

$$\hat{S}_{T_i} = \frac{\frac{1}{2N}\sum_{j=1}^{N}[f(A_j) - f(A_{B,j}^{(i)})]^2}{\hat{D}}. \tag{9.7}$$

Use these Jansen estimators, not the original Saltelli (2002) formulas, which have higher variance. Each numerator is half the mean squared difference of two runs that share some inputs, an unbiased estimate of a variance component; see Jansen [26], Section 2.3. Both estimators are consistent: $\hat{S}_i \to S_i$ and $\hat{S}_{T_i} \to S_{T_i}$ as $N \to \infty$. A comprehensive comparison of eight total-order estimators across a wide range of models, sample sizes, input distributions, and dimensionalities finds that the Jansen estimator consistently outperforms the Saltelli (2002) formula for both factor prioritization and estimation accuracy [45]; for the total-order index, the Jansen formula is the safer default. The key step is that

$$\mathbb{E}\left[\frac{1}{2N}\sum_{j=1}^{N}(f(B_j) - f(A_{B,j}^{(i)}))^2\right] = D - D_i = \mathrm{Var}(Y) - \mathrm{Var}[\mathbb{E}(Y \mid Q_i)],$$

which follows because B_j and $A_{B,j}^{(i)}$ share the *same* value of parameter i (both equal to B's column i) while differing in all other parameters (one from B, the other from A). The shared value of Q_i does not cancel in the difference $f(B_j) - f(A_{B,j}^{(i)})$ unless f is additive in Q_i. For the worked model of Section 9.4 with $i = 1$, $f(b_1, b_2) - f(b_1, a_2) = (b_2 - a_2)(1 + b_1)$. The identity holds because the two evaluations share Q_i while the other parameters are drawn twice, independently. Given Q_i, half the expected squared difference is then the conditional variance $\mathrm{Var}(Y \mid Q_i)$, and its mean over Q_i is $\mathbb{E}[\mathrm{Var}(Y \mid Q_i)] = D - D_i$ by the law of total variance. Dividing by $\hat{D}$ and rearranging recovers $\hat{S}_i \approx D_i/D = S_i$. Its bias at finite N is of order $1/N$, as for any ratio of sample means, and so is small next to the $O(1/\sqrt{N})$ sampling error once N is large.

9.5.3 ▪ Convergence

For IID rows the estimators converge as $O(1/\sqrt{N})$, the standard Monte Carlo rate. With scrambled Sobol' rows they typically converge faster. In practice, $N \geq 1000$ is needed for reliable estimates, and a power of two at or above that is the natural choice. For $m = 4$ parameters (SIR model): $N(m+2) = 6144$ evaluations at $N = 1024$. If each evaluation takes 0.1 seconds: 10 minutes.[4]

Plotting $\hat{S}_i$ and $\hat{S}_{T_i}$ vs. N reveals the convergence: estimates are noisy for $N < 500$ and stable for $N \geq 2000$. This convergence plot is the diagnos-

[4] An alternative path to total-order indices is the *extended FAST* (eFAST), which reaches the same indices via Fourier analysis along a space-filling search curve through the input space [57]. eFAST has different robustness properties at small N but converges to the same values as the Jansen estimator; it is implemented alongside the matrix-based approach in the SALib Python library.

tic that answers the misconception "more samples always help": below a threshold N, additional samples do not produce reliable estimates because the variance of the estimator exceeds the signal being measured.

Confidence intervals. A point estimate $\hat{S}_i$ is useful but incomplete: it gives no information about sampling uncertainty. For a design whose rows are IID, the standard approach is the *bootstrap*: resample the $N(m+2)$ model evaluations with replacement, recompute $\hat{S}_i$ for each resample, and take the 2.5th and 97.5th percentiles of the resulting distribution as a 95% confidence interval.

```
# Bootstrap confidence intervals for Sobol first-order indices
nBoot = 500                              # number of bootstrap replicates
S1_boot = ZEROS(nBoot, m)                # will store replicate estimates

FOR b FROM 1 TO nBoot:
    idx = RANDOM_INTEGERS(N, from=1..N)  # resample the N base rows
    # the tuple (Y_A, Y_B, Y_AB^(1..m)) stays aligned within each row;
    # the model is never re-evaluated
    S1_boot[b, :] = JANSEN_FROM_OUTPUTS(Y_A[idx], Y_B[idx], Y_AB[idx, :]
END

S1_CI = PERCENTILE(S1_boot, [2.5, 97.5])  # 95% CI per index
```

Resampling by row is what makes the replicates valid: the Jansen estimator is a function of the matched tuples $(Y_A, Y_B, Y_{A_B}^{(1)}, \ldots, Y_{A_B}^{(m)})$, and breaking that alignment destroys the pick-freeze structure the estimator relies on. The procedure assumes the rows are independent, which holds for IID sampling. Under any stratified or low-discrepancy design they are not: that covers the scrambled Sobol' sequence and equally the Latin hypercube design of §9.2.2, whose whole purpose is to make the rows depend on one another. In either case the honest analogue is to repeat the whole analysis under independent scrambles, or independent LHS draws, rather than to resample rows.

A confidence interval that straddles zero indicates that the parameter's effect cannot be distinguished from noise at the current sample size. A confidence interval that excludes zero confirms the parameter is genuinely influential. As a practical rule: if the confidence interval width exceeds the index value, double N before drawing conclusions.

Worked numerical example: confidence intervals for the SIR model. Apply the SIR model ($m = 3$ parameters: contact rate c, transmission probability β, infectious period τ_R; QOI: peak infectious fraction $I_{\max}$) with ranges $\pm 50\%$ around nominal values $c = 8$ (a higher-contact setting than

the running example), $\beta = 0.06$ (a probability per contact, dimensionless), and $\tau_R = 10$ days. Generate a Saltelli design with $N = 256$ sample pairs ($N(m+2) = 1280$ model evaluations total), a power of two for the reason given in §9.12. The Jansen estimators applied to the full sample give:

$$\hat{S}_1^{c} = 0.32, \qquad \hat{S}_1^{\beta} = 0.33, \qquad \hat{S}_1^{\tau_R} = 0.33.$$

These are *point estimates*: they carry no information about sampling uncertainty. To quantify that uncertainty, repeat the whole estimation under $R = 32$ independent scrambles, meeting the 30 or more that §9.2.3 asks for, each a fresh design with $N = 256$; the design above is the first. The first four replicates produce:

Scramble r	$\hat{S}_1^{c}$	$\hat{S}_1^{\beta}$	$\hat{S}_1^{\tau_R}$
1	0.32	0.33	0.33
2	0.33	0.34	0.33
3	0.33	0.33	0.32
4	0.34	0.32	0.34
⋮	⋮	⋮	⋮
32	0.35	0.33	0.34

The mean of each column is the estimate. With s the column's standard deviation, the 95% confidence interval is the mean $\pm\, t_{31,\,0.975}\, s/\sqrt{32}$, where $t_{31,\,0.975} \approx 2.04$:

$$\begin{aligned} c: &\quad \hat{S}_1^{c} = 0.332 \quad [0.330,\ 0.335] \\ \beta: &\quad \hat{S}_1^{\beta} = 0.332 \quad [0.329,\ 0.334] \\ \tau_R: &\quad \hat{S}_1^{\tau_R} = 0.332 \quad [0.329,\ 0.335] \end{aligned}$$

All three intervals exclude zero: each parameter is genuinely influential. All three also overlap almost completely, and that is the result, not a failure to resolve one. The three POIs enter this model only through $\mathcal{R}_0 = c\beta\tau_R$, and the $\pm 50\%$ ranges give them the same relative spread, so their first-order indices are equal as population quantities. The small differences among 0.32, 0.33 and 0.33 are Monte Carlo scatter about a common value, and no sample size will turn them into a ranking.

The practical lesson is the one a reader is most likely to need. Faced with three indices near 0.33 and told that they sum to about one, it is tempting to report the largest as the most influential parameter. Here that would be reporting the seed of a random number generator. The

intervals are what stop you: they overlap, so the ordering they suggest is not there. Section 3.5, §8.5 and the Chapter 6 laboratory make the same point from three other directions: when POIs enter only through one combination, sensitivity analysis can rank the combination but not the parameters that form it. All three intervals are narrow: each is under 0.006 wide, less than 2% of its index value. That is what 32 scrambles of the $N = 256$ design buy, at $32 \times 1280 = 40{,}960$ evaluations. More scrambles will not, however, separate the three: they are equal by construction, and a narrower interval only shows that more clearly. The row bootstrap, the fallback for IID rows, gives [0.21, 0.42], [0.20, 0.44] and [0.25, 0.42] on the first design (100 resamples). These are 5 to 8 times as wide as the interval $\pm t_{31,\,0.975}\, s$ that the scrambles assign one design, because its rows are not independent.

When m is large, however, even a single Sobol run at $N = 2048$ may be prohibitively expensive; Morris screening addresses exactly this case.

9.6 ▪ Morris Screening

When m is large (30 or more parameters), running Sobol at $N = 2048$ costs $N(m+2) = 65{,}536$ model evaluations, potentially prohibitive. Morris screening [35] provides a cheap first step: $r(m+1)$ evaluations (typically $r = 10$ trajectories) to identify which parameters are truly unimportant. Campolongo et al. [11] choose the trajectories to spread them across the input space, which gives more uniform coverage at no extra cost in model runs. Campolongo, Saltelli, and Cariboni [12] later found a radial design, which takes one step for each factor from a common random starting point, superior for computing elementary effects.

For each trajectory, one parameter is changed at a time by a fixed fraction Δ of its range. The *elementary effect* of parameter i at point $\vec{q}$ is:

$$EE_i(\vec{q}) = \frac{f(\vec{q} + \Delta \vec{e}_i) - f(\vec{q})}{\Delta}. \tag{9.8}$$

After r trajectories, the mean $\mu_i^* = (1/r)\sum |EE_i|$ and standard deviation σ_i of the elementary effects are computed. Parameters with small μ_i^* are unimportant. Parameters with large σ_i have interactions or nonlinear effects.

The Morris diagnostic plot (σ_i vs. μ_i^*) separates parameters into three groups: unimportant (small μ_i^*, small σ_i); important and linear (large μ_i^*, small σ_i); important and nonlinear or interacting (large μ_i^*, large σ_i).

Worked numerical example: Morris screening for a two-parameter model. To see the computation in full, consider the model $f(u_1, u_2) = 3u_1 + u_2 + 2u_1u_2$ on the unit square $[0,1]^2$, with $\Delta = 1/2$ and $r = 3$

trajectories. The parameter u_1 has a larger main effect (coefficient 3 vs. 1) and interacts with u_2 through the cross term.

Trajectory 1. Start at (0.0, 0.0); change order: u_1, u_2.

$$f(0.0, 0.0) = 0, \quad f(0.5, 0.0) = 1.50, \quad f(0.5, 0.5) = 2.50.$$

$$EE_1 = \frac{1.50 - 0}{0.5} = 3.0, \qquad EE_2 = \frac{2.50 - 1.50}{0.5} = 2.0.$$

Trajectory 2. Start at (0.5, 0.5); change order: u_2, u_1.

$$f(0.5, 0.5) = 2.50, \quad f(0.5, 1.0) = 3.50, \quad f(1.0, 1.0) = 6.00.$$

$$EE_2 = \frac{3.50 - 2.50}{0.5} = 2.0, \qquad EE_1 = \frac{6.00 - 3.50}{0.5} = 5.0.$$

Trajectory 3. Start at (0.5, 0.0); change order: u_1, u_2.

$$f(0.5, 0.0) = 1.50, \quad f(1.0, 0.0) = 3.00, \quad f(1.0, 0.5) = 4.50.$$

$$EE_1 = \frac{3.00 - 1.50}{0.5} = 3.0, \qquad EE_2 = \frac{4.50 - 3.00}{0.5} = 3.0.$$

Collecting and summarizing:

	Elementary effects				
Parameter	Traj. 1	Traj. 2	Traj. 3	μ_i^*	σ_i
u_1	3.0	5.0	3.0	3.67	1.15
u_2	2.0	2.0	3.0	2.33	0.58

where $\mu_i^* = (1/r)\sum_j |EE_i^{(j)}|$ and σ_i is the sample standard deviation of the r values. Both parameters have large μ^*, so both are important. The sample σ differ here, but that difference is an artifact of $r = 3$ and should not be read as structure. The elementary effects are exactly $EE_1 = 3 + 2u_2$ and $EE_2 = 1 + 2u_1$, so over the unit square both have population variance $4\,\mathrm{Var}(u) = 1/3$ and hence the same $\sigma = 1/\sqrt{3} \approx 0.577$. The cross term $2u_1u_2$ is symmetric, and it makes both parameters interacting to the same degree. On a Morris diagnostic plot both belong upper-right; with $r = 3$ one of them will land lower simply by chance. In practice, $r = 3$ is too few trajectories for reliable screening (the standard minimum is $r = 10$); see Figure 9.3 for an application to the SIR model with $r = 50$ and $m = 4$ parameters.

Workflow: run Morris screening first; fix unimportant parameters at their nominal values; run Sobol only on the important subset. Fixing all

but k of the m parameters cuts the Sobol cost from $N(m+2)$ to $N(k+2)$ evaluations. For the $m = 30$ parameters that open this section, screened down to five, the cost at $N = 2048$ falls from 65,536 to 14,336 evaluations, a factor of about 4.5 once the $r(m+1) = 310$ Morris runs at $r = 10$ are counted.

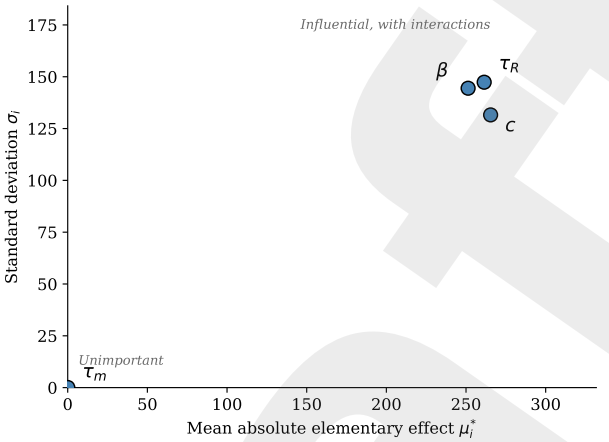


Figure 9.3. *Morris screening diagnostic plot (σ_i vs. μ_i^*) for the SIR model with four parameters: contact rate c, transmission probability β, infectious period τ_R, and mean residence time τ_m. Parameters with small μ_i^* are unimportant (lower left); large μ_i^* and small σ_i indicate a linear effect (lower right); large μ_i^* and large σ_i indicate a nonlinear or interacting effect (upper right). Only the two regions that contain points are labeled on the plot, since naming an empty region invites a search for data that is not there. The three transmission and recovery parameters (c, β, τ_R) are important and exhibit nonlinear behavior at the ranges shown (±50% of nominal). The residence time τ_m has negligible influence on the 90-day epidemic peak and can be fixed at its nominal value before running the Sobol analysis, illustrating the screening workflow: a parameter may govern the model's long-term dynamics yet be irrelevant to the short-horizon question being asked. Computed with $r = 50$ trajectories, $\Delta = 0.5$. The QOI is the peak infectious count in a population of 1000, and each input is scaled to $[0, 1]$ over its range, so μ_i^* and σ_i are in persons per unit of scaled input. These ranges and settings differ from the laboratory's (Part 4 uses $r = 10$ and $\Delta = 0.4$), so the plot is not the laboratory's answer.*

When even reduced Sobol is too expensive: surrogate models. If the model is computationally expensive and Morris screening reduces the parameter set to, say, five influential POIs, running Sobol at $N = 2048$ still requires $N(m+2) = 14{,}336$ model evaluations. For a climate model where each run takes hours, this remains prohibitive.

The practical solution is a *surrogate model* (also called an emulator or meta-model): a cheap statistical approximation to the expensive model, trained on a moderate number of carefully chosen model runs. Concretely, a Gaussian process surrogate fits a probability distribution over functions to the training runs, predicting the model output at any new input along with a confidence interval; a polynomial chaos expansion instead represents

the output as a truncated series in orthogonal polynomials of the uncertain inputs, with coefficients estimated from the training set. Gaussian process emulators and polynomial chaos expansions are the most widely used surrogates. Once trained on roughly 50–200 model evaluations, a surrogate can be evaluated millions of times in seconds, making full Sobol estimation feasible even for computationally expensive simulations.

Surrogate-based global SA does not replace the methods in this chapter; it wraps them. The sampling designs of Section 9.2, the Saltelli construction, and the Jansen estimators all apply to the surrogate exactly as they apply to the original model. The surrogate simply makes the evaluations fast enough to converge. Smith [60] develops Gaussian process, polynomial, and spectral surrogates in Chapters 15–18 of that book, building on the LHS samples and Sobol indices treated in this chapter; readers continuing into surrogate-model construction will build on exactly the kind of training sets generated in this chapter's laboratories.

9.7 ▪ Sobol Indices for Correlated POIs

Standard Sobol decomposition assumes independent POIs. When parameters are correlated, three problems arise. Each index (9.4) still lies in $[0, 1]$, but the first-order indices can sum to more than 1. The standard estimators, applied to correlated inputs sampled jointly, can return estimates outside $[0, 1]$ at any sample size. The ANOVA decomposition (9.2) is no longer unique, because correlation destroys the orthogonality that guarantees uniqueness.

An estimate outside $[0, 1]$ is worth noticing, but it is not by itself evidence of correlation. Monte Carlo error produces the same symptom with perfectly independent inputs whenever a POI has little influence, because the Jansen estimate then scatters about zero and crosses it. The laboratory in §9.12 shows this: its inputs are independent, yet $\hat{S}_1^{\tau_m}$ comes out slightly negative in about half the runs, since τ_m barely moves $I_{\max}$ over a short outbreak. Read a negative or greater-than-one estimate first as a request for more samples, and only after the estimate has stabilized as a sign that the independence assumption deserves scrutiny.

Three approaches address correlated POIs in global SA:

Grouped indices: treat a cluster of correlated parameters as a single group. Compute the group Sobol index (the fraction of variance explained by the entire group jointly). This avoids allocating variance between correlated parameters, which is not well-defined.

Modified estimators: Kucherenko et al. [28] extend the Saltelli algorithm to correlated inputs using copula-based sampling (a copula models the

dependence structure between parameters separately from their marginal distributions). This requires knowledge of the joint distribution of the correlated parameters.

PRCC: reduces the linear effect of all other inputs via rank regression before computing each correlation, making it more robust than simple Pearson correlation (Section F.4) under mild-to-moderate input dependence. It does not, however, fully account for strong nonlinear correlations or capture higher-order dependence structures. Under mild correlation it is often the fastest practical screen.

Practical recommendation: for models with mildly correlated POIs, PRCC is a fast, accessible first screen. When correlation is strong or nonlinear, complement PRCC with grouped Sobol indices or Shapley effects, and report the correlation structure explicitly alongside the SA results.

Table 9.2 summarizes the tradeoffs.

Method	When to use	Key assumption	Implementation
PRCC	Monotonic model; fast screen	Monotonicity	`partialcorr` (MATLAB)
Grouped Sobol	Strong correlation;	Natural grouping	`sobol_jansen.m`
	want variance attribution	known a priori	with grouped inputs
Shapley effects	Strong correlation;	None beyond	`sensitivity` (R)
	no natural grouping	finite variance	or SALib (Python)

Table 9.2. *Comparison of global SA methods for correlated POIs. When POIs are mildly correlated, PRCC is the fastest screen. When correlation is strong and a natural grouping exists (e.g., all transmission parameters), grouped Sobol gives a more honest variance attribution. Shapley effects require no grouping but are more computationally expensive.*

Shapley effects: the principled alternative. For readers seeking a theoretically grounded attribution under correlated inputs, the *Shapley effect* ϕ_j is the unique attribution satisfying four game-theoretic axioms: efficiency (the ϕ_j sum to total output variance), symmetry, dummy (zero importance for irrelevant inputs), and additivity [42]. Owen applied this Shapley value to variance with independent inputs; Song, Nelson, and Staum [62] named it the Shapley effect and analyzed it when inputs are dependent. For independent inputs, the Shapley value of input j reduces to the closed form $\phi_j = \sum_{u \ni j} \sigma_u^2/|u|$: each interaction variance component σ_u^2 is shared equally among all inputs in the interaction set u. Under correlated inputs this closed form no longer holds, and the general game-theoretic definition (averaging marginal contributions over all orderings of the inputs) is required. Normalizing by the total output variance gives the *normalized Shapley effect* $\Phi_j = \phi_j/\mathrm{Var}(Y)$, which lies in $[0, 1]$ and satisfies $\sum_j \Phi_j = 1$. A practical consequence: when inputs are *independent*, the first- and total-order Sobol

indices already computed in this chapter bracket the normalized Shapley effect,

$$S_j \;\le\; \Phi_j \;\le\; S_{T_j},$$

providing useful bounds on the fair attribution without additional cost. Under correlated inputs, the standard Sobol indices themselves become unreliable (as noted earlier in this chapter), so this bracketing should not be relied upon in the correlated case. The Shapley value is computationally more expensive than Sobol indices but is implemented in SALib and the R `sensitivity` package.

9.8 ▪ Beyond Variance: Moment-Independent Sensitivity

Sobol indices decompose output *variance*. For epidemic models with skewed or multimodal output distributions, the distribution of peak infection counts, for instance, is rarely symmetric; variance is a poor summary of uncertainty. A parameter that dominates variance is not necessarily the parameter whose uncertainty most shifts the shape of the epidemic trajectory, and the parameter most critical to a decision-maker who cares about tail events (catastrophic outbreaks) may differ from the one ranked highest by Sobol indices.

Borgonovo (2007) introduced a moment-independent sensitivity measure, the *δ-index* [6], that addresses this directly. Instead of measuring how much fixing Q_i reduces output variance, δ_i measures the *expected shift in the entire output distribution* when Q_i is fixed:

$$\delta_i = \frac{1}{2}\, E_{Q_i}\left[\int \left| f_Y(y) - f_{Y|Q_i}(y) \right| dy\right]. \tag{9.9}$$

Here f_Y is the unconditional output density (Section F.1) and $f_{Y|Q_i}$ is the conditional density given Q_i. The integral $\int |f_Y(y) - f_{Y|Q_i}(y)|\, dy$ is twice the total variation distance between the two densities; the factor $\frac{1}{2}$ outside the expectation normalizes δ_i to lie in $[0, 1]$. The expectation averages over all values Q_i can take. Three properties make δ_i interpretable: it lies in $[0, 1]$; it equals zero when Y is independent of Q_i; and the joint δ of all inputs together equals one.

When do δ_i and Sobol indices disagree? Borgonovo (2007) demonstrates on the Ishigami test function that δ_i and total-order Sobol indices agree on which parameters are *least* important, but can disagree on the most important ones: the parameter that most shifts the output distribution is not always the one that contributes most to output variance. In epidemic models this distinction matters when the decision criterion is not variance but exceedance probability: the probability that $\mathcal{R}_0$ exceeds the critical

threshold of 1, or that peak infections exceed hospital capacity. Lu and Borgonovo [30] compare variance-based indices with a moment-independent index based on the Kuiper distance for a generalized SIR model of COVID-19.

Definition is valid under correlated inputs. Because δ_i requires only the joint density $f_{\mathbf{Q}}$ without assuming independence, it remains well-defined when parameters are correlated [6], a practical advantage over standard Sobol indices in epidemiological models where transmission rate and contact rate are rarely independent.

Computational cost and software. Computing δ_i requires estimating conditional densities, which is more expensive than the matrix construction for Sobol indices. Plischke, Borgonovo and Smith (2013) introduced a substantially more efficient "given-data" estimator. Both the standard and efficient estimators are implemented in SALib (Python) and the `sensitivity` package (R). For a comprehensive graduate-level treatment of moment-independent sensitivity measures, their mathematical properties, and computational algorithms, see Borgonovo [7].

9.9 ▪ Comparison of the Main SA Methods

Table 9.3 consolidates the main SA methods of the book. It omits the σ-normalized index, Shapley effects, the δ-index, eFAST and forward-mode AD. For the choice among local methods, Table 7.3 remains the finer guide; it gives the FSE's condition, $m \leq r$, and its cost, $m+1$ solves. For a broader treatment of global SA methods, see Saltelli et al. [53, 51] and the recent epidemiological applications in Lu and Borgonovo [30].

Table 9.3 settles which *method* to run. A second question survives it: once the indices are in hand, which index answers the question actually being asked? Table 9.4 sets that out. The two failure modes it guards against are not exotic. Quoting S_{T_i} as the variance reduction available from measuring a parameter overstates the gain whenever interactions are present, and reading a variance-based index as a proportional-change prediction attributes to it information it does not carry.

A caution that spans the whole table. These indices rank parameters *of a model*. Feasibility, cost, and whether a lever can be moved at all are not in any of them, and no amount of resampling will put them there. The index identifies what to target; it does not say how, or whether.

Unifying Idea 2 (the adjoint is the transpose; Section 1.5). The adjoint row in Table 9.3 is the payoff for Chapters 5 and 6. The adjoint is the transpose in whatever space you are working in: matrix transpose for linear systems (Chapter 5), reversed ODE for dynamical systems (Chapter 6), and

Method	Chapter	When to use	Assumption	Cost
Local SI (analytic)	2, 4	Differentiable model; small uncertainty	Linearization valid	Cheap
Local SI (FD)	3	Black-box; small uncertainty	Linearization valid	$2m + 1$ evals
Adjoint	5–6	Differentiable; many POIs; few QOIs	Linearization valid	2 solves
AD (reverse)	7	Differentiable code; $m > r$	Linearization valid	2 passes
PRCC	9	Monotone model; moderate uncertainty; corr. POIs OK	Monotone	N evals
Morris	9	Screening; $m \geq 30$	None	$N(m + 1)$ evals
Sobol	9	Nonlinear; large uncertainty; independent POIs	Independent	$N(m + 2)$ evals
Grouped Sobol	9	Correlated POIs; known groups	Known groups	$N(m + 2)$ evals

Table 9.3. *Comparison of the main SA methods across the chapters. The choice of method depends primarily on whether the model is differentiable, whether uncertainty is small or large, and whether* POIs *are independent. No single method is universally best. The adjoint and AD costs are for one* QOI*; r* QOIs *cost $r + 1$ solves or passes.*

reversed chain rule for computation graphs (Chapter 7). Every "2 solves" entry in the adjoint row is the same unifying idea, applied in a different mathematical space.

9.10 ▪ Verifying an SA Result You Did Not Compute

Most of this book teaches you to produce a sensitivity result. This section addresses the complementary task, which you will perform more often: judging one that arrives already finished.

The situation is ordinary. A collaborator reports that the transmission rate dominates. A library returns total-order indices for a model you did

The question	The index	Why not the other one
Which POI should we measure or learn first, to reduce output uncertainty most?	First-order Sobol S_i	Learning Q_i leaves $(1 - S_i)\,\mathrm{Var}(Y)$, so the reduction is S_i. S_{T_i} overstates it whenever interactions are present.
Which POIs can we fix at nominal and stop resolving?	Total-order Sobol S_{T_i}	A small S_{T_i} licenses fixing. A small S_i does not: a purely interacting parameter has $S_i \approx 0$ with S_{T_i} large.
If we change a POI by a given percentage, how far does the QOI move?	Normalized local SI $\mathcal{S}_p^q$ (elasticity), Chapter 2	Variance-based indices carry no proportional-change information; they describe dispersion, not response.
Which POIs can we screen out cheaply before an expensive study?	Morris μ_i^*	Screening only. μ_i^* ranks; it does not decompose variance.
Which POI drives output uncertainty, given how uncertain it actually is?	σ-normalized S_i^σ, or Sobol	The plain local SI scales by the nominal value, not the range, and so ignores how much the POI actually varies (Section 9.1).

Table 9.4. *Which index answers which question. Every entry in the third column is a mistake that appears in the applied literature. The distinction matters most when $S_{T_i} - S_i$ is large; when interactions are weak the rankings often agree, which is exactly why the difference is easy to overlook.*

not write. A paper under your review states a parameter ranking in a single sentence. A computational assistant produces a Sobol decomposition in seconds. Your own analysis from six months ago comes back with the reasoning no longer in memory. In every case the arithmetic is probably right, and that is not the question. The question is whether the quantity computed answers the question being asked, and Table 9.4 exists because the published literature shows that it frequently does not.

Two features of global SA make this failure easy to reach. Computing an index is cheap while choosing one is hard, so nothing in the workflow

forces the choice to be made explicitly. And every method returns a number for every model, whether or not its assumptions hold, so a violated assumption produces a plausible answer rather than an error. A result can be arithmetically correct, confidently reported, and still misleading.

9.10.1 ▪ An audit in six questions

The two tables already assembled in this chapter carry most of what is needed. Table 9.4 maps questions to indices, and Table 9.3 records the assumption each method makes. The following sequence uses them.

1. *What decision does the result inform?* Not what was computed, but what will be done differently depending on the answer. Measure a parameter first, fix one at nominal, or predict how far the QOI moves under a stated change. The literature names the first two factor prioritization and factor fixing; the third asks for a proportional response. They are three different decisions, and Table 9.4 assigns each a different index. Factor mapping, the search for input combinations that drive the output above a threshold, is a separate purpose, and the table has no row for it.

2. *Does the reported index answer that question, or an adjacent one?* The third column of Table 9.4 lists the substitutions that occur in practice. Quoting S_{T_i} as the uncertainty reduction available from measuring a parameter overstates it whenever interactions are present. Reading a variance-based index as a proportional-change prediction attributes to it information it does not carry: it describes dispersion, not response.

3. *What does the method assume?* Read it from the assumption column of Table 9.3. PRCC assumes a monotone response. Standard Sobol estimation assumes independent POIs. Every local method in Chapters 2 through 8 assumes the linearization is valid over the range of interest.

4. *Does this model satisfy that assumption?* This is the step most often skipped, because it requires knowing the model rather than the output. A non-monotone response invalidates a PRCC ranking without producing any symptom in the PRCC value. Correlated inputs invalidate a standard Sobol decomposition while the indices still sum to something near one.

5. *Is the estimate stable enough to support the ranking?* Sobol and Morris estimates carry Monte Carlo error, and a ranking read from point

estimates whose confidence intervals overlap is not a ranking. This is why Section 9.5 gives confidence intervals. A result reported without any indication of sampling error has not established that its order would survive a second run at a different seed.

6. *Is the claim about the model or about the world?* Every index in this book is computed from a model's equations and inherits every simplification in them. A parameter absent from the model returns a sensitivity of exactly zero, which establishes nothing about the system. A result that has traveled from its author to you has usually shed the assumptions that would let you tell the two claims apart.

Unifying Idea 4 (a sensitivity index describes a model, not the world; Section 1.5). Steps 1 and 6 are the same commitment at the two ends of an analysis. The first asks what question the model is being made to answer; the last asks what the answer is a statement about. A finished result that has lost both is a number without a claim attached.

9.10.2 ▪ A worked audit

Return to the example of Section 9.1. A model $Y = 2Q_1 + Q_2$ is reported with the finding that Q_1 is the more important parameter, since its coefficient is twice as large. The arithmetic is correct.

Step 1 asks what will be done with this. Suppose the intent is to decide which parameter to measure more precisely, in order to reduce uncertainty in Y. Step 2 sends us to Table 9.4, whose first row assigns that decision to a first-order Sobol index, or equivalently here to the σ-normalized index, rather than to a comparison of coefficients. Step 3 records what the reported quantity assumes, which is that the two parameters are comparably uncertain. Step 4 checks it, and this is where the reported finding fails: with $\sigma_1 = 1$ and $\sigma_2 = 3$, they are not. Weighting each coefficient by the uncertainty it multiplies gives contributions of 2 and 3, and the ranking reverses. Measuring Q_2 reduces the variance of Y more.

The reported finding is not a computational error. It answers the question of which parameter has the larger coefficient, which was not the question asked. Step 5 does not apply: nothing here was sampled, and the weighted contributions are exact. Step 6 notes that both rankings are properties of this model, which assumes the response is linear and the two parameters independent.

9.10.3 ▪ The referee's version

The same six questions are the ones to ask of a submitted manuscript, and they are worth practicing for that reason alone. A paper reporting a parameter ranking should make its decision context explicit, name the index it used, state the method's assumption, give some evidence the model satisfies it, report sampling error where the estimate is stochastic, and confine its claims to the model it analyzed. Sensitivity analyses in the applied literature routinely omit several of these, which is the observation behind the caption of Table 9.4.

Responsibility here rests with authors rather than reviewers. Publishers have begun to say so explicitly as computational assistance has become routine: SIAM's policy for authors places the burden of ensuring correctness on the author rather than the referee, for mathematical content and for code alike, and holds authors responsible for the accuracy of their figures and citations as well. The audit above is one way to discharge that responsibility on the sensitivity analysis in your own paper before anyone else reads it.

Software for global SA beyond MATLAB. The methods in this chapter are implemented in several open-source libraries. SALib [20] is the dominant Python library for global SA; it implements Morris screening, the Saltelli construction with Jansen estimators, eFAST, and the Borgonovo δ-index under a consistent API and is freely available at `https://salib.readthedocs.io`. The `sensitivity` package for R (Iooss et al.) covers a similar range of methods including Shapley effects. Both are well-documented, actively maintained, and suitable for research-grade applications.

9.11 ▪ Bridge to the Broader UQ Landscape

This book quantifies how model output uncertainty arises from parameter uncertainty, assuming the model structure is correct. A complementary graduate-level treatment of the broader UQ landscape, given in Smith [60], takes up the harder questions that follow: what if the model is wrong (Smith's Chapter 14)? How do we fit models to data (Smith's Chapters 11–12)? How do we construct efficient surrogate models to replace expensive simulations (Smith's Chapters 15–18)?

Smith's Chapters 11–12 develop frequentist and Bayesian parameter estimation, including Markov chain Monte Carlo methods for fitting models to data, the natural sequel to the identifiability analysis introduced in Chapter 3 of this book.

The connections between the two treatments are direct:

Smith's Chapter 8 develops local SA at the graduate level, extending

Chapters 2–7 of this book with functional-analytic rigor.

Smith's Chapter 9 develops global SA and Sobol indices at the graduate level, extending Chapter 9 of this book with measure-theoretic foundations.

Smith's Chapter 7 treats parameter identifiability and influence using the sensitivity matrix, the sensitivity Jacobian of this book (Definition 3.1 in Chapter 3, computed analytically in Chapter 4).

Smith's Chapter 14 develops model discrepancy, and his Chapters 15–18 develop surrogate models including Gaussian process and polynomial chaos methods that reduce the cost of global SA by replacing the expensive model with a cheap approximation. Surrogates are built from space-filling samples of the kind this chapter uses, LHS or scrambled Sobol'.

A complementary modern method between the local indices of Chapters 2–7 and the variance-based global indices of this chapter is *active subspace analysis*, which finds low-dimensional directions in parameter space along which the response varies most. Chapter 10 of Smith [60], on active subspace analysis, develops this technique; it is a natural next step for readers whose models have many parameters but few *effective* ones.

Readers who have completed this book have the foundational vocabulary, the computational intuition, and the mathematical framework to engage productively with the full UQ literature. The path from the sensitivity index in Chapter 2 to the full Bayesian UQ framework in Smith is a single, continuous intellectual development. For an example of SA in a model built to guide mosquito-borne disease control, see Qu, Xue, and Hyman [46]. Their model of Wolbachia spread in mosquitoes uses local sensitivity indices and one-at-a-time sweeps. The analysis finds the fitness costs of infection and the maternal transmission rate to be key factors in whether the infection can be established.

The reader who has reached this point knows how to ask which parameters matter, how to compute that answer efficiently and honestly, and, just as importantly, when not to trust the answer. Every mathematical model is an argument; sensitivity analysis is the discipline that reveals which assumptions the argument depends on.

Good sensitivity analysis does not end the discussion. It tells you where to begin it.

9.12 ▪ Computational Laboratory: Complete Global SA Pipeline for SIR

This laboratory implements the full global SA pipeline for the SIR model and compares all four methods: sigma-normalized, PRCC, Sobol, and Morris.

Setup. SIR model with four POIs $(c, \beta, \tau_R, \tau_m)$ and response $J = I_{\max}$ (peak infectious count). Use these uncertainty ranges, chosen for this laboratory: $c \in [2, 8]$, $\beta \in [0.02, 0.12]$, $\tau_R \in [4, 14]$ days, $\tau_m \in [1000, 100000]$ days.

Part 1: LHS and sigma-normalized indices. Draw $N = 500$ LHS samples using the `lhs_sample` function. Compute σ_J from the samples. Compute the sigma-normalized index for each POI using the local SI from Chapter 2 and the sample standard deviations. Report the ranking.

Part 2: PRCC. Run the SIR model at all 500 LHS samples. Rank-transform all inputs and the output. Before reading the result, note where the box sits relative to the epidemic threshold. About 10.8% of it has $\mathcal{R}_0 \leq 1$, and across that region $I_{\max}$ is pinned at I_0 and does not respond to the POIs at all. PRCC assumes a monotonic input-output relationship. A flat stretch followed by a rise is still monotone, so the flat region does not break that assumption. It does fill the ranks of $I_{\max}$ with ties (§8.2.2 describes the threshold behind it). Report the PRCC on the full sample and again on the super-threshold subsample, and compare. On this box expect the two to agree closely and to rank the POIs in the same order. Compute PRCC using the `partialcorr` function in your implementation's Statistics and Machine Learning Toolbox (or download Marino et al.'s implementation from the supplementary material of [32]). Produce a PRCC tornado plot and report p-values.

Part 3: Sobol estimation. Draw the sample from a scrambled Sobol' sequence (MATLAB: `sobolset`; Python: `scipy.stats.qmc.Sobol`), and take $N = 2048$. Sobol sequences are balanced in blocks of 2^k, so truncating mid-block degrades the low-discrepancy property that makes quasi-random sampling worth using; most libraries will warn you when N is not a power of two. The same reasoning does not apply to a Halton sequence, which is not built in binary blocks, so a Halton design may use any N. Build the A, B, and m matrices $A_B^{(i)}$: $2048 \times 6 = 12{,}288$ total evaluations. Compute $\hat{S}_i$ and $\hat{S}_{T_i}$ using the Jansen estimators.

```
# Sobol variance decomposition (Jansen estimator)
# See the companion notebooks (Python) and reference library (MATLAB)

set_random_seed(42)                              # reproducibility
A, B        = SALTELLI_SAMPLE(N, m, lb, ub) # two N-by-m quasi-random ma

S1, ST, CI = SOBOL_JANSEN(model, A, B,
                          bootstrap_reps=500, alpha=0.05)
# S1[j]     = first-order Sobol index for parameter j
# ST[j]     = total-order Sobol index for parameter j
```

```
# CI[j,:]   = [lower, upper] 95% confidence interval for S1[j]
```

The call's `bootstrap_reps` option resamples rows, so its intervals apply to IID rows only. For this scrambled Sobol' design, take the intervals from independent scrambles instead, as in §9.5. Each scramble repeats all 12,288 evaluations, so 30 scrambles cost 368,640.

Part 4: Morris screening. Implement $r = 10$ Morris trajectories with $\Delta = 0.4$. Compute μ_i^* and σ_i for each parameter. Produce the Morris diagnostic plot (σ_i vs. μ_i^*). Which parameter(s) could be fixed without significant loss of accuracy?

Part 5: Convergence check. Plot $\hat{S}_\beta$ and $\hat{S}_{T_\beta}$ vs. N for $N \in \{128, 256, 512, 1024, 2048, 4096\}$, staying on powers of two for the reason given in Part 3. At what N do the estimates stabilize to within ± 0.05?

Part 6: Full comparison table. Fill in the following table from your results:

Parameter	Local SI	S_i^σ	PRCC	S_i (Sobol)
c				
β				
τ_R				
τ_m				

Where do the four methods agree? Where do they disagree? What accounts for the differences?

9.13 ▪ Exercises

Exercise 9.1 *(Bloom L3).* The local SIs and $\mathcal{R}_0^*$ in this exercise are illustrative. They do not follow from the SEIR model of Exercise 2.5, which uses the same nominal c, β, τ_E and τ_I. For the SEIR model with local SIs $\mathcal{S}_{\tau_E}^{\mathcal{R}_0} = 0.6$, $\mathcal{S}_{\tau_I}^{\mathcal{R}_0} = 1.2$, $\mathcal{S}_\beta^{\mathcal{R}_0} = 1.0$, $\mathcal{S}_c^{\mathcal{R}_0} = 1.0$ and standard deviations $\sigma_{\tau_E} = 2$ days, $\sigma_{\tau_I} = 3$ days, $\sigma_\beta = 0.01$, $\sigma_c = 0.8$, at nominal parameter values $\tau_E^* = 5.5$ days, $\tau_I^* = 7$ days, $\beta^* = 0.06$, $c^* = 5$, $\mathcal{R}_0^* = 2.0$:

(a) Compute the output standard deviation $\sigma_{\mathcal{R}_0}$ using the delta method, at nominal $\mathcal{R}_0^* = 2.0$.

(b) Compute all four sigma-normalized indices.

(c) Produce a tornado plot of the sigma-normalized indices.

(d) Compare the ranking with the local SI ranking. Which parameter changed rank most dramatically? Why?

Exercise 9.2 *(Bloom L3).*

(a) Implement the `lhs_sample` function. For $N = 100$, $m = 3$, $[0,1]^3$, generate an LHS and a random sample of the same size.

(b) Produce scatter plots of all three pairs of parameters for both samples (6 plots total).

(c) Compute the mean and standard deviation of the sample along each dimension. Which method is closer to the true uniform mean and standard deviation?

(d) For one parameter, plot the histogram of LHS vs. random sampling. Which covers the range more evenly?

Exercise 9.3 *(Bloom L3).* For the SIR model, with each of the four parameters ranging over $\pm 50\%$ about its nominal value:

(a) Generate $N = 200$ LHS samples and run the SIR model at each.

(b) Compute PRCC for all four parameters with respect to $I_{\max}$.

(c) Produce a PRCC bar chart sorted by $|\text{PRCC}|$.

(d) Does any parameter have a reliably negative PRCC? Say in plain language what a negative PRCC would mean, and what a PRCC near zero means.

(e) Compare the PRCC ranking with the local SI tornado plot from Chapter 2. Do they agree?

Exercise 9.4 *(Bloom L4).* For $Y = 2Q_1^2 + Q_2$ with $Q_1, Q_2 \sim \mathcal{U}(0,1)$, independent:

(a) Compute $\mathbb{E}[Y]$, $\mathrm{Var}(Y)$, $D_1 = \mathrm{Var}[\mathbb{E}(Y \mid Q_1)]$, $D_2 = \mathrm{Var}[\mathbb{E}(Y \mid Q_2)]$.

(b) Compute S_1, S_2. Do they sum to 1? Explain.

(c) Compute the total indices S_{T_1}, S_{T_2}.

(d) Now add the product $Q_1 Q_2$ to get $Y' = 2Q_1^2 + Q_2 + Q_1 Q_2$ (an interaction term). Without computing: does adding the interaction term change S_1, S_2, or both? In which direction? Then verify by computing the Sobol indices for Y' analytically.

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Exercise 9.5 *(Bloom L4).* Using the Jansen estimator implementation from the laboratory, with each of the four SIR parameters ranging over ±50% about its nominal value:

(a) Compute $\hat{S}_i$ and $\hat{S}_{T_i}$ for all four SIR parameters at $N = 2048$.

(b) For which parameters is $S_{T_i} \gg S_i$? What does this indicate about their role in the model?

(c) Compute $\sum_i S_i$ and $\sum_i S_{T_i}$. What do you expect these to sum to for an additive model? Are your results consistent with this expectation?

(d) If you could measure exactly one parameter very precisely (reducing its uncertainty to zero), which parameter would reduce $\mathrm{Var}(I_{\max})$ the most? Justify using the Sobol indices.

Exercise 9.6 *(Bloom L4).* For the SIR model, implement $r = 15$ Morris trajectories with $\Delta = 1/3$:

(a) Compute μ_i^* and σ_i for each parameter.

(b) Produce the Morris diagnostic plot.

(c) Based on the Morris results, which parameter appears most influential? Which appears least influential?

(d) Run a reduced Sobol analysis using only the three most important parameters identified by Morris (fixing τ_m at its nominal value). Compare the cost and results with the full four-parameter Sobol.

Exercise 9.7 *(Bloom L5).* Choose a published epidemic model from the literature with at least 8 parameters. Design a complete global SA study:

(a) State the model, its parameters, and a justification for the uncertainty ranges you will use (with citations).

(b) State the QOI or QOIs and explain why they are the most relevant for the model's purpose.

(c) Choose between PRCC and Sobol as the primary global SA method. Justify your choice based on the model's properties.

(d) Specify the sample size N and justify it based on convergence.

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(e) Describe how you would present the results to a non-technical audience (e.g., a public health committee or funding agency).

(f) Identify one situation where your global SA results could still be misleading, and describe what additional analysis you would recommend.

Exercise 9.8 *(Bloom L4). Audit exercise.* A collaborator reports total-order Sobol indices for a four-parameter model and writes: "$S_{T_\beta} = 0.62$, so measuring β more precisely would remove about 62% of the uncertainty in the peak infection count. That is the single most valuable measurement we can make." The estimator was implemented and run correctly.

(a) Apply the audit of Section 9.10.1. Identify which decision the collaborator is making and which row of Table 9.4 governs it.

(b) State the correct index for that decision and explain, in terms of the variance decomposition, why the reported one overstates the gain.

(c) Under what condition on the model would the two indices give effectively the same answer? Explain why that condition makes the error easy to overlook in practice.

(d) The final sentence contains a second claim that no sensitivity index of any kind can support. Identify it.

Exercise 9.9 *(Bloom L5). Audit exercise.* Two analyses of the same model are reported. The first uses PRCC on $N = 500$ Latin hypercube samples and ranks the parameters (p_1, p_2, p_3) in that order. The second uses first-order Sobol indices on $N = 2048$ samples and ranks them (p_3, p_1, p_2). Both are computed correctly. The model's response to p_3 is known to rise and then fall over the sampled range.

(a) Using Table 9.3, state the assumption each method makes and determine which is violated here.

(b) Explain how a violated monotonicity assumption can produce a low PRCC for a parameter that in fact drives a large share of the output variance.

(c) Neither report gives confidence intervals. Describe how you would obtain them for each analysis, and state what result would lead you to conclude that the two rankings do not actually disagree.

(d) Suppose the intervals confirm the disagreement. Write the one-sentence finding you would report, taking care that it is a claim about the model and not about the system.

Exercise 9.10 *(Bloom L6).* A published SA study reports Sobol first-order and total indices for a 15-parameter water-quality model. The authors note in the data section that three parameters, phosphorus loading rate, nitrogen loading rate, and water residence time, are positively correlated across the sampled watersheds. The SA section reports standard Sobol indices computed using independent sampling for all 15 parameters, without addressing the correlation.

(a) Identify the specific mathematical problem that correlation creates for standard Sobol indices.

(b) For each of the three correlated parameters, state whether the reported first-order index is likely over-estimated, under-estimated, or unchanged, and explain why.

(c) Propose the minimal methodological correction that would make the analysis valid.

(d) The authors justify their approach by noting that their Sobol indices sum to approximately 1.0. Explain why this is not a valid check for independence.

Notes

The ANOVA decomposition of output variance for functions of independent uniform inputs, the theoretical foundation of Sobol indices, was established by Sobol' [61]. The matrix-based sampling design adopted in this chapter is due to Saltelli [48, 50]. The squared-difference estimators used with it are older: Jansen [26] estimated variance components by half the mean squared difference of two runs that share some inputs. For the total-order index, the benchmark of Puy et al. [45] ranks Jansen's estimator among the most accurate, while the Saltelli estimator of the Primer [53] lags behind.

The Latin hypercube sampling design (Section 9.2.2) was introduced by McKay, Beckman, and Conover [33]; Iman and Conover [25] developed the rank-correlation variant that is standard in practice. Helton and Davis [19] give a comprehensive assessment of LHS performance for SA applications.

Morris screening was introduced in Morris [35]; the sampling strategy that spreads the trajectories for more uniform coverage is due to Campolongo, Cariboni, and Saltelli [11], and the radial design to Campolongo, Saltelli, and Cariboni [12].

The moment-independent δ-index was introduced by Borgonovo [6] and is given a comprehensive mathematical treatment in Borgonovo [7]. The extension of Sobol indices to correlated inputs uses copula-based sampling as described by Kucherenko et al. [28]. The Shapley value was brought to variance-based sensitivity analysis by Owen [42], for independent inputs; Song, Nelson, and Staum [62] named it the Shapley effect and analyzed it under dependent inputs. Lu and Borgonovo [30] apply global SA to a generalized SIR model of COVID-19; with correlated inputs, their first-order and Kuiper-based rankings change while the total indices barely move. Puy et al. [45] benchmark eight estimators of the total-order index over wide ranges of sample size, dimension, and model form.

Answers to the self-checks

9.1. Delta method: $\mathrm{Var}(\mathcal{R}_0) \approx \sum_i (\partial \mathcal{R}_0/\partial p_i)^2 \sigma_i^2$. $\partial \mathcal{R}_0/\partial c = \beta\tau_R = 0.42$; $\partial \mathcal{R}_0/\partial \beta = c\tau_R = 35$; $\partial \mathcal{R}_0/\partial \tau_R = c\beta = 0.3$. $\mathrm{Var}(\mathcal{R}_0) \approx 0.42^2(0.25) + 35^2(0.0001) + 0.3^2(4) = 0.0441 + 0.1225 + 0.36 = 0.527$. $\sigma_{\mathcal{R}_0} \approx 0.726$.

Compute each index directly from $S_i^\sigma = (\partial \mathcal{R}_0/\partial p_i)\sigma_i/\sigma_{\mathcal{R}_0}$. $S_c^\sigma = 0.42{\times}0.5/0.726 \approx 0.29$. $S_\beta^\sigma = 35 \times 0.01/0.726 \approx 0.48$. $S_{\tau_R}^\sigma = 0.3 \times 2/0.726 \approx 0.83$.

Ranking: $\tau_R > \beta > c$. The mean infectious period drives the most uncertainty in $\mathcal{R}_0$, despite all three having equal local SIs. The CV form of Unifying Idea 1 above gives the same numbers in one line. With all three local SIs equal to 1, $S_i^\sigma = \mathrm{CV}_i/\mathrm{CV}_{\mathcal{R}_0}$, where $\mathrm{CV}_c = 0.10$, $\mathrm{CV}_\beta = 0.17$, $\mathrm{CV}_{\tau_R} = 0.29$ and $\mathrm{CV}_{\mathcal{R}_0} = 0.726/2.1 = 0.346$, so the ranking is the ranking of the input CVs.

9.2. (a) Yes, random sampling can place all 4 points in $[0, 0.5]$ (probability $(1/2)^4 = 1/16$). With LHS, this is impossible: each quarter-interval $[0, 0.25)$, $[0.25, 0.5)$, $[0.5, 0.75)$, $[0.75, 1]$ must contain exactly one sample.

(b) One possible LHS: one point in each of $[0, 0.25)$, $[0.25, 0.5)$, $[0.5, 0.75)$, $[0.75, 1]$, with uniform jitter within each. Example: $X = \{0.18, 0.43, 0.62, 0.91\}$, one value per stratum, randomly positioned within each.

9.3. (a) $\mathbb{E}[Y \mid Q_1] = c_1 Q_1 + c_2/2$. $\mathrm{Var}[\mathbb{E}(Y \mid Q_1)] = c_1^2 \mathrm{Var}(Q_1) = c_1^2/12$.

(b) $\mathrm{Var}(Y) = c_1^2 \mathrm{Var}(Q_1) + c_2^2 \mathrm{Var}(Q_2) = (c_1^2 + c_2^2)/12$.

(c) $S_1 = D_1/D = (c_1^2/12)/((c_1^2 + c_2^2)/12) = c_1^2/(c_1^2 + c_2^2)$. Similarly S_2.

(d) $S_1 + S_2 = (c_1^2 + c_2^2)/(c_1^2 + c_2^2) = 1$. For linear additive models, there are no interaction terms and first-order indices sum to 1. Part (c) is Unifying Idea 1 at work: both inputs have $\sigma = 1/\sqrt{12}$, so $S_i^\sigma = c_i/\sqrt{c_1^2 + c_2^2}$ and $S_i = (S_i^\sigma)^2$.

Appendices

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Appendix A

Inner Products, Norms, and Function Spaces

This appendix provides the minimum background in functional analysis needed to read Appendix B and the related material in Smith [60]. Readers who are comfortable with inner products and the L^2 space can proceed directly to Appendix B.

A.1 ▪ Inner Products and Norms

An *inner product* on a vector space V over $\mathbb{R}$ is a function $\langle \cdot, \cdot \rangle : V \times V \to \mathbb{R}$ satisfying:

1. *Symmetry*: $\langle u, v \rangle = \langle v, u \rangle$.
2. *Linearity in the first argument*: $\langle \alpha u + \beta w, v \rangle = \alpha \langle u, v \rangle + \beta \langle w, v \rangle$. With symmetry, this makes the inner product linear in the second argument as well.
3. *Positive definiteness*: $\langle u, u \rangle \geq 0$, with equality only when $u = 0$.

The *induced norm* is $\|u\| = \sqrt{\langle u, u \rangle}$. The Cauchy-Schwarz inequality states $|\langle u, v \rangle| \leq \|u\| \, \|v\|$.

Example A.1 (Euclidean inner product). For $\vec{u}, \vec{v} \in \mathbb{R}^n$:

$$\langle \vec{u}, \vec{v} \rangle = \vec{u}^T \vec{v} = \sum_{i=1}^{n} u_i v_i.$$

This is the dot product. The induced norm is the Euclidean norm $\|\vec{u}\|_2 = \sqrt{\sum u_i^2}$.

Example A.2 (Weighted inner product). For a symmetric positive definite matrix $W \in \mathbb{R}^{n \times n}$: $\langle \vec{u}, \vec{v} \rangle_W = \vec{u}^T W \vec{v}$. The induced norm $\|\vec{u}\|_W = \sqrt{\vec{u}^T W \vec{u}}$ is the W-weighted Euclidean norm. SA computations sometimes use this when parameters have very different scales; the chapters of this book do not need it.

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A.2 ▪ The L^2 Inner Product

For functions $f, g : [a, b] \to \mathbb{R}$ that are square-integrable (i.e., $\int_a^b f(t)^2\,dt < \infty$), the L^2 *inner product* is:

$$\langle f, g\rangle_{L^2} = \int_a^b f(t)\,g(t)\,dt. \tag{A.1}$$

The induced norm is $\|f\|_{L^2} = \sqrt{\int_a^b f(t)^2\,dt}$.

The space $L^2([a, b])$ of all square-integrable functions on $[a, b]$ with inner product (A.1) is a *Hilbert space*: a complete inner product space. Completeness means that every Cauchy sequence converges to a function in $L^2([a, b])$. It matters for proofs in the literature; nothing in this book uses it beyond the inner product itself.

For Sobol decomposition (Chapter 9), the relevant space is $L^2([0, 1]^m)$ with inner product $\langle f, g\rangle = \int_{[0,1]^m} f(\vec{q})g(\vec{q})\,d\vec{q}$.

A.3 ▪ Integration by Parts as Transposition

The key connection between adjoint sensitivity analysis and linear algebra is this: integration by parts is the function-space analog of the matrix transpose.

In $\mathbb{R}^n$, for vectors $\vec{u}$, $\vec{v}$ and a matrix A:

$$\langle A\vec{u}, \vec{v}\rangle = (A\vec{u})^T\vec{v} = \vec{u}^T(A^T\vec{v}) = \langle \vec{u}, A^T\vec{v}\rangle.$$

So A^T is the operator satisfying $\langle Au, v\rangle = \langle u, A^T v\rangle$.

In $L^2([0, T])$, for functions u, v and the differentiation operator $L = d/dt$:

$$\langle Lu, v\rangle_{L^2} = \int_0^T \dot{u}(t)\,v(t)\,dt = \big[u(t)v(t)\big]_0^T - \int_0^T u(t)\,\dot{v}(t)\,dt = \big[uv\big]_0^T + \langle u, L^*v\rangle_{L^2},$$

where $L^* = -d/dt$ (with appropriate boundary conditions) is the *adjoint operator* of $L = d/dt$. The boundary term $[uv]_0^T$ gives rise to the terminal condition $\lambda(T) = +h'(u(T))$ in the adjoint ODE derivation of Chapter 6.

This correspondence makes precise what we stated informally throughout the book: the adjoint ODE is the ODE whose differential operator is the transpose of the forward ODE's operator, exactly as A^T is the transpose of A.

A.4 ▪ Orthogonality and the ANOVA Decomposition

Two elements u, v of an inner product space are *orthogonal* if $\langle u, v\rangle = 0$.

In $L^2([0,1]^m)$, with the inner product integrated in $d\vec{q}$ over the unit cube (the uniform measure), the functions appearing in the Sobol ANOVA decomposition (Chapter 9) are mutually orthogonal. Specifically, for the decomposition $f = f_0 + \sum_i f_i + \sum_{i<j} f_{ij} + \cdots$, each term is orthogonal to every other term: $\langle f_u, f_v \rangle_{L^2} = 0$ when $u \neq v$ (as index sets).

This orthogonality is why the output variance decomposes as a sum. For independent uniform inputs, $\mathrm{Var}(Y) = \int (f - f_0)^2 \, d\vec{q} = \|f - f_0\|^2$. Write $f - f_0$ as the sum of the other terms and square it: each term contributes its own $\|f_u\|^2$, and every cross term $2\langle f_u, f_v \rangle$ is zero. Each $\|f_u\|^2$ is the variance D_u of its term, so $\mathrm{Var}(Y) = \sum_i D_i + \sum_{i<j} D_{ij} + \cdots$. It also explains why the decomposition requires independent parameters: independence of the Q_i (their joint density is the product of their separate densities, Section F.1) makes the integration over $[0,1]^m$ factorize, which is what produces the orthogonality.

A.5 ▪ Exercises

Self-check A.1. Let $f(t) = e^{-t}$ and $g(t) = \cos(\pi t)$ on $[0,1]$. (a) Compute $\langle f, g \rangle_{L^2}$ numerically (use a numerical integration routine). (b) Are f and g orthogonal in $L^2([0,1])$? (c) Compute $\|f\|_{L^2}$ and $\|g\|_{L^2}$, and verify the Cauchy–Schwarz inequality $|\langle f, g \rangle| \leq \|f\| \, \|g\|$.

Self-check A.2. The integration-by-parts identity in Section A.3 establishes that $L^* = -d/dt$ (with zero terminal condition) is the adjoint of $L = d/dt$ (with zero initial condition) in $L^2([0,T])$. The function v below breaks the zero terminal condition on purpose ($v(1) = 1$), so that the boundary term is nonzero and can be seen. (a) Write out $\langle Lu, v \rangle_{L^2}$ explicitly for $u(t) = t^2$ and $v(t) = t$ on $[0,1]$. (b) Compute $\langle u, L^* v \rangle_{L^2}$ for the same functions. (c) Verify that the two results differ by the boundary term $[u(t)v(t)]_0^1$.

Self-check A.3. The ANOVA decomposition in $L^2([0,1]^2)$ for the model $f(Q_1, Q_2) = Q_1 + 2Q_1Q_2$ (with $Q_i \sim \mathcal{U}(0,1)$ independent) is: $f = f_0 + f_1(Q_1) + f_2(Q_2) + f_{12}(Q_1, Q_2)$. Compute each term and verify that all terms are mutually orthogonal in $L^2([0,1]^2)$.

Answers to the self-checks

A.1. (a) $\langle f, g \rangle_{L^2} = \int_0^1 e^{-t} \cos(\pi t) \, dt$. Adaptive quadrature (MATLAB's `integral` or SciPy's `quad`) returns 0.12584. Integrating by parts twice with $a = -1$, $b = \pi$ gives the closed form $(e^{-1} + 1)/(1 + \pi^2) \approx 0.126$ as a check. (b) No; the inner product is nonzero, so f and g are not orthogonal. (c) $\|f\|_{L^2}^2 = \int_0^1 e^{-2t} \, dt = (1 - e^{-2})/2 \approx$

0.432, so $\|f\|_{L^2} \approx 0.658$. $\|g\|_{L^2}^2 = \int_0^1 \cos^2(\pi t)\,dt = 1/2$, so $\|g\|_{L^2} = 1/\sqrt{2} \approx 0.707$. Cauchy–Schwarz: $|0.126| \le 0.658 \times 0.707 \approx 0.465$. ✓

A.2. (a) $Lu = 2t$. $\langle Lu, v\rangle = \int_0^1 2t \cdot t\,dt = \int_0^1 2t^2\,dt = 2/3$. (b) $L^*v = -dv/dt = -1$. $\langle u, L^*v\rangle = \int_0^1 t^2 \cdot (-1)\,dt = -1/3$. (c) Boundary term: $[u(t)v(t)]_0^1 = 1 \cdot 1 - 0 \cdot 0 = 1$. Check: $\langle Lu, v\rangle = \langle u, L^*v\rangle + [uv]_0^1 \Rightarrow 2/3 = -1/3 + 1 = 2/3$. ✓

A.3. $f_0 = \mathbb{E}[f] = 1/2 + 2(1/2)(1/2) = 1$. $f_1(Q_1) = \mathbb{E}[f|Q_1] - f_0 = Q_1 + Q_1 - 1 = 2(Q_1 - 1/2)$. $f_2(Q_2) = \mathbb{E}[f|Q_2] - f_0 = 1/2 + Q_2 - 1 = Q_2 - 1/2$. $f_{12} = f - f_0 - f_1 - f_2 = 2Q_1Q_2 - Q_1 - Q_2 + 1/2 = 2(Q_1 - 1/2)(Q_2 - 1/2)$. Every term but f_0 integrates to zero in each of its own variables, since $\int_0^1 (Q - 1/2)\,dQ = 0$, and that gives every orthogonality: $\langle f_0, f_u\rangle = f_0 \int f_u = 0$ for each u; $\langle f_1, f_2\rangle = 2\int_0^1 (Q_1 - 1/2)\,dQ_1 \int_0^1 (Q_2 - 1/2)\,dQ_2 = 0$; $\langle f_1, f_{12}\rangle = 4\int_0^1 (Q_1 - 1/2)^2\,dQ_1 \int_0^1 (Q_2 - 1/2)\,dQ_2 = 4 \cdot \frac{1}{12} \cdot 0 = 0$; $\langle f_2, f_{12}\rangle = 2\int_0^1 (Q_1 - 1/2)\,dQ_1 \int_0^1 (Q_2 - 1/2)^2\,dQ_2 = 0$. ✓

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Appendix B
The Adjoint Formalism

This appendix develops the abstract operator framework that unifies the matrix adjoint (Chapter 5) and the ODE adjoint (Chapter 6). It is intended for students who want a rigorous foundation, not just the computational recipe.

B.1 ▪ Linear Operators and Their Adjoints

Let H be a Hilbert space with inner product $\langle \cdot, \cdot \rangle$. A *linear operator* $L : H \to H$ satisfies $L(\alpha u + \beta v) = \alpha Lu + \beta Lv$.

The *adjoint operator* $L^\dagger$ is defined by:

$$\langle Lu, v \rangle = \langle u, L^\dagger v \rangle \qquad \text{for all } u, v \in H. \tag{B.1}$$

The adjoint $L^\dagger$ is the unique operator satisfying (B.1) (when it exists).

Example B.1 (Matrix adjoint). In $\mathbb{R}^n$ with the Euclidean inner product, $L = A$ (left multiplication). Then $\langle Au, v \rangle = (Au)^T v = u^T (A^T v) = \langle u, A^T v \rangle$, so $L^\dagger = A^T$. This is the matrix transpose: Chapter 5's adjoint equation $A^T \vec{v} = \vec{c}$ is $L^\dagger \vec{v} = \vec{c}$.

Example B.2 (Differential operator adjoint). In $L^2([0, T])$, let $Lu = du/dt$ with $u(0) = 0$ (zero initial condition). By integration by parts (Appendix A.3):

$$\langle Lu, v \rangle = \int_0^T \dot{u} v \, dt = [uv]_0^T - \int_0^T u \dot{v} \, dt = u(T)v(T) - \int_0^T u \dot{v} \, dt.$$

For the adjoint to satisfy $\langle Lu, v \rangle = \langle u, L^\dagger v \rangle$, we need $L^\dagger v = -\dot{v}$ with $v(T) = 0$ (zero terminal condition). So $L^\dagger = -d/dt$ with the boundary condition shifted from $t = 0$ to $t = T$. This is exactly the structure of the adjoint ODE in Chapter 6.

B.2 ▪ The Lagrange Identity

For the ODE operator $L = d/dt - F_u$ (the linearization of $\dot{u} = F(u;p)$), the Lagrange identity states:

$$\frac{d}{dt}\big[\vec{\lambda}^T \vec{w}\big] = \vec{\lambda}^T (L\vec{w}) - (L^\dagger \vec{\lambda})^T \vec{w}, \tag{B.2}$$

where $\vec{w} = \partial\vec{u}/\partial p$ satisfies the FSE $L\vec{w} = F_p$ and $\vec{\lambda}$ satisfies the adjoint ODE $L^\dagger\vec{\lambda} = g_u$.

Integrating (B.2) over $[0,T]$, with $L\vec{w} = F_p$ and $L^\dagger\vec{\lambda} = g_u$, gives

$$\vec{\lambda}(T)^T \vec{w}(T) - \vec{\lambda}(0)^T \vec{w}(0) = \int_0^T \vec{\lambda}^T F_p \, dt - \int_0^T g_u \, \vec{w} \, dt.$$

For $J = \int_0^T g(\vec{u})\,dt + h(\vec{u}(T))$ the chain rule gives $dJ/dp = \int_0^T g_u \, \vec{w}\, dt + \nabla h \, \vec{w}(T)$. The terminal condition $\vec{\lambda}(T) = \nabla h^T$ and the initial value $\vec{w}(0) = \partial\vec{u}_0/\partial p$ turn the display into

$$\frac{dJ}{dp} = \int_0^T \vec{\lambda}^T F_p \, dt + \vec{\lambda}(0)^T \frac{\partial \vec{u}_0}{\partial p},$$

the sensitivity formula (6.18) of Chapter 6, whose scalar form is (6.14). This is the rigorous version of the "integration by parts" step that was announced but not fully proved in the main text.

B.3 ▪ Solvability Conditions

For the equation $Lu = f$ to have a solution when L is not invertible, f must be orthogonal to the kernel of $L^\dagger$. Both statements below hold for the matrix operators of Chapter 5 and for the ODE operators of Chapter 6 with their boundary conditions, which is the only setting we use. For general operators on spaces of functions they need a further technical condition (a *closed range*), which these operators meet; without it the alternative can fail.

SC1 (Fredholm Alternative). $Lu = f$ has a solution if and only if $\langle f, v\rangle = 0$ for every v satisfying $L^\dagger v = 0$.

SC2 (Uniqueness). If $L^\dagger v = 0$ has only the trivial solution $v = 0$, then $Lu = f$ has a unique solution for every f.[5]

[5]For the square, closed-range operators used here, $\ker L^\dagger = \{0\}$ forces $\ker L = \{0\}$, so existence and uniqueness coincide; for a non-square operator uniqueness would require $\ker L = \{0\}$ as a separate hypothesis.

In the SA context, SC2 applies in two ways. The time-dependent FSE of Section 4.3 has $L = d/dt - J_F$ with an initial condition. Then $\ker L^\dagger = \{0\}$ for every Jacobian, so the FSE is a linear initial-value problem with a unique solution, near a bifurcation or not (Self-check B.3 works the scalar case). The equilibrium FSE of Theorem 4.1 has $L = D_{\vec{u}}[\vec{f}]$. There SC2 is the assumption that this Jacobian is nonsingular: the FSE then has a unique solution, and the adjoint equation has a unique adjoint variable. That assumption fails at a fold or transcritical bifurcation, where an eigenvalue reaches zero, not near it. Near a fold the Jacobian approaches singularity while $\partial\vec{f}/\partial p$ does not, so the raw derivative grows without bound. Near a transcritical threshold $\partial\vec{f}/\partial p$ vanishes as the Jacobian becomes singular, the raw derivative stays finite, and the normalized index diverges because the equilibrium goes to zero. Section 8.4.1 sorts these cases.

B.4 ▪ Connection to Chapters 5 and 6

The following table makes the connection between Chapters 5 and 6 explicit in the language of this appendix:

Concept	Chapter 5 (matrices)	Chapter 6 (ODEs)
Space H	$\mathbb{R}^n$	$L^2([0,T])^n$
Operator L	A (matrix mult.)	$d/dt - J_F$
Adjoint $L^\dagger$	A^T	$-d/dt - J_F^T$
Adjoint equation	$A^T\vec{v} = \vec{c}$	$-\dot{\vec{\lambda}} = J_F^T\vec{\lambda} + g_u$
Boundary cond.	none (algebraic)	$\vec{\lambda}(T) = +\nabla h^T$
Sensitivity formula	$\vec{v}^T\vec{r}_j$	$+\int_0^T \vec{\lambda}^T F_{p_j}\, dt$
Key operation	matrix transpose	integration by parts

The formal apparatus of Hilbert spaces and adjoint operators shows that Chapters 5 and 6 are not two separate methods; they are the same method applied to two different Hilbert spaces.

B.5 ▪ Exercises

Self-check B.1. Let $A = \begin{pmatrix} 3 & 1 \\ 0 & 2 \end{pmatrix}$ and response $\vec{c} = (1,0)^T$. (a) Write the adjoint equation $A^T\vec{v} = \vec{c}$ and solve for $\vec{v}$. (b) For the right-hand side $\vec{b} = (b_1, b_2)^T$ and solution $\vec{u} = A^{-1}\vec{b}$, compute $J = \vec{c}^T\vec{u}$ directly. (c) Verify

using the adjoint that $\partial J/\partial b_i = v_i$, consistent with the sensitivity formula $dJ/d\vec{b} = \vec{v}$.

Self-check B.2. The Lagrange identity (B.2) for the operator $L = d/dt - F_u$ (the ODE linearization) states: $\frac{d}{dt}[\vec{\lambda}^T\vec{w}] = \vec{\lambda}^T(L\vec{w}) - (L^\dagger\vec{\lambda})^T\vec{w}$. (a) For the scalar case $L = d/dt - a(t)$ (constant a), write out both sides of the Lagrange identity explicitly and verify it holds. (b) If w satisfies $Lw = r(t)$ (the FSE with forcing r) and λ satisfies $L^\dagger\lambda = 0$, integrate the Lagrange identity over $[0, T]$ to obtain a formula for $\lambda(T)w(T) - \lambda(0)w(0)$ in terms of $\int_0^T \lambda r\, dt$.

Self-check B.3. The Fredholm alternative (Section B.3) states that $Lu = f$ has a solution if and only if $f \perp \ker(L^\dagger)$. For $L = d/dt - a$ on $[0, T]$ with $u(0) = 0$ (zero initial condition), find $\ker(L^\dagger)$ and state the solvability condition for $Lu = f$ in terms of an integral of f, or explain why none is needed.

Answers to the self-checks

B.1. (a) $A^T = \begin{pmatrix} 3 & 0 \\ 1 & 2 \end{pmatrix}$. Solving $A^T\vec{v} = (1, 0)^T$: $3v_1 = 1$, $v_1 + 2v_2 = 0$, giving $\vec{v} = (1/3, -1/6)^T$. (b) $A^{-1} = \frac{1}{6}\begin{pmatrix} 2 & -1 \\ 0 & 3 \end{pmatrix}$. $\vec{u} = A^{-1}\vec{b} = \frac{1}{6}(2b_1 - b_2,\, 3b_2)^T$. $J = u_1 = (2b_1 - b_2)/6$. (c) $\partial J/\partial b_1 = 2/6 = 1/3 = v_1$. ✓ $\partial J/\partial b_2 = -1/6 = v_2$. ✓ The adjoint gives all sensitivities with one solve.

B.2. (a) $L = d/dt - a$, so $L^\dagger = -d/dt - a$ (from integration by parts). LHS: $d(\lambda w)/dt = \dot\lambda w + \lambda\dot w$. RHS: $\lambda(Lw) - (L^\dagger\lambda)w = \lambda(\dot w - aw) - (-\dot\lambda - a\lambda)w = \lambda\dot w - a\lambda w + \dot\lambda w + a\lambda w = \dot\lambda w + \lambda\dot w$. LHS = RHS. ✓ (b) Integrating: $[\lambda w]_0^T = \int_0^T \lambda^T(Lw)\, dt - \int_0^T (L^\dagger\lambda)^T w\, dt = \int_0^T \lambda r\, dt - 0$. So $\lambda(T)w(T) - \lambda(0)w(0) = \int_0^T \lambda(t)r(t)\, dt$.

B.3. $L^\dagger = -d/dt - a$ with terminal condition $v(T) = 0$. $L^\dagger v = 0$ means $-\dot v = av$, so $v(t) = Ce^{-a(t-T)}$. Applying $v(T) = 0$: $Ce^0 = 0$, so $C = 0$ and $\ker(L^\dagger) = \{0\}$. Since the kernel is trivial, the Fredholm alternative says $Lu = f$ has a unique solution for every f (with $u(0) = 0$), namely $u(t) = \int_0^t e^{a(t-s)} f(s)\, ds$, and no integral condition on f is needed. This holds for every a, $a = 0$ included: a zero Jacobian, the scalar analogue of a bifurcation, does not make the kernel nontrivial. L has no eigenvalues at all, since $\dot u - au = \mu u$ with $u(0) = 0$ forces $u \equiv 0$ for every μ. The time-dependent FSE therefore always has a unique solution; why SA can still fail near a bifurcation is the subject of §8.4.1.

Appendix C

Eigenvalue Sensitivity and Singular Value Decomposition

This appendix covers two topics moved from Chapter 5 to keep the main text focused: sensitivity of eigenvalues and eigenvectors (important for stability analysis), and the singular value decomposition (important for understanding ill-conditioning).

C.1 ▪ Sensitivity of Eigenvalues

Let $A(\varepsilon) = A_0 + \varepsilon B$ be a matrix depending on a scalar parameter ε. The eigenvalue $\lambda_k(\varepsilon)$ and eigenvector $\vec{x}_k(\varepsilon)$ satisfy $A(\varepsilon)\vec{x}_k = \lambda_k \vec{x}_k$.

C.1.1 ▪ Symmetric Matrices

For symmetric A_0 with distinct eigenvalues, differentiating $A\vec{x}_k = \lambda_k \vec{x}_k$ at $\varepsilon = 0$ gives $B\vec{x}_k + A_0\vec{x}_k' = \lambda_k'\vec{x}_k + \lambda_k\vec{x}_k'$, where a prime denotes $d/d\varepsilon$. Multiply on the left by $\vec{x}_k^T$. Since A_0 is symmetric, $\vec{x}_k^T A_0 = \lambda_k \vec{x}_k^T$, and the two $\vec{x}_k'$ terms cancel. With the eigenvector normalized, $\vec{x}_k^T\vec{x}_k = 1$, what remains is

$$\left.\frac{d\lambda_k}{d\varepsilon}\right|_{\varepsilon=0} = \vec{x}_k^T B \vec{x}_k. \tag{C.1}$$

The eigenvalue sensitivity is a quadratic form in the perturbation matrix and the normalized eigenvector.

C.1.2 ▪ Non-Symmetric Matrices

For non-symmetric A_0, both left eigenvectors $\vec{y}_k$ (satisfying $\vec{y}_k^T A = \lambda_k \vec{y}_k^T$) and right eigenvectors $\vec{x}_k$ are needed:

$$\left.\frac{d\lambda_k}{d\varepsilon}\right|_{\varepsilon=0} = \frac{\vec{y}_k^T B \vec{x}_k}{\vec{y}_k^T \vec{x}_k}. \tag{C.2}$$

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The formula is well-defined when λ_k is a simple (distinct) eigenvalue, because in that case $\vec{y}_k^T \vec{x}_k \neq 0$.

Proposition C.1. *If A_0 has distinct eigenvalues $\lambda_k \neq \lambda_j$, then the corresponding left and right eigenvectors satisfy $\vec{y}_k^T \vec{x}_j = 0$ for $k \neq j$ (*biorthogonality*) and $\vec{y}_k^T \vec{x}_k \neq 0$.*

Proof. Let $A\vec{x}_j = \lambda_j \vec{x}_j$ and $\vec{y}_k^T A = \lambda_k \vec{y}_k^T$. Multiply the first equation on the left by $\vec{y}_k^T$ and the second on the right by $\vec{x}_j$:

$$\vec{y}_k^T A \vec{x}_j = \lambda_j (\vec{y}_k^T \vec{x}_j) \qquad \text{and} \qquad \vec{y}_k^T A \vec{x}_j = \lambda_k (\vec{y}_k^T \vec{x}_j).$$

Subtracting: $(\lambda_k - \lambda_j)(\vec{y}_k^T \vec{x}_j) = 0$. Since $\lambda_k \neq \lambda_j$, we conclude $\vec{y}_k^T \vec{x}_j = 0$.

To see that $\vec{y}_k^T \vec{x}_k \neq 0$: if it were zero, then $\vec{x}_k$ would be orthogonal to every $\vec{y}_j$ (via biorthogonality for $j \neq k$ and the assumed zero for $j = k$). But the left eigenvectors $\{\vec{y}_j\}$ span $\mathbb{R}^n$ when eigenvalues are distinct [23, Theorem 1.4.5], so no nonzero vector can be orthogonal to all of them. This contradiction shows $\vec{y}_k^T \vec{x}_k \neq 0$. □

For a real matrix with complex eigenvalues the same argument applies with the conjugate transpose $\vec{y}_k^*$ in place of $\vec{y}_k^T$ and $\mathbb{C}^n$ in place of $\mathbb{R}^n$.

Repetition alone does not break the formula; defectiveness does. If λ_k is repeated but *semisimple*, its algebraic and geometric multiplicities agree, bases can still be chosen with $\vec{y}_k^T \vec{x}_k \neq 0$, and what fails is uniqueness: the individual eigenvectors are no longer determined, so the eigenvalues are in general only directionally differentiable: at $\varepsilon = 0$ the one-sided derivatives exist but can differ (in the example below the larger eigenvalue is $|\varepsilon|$, with slopes $+1$ and -1). In this case formula (C.2) must be read with care. Evaluated on an arbitrary basis of the eigenspace it need not return a branch derivative at all: for $A(0) = \mathbf{0}$ with $A'(0) = \left[\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right]$ the two branch derivatives are ± 1, yet the quotient evaluated at $\vec{x} = \vec{e}_1$ returns 0. The branch derivatives are recovered only by restricting the perturbation to the eigenspace and taking the eigenvalues of that restriction; the single-vector quotient applies unambiguously only when λ_k is simple. If λ_k is *defective*, its geometric multiplicity falls below its algebraic multiplicity, $\vec{y}_k^T \vec{x}_k = 0$, and formula (C.2) fails outright: for a Jordan block of size m (an $m \times m$ matrix with λ_k on the diagonal, ones just above it and zeros elsewhere) the eigenvalues respond to a perturbation of size ϵ like $\epsilon^{1/m}$ rather than linearly. In applications it is defectiveness, not repetition, that marks the breakdown.

When $\vec{y}_k^T \vec{x}_k$ is small but nonzero (for unit vectors), the eigenvalue is highly sensitive to perturbations. This occurs near a defective eigenvalue of a non-normal matrix, where two eigenvalues are also close. Closeness alone does not produce it: a symmetric matrix keeps $\vec{y}_k^T \vec{x}_k = 1$ however close its eigenvalues are.

For the SIR Jacobian at the endemic equilibrium near $\mathcal{R}_0 = 1$, one eigenvalue stays at $-\nu_m$ and the other two form a complex pair over the range of Chapter 8's laboratory. As $\mathcal{R}_0 \to 1$ the pair shrinks toward zero and its left and right eigenvectors turn toward orthogonal; the pair coalesces, defectively, a little above $\mathcal{R}_0 = 1$. Both effects shrink the smallest singular value (Section C.2), and together they produce the condition-number spike observed in Chapter 8. The spike is a property of J_F; why the indices themselves diverge at the threshold is a separate matter (§8.4.1).

C.2 ▪ Singular Value Decomposition

Every matrix $A \in \mathbb{R}^{m\times n}$ has the *singular value decomposition* (SVD):

$$A = U\Sigma V^T, \tag{C.3}$$

where $U \in \mathbb{R}^{m\times m}$ and $V \in \mathbb{R}^{n\times n}$ are orthogonal and $\Sigma \in \mathbb{R}^{m\times n}$ is diagonal with nonnegative entries $\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_r > 0 = \sigma_{r+1} = \cdots$ (the singular values; r is the rank of A). To compute them, note that $A^TA = V\Sigma^T\Sigma V^T$: the σ_i^2 are the eigenvalues of A^TA, the columns $\vec{v}_i$ of V are its eigenvectors, and $A\vec{v}_i = \sigma_i\vec{u}_i$. For a symmetric matrix the σ_i are the absolute values of its eigenvalues; for a non-symmetric matrix the eigenvalues do not give them. MATLAB's `svd` returns the σ_i, and `cond` returns κ_2.

C.2.1 ▪ SVD and the Condition Number

For a square matrix of full rank, $r = n$ and the condition number from Chapter 8 is $\kappa_2(A) = \sigma_1/\sigma_n$. If A is rank deficient then $\sigma_n = 0$ and $\kappa_2(A) = \infty$; the finite ratio σ_1/σ_r over the nonzero singular values conditions the pseudoinverse problem on the range of A, not A itself. The SVD makes clear why: the smallest singular value σ_n measures how close A is to being rank-deficient. When $\sigma_n \approx 0$, A is nearly singular and tiny perturbations to the right-hand side $\vec{b}$ can produce large changes in the solution $\vec{u} = A^{-1}\vec{b}$.

C.2.2 ▪ SVD and SA

The sensitivity Jacobian $\mathbf{S}$ from Chapter 3 can be analyzed using its SVD. For $\mathbf{S}$ shaped as (QOIs) $\times$ (POIs), the right singular vectors of $\mathbf{S}$ identify the combinations of parameters (POIs) that most strongly affect the model output, and the left singular vectors identify the combinations of outputs (QOIs) most strongly driven by those parameter combinations.

Readers who know Principal Component Analysis (PCA) from statistics will recognize a connection: the right singular vectors of the design matrix X in a regression are the principal components. The analogous object in

SA is the sensitivity Jacobian $\mathbf{S}$ above, not the sampled parameter matrix. PCA of an LHS input sample (Chapter 9) describes the sampling design, which is close to isotropic by construction, and says nothing about which parameter directions drive the output.

C.2.3 ▪ SVD and the Pseudoinverse

When A is not square or is rank-deficient, the Moore-Penrose pseudoinverse $A^+ = V\Sigma^+U^T$ provides the minimum-norm least-squares solution: $\vec{u} = A^+\vec{b}$. The pseudoinverse appears in parameter estimation from sensitivity data and in regularized inversion methods.

C.3 ▪ Exercises

Self-check C.1. The sensitivity Jacobian $\mathbf{S}$ for the SIR model at $T = 60$ days (from the Chapter 3 computational laboratory) is a 3×4 matrix. (a) Explain what the SVD of $\mathbf{S}$ reveals about the SA problem that the individual sensitivity indices do not. (b) If the largest singular value of $\mathbf{S}$ is $\sigma_1 = 2.3$ and the smallest *nonzero* singular value is $\sigma_3 = 0.02$ (recall a 3×4 matrix has at most three nonzero singular values), what is the condition number $\kappa_2(\mathbf{S})$? Interpret this number in plain language for the SA context. (c) The right singular vector $\vec{v}_1$ corresponding to σ_1 identifies the most "influential" linear combination of POIs. If $\vec{v}_1 \approx (0.7, 0.7, 0.1, 0.0)^T$ for POIs $(c, \beta, \tau_R, \nu_m)$, what does this tell you about the SA of the SIR model?

Self-check C.2. For a symmetric 2×2 matrix $A = \begin{pmatrix} a & b \\ b & d \end{pmatrix}$ with $a > d > 0$ and b small, the eigenvalue sensitivity formula (C.1) gives $d\lambda_1/d\varepsilon = \vec{x}_1^T B \vec{x}_1$ where B is the perturbation matrix. (a) Find the eigenvectors of A for $b = 0$. (b) For the perturbation $B = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ (an off-diagonal perturbation), compute $d\lambda_1/d\varepsilon$ at $b = 0$. (c) Near the threshold $\mathcal{R}_0 = 1$ of the SIR model, eigenvalues of the (non-symmetric) Jacobian come close together. Does closeness alone make formula (C.2) unreliable? If not, what does?

Answers to the self-checks

C.1. (a) The SVD reveals which *combinations* of parameters drive the largest output changes. Individual SIs only measure one-at-a-time effects; the SVD captures the directions in parameter space that most amplify perturbations to the model output. (b) For a full-rank rectangular matrix, κ_2 is the ratio of the largest to the smallest nonzero singular value, the σ_1/σ_r of Section C.2; MATLAB's `cond` returns the

same ratio. Here $\kappa_2 = 2.3/0.02 = 115$. This means that a perturbation in the direction $\vec{v}_3$ (least influential nonzero parameter combination) has only 1/115 the output effect of a perturbation in direction $\vec{v}_1$. The SA result is dominated by a small subspace of parameter space. A 3×4 matrix also has a fourth right singular vector $\vec{v}_4$ with no singular value: a direction of parameter change that the three outputs do not register. For the normalized Jacobian of Definition 3.1 the c and β columns are equal, because the model uses only $c\beta$, so $\vec{v}_4 = (1, -1, 0, 0)^T/\sqrt{2}$: raise c and lower β in proportion. (c) The most influential combination weights c and β almost equally ($0.7 \approx 0.7$) with little contribution from τ_R and essentially none from ν_m. This confirms the structural correlation identified in Chapter 3: c and β appear only as the product $c\beta$ in the SIR model, so they always have equal influence on any output.

C.2. (a) For $b = 0$: eigenvectors are $\vec{e}_1 = (1, 0)^T$ ($\lambda_1 = a$) and $\vec{e}_2 = (0, 1)^T$ ($\lambda_2 = d$). (b) $d\lambda_1/d\varepsilon = \vec{x}_1^T B \vec{x}_1 = (1, 0) \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} (1, 0)^T = 0$. The leading eigenvalue has zero first-order sensitivity to this off-diagonal perturbation (by symmetry; it appears at second order). (c) Closeness alone does not. For a symmetric matrix $\vec{y}_k = \vec{x}_k$, so $\vec{y}_k^T \vec{x}_k = 1$ however close the eigenvalues are; $\mathrm{diag}(1, 1 + \delta)$ is an example. Formula (C.2) fails as an eigenvalue becomes defective, where $\vec{y}_k^T \vec{x}_k \to 0$ for unit vectors (Section C.1). That needs a non-normal matrix: $\left[\begin{smallmatrix} 0 & 1 \\ \delta & 0 \end{smallmatrix}\right]$ has eigenvalues $\pm\sqrt{\delta}$ with $|\vec{y}_k^T \vec{x}_k| = 2\sqrt{\delta}/(1 + \delta)$, which is 0.063 at $\delta = 10^{-3}$. The SIR Jacobian is non-symmetric, and its complex pair coalesces defectively a little above $\mathcal{R}_0 = 1$; the formula fails there. That point is not the bifurcation. At $\mathcal{R}_0 = 1$ a single simple eigenvalue reaches zero, and the eigenvalue $-\nu_m$ becomes double without becoming defective. The divergence of the indices at the threshold has another cause (§8.4.1).

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Appendix D
Periodic Solutions and Floquet Theory

For models with periodic forcing or limit-cycle behavior, SA of the periodic orbit requires tools beyond those developed in Chapter 4. This appendix introduces Floquet theory and the sensitivity of periodic solutions, based on the 2009 monograph [3].

D.1 ▪ Floquet Theory

Consider the linear, T-periodic ODE $\dot{\vec{u}} = A(t)\vec{u}$ where $A(t+T) = A(t)$. The *fundamental matrix solution* $\Phi(t)$ satisfies $\dot{\Phi} = A(t)\Phi$, $\Phi(0) = I$.

The *monodromy matrix* is $M = \Phi(T)$: the fundamental matrix evaluated at the end of one period. Its eigenvalues $\mu_1, \ldots, \mu_n$ are the *Floquet multipliers*.

Floquet's Theorem. The fundamental matrix has the form $\Phi(t) = P(t)e^{Bt}$ where $P(t+T) = P(t)$ is periodic and B is a constant matrix satisfying $e^{BT} = M$. Because $A(t)$ repeats every period, $\Phi(t+T) = \Phi(t)M$, so after k periods a disturbance $\vec{u}(0)$ has become $M^k\vec{u}(0)$. The theorem packages the same fact: $P(t)$ repeats, and e^{Bt} carries all the growth or decay from one period to the next. The stability statements below follow from M^k.

The Floquet multipliers determine stability, and the statement takes two forms that are easily conflated.

For the linear T-periodic system above, the zero solution is asymptotically stable when every $|\mu_k| < 1$, and unstable when some $|\mu_k| > 1$. Nothing forces any multiplier to equal one.

For a limit cycle of an *autonomous* system $\dot{\vec{u}} = \vec{F}(\vec{u})$, the variational equation about the orbit always carries the multiplier $\mu_1 = 1$, with eigenvector $\vec{F}(\vec{u}^*(0))$ tangent to the orbit: sliding the orbit along itself maps it to itself. That multiplier reflects time translation and carries no stability in-

formation. The orbit is orbitally asymptotically stable when the remaining $n-1$ multipliers satisfy $|\mu_k| < 1$, and unstable when any of them exceeds one in modulus.

D.2 ▪ Sensitivity of the Period

For a nonlinear system $\dot{\vec{u}} = \vec{F}(\vec{u};\vec{p})$ with a periodic orbit $\vec{u}^*(t;\vec{p})$ of period $T(\vec{p})$, the period itself is a QOI. Its sensitivity is the Chapter 6 adjoint construction carried out on a closed orbit instead of a fixed interval.

Because the orbit closes on itself, there is no terminal condition to impose. The adjoint is instead the T-periodic solution $\vec{z}(t)$ of

$$\dot{\vec{z}} = -J_F^T(\vec{u}^*(t))\,\vec{z}, \qquad \vec{z}(t+T) = \vec{z}(t), \tag{D.1}$$

made unique by the phase normalization $\vec{z}(t)^T\vec{F}(\vec{u}^*(t);\vec{p}) = 1$. This product is constant along the orbit for any solution of (D.1), so fixing it at one instant fixes it at every instant. The period sensitivity is then

$$\frac{dT}{dp_j} = -\int_0^T \vec{z}(t)^T \frac{\partial \vec{F}}{\partial p_j}\big(\vec{u}^*(t);\vec{p}\big)\,dt. \tag{D.2}$$

The integration is essential. Perturbing p_j changes the vector field at every point of the orbit, and the period responds to the phase shift accumulated over the whole circuit. The sign is the expected one: where the perturbation pushes the state along the direction of travel, the orbit closes sooner and T decreases. Periodicity also forces $M^T\vec{z}(0) = \vec{z}(0)$. For any solution $\vec{w}$ of the variational equation $\dot{\vec{w}} = J_F\vec{w}$, the product $\vec{z}^T\vec{w}$ is constant, because its derivative is $\dot{\vec{z}}^T\vec{w} + \vec{z}^T J_F\vec{w} = 0$. Taking $\vec{w}(t) = \Phi(t)\vec{w}(0)$, with Φ the fundamental matrix of the variational equation and $M = \Phi(T)$, gives $\vec{z}(T)^T M = \vec{z}(0)^T$, and $\vec{z}(T) = \vec{z}(0)$ turns this into $M^T\vec{z}(0) = \vec{z}(0)$. So $\vec{z}(0)$ is the left eigenvector of the monodromy matrix belonging to the unit multiplier, scaled by the normalization. That eigenvector fixes the direction in which the orbit responds; only the integral supplies the magnitude.

Unifying Idea 2 (the adjoint is the transpose; Section 1.5). Equation (D.1) is the adjoint ODE (6.16) with $g_u = 0$, and (D.2) pairs that adjoint against $\partial\vec{F}/\partial p_j$ just as the sensitivity formula (6.18) does. Only the boundary condition is new: periodicity and a phase normalization replace the terminal condition (6.17), and the pairing enters with the opposite sign because T is defined implicitly by the return condition rather than read off a cost functional.

D.3 ▪ The van der Pol Example

The van der Pol oscillator $\ddot{u} - \epsilon(1-u^2)\dot{u} + u = 0$ (written as a system: $\dot{u}_1 = u_2$, $\dot{u}_2 = \epsilon(1-u_1^2)u_2 - u_1$) has a stable limit cycle for $\epsilon > 0$.

For small ϵ, the period is approximately $T \approx 2\pi(1+\epsilon^2/16)$. Differentiating: $dT/d\epsilon = \pi\epsilon/4$. The normalized period sensitivity is therefore

$$\mathcal{S}_\epsilon^T = \frac{\epsilon}{T}\frac{dT}{d\epsilon} = \frac{\epsilon}{2\pi(1+\epsilon^2/16)} \cdot \frac{\pi\epsilon}{4} = \frac{\epsilon^2}{8(1+\epsilon^2/16)},$$

which is small for small ϵ (approximately $\epsilon^2/8$ for $\epsilon \ll 1$) but grows with increasing nonlinearity.

For larger ϵ, the limit cycle becomes increasingly asymmetric, the asymptotic period expansion fails, and $dT/d\epsilon$ must be computed numerically from (D.2), in five steps. (1) Find the cycle: integrate from any start until the transient has died out, take $\vec{u}^*(0)$ at a crossing of the line $u_2 = 0$, and take T as the time to the next crossing in the same direction. (2) Integrate the variational equation $\dot{\Phi} = J_F(\vec{u}^*(t))\Phi$, $\Phi(0) = I$, over one period to get $M = \Phi(T)$. (3) Take $\vec{z}(0)$ as the left eigenvector of M for the unit multiplier, scaled so that $\vec{z}(0)^T \vec{F}(\vec{u}^*(0)) = 1$ (Section D.2). (4) Integrate (D.1) backward over one period from $\vec{z}(T) = \vec{z}(0)$; for a stable cycle the backward direction is the stable one. (5) Evaluate the integral (D.2). The answer to Self-check D.1(c) gives values to check against at $\epsilon = 2$: $T = 7.630$ and $dT/d\epsilon = 1.138$.

D.4 ▪ Sensitivity of Floquet Multipliers

The sensitivity of the Floquet multiplier μ_k with respect to parameter p_j uses the eigenvalue sensitivity formula from Appendix C.1 applied to the monodromy matrix:

$$\frac{d\mu_k}{dp_j} = \frac{\vec{y}_k^T \frac{\partial M}{\partial p_j}\vec{x}_k}{\vec{y}_k^T \vec{x}_k}, \tag{D.3}$$

where $\vec{x}_k$ and $\vec{y}_k$ are the right and left eigenvectors of M for eigenvalue μ_k, and $\partial M/\partial p_j$ is computed by integrating the variational equation for $\partial\Phi/\partial p_j$ over one period.

When $|\mu_k| \to 1$ for some $k \neq 1$, the periodic orbit approaches a bifurcation, where its stability changes. A real multiplier can pass through -1 (a period-doubling: an orbit of twice the period appears) or through $+1$. A small orbit can also shrink onto an equilibrium (a Hopf point), its multiplier μ_k tending to 1. At a period-doubling, and as an orbit shrinks to a Hopf point, the multiplier typically crosses at a finite rate, so (D.3) stays finite.

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By (D.3), $d\mu_k/dp_j$ can diverge only if $\vec{y}_k^T\vec{x}_k \to 0$ or $\partial M/\partial p_j$ grows without bound. That happens at a fold of cycles, where two periodic orbits merge and vanish as a multiplier reaches $+1$. None of this is the mechanism of Chapter 8's transcritical threshold, where the index diverges because the equilibrium goes to zero. A multiplier tends to a point on the unit circle, not to zero. Section 8.4.1 sorts these cases.

Connection to Chapter 5 and Chapter 6. The computation of $\partial M/\partial p_j$ requires integrating the variational equation $\partial\dot{\Phi}/\partial p_j = J_F(t)\,\partial\Phi/\partial p_j + \partial J_F/\partial p_j \cdot \Phi$ over one period, the same forward sensitivity equation developed in Chapter 4 and analyzed in Chapter 5. The left eigenvector $\vec{y}_k$ in equation (D.3) plays the role of the adjoint variable: it weights the state perturbation $\partial\Phi/\partial p_j \cdot \vec{x}_k$ to extract the component that matters for eigenvalue μ_k, mirroring exactly the role $\vec{\lambda}$ plays in Chapter 6 for extracting the gradient of J with respect to p. Students who have worked through Chapters 5–6 will recognize this as another instance of the same duality: the forward computation propagates perturbations forward in time; the left eigenvector (or adjoint variable) weights them by their downstream impact.

D.5 ▪ Exercises

Self-check D.1. The van der Pol oscillator with $\epsilon = 0.5$ has a stable limit cycle. (a) Using the approximation $T \approx 2\pi(1 + \epsilon^2/16)$, compute the period and the normalized sensitivity $\mathcal{S}_\epsilon^T$. (b) Is the period more or less sensitive to ϵ than a linear oscillator's period is to its natural frequency? Explain. (c) As ϵ increases from 0.5 to 2.0, does the period sensitivity increase or decrease? What does this imply about using local SA at large ϵ?

Self-check D.2. A compartmental epidemic model exhibits a periodic outbreak pattern with period $T^* = 365$ days (annual epidemic) under seasonal forcing. The model is forced, hence non-autonomous, so the first form of Section D.1 applies and no multiplier is fixed at 1. The Floquet multiplier of largest modulus is $\mu = 0.85$. (a) Is this periodic orbit stable or unstable? How do you know? (b) If a parameter perturbation changes μ to 1.02, what happens to the epidemic dynamics? (c) The sensitivity of μ to the transmission rate β is $d\mu/d\beta = 3.2$ (computed via (D.3)), and the nominal transmission rate is $\beta^* = 0.3$. If β increases by 5%, estimate the new value of μ. Is the periodic outbreak still stable?

Answers to the self-checks

D.1. (a) $T \approx 2\pi(1+0.25/16) = 2\pi \times 1.01563 \approx 6.381$. $dT/d\epsilon = 2\pi(\epsilon/8) = 2\pi(0.0625) \approx 0.393$. $\mathcal{S}_\epsilon^T = (\epsilon/T)(dT/d\epsilon) = (0.5/6.381)(0.393) \approx 0.031$. (b) Less sensitive, by a factor of about thirty. A linear oscillator's period $T = 2\pi/\omega$ is a power law in its natural frequency, so $\mathcal{S}_\omega^T = -1$ (Chapter 2: a monomial's index is its exponent). At $\epsilon = 0.5$ the van der Pol index is 0.031: a 1% change in ϵ moves the period by about 0.03%, against 1% for a 1% change in ω. The two parameters play different roles, since ϵ is the nonlinear damping coefficient and the natural frequency here is fixed at 1, but both indices are dimensionless and can be compared. (c) The sensitivity increases. Computed from (D.2) with the periodic adjoint, the index is 0.030 at $\epsilon = 0.5$ (part (a)'s estimate gave 0.031) and 0.298 at $\epsilon = 2$, about tenfold. Beyond the weakly nonlinear regime the small-ϵ estimate overstates both the period and its slope: at $\epsilon = 2$ the computed values are $T = 7.630$ and $dT/d\epsilon = 1.138$, against 7.854 and $\pi\epsilon/4 = 1.571$. The formula's index there, 0.400, overstates the true 0.298. What fails at large ϵ is the expansion, not local SA: (D.2) still gives the exact local derivative. Global SA (Chapter 9) would add something only if ϵ were uncertain over a range.

D.2. (a) Stable: the largest multiplier has $|\mu| = 0.85 < 1$, so every multiplier lies inside the unit circle and perturbations from the periodic orbit decay over successive periods. The orbit is an attractor. (b) If $\mu = 1.02 > 1$, the periodic orbit becomes unstable: nearby trajectories drift away from the annual cycle toward some other attractor, which the multiplier alone does not identify. Two outcomes are ruled out. A multiplier passing through $+1$ does not produce period-doubling, which needs a multiplier through -1. A seasonally forced model has no endemic equilibrium to settle on: a constant state with $I > 0$ cannot keep $\dot{I} = 0$ while β varies. (c) $\Delta\mu \approx (d\mu/d\beta)\,\Delta\beta = 3.2 \times (0.05 \times \beta)$. With $\beta^* = 0.3$, $\Delta\beta = 0.015$ and $\Delta\mu \approx 3.2 \times 0.015 = 0.048$. New $\mu \approx 0.85 + 0.048 = 0.898 < 1$: still stable. The first-order approximation suggests stability is preserved for this perturbation size.

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Appendix E
The Adjoint in Optimization

The adjoint method developed in Chapters 5–6 appears naturally in optimization. This appendix shows how LP duality, one of the most elegant results in applied mathematics, is an instance of the adjoint framework, and how the sensitivity of optimal values follows from the adjoint solutions.

E.1 ▪ Linear Programming Duality

The *primal linear program* is:

$$\min_{\vec{x}} \ \vec{c}^T\vec{x} \quad \text{subject to} \quad A\vec{x} \geq \vec{b},\ \vec{x} \geq 0. \tag{E.1}$$

Its *dual* is:

$$\max_{\vec{y}} \ \vec{b}^T\vec{y} \quad \text{subject to} \quad A^T\vec{y} \leq \vec{c},\ \vec{y} \geq 0. \tag{E.2}$$

The dual constraint $A^T\vec{y} \leq \vec{c}$ involves the transposed matrix A^T, exactly the structure of the adjoint equation $A^T\vec{v} = \vec{c}$ from Chapter 5. The dual variables $\vec{y}$ play the role of the adjoint vector $\vec{v}$. The inequality becomes an equation where it matters. At the optimum, *complementary slackness* holds: a dual constraint whose primal variable is positive, $x_j^* > 0$, is met with equality, and a dual variable whose primal constraint is slack is zero. On those columns $\vec{y}^*$ solves an adjoint equation of Chapter 5's form; Self-check E.2 uses this.

Strong Duality Theorem. If both the primal (E.1) and dual (E.2) are feasible, their optimal values are equal:

$$\min_{\text{primal}} \vec{c}^T\vec{x}^* = \max_{\text{dual}} \vec{b}^T\vec{y}^*.$$

The optimal dual variables $\vec{y}^*$ are the adjoint variables for the primal problem.

E.2 ▪ Sensitivity of the Optimal Value

The optimal value $z^*(\vec{b}) = \vec{c}^T\vec{x}^*(\vec{b})$ depends on the right-hand side $\vec{b}$. The *shadow price* (or *dual price*) of constraint i is:

$$\frac{\partial z^*}{\partial b_i} = y_i^*, \tag{E.3}$$

where y_i^* is the i-th component of the optimal dual vector. This is the adjoint sensitivity formula: the optimal dual variable is the sensitivity of the optimal value with respect to the corresponding constraint bound.

The normalized version gives a sensitivity index: $\mathcal{S}_{b_i}^{z^*} = (b_i/z^*)y_i^*$, directly analogous to the $\mathcal{S}_{p_j}^J = (p_j/J)(dJ/dp_j)$ formula from Chapter 2.

Example E.1. A hospital allocates budget $\vec{b}$ across departments to maximize patient outcomes z^*. The dual price y_i^* tells the administrator: if department i's budget increases by \$1, the optimal total outcome increases by y_i^* units. This is SA of the optimal value with respect to the budget constraints.

E.3 ▪ Quadratic Programming

For the equality-constrained quadratic program $\min_{\vec{x}} \frac{1}{2}\vec{x}^T H\vec{x} + \vec{c}^T\vec{x}$ subject to $A\vec{x} = \vec{b}$, the Lagrangian is $\mathcal{L} = \frac{1}{2}\vec{x}^T H\vec{x} + \vec{c}^T\vec{x} - \vec{\lambda}^T(A\vec{x} - \vec{b})$, and stationarity in $\vec{x}$ gives the adjoint equation $H\vec{x} + \vec{c} = A^T\vec{\lambda}$. The envelope theorem (the derivative of the optimal value with respect to a parameter equals the partial derivative of the Lagrangian with respect to that parameter, here $\partial\mathcal{L}/\partial b_i = \lambda_i$) then yields

$$\frac{\partial z^*}{\partial b_i} = \lambda_i^*, \tag{E.4}$$

so a unit increase in b_i changes the optimal value by λ_i^*, improving or worsening it according to the sign of λ_i^*. This is the LP shadow price (E.3) again, written in the same sign convention: $\vec{\lambda}^* = \vec{y}^*$, one object with one sign throughout. Bound constraints $\vec{x} \geq 0$ can be added; they carry their own multipliers, each zero unless its bound is active. Under strict complementarity (each active bound has a positive multiplier) they leave (E.4) unchanged.

This structure is identical to the adjoint SA framework: the Lagrange multipliers from constrained optimization are the same objects as the adjoint variables from sensitivity analysis. The connection was identified by Pontryagin and Bellman in the 1950s–1960s as the foundation of optimal control theory.

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E.4 ▪ Connection to Optimal Control

The adjoint ODE from Chapter 6 is not unique to SA; it is the fundamental object in Pontryagin's minimum principle for optimal control.

Given a controlled dynamical system $\dot{\vec{u}} = \vec{F}(\vec{u}, \vec{p}, t)$ and an objective $J = \int_0^T g(\vec{u}, \vec{p})\, dt + h(\vec{u}(T))$, the Pontryagin minimum principle states that, along an optimal trajectory, the costate $\vec{\lambda}(t)$ satisfies the same adjoint ODE (6.16) derived in Chapter 6, while the optimal control $\vec{p}^*(t)$ minimizes the Hamiltonian $\mathcal{H} = g + \vec{\lambda}^T \vec{F}$ at each time.

In SA, we compute the adjoint to find how J changes with parameters. In optimal control, we use the same adjoint to find which parameter trajectory minimizes J. The mathematical structure is identical.

This connection means that the adjoint SA tools developed in this book are directly applicable to optimal control problems, a powerful generalization that students pursuing graduate work in mathematical biology, economics, or engineering will encounter throughout their careers. Smith [60] develops parameter identifiability and influence in Chapter 7 of that book, and parameter subset selection in its Chapter 8, on local sensitivity analysis.

E.5 ▪ Exercises

Self-check E.1. A hospital network allocates budget across three departments: emergency ($b_1 = \$400$K), ICU ($b_2 = \$300$K), surgery ($b_3 = \$250$K). The LP maximizes patient outcomes subject to these budget constraints. It has the form $\max \vec{c}^T \vec{x}$ subject to $A\vec{x} \le \vec{b}$, $\vec{x} \ge 0$, the mirror image of (E.1). Its dual is $\min \vec{b}^T \vec{y}$ subject to $A^T \vec{y} \ge \vec{c}$, $\vec{y} \ge 0$, and the Strong Duality Theorem and (E.3) hold for this pair as well, with $y_i^* \ge 0$. The optimal dual variables are $\vec{y}^* = (1.8, 2.4, 1.1)$ (outcomes per \$1000 additional budget). (a) Compute the normalized sensitivity index $\mathcal{S}_{b_i}^{z^*}$ for each department, assuming $z^* = 1715$ outcomes. (b) Which department has the highest shadow price? What does this mean for hospital policy? (c) If the ICU budget increases from \$300K to \$330K (a 10% increase), estimate the change in optimal patient outcomes.

Self-check E.2. Consider the primal LP $\min\{2x_1 + 3x_2 : x_1 + x_2 \ge 4,\ 2x_1 + x_2 \ge 6,\ x_1, x_2 \ge 0\}$. (a) Find the optimal primal solution $\vec{x}^*$ and optimal value z^* by checking the corner points of the feasible region. (b) Write the dual LP and find $\vec{y}^*$ (the dual optimal solution) using complementary slackness. (c) Verify strong duality: $\vec{c}^T \vec{x}^* = \vec{b}^T \vec{y}^*$. (d) Use the shadow price y_1^* to estimate the new optimal value if the first constraint's right-hand side changes from $b_1 = 4$ to $b_1 = 5$. Verify by re-solving.

Self-check E.3. The optimal control connection (Section E.4) states that the adjoint ODE from Chapter 6 is identical to the costate equation in Pontryagin's minimum principle. (a) For the simple epidemic control problem: minimize total infections $J = \int_0^T I(t)\,dt$ for the SIR model (4.15)–(4.17) with a control $u(t)$ that scales the incidence $c\beta SI/N$ by $1-u$ in both $\dot{S}$ and $\dot{I}$, so that $\dot{I} = (c\beta(1-u)S/N - \gamma_R - \nu_m)I$, write the Hamiltonian $\mathcal{H}(\vec{u}, \vec{p}, \vec{\lambda}, t)$. (b) Write the costate (adjoint) equations $-\dot{\lambda}_i = \partial\mathcal{H}/\partial u_i$, one for each *state* $u_i \in \{S, I, R\}$. Note that the scalar control $u(t)$ in part (a) is a different object from the state vector $\vec{u}$; stationarity in the control, $\partial\mathcal{H}/\partial u = 0$, is a separate condition. (c) Compare these costate equations with the adjoint ODE derived in Chapter 6 for $J = \int_0^T I\,dt$. Are they the same?

Answers to the self-checks

E.1. (a) $\mathcal{S}_{b_i}^{z^*} = (b_i/z^*)\,y_i^*$: $\mathcal{S}_{b_1}^{z^*} = (400/1715)\times 1.8 = 0.42$. $\mathcal{S}_{b_2}^{z^*} = (300/1715)\times 2.4 = 0.42$. $\mathcal{S}_{b_3}^{z^*} = (250/1715)\times 1.1 = 0.16$. The three indices sum to 1, as they must when the budgets are the only constraints: then $z^* = \vec{b}^T\vec{y}^* = 720 + 720 + 275 = 1715$ by the Strong Duality Theorem. (b) The ICU has the highest shadow price ($y_2^* = 2.4$): each additional \$1000 in ICU budget yields 2.4 more patient outcomes: the highest marginal return. Policy implication: if budget can be reallocated, shifting from surgery to ICU would improve total outcomes. (c) $\Delta z^* \approx y_2^*\,\Delta b_2 = 2.4 \times 30 = 72$ additional outcomes. The 10% budget increase in ICU produces a $72/1715 \approx 4.2\%$ improvement in total outcomes: $\mathcal{S}_{b_2}^{z^*} = 0.42$, so $10\% \times 0.42 = 4.2\%$. ✓

E.2. (a) The feasible region is defined by $x_1 + x_2 \geq 4$, $2x_1 + x_2 \geq 6$, $x_1, x_2 \geq 0$. The region is unbounded, but both costs are positive and $\vec{x} \geq 0$, so the objective grows along every unbounded direction and the minimum sits at a corner. The corner points of the feasible region are: $(4, 0)$ (intersection of $x_1 + x_2 = 4$ and $x_2 = 0$), $(0, 6)$ (intersection of $2x_1 + x_2 = 6$ and $x_1 = 0$), and $(2, 2)$ (intersection of the two constraints). Objective values: $2(4) + 3(0) = 8$; $2(0) + 3(6) = 18$; $2(2) + 3(2) = 10$. The minimum is $z^* = 8$ at $\vec{x}^* = (4, 0)^T$.

(b) The dual of $\min\{\vec{c}^T\vec{x} : A\vec{x} \geq \vec{b},\ \vec{x} \geq 0\}$ is $\max\{\vec{b}^T\vec{y} : A^T\vec{y} \leq \vec{c},\ \vec{y} \geq 0\}$. Here $A = \begin{pmatrix} 1 & 1 \\ 2 & 1 \end{pmatrix}$, $\vec{b} = (4, 6)^T$, $\vec{c} = (2, 3)^T$, giving:

$$\max\{4y_1 + 6y_2\} \ \text{ s.t. } \ y_1 + 2y_2 \leq 2, \ \ y_1 + y_2 \leq 3, \ \ y_1, y_2 \geq 0.$$

At $\vec{x}^* = (4, 0)$: constraint 1 is active ($x_1 + x_2 = 4$) and constraint 2 is inactive ($2x_1 + x_2 = 8 > 6$). By complementary slackness, $y_2^* = 0$ (since constraint 2 is not active). The dual constraint corresponding to $x_1 > 0$ must hold with equality: $y_1^* + 2(0) = 2$, so $y_1^* = 2$. Thus $\vec{y}^* = (2, 0)^T$.

(c) Strong duality: $\vec{c}^T\vec{x}^* = 2(4) + 3(0) = 8$ and $\vec{b}^T\vec{y}^* = 4(2) + 6(0) = 8$. Equal. ✓

(d) The shadow price $y_1^* = 2$ means each unit increase in b_1 increases z^* by 2: for a minimization problem with $\geq$ constraints, raising b_1 *tightens* the constraint and shrinks the feasible region, so the optimal value can only rise. Estimate: $z^* \approx 8 + 2 \times (5 - 4) = 10$. Verification: with $b_1 = 5$ the relevant corners are

$(5,0)$ with $z = 10$, $(1,4)$ with $z = 14$, and $(0,6)$ with $z = 18$, so $(5,0)$ is optimal. Note that $(5,0)$ lies on $x_1 + x_2 = 5$ and $x_2 = 0$; the two constraint boundaries $x_1 + x_2 = 5$ and $2x_1 + x_2 = 6$ meet at $(1,4)$. Exact value: $z^* = 10$. ✓The shadow price approximation is exact here because the optimal basis (constraint 1 active, $x_2 = 0$) is unchanged by this perturbation.

E.3. (a) Hamiltonian: $\mathcal{H} = I + \lambda_S \dot{S} + \lambda_I \dot{I} + \lambda_R \dot{R} = I + \lambda_S(\nu_m N - c\beta(1-u)SI/N - \nu_m S) + \lambda_I(c\beta(1-u)SI/N - (\gamma_R + \nu_m)I) + \lambda_R(\gamma_R I - \nu_m R)$. (b) Costate equations: $-\dot{\lambda}_S = \partial\mathcal{H}/\partial S = c\beta(1-u)I/N(\lambda_I - \lambda_S) - \nu_m\lambda_S$, etc. (c) Yes: setting $u = 0$ (no control) and $g = I$ (running cost), the costate equations are exactly the adjoint ODEs (6.21)–(6.23) from Chapter 6. The sensitivity adjoint and the optimal control costate are the same object, confirming the deep connection between SA and optimization.

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Appendix F
Probability Toolkit

Most of this book treats each parameter as a number with a nominal value; Chapter 9 treats it instead as a draw from a range, and asks how the spread of the inputs turns into spread in the output. That question needs five tools: expectation, variance, the uniform distribution, conditional expectation and correlation. This appendix supplies those five and nothing more. Chapter 3 needs only Section F.4, Chapter 8 needs Section F.2, and Chapter 9 needs all of it.

F.1 ▪ Random Variables and Expectation

In the worked example of Section 9.5, the transmission probability is not fixed. It may take any value within 50% of the nominal $\beta = 0.06$, so it lies somewhere in $[0.03, 0.09]$, with no part of the range favored over another. Each model run draws a fresh value of β from that interval. A quantity whose value is set by such a draw is called a *random variable*. We write the random variable with a capital letter, Q, and a value it takes with a lower-case letter, q.

The rule that says how the draws spread out is the *distribution* of Q. For a continuous Q it is given by a density $f_Q(q)$, which is never negative and integrates to one, so that the chance of a draw landing in a short interval of width dq is $f_Q(q)\,dq$. When every part of $[a, b]$ is equally likely, the density is the constant $1/(b-a)$ on that interval and zero outside it. This is the *uniform distribution*, written $Q \sim \mathcal{U}(a, b)$, and the worked examples of Chapter 9 use it throughout.

The *expectation* of Q is its average over the distribution, with each value weighted by how likely it is:

$$\mathbb{E}[Q] = \int q\, f_Q(q)\, dq, \qquad \mathbb{E}[g(Q)] = \int g(q)\, f_Q(q)\, dq. \tag{F.1}$$

The second form averages any function g of Q in the same way. The expectation is also called the *mean* and is often written μ; for $Q \sim \mathcal{U}(0,1)$ the density is 1 on $[0,1]$, so $\mathbb{E}[Q] = \int_0^1 q\,dq = 1/2$ and $\mathbb{E}[Q^2] = \int_0^1 q^2\,dq = 1/3$.

Two rules do most of the work in Chapter 9. The first is that expectation is linear, so $\mathbb{E}[aX + bY] = a\mathbb{E}[X] + b\mathbb{E}[Y]$ for any random variables X and Y and any constants a and b. The second is that when X and Y are independent, the mean of a product is the product of the means, $\mathbb{E}[XY] = \mathbb{E}[X]\,\mathbb{E}[Y]$. Two inputs are *independent* when knowing the value of one tells us nothing about the other. Their joint density is then the product of their separate densities. An average over both can then be taken one input at a time, and the ANOVA decomposition of Section 9.4 depends on exactly this factoring.

F.2 ▪ Variance and Standard Deviation

Two inputs can share a mean and still differ in how far their draws stray from it. The *variance* measures that spread as the mean squared distance from the mean:

$$\mathrm{Var}(Q) = \mathbb{E}\big[(Q-\mu)^2\big] = \mathbb{E}[Q^2] - \mu^2, \qquad \mu = \mathbb{E}[Q]. \tag{F.2}$$

The second form follows from expanding the square and using linearity, and the *standard deviation* $\sigma = \sqrt{\mathrm{Var}(Q)}$ carries the units of Q. It is the σ_i of Chapters 8 and 9.

Example F.1 (The variance of $\mathcal{U}(0,1)$). For $Q \sim \mathcal{U}(0,1)$ we found $\mathbb{E}[Q] = 1/2$ and $\mathbb{E}[Q^2] = 1/3$, so

$$\mathrm{Var}(Q) = \mathbb{E}[Q^2] - \big(\mathbb{E}[Q]\big)^2 = \frac{1}{3} - \frac{1}{4} = \frac{1}{12}, \qquad \sigma = \frac{1}{\sqrt{12}} \approx 0.289.$$

This is the $1/12$ that appears in every variance of Section 9.4.

A uniform draw on $[a,b]$ is a stretched and shifted draw on $[0,1]$, since $Q = a + (b-a)U$ with $U \sim \mathcal{U}(0,1)$. A shift moves the mean but leaves the spread alone, while a stretch by a factor k multiplies the variance by k^2; in symbols, $\mathrm{Var}(kU + a) = k^2\,\mathrm{Var}(U)$. Therefore

$$Q \sim \mathcal{U}(a,b) \quad\Longrightarrow\quad \mathbb{E}[Q] = \frac{a+b}{2}, \qquad \mathrm{Var}(Q) = \frac{(b-a)^2}{12}. \tag{F.3}$$

For $\beta \sim \mathcal{U}(0.03, 0.09)$ the mean is 0.06, the variance is $(0.06)^2/12 = 0.0003$, and $\sigma_\beta \approx 0.0173$, so the coefficient of variation $\mathrm{CV} = \sigma/\mu$ is about 0.289. That is $1/\sqrt{12}$ again: every uniform range of $\pm 50\%$ about its mean has the same relative spread, whatever the parameter.

Sums of independent inputs. For independent X and Y and constants s_1 and s_2,

$$\mathrm{Var}(s_1X + s_2Y) = s_1^2\,\mathrm{Var}(X) + s_2^2\,\mathrm{Var}(Y). \tag{F.4}$$

Variances add, but standard deviations do not. This is the step behind the portfolio of Sections 8.5 and 9.1, where $s_1 = 2$, $s_2 = 1$, $\sigma_1 = 1$ and $\sigma_2 = 3$ give an output variance of $4 + 9 = 13$ and so $\sigma_Y = \sqrt{13}$. Equation (F.4) fails when the inputs move together, and Section F.4 supplies the term that then goes missing.

Two consequences of (F.4) appear in Chapter 9. The first concerns two independent draws Y and Y' of the same output, whose difference has mean zero and variance $2\,\mathrm{Var}(Y)$, so that

$$\tfrac{1}{2}\,\mathbb{E}\big[(Y - Y')^2\big] = \mathrm{Var}(Y). \tag{F.5}$$

For two $\mathcal{U}(0,1)$ draws the mean squared difference is $1/6$, twice $1/12$, and the Jansen estimators (9.6) and (9.7) are built on (F.5).

The second concerns the average $\bar{Y}$ of N independent draws, which has variance $\mathrm{Var}(Y)/N$; its standard deviation $\sigma/\sqrt{N}$ is the *standard error* of the mean, so the average of 100 draws from $\mathcal{U}(0,1)$ has standard error $0.289/10 \approx 0.029$. This is the $O(1/\sqrt{N})$ Monte Carlo rate of Section 9.5. Doubling N shrinks the error only by the factor $1/\sqrt{2} \approx 0.71$, and halving it takes four times the samples.

The delta method. Most models are not sums, but for a smooth $Y = g(Q_1, \ldots, Q_m)$ with independent inputs and small spreads, we replace g by its tangent plane at the means and then apply (F.4) to that linear model:

$$\mathrm{Var}(Y) \approx \sum_{i=1}^{m} \left(\frac{\partial g}{\partial Q_i}\right)^2 \sigma_i^2. \tag{F.6}$$

This is the delta method of Chapters 8 and 9. Because it is a linearization, it inherits the limits of local SA. In the Chapter 8 self-check, $\mathcal{R}_0 = c\beta\tau_R$ with $c = 5$ fixed, while β has mean 0.06 and $\sigma_\beta = 0.02$ and τ_R has mean 7 and $\sigma_{\tau_R} = 3$, the two independent. The delta method gives $\mathrm{Var}(\mathcal{R}_0) \approx 1.30$, while the exact variance of the product is $c^2\big(\mathbb{E}[\beta^2]\,\mathbb{E}[\tau_R^2] - (\mathbb{E}[\beta]\,\mathbb{E}[\tau_R])^2\big) = 25\,(0.004 \times 58 - 0.42^2) = 1.39$. The gap of $c^2\sigma_\beta^2\sigma_{\tau_R}^2 = 0.09$ is the curvature that the tangent plane leaves out, and it grows with the input spreads.

F.3 ▪ Conditional Expectation and the Law of Total Variance

We use the worked model of Section 9.4, $Y = Q_1 + Q_2 + Q_1Q_2$ with $Q_1, Q_2 \sim \mathcal{U}(0,1)$ independent, whose mean is $\mathbb{E}[Y] = 1/2 + 1/2 + 1/4 = 5/4$.

Suppose we learn that $Q_1 = q_1$ while Q_2 is still unknown. The model then becomes $Y = q_1 + Q_2(1 + q_1)$, and averaging over Q_2 alone gives

$$\mathbb{E}[Y \mid Q_1 = q_1] = q_1 + \tfrac{1}{2}(1 + q_1) = \tfrac{1}{2} + \tfrac{3}{2}\, q_1,$$

which is 0.5 at $q_1 = 0$, 1.25 at $q_1 = 1/2$ and 2 at $q_1 = 1$. The *conditional expectation* $\mathbb{E}[Y \mid Q_1]$ is this average over the other inputs, read as a function of the input held fixed. Chapter 9 writes it the same way, and once as $\mathbb{E}_{Q_{-1}}(Y \mid Q_1)$, where the subscript Q_{-1} names the inputs averaged: every input except Q_1. Because Q_1 is itself a draw, $\mathbb{E}[Y \mid Q_1]$ is a random variable, and its own mean is $\mathbb{E}[Y]$.

If Q_1 matters, this average moves a great deal as q_1 moves, and its variance measures how much; Example F.1 supplies the arithmetic:

$$\mathrm{Var}\big[\mathbb{E}(Y \mid Q_1)\big] = \mathrm{Var}\big(\tfrac{1}{2} + \tfrac{3}{2}Q_1\big) = \tfrac{9}{4} \cdot \tfrac{1}{12} = \tfrac{3}{16} = \tfrac{27}{144},$$

which is D_1 in (9.3).

Knowing Q_1 does not settle Y. With q_1 fixed, the spread left over still comes from Q_2, and that leftover spread is the *conditional variance* $\mathrm{Var}(Y \mid Q_1 = q_1) = (1 + q_1)^2\, \mathrm{Var}(Q_2) = (1 + q_1)^2/12$, which runs from $1/12$ at $q_1 = 0$ to $1/3$ at $q_1 = 1$. Its average over Q_1 uses $\mathbb{E}[(1 + Q_1)^2] = 1 + 1 + 1/3 = 7/3$, so $\mathbb{E}[\mathrm{Var}(Y \mid Q_1)] = (7/3)/12 = 7/36 = 28/144$.

The two pieces add up to the whole, since $27/144 + 28/144 = 55/144 = \mathrm{Var}(Y)$, and this is no accident. For any output Y and any input Q_i,

$$\mathrm{Var}(Y) = \mathrm{Var}\big[\mathbb{E}(Y \mid Q_i)\big] + \mathbb{E}\big[\mathrm{Var}(Y \mid Q_i)\big], \tag{F.7}$$

which is the *law of total variance*. To see why it holds, write $m(Q_i) = \mathbb{E}[Y \mid Q_i]$ and split $Y - \mathbb{E}[Y]$ into $\big(Y - m(Q_i)\big) + \big(m(Q_i) - \mathbb{E}[Y]\big)$, then square and average. The first square averages to $\mathbb{E}[\mathrm{Var}(Y \mid Q_i)]$ and the second to $\mathrm{Var}[m(Q_i)]$. The cross term vanishes, because with Q_i held fixed the first factor averages to zero and the second is a constant.

Equation (F.7) splits the output variance into two parts. The part *explained* by Q_i is the spread of its conditional mean, and the part *left over* is the spread that remains, on average, once Q_i is known. Chapter 9 reads both parts as indices. For the worked model they are

$$S_1 = \frac{\mathrm{Var}[\mathbb{E}(Y \mid Q_1)]}{\mathrm{Var}(Y)} = \frac{27}{55} \approx 0.491, \qquad \frac{\mathbb{E}[\mathrm{Var}(Y \mid Q_1)]}{\mathrm{Var}(Y)} = \frac{28}{55} \approx 0.509.$$

The first number is the first-order index (9.4): the share of the variance that learning Q_1 exactly would remove, on average. The second is the share that would remain, and it belongs to Q_2. With two inputs $\vec{Q}_{-2}$ is just Q_1, so

by (9.5) the second number is the total index S_{T_2}. By symmetry $S_{T_1} = 28/55$ as well, which equals $S_1 + S_{12} = 27/55 + 1/55$ because the total index adds the interaction to the direct effect. The second form of (9.5) is (F.7) applied with $\vec{Q}_{-i}$ in place of Q_i, and the step that the Jansen estimator rests on (Section 9.5) is (F.5) applied with Q_i held fixed, followed by (F.7).

F.4 ▪ Covariance and Correlation

Return to the portfolio $Y = 2Q_1 + Q_2$ with $\sigma_1 = 1$ and $\sigma_2 = 3$. Suppose now that Q_1 and Q_2 tend to be high together and low together, so that their deviations reinforce each other and Y spreads more than (F.4) allows. The extra term is the *covariance* $\mathrm{Cov}(X, Y) = \mathbb{E}[(X - \mu_X)(Y - \mu_Y)] = \mathbb{E}[XY] - \mu_X \mu_Y$, and the general rule for a weighted sum is

$$\mathrm{Var}(s_1 X + s_2 Y) = s_1^2 \,\mathrm{Var}(X) + s_2^2 \,\mathrm{Var}(Y) + 2 s_1 s_2 \,\mathrm{Cov}(X, Y). \tag{F.8}$$

The covariance carries the units of X times those of Y, and dividing it by both standard deviations gives a pure number, the *correlation coefficient*

$$\rho_{XY} = \frac{\mathrm{Cov}(X, Y)}{\sigma_X \, \sigma_Y}, \qquad -1 \le \rho_{XY} \le 1, \tag{F.9}$$

which reaches $+1$ or -1 only when one variable is an exact linear function of the other. The Pearson correlation of Section 3.5 is this same ρ estimated from pairs of observed values.

For the portfolio with $\rho = 0.5$ the covariance is $0.5 \times 1 \times 3 = 1.5$, and (F.8) gives $\mathrm{Var}(Y) = 13 + 2 \times 2 \times 1 \times 1.5 = 19$, so the output standard deviation is $\sqrt{19} \approx 4.36$ rather than $\sqrt{13} \approx 3.61$. Assuming independence understates the output spread by about 17%, and the sigma-normalized indices inherit that error; this is the warning of Section 8.5. Constraint correlation (Section 3.5) is the extreme case: if two shares must sum to one, then $Q_2 = 1 - Q_1$, and for $Q_1 \sim \mathcal{U}(0, 1)$ the covariance is $-1/12$ and $\rho = -1$. Then $\mathrm{Var}(Q_1 + Q_2) = 1/12 + 1/12 - 2/12 = 0$, as it must be for a sum that never changes, while the independence formula would report $1/6$.

Three cautions keep correlation in its place.

Zero correlation is not independence. Independent inputs have zero covariance, but the converse fails. For $Q \sim \mathcal{U}(0, 1)$ the variable $(Q - 1/2)^2$ is completely fixed by Q, yet its covariance with Q is zero, because correlation sees only straight-line association. The orthogonality behind the ANOVA decomposition needs full independence, not just zero correlation.

A correlation is not a coupling coefficient. Equation (3.12) needs ω_{ij}, the proportional change in p_j per proportional change in p_i. The least-squares slope of Y on X is $\mathrm{Cov}(X, Y)/\,\mathrm{Var}(X) = \rho_{XY}\, \sigma_Y / \sigma_X$. On the relative scale

the slope is therefore ρ times the ratio of the two coefficients of variation, as the Chapter 3 laboratory states. With $\rho_{c\beta} = 0.6$ and β half as variable as c in relative terms, the slope is 0.3, not 0.6. A slope fitted to field data also describes association rather than cause, and the caution in Section 3.5 applies to it.

Correlated inputs break the Sobol decomposition. If we take the worked model of Section F.3 and set $Q_2 = Q_1$, then knowing Q_1 fixes Y, so $\mathbb{E}[Y \mid Q_1] = Y$ and $S_1 = 1$; the same holds for Q_2, and the first-order indices sum to 2. Each index is still a fair answer to its own question. What fails is the split: the variance can no longer be divided into shares, because the inputs are no longer separate sources. Section 9.7 gives the tools that replace it.

F.5 ▪ Exercises

Self-check F.1. The mean infectious period is drawn as $\tau_R \sim \mathcal{U}(3.5, 10.5)$ days, which is ±50% about the nominal 7 days; find the mean $\mathbb{E}[\tau_R]$, the variance $\mathrm{Var}(\tau_R)$, the standard deviation σ_{τ_R} and the coefficient of variation.

Self-check F.2. Let $Y = Q_1 Q_2$ with $Q_1, Q_2 \sim \mathcal{U}(0, 1)$ independent. (a) Find $\mathbb{E}[Y \mid Q_1]$ and its variance, then (b) find $\mathrm{Var}(Y \mid Q_1)$ and its mean over Q_1. (c) Check that the two parts add up to $\mathrm{Var}(Y)$ computed directly, as the law of total variance says they must. (d) Find S_1 and S_{T_1}, and say what share of the variance comes from the interaction of the two inputs.

Self-check F.3. In the portfolio $Y = 2Q_1 + Q_2$ with $\sigma_1 = 1$ and $\sigma_2 = 3$, suppose the inputs tend to move in opposite directions, with $\rho = -0.5$. Find σ_Y and compare it with the value for independent inputs.

Answers to the self-checks

F.1. By (F.3), $\mathbb{E}[\tau_R] = (3.5 + 10.5)/2 = 7$ days and $\mathrm{Var}(\tau_R) = 7^2/12 = 49/12 \approx 4.08$ days^2, so $\sigma_{\tau_R} \approx 2.02$ days and CV $= 2.02/7 \approx 0.289$. This is $1/\sqrt{12}$, the same value we found for β in Section F.2, since every ±50% uniform range has it.

F.2. (a) Holding Q_1 fixed and averaging over Q_2 gives $\mathbb{E}[Y \mid Q_1] = Q_1/2$, so $\mathrm{Var}[\mathbb{E}(Y \mid Q_1)] = \frac{1}{4} \cdot \frac{1}{12} = 1/48$. (b) With Q_1 fixed, Y is Q_1 times a $\mathcal{U}(0, 1)$ draw, so $\mathrm{Var}(Y \mid Q_1) = Q_1^2/12$, whose mean over Q_1 is $\frac{1}{3} \cdot \frac{1}{12} = 1/36$. (c) The two parts sum to $1/48 + 1/36 = 3/144 + 4/144 = 7/144$, and directly $\mathbb{E}[Y^2] = \mathbb{E}[Q_1^2]\,\mathbb{E}[Q_2^2] = 1/9$ and $\mathbb{E}[Y] = 1/4$, so that $\mathrm{Var}(Y) = 1/9 - 1/16 = 7/144$ as well. ✓(d) The first-order index is $S_1 = (3/144)/(7/144) = 3/7 \approx 0.429$, and by symmetry $\mathbb{E}[\mathrm{Var}(Y \mid Q_2)] = 4/144$, so $S_{T_1} = 4/7 \approx 0.571$. The interaction share is $S_{T_1} - S_1 = 1/7 \approx 0.143$, and $S_1 + S_2 + S_{12} = 3/7 + 3/7 + 1/7 = 1$.

F.3. The covariance is $-0.5\times1\times3 = -1.5$, so by (F.8) $\mathrm{Var}(Y) = 4+9+2\times2\times1\times(-1.5) = 7$ and $\sigma_Y = \sqrt{7} \approx 2.65$, against $\sqrt{13} \approx 3.61$ for independent inputs: a negative correlation cancels part of the spread.

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Glossary

This glossary is a finding aid for the reader who meets a term away from where it was first defined. Each entry states what the term means and points to the chapter, appendix or section that develops it. Symbols are collected separately in the Notation and Conventions section; the choice among the sensitivity indices is laid out in Table 9.4.

Adjoint method A technique for computing all parameter sensitivities of a scalar response at a cost independent of the number of parameters, by solving one transposed (adjoint) system, which for an ODE model runs backward in time. The adjoint is the transpose of the forward sensitivity problem. See Chapters 5 and 6.

Adjoint operator The generalization of the matrix transpose to a linear operator L on an inner product space: the operator $L^\dagger$ defined by $\langle Lu, v\rangle = \langle u, L^\dagger v\rangle$ for all u, v. The adjoint method rests on this identity. See Appendix B.

Algorithmic (automatic) differentiation The evaluation of exact derivatives of a function specified by a program, by applying the chain rule to its elementary operations. Reverse-mode algorithmic differentiation is the adjoint method applied to a computation. See Chapters 5 and 7.

ANOVA decomposition The decomposition of a function's variance into contributions from each parameter and each combination of parameters; the basis of the Sobol indices. See Chapter 9 and Section A.4.

Augmented system The model's ODE together with its forward sensitivity equations, solved as one system; for n states and m parameters it has $n(1 + m)$ equations. See Section 4.3.

Backpropagation The name used in machine learning for reverse-mode algorithmic differentiation, the method by which neural networks are trained. It is the adjoint method under another name: a reader who

knows backpropagation already knows the book's central computational tool. See Chapter 5.

Backward bifurcation A bifurcation in which the endemic branch bends back below $\mathcal{R}_0 = 1$, so for some $\mathcal{R}_0 < 1$ a stable endemic equilibrium coexists with the stable disease-free state. The threshold is then not simply $\mathcal{R}_0 = 1$, and each stable equilibrium needs its own SA. See Section 8.2.

Basic reproduction number ($\mathcal{R}_0$) The expected number of secondary infections produced by one infectious individual in a fully susceptible population; for the running SIR model, $\mathcal{R}_0 = c\beta\tau_R$. See Chapter 2.

Bifurcation A parameter value at which the qualitative behavior of a model changes: equilibria merge, or an equilibrium changes stability. How local sensitivity analysis fails at a bifurcation depends on what is differentiated. The book distinguishes two kinds: at a *fold* bifurcation two equilibria collide and annihilate, and the raw derivative of the equilibrium diverges; at a *transcritical* bifurcation two equilibria exchange stability, the raw derivative stays finite, and only the normalized index diverges. The epidemic threshold is transcritical. See Chapter 8.

Bootstrap A resampling procedure for placing confidence intervals on estimated sensitivity indices without assuming a distributional form. See Chapter 9.

Checkpointing A strategy for the adjoint method that trades computation for memory, storing the forward solution at selected times and recomputing the rest, rather than storing the entire trajectory. See Chapter 7.

Complementary slackness The condition that holds at a linear-programming optimum: a dual constraint whose primal variable is positive is met with equality, and a dual variable whose primal constraint is slack is zero. On those columns the dual solution solves an adjoint equation. See Section E.1.

Computation graph A directed acyclic graph that represents a differentiable computation: each node holds an intermediate value and each edge an elementary operation. Forward-mode differentiation runs through it from the inputs; reverse mode runs back from the output. See Section 5.6.

Condition number A measure, $\kappa(A) = \|A\|\,\|A^{-1}\|$ with $\|A\|$ the *matrix norm*, of how much a problem's output can change relative to a change

in its input; large κ signals ill-conditioning. See Chapter 8 and Appendix C.

Conditional expectation The mean of an output with some inputs held at given values, $\mathbb{E}[Y \mid Q_i]$. The *law of total variance* splits $\mathrm{Var}(Y)$ into the variance of this conditional mean and the mean of the conditional variance, the split that the Sobol indices read. See Section F.3.

Continuation Tracing a solution branch as a parameter changes, starting each solve from the solution at the previous parameter value. Packages such as MATCONT do this and flag the bifurcations they pass. See Section 8.7.

Correlated POIs Parameters whose effects or values are linked: they enter the model only in a fixed combination (structural correlation), the data that estimate them link their values (empirical correlation), or a physical or biological requirement ties them together (constraint correlation). Indices computed one parameter at a time can then mislead. See Section 3.5.

Covariance and correlation Measures of how two random inputs move together. The correlation ρ is the covariance divided by the two standard deviations, so $-1 \le \rho \le 1$. See Section F.4.

Delta index A moment-independent global sensitivity measure (Borgonovo's δ) based on the shift a parameter induces in the entire output distribution, not only its variance. See Chapter 9.

Delta method The approximation of an output's variance by linearizing the model at the input means: for independent inputs, $\mathrm{Var}(Y) \approx \sum_i (\partial g/\partial Q_i)^2 \sigma_i^2$. Because it is a linearization, it inherits the limits of local SA. See Section F.2.

Directed acyclic graph A graph whose arrows form no loops, so each value is computed only from values before it. Every differentiable computation can be drawn as one (see *computation graph*). See Section 5.6.

Discretize-then-adjoint /adjoint-then-discretize The two orderings of the operations that produce a numerical adjoint. Discretize-then-adjoint forms the adjoint of the discretized problem; adjoint-then-discretize discretizes the continuous adjoint. The two need not agree, and the distinction matters for gradient accuracy. See Chapter 7.

eFAST The extended Fourier Amplitude Sensitivity Test, a spectral method for estimating first-order and total Sobol indices. See Chapter 9.

Eigenvalue sensitivity The derivative of an eigenvalue of a matrix $A(\varepsilon)$ with respect to a parameter, given for a simple eigenvalue by a ratio of left and right eigenvectors. The formula requires care when the eigenvalue is repeated (see *semisimple / defective eigenvalue*). See Appendix C.

Elasticity A synonym for the normalized sensitivity index, borrowed from economics: the fractional change in the response per fractional change in the parameter. An elasticity of one means a one-percent change in the parameter produces a one-percent change in the response, not that the relationship is linear. See Chapter 2.

Envelope theorem The statement that the derivative of the optimal value with respect to a parameter equals the partial derivative of the Lagrangian with respect to that parameter. For a constraint right-hand side b_i it gives $\partial z^*/\partial b_i = \lambda_i^*$, the shadow price. See Section E.3.

Expectation and variance The mean of a random quantity, $\mathbb{E}[Y]$, and its mean squared deviation from that mean, $\mathrm{Var}(Y)$; the standard deviation σ is the square root of the variance. See Appendix F.

Factor prioritization, fixing, and mapping The three settings in which a global sensitivity analysis is read: which parameters to measure more precisely (prioritization), which can be fixed without loss (fixing), and which drive the output into a region of interest (mapping). See Chapter 9.

Finite difference An approximation of a derivative from model evaluations at nearby parameter values. It needs only the ability to run the model; the centered difference has truncation error $O(h^2)$ in the step h. See Section 3.2.

Floquet multipliers The eigenvalues of the monodromy matrix of a periodic system, governing the stability of a periodic orbit and the sensitivity of its period. See Appendix D.

Forward sensitivity equation (FSE) The differential equation obtained by differentiating the model with respect to a parameter and solving alongside it; it yields the sensitivity of the full trajectory to one parameter. See Chapter 4.

Fredholm alternative The statement that a linear equation $Lu = f$ is solvable exactly when f is orthogonal to the kernel of the adjoint operator $L^\dagger$. See Appendix B.

Gaussian process emulator (surrogate model) A cheap statistical approximation to an expensive model, fitted to a set of runs and used in its place for sensitivity analysis. See Chapter 9.

Global sensitivity analysis The study of how a model's output responds to parameters varied over their whole plausible ranges simultaneously, as opposed to local analysis at a single nominal point. See Chapter 9.

Hilbert space A complete inner product space, one in which every Cauchy sequence converges within the space. The function space $L^2([a,b])$ is the example the book uses. Completeness matters for proofs in the literature; nothing in this book uses it beyond the inner product itself. See Appendix A.

Inner product A function $\langle \cdot, \cdot \rangle$ on a vector space that is symmetric, linear in each argument, and positive-definite, generalizing the dot product; on a function space it is an integral, $\langle f, g \rangle = \int fg$. Its *induced norm* is $\|u\| = \sqrt{\langle u, u \rangle}$. Orthogonality, the adjoint, and the Fredholm alternative are all stated through it. See Appendix A.

Jansen estimator An efficient variance-based estimator of Sobol indices that reuses model evaluations across the first-order and total-effect calculations. See Chapter 9.

Jordan block An $m \times m$ matrix with one eigenvalue λ on the diagonal, ones just above it and zeros elsewhere. It is the case in which the eigenvalue-sensitivity formula fails outright: under a perturbation of size ϵ its eigenvalues move like $\epsilon^{1/m}$ rather than linearly (see *semisimple / defective eigenvalue*). See Section C.1.

Lagrange identity The identity $\frac{d}{dt}[\vec{\lambda}^T \vec{w}] = \vec{\lambda}^T (L\vec{w}) - (L^\dagger \vec{\lambda})^T \vec{w}$ for the linearized ODE operator L and its adjoint $L^\dagger$; integrated over $[0, T]$ it gives the adjoint sensitivity formula. See Section B.2.

Latin hypercube sampling (LHS) A stratified sampling scheme that spreads points evenly across each parameter's range, improving coverage over independent uniform sampling. See Chapter 9.

Low-discrepancy sequence A deterministic sequence (such as a Sobol' sequence) that fills space more evenly than random points, reducing the variance of a sensitivity estimate at a given sample size. See Chapter 9.

MATCONT A MATLAB package for numerical continuation, used in the Chapter 8 laboratory to trace the endemic equilibrium and detect the bifurcation at $\mathcal{R}_0 = 1$. See Section 8.7.

Matrix norm The number $\|A\| = \max_{\vec{x}\neq 0} \|A\vec{x}\|/\|\vec{x}\|$, the largest factor by which A stretches a vector. It enters the condition number. See Section 8.3.

Monodromy matrix The fundamental matrix of a linear periodic system evaluated after one period, $M = \Phi(T)$; its eigenvalues are the Floquet multipliers. See Appendix D.

Morris screening (elementary effects) A low-cost method for ranking many parameters by importance, using a small number of one-at-a-time perturbations to estimate the mean and spread of each parameter's effect. See Chapter 9.

Parameter of interest (POI) /Quantity of interest (QOI) The input under study (POI) and the output being analyzed (QOI). A sensitivity index always relates one QOI to one POI. See Chapter 1.

Permutation matrix The matrix P in the factorization $A = P^T LU$ that many LU routines return; it records the row swaps (pivoting) that the routine makes to keep the elimination stable. See Section 5.8.

Polynomial chaos expansion A representation of a model's output as a series in orthogonal polynomials of its random inputs, from which sensitivity indices can be read off analytically. See Chapter 9.

PRCC (partial rank correlation coefficient) A sampling-based sensitivity measure that ranks parameters by their monotonic influence on the output after removing the effect of the others. See Chapter 9.

Proliferation problem The growth of forward sensitivity cost with the number of parameters: each parameter adds a block of sensitivity equations, so for n states and m parameters the augmented system has $n(1+m)$ equations. It is the bottleneck the adjoint method removes. See Section 4.4.

Pseudoinverse The generalization of the matrix inverse to non-square or rank-deficient matrices, built from the singular value decomposition; it solves a least-squares problem on the range of the matrix. A near-zero smallest singular value signals ill-conditioning. See Appendix C.

Response functional A scalar quantity of interest computed from the model's solution, such as $J = \vec{c}^{\,T}\vec{u}$ or $J = \int_0^T I\,dt$; the adjoint method computes the sensitivities of one such quantity at a time. See Chapter 5.

Rounding and truncation error The two errors in a finite difference. Truncation error comes from the Taylor series and shrinks with the

step h; rounding error comes from finite machine precision and grows as h shrinks. Their balance gives the U-curve and the optimal step. See Section 3.3.

Saltelli construction A sampling design that arranges model evaluations so that first-order and total Sobol indices can be estimated together with minimal cost. See Chapter 9.

Semisimple / defective eigenvalue A repeated eigenvalue is *semisimple* when its geometric and algebraic multiplicities agree and *defective* when the geometric multiplicity is smaller. The simple eigenvalue-sensitivity formula must be read with care in the semisimple case and fails outright in the defective case, where eigenvalues respond to a perturbation like a fractional power rather than linearly. See Appendix C.

Sensitivity index (SI) The central object of the book: the normalized (relative) derivative $\mathcal{S}_p^q = (p/q)\,\partial q/\partial p$, the fractional change in a response q per fractional change in a parameter p. A sensitivity index describes a *model*, not the system the model represents; it inherits every assumption built into that model. See Definition 2.1.

Sensitivity Jacobian The matrix of sensitivity indices of every QOI with respect to every POI, one row per QOI and one column per POI. See Definition 3.1.

Separatrix A boundary in state space that trajectories cannot cross, separating regions of qualitatively different behavior. An index computed on one side describes behavior that does not hold on the other. See Section 8.4.

Shapley effects A global sensitivity measure that allocates the output variance among parameters by averaging each parameter's marginal contribution over all orderings, giving a fair share even under correlation. See Chapter 9.

Sigma-normalized sensitivity index The derivative scaled by the input and output standard deviations, $S_i^\sigma = (\sigma_i/\sigma_Y)\,\partial Y/\partial Q_i$, so that it weighs both how strongly the output responds and how uncertain the input is. It bridges local and global SA. See Section 9.1.

Sobol indices Variance-based global sensitivity measures. The first-order index is the fraction of output variance explained by a parameter alone; the total-effect index adds every interaction that parameter participates in. See Chapter 9.

Standardized (rank) regression coefficient A sensitivity measure from a linear (or rank-linear) fit of output to inputs, valid when that fit captures most of the output variance. See Chapter 9.

Stiff ODE An ODE with a component that decays, at least in some region, on a time scale far shorter than the one of interest. An explicit solver must then take very small steps to stay stable, even where the solution varies slowly; an implicit (stiff) solver can take steps sized to the solution. It forms F_u as it steps, and the forward sensitivity equations can reuse that work. See Chapter 4 and Section 7.4.

Structural identifiability Whether a model's parameters can, even in principle, be determined from perfect output data; parameters that appear only in a fixed combination are not separately identifiable. See Chapter 3.

Tape The record kept by the forward pass of reverse-mode differentiation: it holds each operation and its intermediate value, which the backward pass reads back to accumulate derivatives. See Section 7.1.

Tornado plot A horizontal bar chart of sensitivity indices, one bar per POI, sorted by magnitude with the largest at the top; positive indices extend to the right and negative to the left. See Section 2.7.

Total-derivative sensitivity index The index of a QOI with respect to one POI when correlated POIs move with it: the standard index plus the indices of the others weighted by the coupling coefficients ω_{ij}. See Equation (3.12).

Unifying ideas The four ideas the book returns to: the sensitivity index is a normalized derivative; the adjoint is the transpose; local SA is a linearization; a sensitivity index describes a model, not the world. Callbacks in the text cite them by number. See Section 1.5.

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